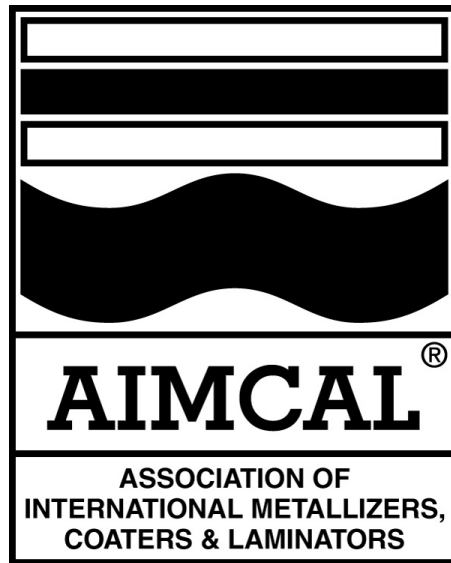




Metallizing Technical Reference

Fifth Edition

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Edited by

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and
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TABLE OF CONTENTS

Section 1: Test Methods and Procedures

- 1.1 AIMCAL Test Procedure TP-101-78 for Determining Optical Density
- 1.2 AIMCAL Test Procedure TP-102-78 for Determining Surface Resistance
- 1.3 AIMCAL Standard Methods TP-103-82 for Determining and Reporting Solar Control Film Parameters
- 1.4 AIMCAL Test Procedure TP-104-87 for Qualitatively Determining Metal Adhesion
- 1.5 AIMCAL Test Procedure TP-105-92 for Qualitatively Determining Metal Adhesion of Metallized Films
- 1.6 AIMCAL Typical Test Conditions for Determining the Barrier Properties of Metallized Films
- 1.7 Index of ASTM Test Methods Applicable to Metallized Films and the Plain Substrates
- 1.8 Eddy Current Monitoring
- 1.9 Roll Hardness Testing
- 1.10 Roll Density Measurement
- 1.11 Section 1 References Cited

Section 2: Electrical, Optical, and Metal Thickness Relationships

- 2.1 Metal Thickness vs. Surface Resistance
- 2.2 Relationship of Aluminum Thickness to Optical Density
- 2.3 Relationship of Resistivity to Optical Density and Oxygen and Water Vapor Barrier and Optical Density for Aluminum Metallized Polyester Film
- 2.4 Light Transmission vs. Optical Density
- 2.5 Electrical, Optical, and Barrier Properties of Various Metallized Oriented Polypropylene Films
- 2.6 Emmissivity of Polyester and Kapton H and HN Films
- 2.7 Surface Reflectance of a Vacuum-deposited Aluminum Layer
- 2.8 Surface Reflectance of a Vacuum-deposited Silver Layer
- 2.9 Surface Reflectance of a Vacuum-deposited Gold and Copper Layer
- 2.10 Section 2 References Cited

Section 3: Barrier Properties of Metallized Films

- 3.1 Plot of Water Vapor Barrier vs. Oxygen Barrier for Various Barrier Films
- 3.2 Relationship of Oxygen and Water Vapor Transmission Rate to Optical Density for Metallized 48-gauge Polyester
- 3.3 Relationship of Oxygen Transmission Rate to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate
- 3.4 Relationship of Water Vapor Transmission Rate to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate
- 3.5 Oxygen Barrier vs. Optical Density for Metallized OPP, MET-HB
- 3.6 $\log_{10}$ (Oxygen Barrier) vs. Optical Density for Metallized OPP, MET-HB
- 3.7 Average Water Vapor Transmission vs. Oxygen Transmission for Metallized Production, MET-HB
- 3.8 Barrier Properties for Metallized Films- 1
- 3.9 Barrier Properties for Metallized Films- 2
- 3.10 Oxygen Permeability of Various Films
- 3.11 Water Vapor Transmission of Various Films
- 3.12 Effect of Creasing on Oxygen Transmission of Various Metallized Films
- 3.13 Effect of Creasing on Water Vapor Transmission of Various Metallized Films
- 3.14 Comparison of Barrier, Electrical, and Optical Properties of Various Metallized Films
- 3.15 Section 3 References Cited

Addendum – Paper: Basic Principles of Thin Film Barrier Coatings.

Section 4: Useful Relationships for Metallized Films

- 4.1 Typical Area Yield for Various Films of Various Thickness
- 4.2 Typical Area Yield for Polyester Film of Various Thickness
- 4.3 Formula for Calculating Length or Outer Diameter of a Roll of Polyester Film
- 4.4 Relationship of Roll Length to Roll Diameter for Various Thickness Polyester Films Wound on a 6-inch Core
- 4.5 Relationship of Roll Length to Roll Diameter for 48-gauge Polyester Films Wound on a 10-inch Core
- 4.6 Formula for Calculating Roll Weight of Polyester, Nylon, and Propylene Film
- 4.7 Formula for Calculating MSI (1,000 Square Inches) Cost of Polyester, Nylon, and Propylene Film
- 4.8 Factors for Converting Various Basis Weights of Metallized Paper
- 4.9 Formulas for Calculating Wound Roll Density Characteristics
- 4.10 Density for Various Polymer Films
- 4.11 Conversion Table for Commonly Used English/Metric Units of Measure
- 4.12 Section 4 References Cited

Section 5: Metal Evaporation and Heated Boat Data

- 5.1 Some Factors Influencing Boat Life
- 5.2 Recommendations for the Use of Evaporation Boats
- 5.3 Boat Temperature vs. Approximate Boat Life
- 5.4 Boat Resistivity (HT) and Approximate Service Life at Different Temperatures for Aluminum Evaporation
- 5.5 Aluminum Wire Feed Rates
- 5.6 Evaporation Rate vs. Temperature for Aluminum Metal
- 5.7 Calculation of Evaporation Rates
- 5.8 Evaporation rate G , $\text{gm/cm}^2 \text{ sec}$, Constants for Various Metals
- 5.9 Vapor Pressure Constants of Nesmeyanov for Various Elements Where the Condensed Phase of the Element Is a Solid
- 5.10 Vapor Pressure Constants of Nesmeyanov for Various Elements Where the Condensed Phase of the Element Is a Liquid
- 5.11 Section 5 References Cited

Section 6: Aluminum Wire Chemistry, Temper, and Oxidation

- 6.1 Aluminum Wire Alloys
- 6.2 Chemical and Physical Properties of Aluminum
- 6.3 Oxidation of Aluminum
- 6.4 Wire Oxidation, Spitting, and Pinholes
- 6.5 Wire Feeding
- 6.6 Oxidation Hazard
- 6.7 Section 6 References Cited

Section 7: Surface Treatment of Webs for Aluminum Vapor Deposition

- 7.1 Discussion of Surface Treatment Technologies
- 7.2 Table of Dyne Level Testing Solution Compositions
- 7.3 Table of High Surface Energy Test Mixtures
- 7.4 Table of Surface Free Energy of Various Liquids at 20 C
- 7.5 Table of Components of Surface Energies for Various Polymers
- 7.6 Table of Inherent Surface Energy Levels of Polymer Substrates
- 7.7 Table of Critical Surface Tensions of Polymers
- 7.8 Schematic of Corona Treatment
- 7.9 Schematic of Flame Treatment System
- 7.10 Schematic of Atmospheric Plasma System
- 7.11 Schematic of Magnetron Plasma Treater
- 7.12 Schematic of Hollow-Cathode Plasma Treater
- 7.13 Figure of Flame-treated Polypropylene Film Surface Oxygen Level as Function of Treatment Level for One-pass Treatment
- 7.14 Section 7 References Cited

Section 8: Substrate Materials for Metallization

- 8.1 Polymer-based Substrates
- 8.2 Paper and Other Substrates
- 8.3 Section 8 References Cited

Section 9: Substrate Manufacturing Technologies

- 9.1 Single-screw Extruder
- 9.2 Cast Films
- 9.3 Solution Casting
- 9.4 Melt Casting
- 9.5 Blown Films
- 9.6 Coextrusion
- 9.7 Section 9 References Cited

Section 10: Contamination, Web Cleaning, Planarization, and Heat Stabilization

- 10.1 Contamination
- 10.2 Web Cleaning
- 10.3 Planarization
- 10.4 Polymer Substrate Heat Stabilization
- 10.5 Section 10 References Cited

Section 11: Winding in Vacuum

- 11.1 Vacuum Winding Systems
- 11.2 Tension Measurements
- 11.3 Rolls and Roll Alignment
- 11.4 Heat-load Considerations
- 11.5 Tramlines (Railroad Tracks)
- 11.6 Polymer Substrates and Winding
- 11.7 Formulating for Winding
- 11.8 Section 11 References Cited

Section 12: Converting Metallized Films

- 12.1 Extrusion Coating
- 12.2 Adhesive Lamination
- 12.3 Extrusion Lamination
- 12.4 Printing
- 12.5 Clear Window Formation
- 12.6 Thermoforming Metallized Films
- 12.7 Section 12 References Cited

Section 13: Pattern Metallization

- 13.1 Mask-based Methods
- 13.2 Printing-based Methods
- 13.3 In-Register, In-vacuum Pattern Metallization
- 13.4 Section 13 References Cited

Section 14: Non-aluminum Deposition Materials and Techniques

- 14.1 Resistance-heated Sources
- 14.2 Slot Sources and Jet Vapor Sources
- 14.3 Electron-beam Deposition
- 14.4 Magnetron Sputtering
- 14.5 Chemical Vapor Deposition
- 14.6 Comparison of Sources
- 14.7 Atomic Layer Deposition
- 14.8 Atmospheric Deposition Techniques
- 14.9 Stress in Films
- 14.10 Hazards
- 14.11 Section 14 References Cited

Section 15: Water in Vacuum

- 15.1 Water in Vacuum
- 15.2 Substrate Outgassing
- 15.3 Condensation of Water onto Cryosurfaces
- 15.4 Paper Substrate
- 15.4 Section 15 References Cited

Section 16: Troubleshooting Vacuum Metallizing

- 16.1 Basic Troubleshooting Methodology
- 16.2 Adhesion
- 16.3 Vacuum
- 16.4 Process
- 16.5 Winding
- 16.6 Measurements
- 16.7 Section 16 References Cited

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AIMCAL

The Association of International Metallizers, Coaters and Laminators (AIMCAL) is a global nonprofit trade association representing manufacturers and converters of coated and laminated flexible substrates. Founded in 1970, membership includes manufacturers, suppliers, trade press and universities.

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Section 1: Test Methods and Procedures

1.1 AIMCAL Test Procedure for Determining Optical Density

TP-101-78

1) INTRODUCTION

The amount of metal deposited on the surface of a film will govern the oxygen (O₂), water vapor transmission (WVTR), and light barrier properties of that film. Hence a means to measure this coating is needed so that one can determine the amount of metal that will meet the particular requirement. A very rapid determination can be made with excellent reproducibility on a uniform sheet. USA Standard Number PH2-19-1959^a provides the basis for the concept of diffuse optical density.

2) SCOPE

This procedure gives a means for determining the amount of metal as a function of the diffuse optical density.

3) APPARATUS

Densitometer: Several types of testing apparatus are available for use in this procedure. See Item 6 below for characteristics.

4) PROCEDURE

4.1 Refer to the operator instruction manual of any given instrument for calibration.

4.2 Mark the sample at 10 evenly spaced intervals across the width of the material with the metal side up.

4.3 Place the densitometer in the diffuse mode and set for visual light transmittance.

4.4 Test the sample with the metal side up. Special care should be given in handling the material to avoid scratches, pinholes, etc.

Note: This procedure may be used for virgin metallized film and also laminates providing the additional opacity of the second film is calculated into the reading.

5) CALCULATIONS

Average the measured values of each interval and record as the average diffuse optical density.

6) DENSITOMETERS

A densitometer is a device designed to measure the optical density of an exposed photographic film. Such an instrument can be utilized as a working part of the Processor Quality Assurance Program to determine the density of the individual steps that were obtained by using the densitometer.

There are several styles of densitometers available, with the greatest difference being the type of readout. The two types of readout are dial and meter with the latter being generally easier to use.

Note: Extreme caution should be exercised in the use of units that do not necessarily correlate to NIST measurements across the entire light spectrum. Equipment measuring light transmission using photo multiplier technology has severe limitations of tenacity due to saturation at the high intensity end and dark current noise at the low intensity level.

^aUSA Standard Number PH2-19-1959, Society of Motion Picture and Television Engineers

1) INTRODUCTION

This method is valuable when testing metallized films in which the amount of metal must be kept constant. It is recommended when measurement by the optical density method is not available. This method also saves testing time when compared to coating weight tests. In summary, a flat, non-creased strip of metallized film is placed on the electrode unit, either by holding it taut over the electrodes or placing it on a flat surface with the electrodes on the sample. The ohms-per-square reading is made directly from an ohmmeter. The thickness is related to ohms per square.

2) SCOPE

A method and test apparatus are described for measuring the relative ohms per square of a vacuum metallized film.

3) APPARATUS

3.1 Sensitive ohmmeter of the multi-scale type with leads ending with alligator clips.

3.2 A unit having electrodes precisely 10in apart mounted on a 2in-wide suitably insulated rigid sheet (*see Figure 1.2-1*).

3.3 A cutting apparatus can be used to precisely cut a sample 1in-wide by 12in long. (The strip also may be cut by using two razor blades spaced 1in apart in a fixed width block.)

3.4 One sheet of semi-hard rubber (50 durometer hardness).

4) PROCEDURE

4.1 Samples from a roll should be taken across the width of the metallized film for testing and should be at least 12in long. The sample must be free of dirt, grease, dust or fingerprints. The metallized surface also must be free of abrasions, scars or blemishes. The strips should be cut 1in-wide by 12in long in the machine direction at 10% width intervals.

4.2 The ohmmeter leads should be connected to the twin electrodes. Short the circuit by placing a piece of copper wire between the electrodes. Turn on the ohmmeter and adjust it to zero on the 0-100 ohm scale. Remove the jumper. If the needle goes to the other side of the scale, the unit is ready for testing.

4.3 Either take the strip and press it with the thumbs onto the electrodes OR place the strip on the rubber backing and press the terminals firmly on the strip.

4.4 Read the resistance on the ohmmeter scale as ohms per square.

5) CALCULATIONS

5.1 Average the readings of the measurements on each sample.

5.2 Divide the ohm reading by 10 in^2 to obtain ohms/square

Figure 1.2-1 Electrode Unit for Measuring Resistance

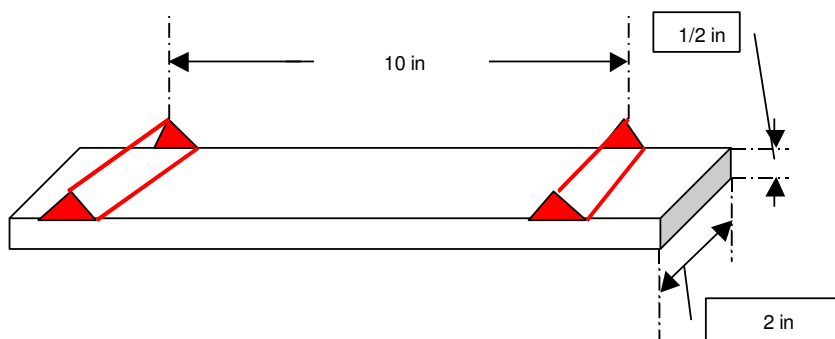


Figure 1.2-2 Homemade Example of TP-102-78 Electrode, Mat Has 1-in Markings

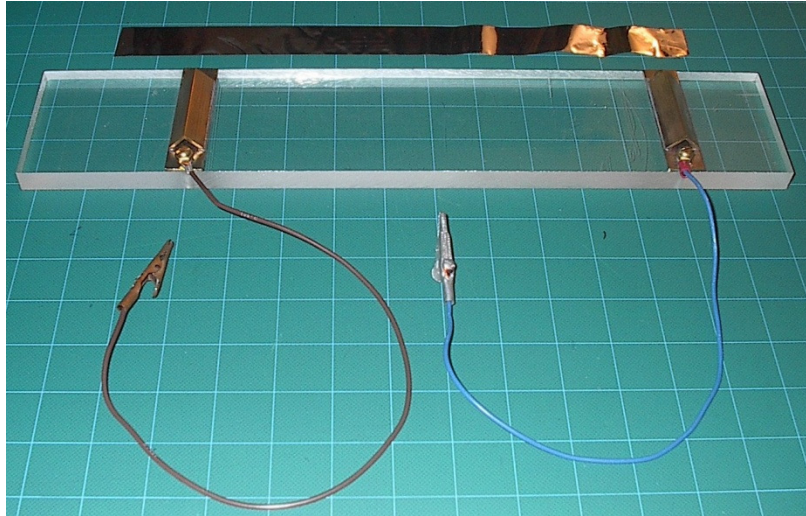
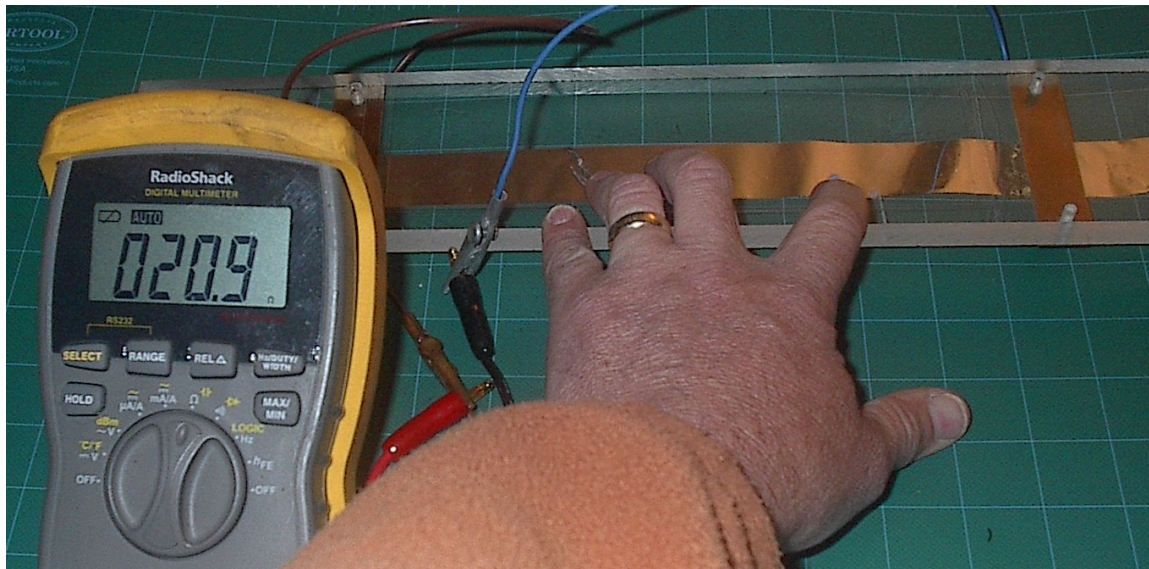


Figure 1.2-3 Measuring Per Instructions in TP-102-78 Procedure 4.3, Divide Reading by 10 in^2 to Obtain Ohms/Square. (Employ care to minimize/eliminate wrinkles in sample.)



1.3 AIMCAL Standard Methods for Determining and Reporting Solar Control Film Parameters

TP-103-82

The Association of International Metallizers, Coaters and Laminators (AIMCAL) suggests the following practices for determining and reporting solar control film performance criteria:

- 1) **WINTER DESIGN CRITERIA** (peak load conditions, night) shall be calculated at 70F (21.1C) indoors, 0F (-17.8C) outdoors, 15-mile-per-hour (MPH) wind, and indoor dew point below the glass temperature. Calculations shall be done in accordance with the Window 4.1 Window Thermal Performance software published by the Windows and Daylighting Group of Lawrence Berkeley Laboratories.
- 2) **SUMMER DESIGN CRITERIA** shall be calculated using 75F (23.9C) indoors, 89F (31.7C) outdoors, 7.5 MPH wind.
- 3) **SHADING COEFFICIENTS** shall be calculated in accordance with ASHRAE, Chapter 26, 1977, using Item 2 conditions above. For simplicity and since calculations have shown no significant difference in results, shading coefficients shall be calculated using summer nighttime U value.
- 4) Other values can be used to calculate energy savings at specific conditions as long as those conditions are stated.
- 5) If reported, ultraviolet (UV) transmittance will be determined in accordance with ASTM E424-71 method A, with the following modifications:
 - 5.1 Solar UV will be defined as 300-380nm (agrees with ASTM solar group).
 - 5.2 Solar UV transmittance will be measured on double-strength clear glass or, if measured on film alone, weighted to adapt the results to include glass. The measurements will be made as described in ASTM method E424 or the ASTM techniques manual, and the results will be weighted using the following solar spectrum weighting factors:

WEIGHTING FACTORS

Wavelength	Measured on Glass	Measured on Film Alone*
380nm	.142	.12
370	.235	.205
360	.197	.17
350	.159	.13
340	.127	.10
330	.085	.05
320	.046	.01
310	.009	0
300	.000	0
	100%	78.5%

*Corrects to double-strength clear glass

- 6) Total solar reflectance and transmittance will be determined in accordance with ASTM E424, by spectrophotometer, or pyranometer (sample and instrument at 20° angle).

1.4 AIMCAL Procedure for Qualitatively Determining Metal Adhesion TP-104-87

1) INTRODUCTION

This test method quickly and easily identifies whether a vacuum-deposited thin metal layer has any substantial level of adhesion to a flexible substrate. It is intended to measure adhesion qualitatively, providing a check that the metallized substrate is acceptable or alternatively identifying a potential problem. Few precautions and minimal test equipment are required. If a more detailed qualitative measure of adhesion is required, then the “AIMCAL Procedure for Qualitatively Determining the Metal Adhesion of Metallized Films” should be used (*see Section 1.5*).

2) SCOPE

A simple test method is described for measuring whether a metallized layer on a flexible substrate has any substantial adhesion to that substrate.

3) APPARATUS

Tape dispenser containing a reel of 1-in-wide Scotch™ 610 adhesive tape.

4) PROCEDURE

4.1 Place a sample of the metallized substrate on a smooth, flat surface with the metallized layer facing up.

4.2 Apply approximately 6in of Scotch™ 610 adhesive tape to the surface and smooth with thumb and forefinger to ensure good contact. To facilitate separation, one end of the tape should be doubled over.

4.3 Hold the sample down with one hand. With the other hand, peel the tape back at approximately 180° in one smooth movement (*see Figure 1.4-1*).

5) INTERPRETATION

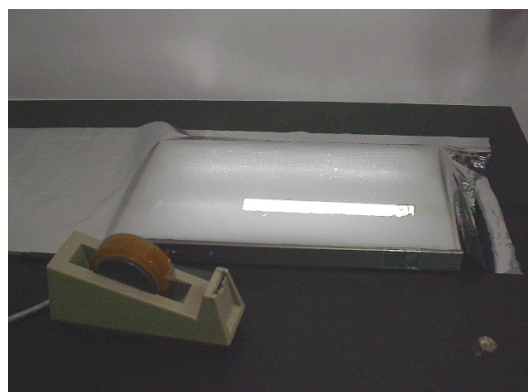
5.1 Examine the test substrate for metal removal, and the used tape for metal transfer, using a strong backlight (*see Figure 1.4-2*).

5.2 The significance of the visual assessment depends upon the application for which the metallized substrate is intended. If any metal is removed then either (a) visual comparisons must be made with specified visual standards or (b) the more quantitative AIMCAL procedure must be performed and the data obtained compared with numeric and/or visual standards (*see Section 1.5*).

Figure 1.4-1



Figure 1.4-2



1.5 AIMCAL Procedure for Qualitatively Determining Metal Adhesion of Metallized Films*

TP-105-92

1) SCOPE

This test is a means for comparing metal adhesion properties of metallized polyester films.

2) APPARATUS

- 2.1 A tensile tester capable of running at a crosshead speed of 12in per minute.
- 2.2 Precision cutters, templates, or other means to cut samples accurately.
- 2.3 A heat sealer (*e.g.*, Sentinel) with controllable temperature, pressure, and dwell time and a 1in-wide jaw.
- 2.4 Silicone release-coated paper or wax paper, 50-lb basis weight. Release coating on both sides.
- 2.5 Dow Chemical DAF 899**, 1-mil film of ethylene acrylic acid (EAA) resin.
- 2.6 A densitometer to measure the optical density (OD) of the metallized film.
- 2.7 A controlled environment; 50% RH, 72F (22.2C).
- 2.8 Scotch™ 610 adhesive tape, 2in wide, to support the metallized film side of each test replicate.

* A kit containing the adhesive tape, silicone release paper and DAF 899 film is now available from AIMCAL. Go to www.aimcal.org and look under publications.

** Available from Dielectric Polymers, Inc.; 218 Race St., Holyoke, MA 01040; phone +1 413-532-3288

3) PROCEDURE

- 3.1 Condition the metallized film samples at 50% RH, 72F (22.2C) before testing. Test replicates also must be prepared and tested in this environment.
- 3.2 Measure and record the optical density range of the test films.
- 3.3 Cut a sample of metallized film approximately 12in (machine direction) by 4in (transverse direction). Place the sample on a smooth working surface with the metallized side down and secure to the work surface with tape in such a manner that no wrinkles are present in the film.
- 3.4 Attach a 2-in-wide strip of Scotch™ 610 adhesive tape to the nonmetal side of the sample. This is done to prevent tearing the polyester film during the peeling procedure. Locate the tape at the approximate center of the film sample and apply for the full length of the sample. A hand roller is useful in providing a smooth, wrinkle-free adhesion. The tape is applied in the machine direction (MD) of the test sample (*Figure 1.5-1*).
- 3.5 Remove the tape/metallized film sample from the work surface.
- 3.6 Cut the tape/film sample into approximately 2.5in lengths (*Figure 1.5-2*).
- 3.7 Invert the tape/film sample and place an approximately 3.5in-long-by-3in-wide piece of DAF 899 film on top (metallized surface of film sample). The treated surface of 1-mil DAF 899 (1-mil EAA film) should be in contact with metallized surface of tape/film sample (*Figure 1.5-3*).
- 3.8 Heat sealer conditions for the test are 220F (104.4C), 15 sec., 15 psi.
- 3.9 Place the tape/film/DAF 899 sandwich on silicone-release-coated paper and cover with a second piece of silicone paper. The tape/film/DAF 899 sample should be located at top (leading) edge of slip sheets. This allows the entire layered sample to be inserted into the jaws of the heat sealer. (*Figure 1.5-4 and Figure 1.5-5*).

- 3.10 When the sealed sample is removed from the sealer, the slip-sheets are removed and a 1in-deep-by-3in-wide seal exists between the DAF 899 film and the metallized surface of the test film. The 1in dimension of the seal area results from the shorter dimension of the seal bar.
- 3.11 A center strip of the test specimen is now cut through its full length at a 1in width on a precision cutter.
- 3.12 For metal adhesion determination, the sample is mounted in the jaws of a tensile tester in such a manner as to affect a 180° peel of the DAF 899 from the metallized surface of the film. A pencil may be used to hold the tape/film/DAF 899 tail down to ensure the 180° peel, if necessary (*see Figure 1.5-6*).
- 3.13 The tape/film tail of the sample is secured in the stationary jaw. The DAF 899 film is secured in the moving jaw and the rate of peel is set at 12in/min.
- 3.14 Record the peel force for each of the five replicates.
- 3.15 An estimate of the percent metal removed in the 1-by-1in seal area can be made for each of the replicates.

Figure 1.5-1

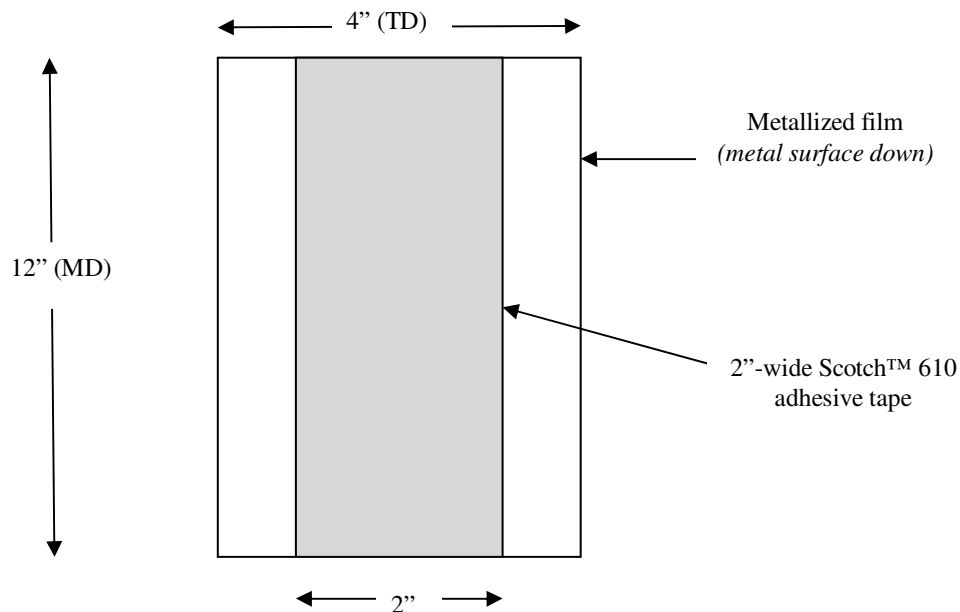


Figure 1.5-2

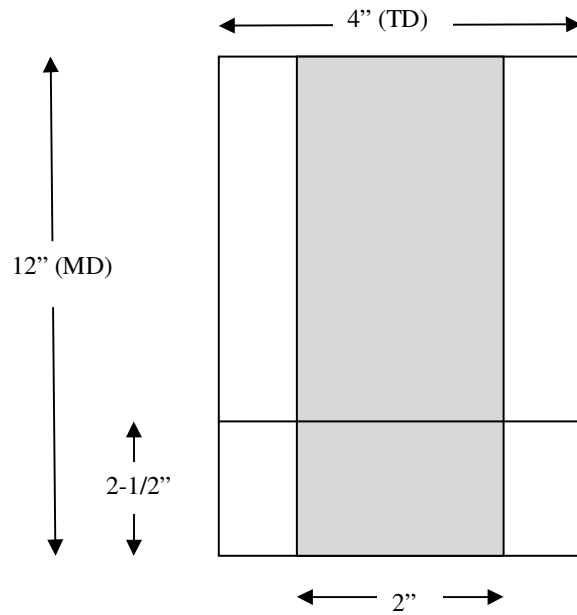
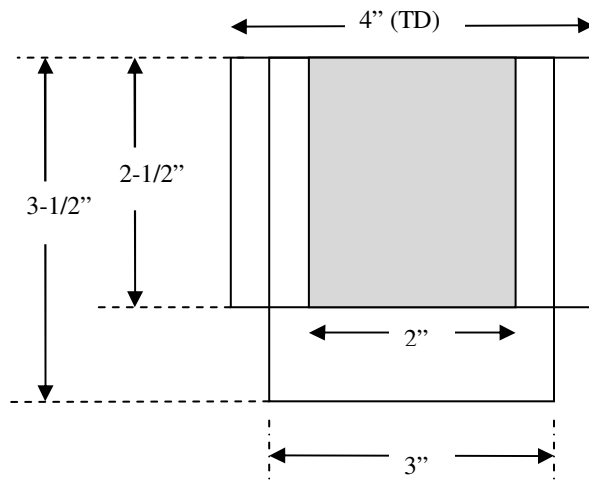


Figure 1.5-3



Inverted sample with metallized surface up and
overlay of DAF 899 film (3.5 x 3in)

Figure 1.5-4

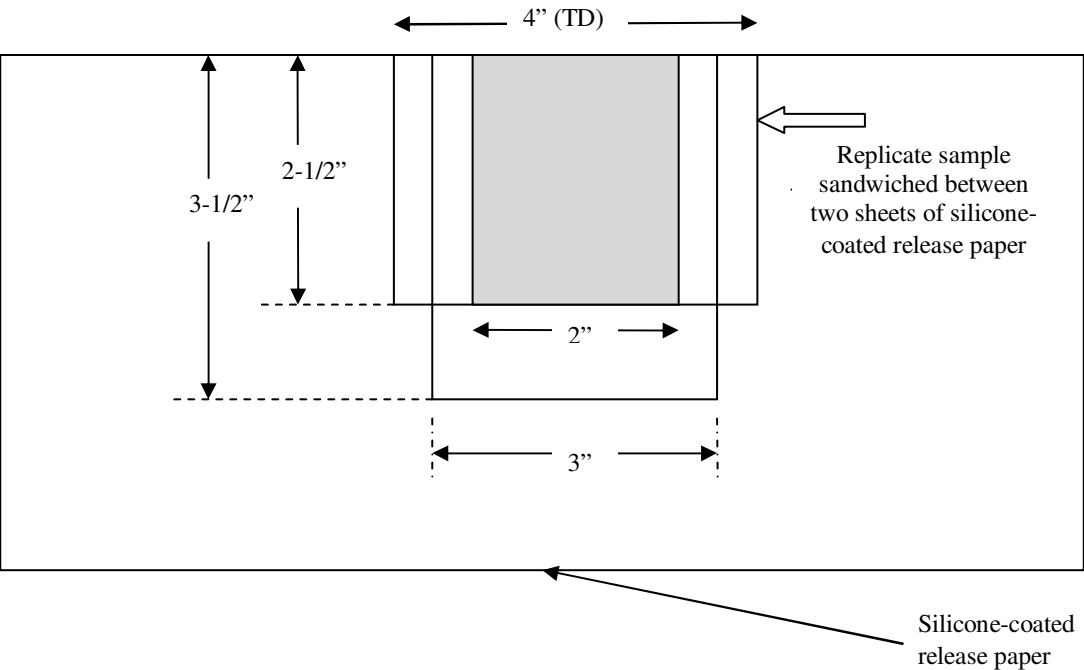


Figure 1.5-5

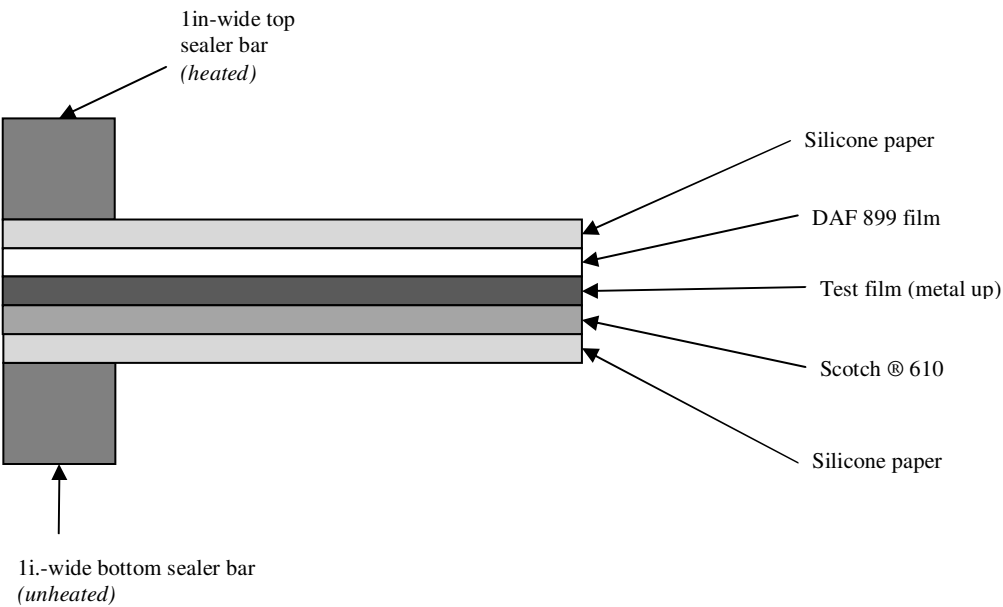


Figure 1.5-6



Purpose and Use of TP-105-92 for Adhesion Testing

AIMCAL developed and published two standard procedures for measuring the metal adhesion of vacuum-metallized films. At the time of the development, the test development was performed using metallized polyester substrates. The tests are TP-104-87 “Procedure for Qualitatively Determining Metal Adhesion” and TP-105-92 “Procedure for Qualitatively Determining the Metal Adhesion of Metallized Films” both described above. The first test is a go/no-go test to check metal adhesion relative to Scotch™ 610 adhesive tape, and the second prepares a test specimen by thermal lamination for measuring peel strength.

Since the tests were developed, the number of commercial metallizers has increased as well as the number of vacuum metallized substrates (*see Section 8.1*). There is a great deal of interest in measuring metal adhesion relative to various customer specifications, knowing that the metallized product is uniform lot-to-lot and that it is fit for use in the various applications for metallized films. This section will explore what the tests actually measure so that it is better understood how to use the test and how to compare the test results to results obtained in the field.

Discussion

The adhesion test TP-104-87 is a simple test to determine if the metal can be readily removed from a substrate using Scotch™ 610 adhesive tape. As such, it is easy to perform and it has come to represent a minimum level of metal adhesion on a substrate. It is a go/no-go test. The test tells you if the metal adheres to the substrate better than to the adhesive of the tape. The tape test can be an early warning test to indicate that something has changed in the metallization process. If the metal is easily removed, either the evaporation process has deteriorated, causing poor metal adhesion, or the substrate has changed, and it needs to be replaced. As such, TP-104-87 has come to be considered an “adhesion” test.

However, this test may not tell you how successfully the film will perform in a lamination or what the peel strength of a lamination will be, or the behavior of the metal layer during the peel test of the lamination. In actual use, many metallized films are incorporated into a lamination to create a composite structure usually with the metal layer buried in the lamination (*see Sections 8.5 and 8.6*). When the lamination is peeled apart to test the quality of the adhesive bond, it is comforting if the metal layer stays on the surface of the film it was placed on before lamination. If in the peeling of the lamination, the metal layer lifts off the substrate, it creates a suspicion that the metal layer may not have been formed properly. If the peel strength is high, or above a specified value, the product has passed the test. However, if the peel strength is low and 100% of the metal transfers to the adhesive then it is assumed that the metal layer is inferior. It is this situation which creates the need for an independent adhesion test to test the metal layer adhesion directly.

The difficulty in establishing an adhesion test is in the balance of sample preparation and testing time (expense) and the need to test frequently. In a true adhesion test, the sample will be tested with a system which pulls the metal off without distorting the substrate so the actual strength of the metal bond can be measured. This procedure requires careful sample preparation and special test fixtures, and it is slow. In comparison, peel tests can be performed

relatively quickly with minimal or straightforward sample preparation, or by simply cutting the finished converted product for direct testing. However, a peel test measures several forces, as well as the metal adhesion so the value obtained is very specific to the structure being tested.

Consequently, while it is possible to develop standard laboratory test structures for peel testing of metallized films, they may not give the same value when compared to a product lamination, due to differences in substrate combinations, adhesives, etc. Therefore, standard peel test structures can only be used for in-process quality control to insure the uniformity of a particular metallized product or the process used to produce it. The peel strength will then show differences if the metallized product and/or process changes because the rest of the test structure has been standardized. Variations in the peel strength will highlight changes in the metallized film coming from a particular manufacturing process.

Film “quality” or Fitness for Use will need to be determined by testing the product in actual applications. The laboratory peel test can then be used to distinguish acceptable film from unacceptable film within that application. For example, the peel strength test behavior needs to be correlated to various end-use structures. The actual numerical value obtained in the laboratory test and in the application may very well differ due to differences in the structures of the test sample and the actual lamination. However, there should be a correlation between the test performance and the final structure. Thus, the laboratory peel test method is a means of testing the adhesion performance of a specific component of the peel test specimen. This is why the sample preparation and components have been standardized. Variation in the components will be expected to change the value of the peel force obtained.

Therefore, comparison of peel force results between two different metallized films or two different metallization processes, can only be made if the peel test samples have been made using the same components and process settings. This means that the film-to-film or process-to-process comparisons are performed on the same laminator using identical components and process settings (sometimes difficult to control) or by the use of a standard test specimen as in the TP-105-92 description. In the absence of a laminator with which to make standardized test materials, the laboratory sample preparation permits the development of control charts for the metallization process and/or control of the film supply. This is how the laboratory test procedure TP-105-92 should be used.

The value of standardized test structures lies in the ability to directly compare metal adhesion quality between various sources of metallized films or between different manufacturing locations in the preparation of the laminations. Without a standardized test structure, it is not possible to directly compare results obtained with a specific metallized film and the various product structures of that film, or processes where it is used, without extensive correlations being developed. Consequently, the ability to have a standardized test for the routine evaluation of metal adhesion at various stages of product production has an inherent value to the industry.

1.6 AIMCAL Typical Test Conditions for Determining the Barrier Properties of Metallized Films

1) WATER VAPOR TRANSMISSION RATE

Measurements should be made in accordance with ASTM 1249, which defines a procedure using an infrared diffusometer. Test conditions are normally 100F (37.7C) and 90% RH, or 73.4F (23C) and 50% RH. An alternative procedure is ASTM E96 (sometimes called the “dish” or “cup” method).

2) OXYGEN PERMEABILITY

Measurements should be made in accordance with ASTM D3985, which defines a procedure using a coulombmetric detector. Test conditions are normally 73.4F (23C) at 0% and 50% RH, respectively. An alternative procedure is ASTM D1434, which describes “volumetric” and “manometric” test methods.

1.7 Index of ASTM Test Methods Applicable to Metallized Films and the Plain Substrates

<u>Property</u>	<u>Test Method</u>	<u>Plain</u>	<u>Metallized</u>
Clarity Transparency	ASTM D1746	X	*
Distinctness of Image	Special	X	X
Gloss (Gardner)	ASTM D2457	X	X
Haze	ASTM D1003	X	*
Transmittance	ASTM D1003	X	*
Solar Energy:	ASTM E424	X	X
Transmittance/Reflectance			
Optical Density	ASTM D1003	X	X
Surface Resistivity	ASTM D257		X
Surface Roughness	Profilometer	X	X
	Interference Microscopy		X
Metal Adhesion	Special		X
Coating Adhesion	ASTM D3359	X	X
Thickness (weight/area)	ASTM E252	X	X
Coefficient of Friction	ASTM D1894	X	X
Tensile Strength	ASTM D882	X	X
Tear, Initial, Graves	ASTM D1004	X	X
Running, Elmendorf	ASTM D1922	X	X
Mullen Burst	ASTM D774	X	X
Shrinkage - Oven	ASTM D1204	X	X
- Dynamic	TMA	X	X
Density	ASTM D1505	X	X
WVTR - Cup or Dish	ASTM E96-E	X	X
- Instrument	ASTM F1249	X	X
Oxygen Permeability	ASTM D3985	X	X
Surface Tension	ASTM D2578	X	X
Melting Point	ASTM D2117	X	X

*Suitable only for transparent samples

1.8 Eddy Current Monitoring

The use of an eddy current monitor has been traced to the invention of the telephone. Hughes used a couple of identical coils from one of the recently invented telephones and was able to distinguish between metals and different thickness samples of the same metal¹.

This monitoring technique uses the principle that a conducting coating can have an induced current if it is placed in a magnetic field. The construction of this type of monitor can vary, but essentially there is a coil that is used to generate a magnetic field that will produce the eddy current in the conducting coating. A second coil is then used to monitor the current that has been produced^{2,3}.

There are two ways of arranging the source and the detector. The first is to have the coils placed directly opposite each other with the web between the transmitting source and the receiving sensor coils. This arrangement is known as a transmission monitor. The second is to have the transmitting and receiving coils on the same side of the web, and this is referred to as a reflection monitor. The single-side reflection monitor can usually be arranged so the transmitting and receiving coils are wound concentrically in the monitoring head, making the unit quite compact.

The air gap between the metal coating and the source or sensor is a critical parameter and important to the signal strength. The induced current falls rapidly with increasing source distance. Similarly the sensor signal falls rapidly with an increasing air gap. The transmitter unit does not require the web to be any precise distance away from the source or sensor. It simply has to pass somewhere between them. If the web is close to the source coil, the air gap to the sensor will be increased so while the strength of the induced current is increased, there is a corresponding fall in the detected signal. Therefore, this transmitting arrangement is often regarded as being tolerant of the web position. However, it is still advantageous to have both the source and the sensor close to the web. However, if the web vibrates, it can catch the monitoring head and scratch the film.

The single-side sensor is much more sensitive to the distance between the web and the head. With this reflection arrangement, any change of distance will affect both the source and the sensor. To avoid this problem, it is usual to site the monitoring unit at a position where the web is passing around a roller. The roller precisely defines the web and the coating position and makes it easier to position the monitoring head close to the web, giving a much higher signal. In order to not confuse the signal by the conductivity of a metal roller, these rolls usually are made of a non-conducting material such as fiberglass. The source coil is powered by an RF signal, and any change in frequency and intensity monitored by the sensor is dependent upon the coating material. The thickness and the crystal structure of the coating are what define the conductivity.

Eddy current monitors cannot measure a wide range of coating conductivity so it is common for the customer to provide information on the desired conductivity monitoring range to be produced. The coils are designed specifically to suit the particular coating and thickness range required. Changing the material or thickness of the coating can reduce the accuracy of the measurement significantly. Thus, the same monitor may not be used to measure the thickness of a metal and a transparent conducting oxide. This is not to say that there will not be an output from the sensor, but it may be relatively insensitive to changes in coating conductivity or thickness beyond the specified operating conductivity range.

Eddy current monitors can be susceptible to changing thermal conditions, and this can cause the signal to vary for materials of the same conductivity. To eliminate this source of error, these monitors need to have a built-in temperature compensation system⁴. It is common for metallizers to increase in temperature by several degrees over the course of the metallizing process.

The significant parameters used for designing the coil are:

- δ = Thickness of layer
- σ = Conductivity of the layer
- μ = Magnetic permeability of the layer
- ω = Frequency of the oscillator
- h = Spacing between the head and the web

With the above parameters known, the variables are the number and diameter of turns in the coil⁵⁻⁸.

Each eddy current monitor will measure the average conductivity from an area under the monitoring head. The size of the head and the area it takes the signal from can be varied somewhat. Typically, one monitor is used per resistance-heated evaporation boat so that the performance of each boat can be controlled accurately. The high frequency used for the monitors is more than adequate to provide a control signal for the source boat, as these boats often have a response time of a few seconds. The usual arrangement is to have all the monitors mounted in a single bar fitted in close proximity to one of the idler rolls.

It is common for calibration graphs to relate conductivity to optical density or to the reflectance or transmittance. It is important to calibrate the monitors. This can be done against standard materials but is more commonly done using techniques such as an off-line, four-point probe or Van der Pauw measurement technique⁹⁻¹⁶. If there is more than one production machine, it is important to calibrate both machines and compare them so there is an increased confidence in measurements of thickness and how they relate to other properties such as optical performance. Once the calibration has been made for a particular machine and process deposition condition, it is often accurate enough to reach a desired resistance coating simply by monitoring the optical characteristics or to achieve an optical performance by measuring the resistivity.

This can be simple for single metals such as gold, silver, copper or aluminum. For alloys a matrix of calibration curves is required in order to have an understanding of the sensitivity of the conductivity to alloy changes and how tolerant the process window is.

Figure 1.8-1 A Schematic of a Same-side Eddy Current Monitor

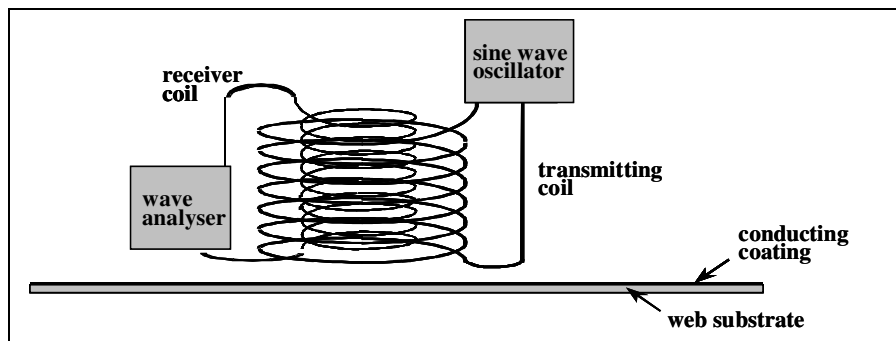
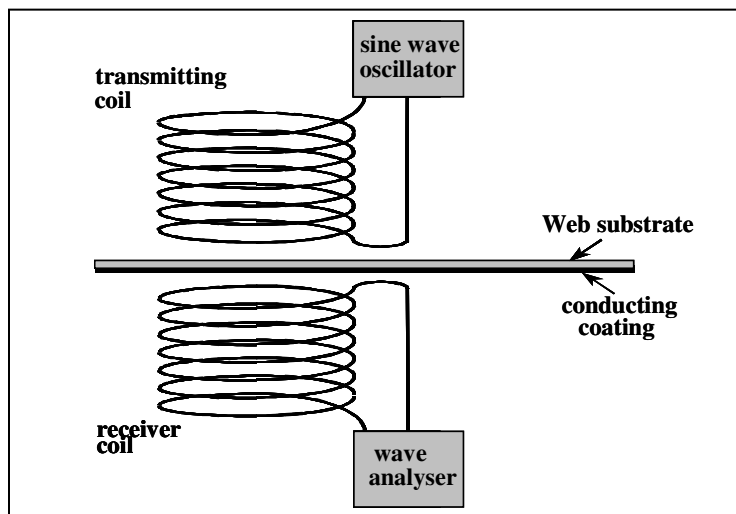


Figure 1.8-2 A Schematic of a Transmission Eddy Current Monitor



1.9 Roll Hardness Testing

The hardness of rolls can be quite important. If a roll arrives too loosely wound it can easily telescope during handling or during the initial vacuum pump-down process. If the rolls are wound hard they can highlight profile variations, so a softly wound roll may immediately raise suspicions of either poor handling or poor quality film. If there is any profile variation, such as an area of over-thickness, this will show up as a gauge band. If the thickness variation is large and the hardness too high, this can stretch the film along the gauge band and permanently deform the film.

No matter what state of hardness the film arrives in, by the time it has been rewound in the vacuum system it will be a hard roll, unless it has been formulated for winding in vacuum. This is because there is no air between the film layers as the film is rewound in the vacuum chamber. The hardness is, in effect, a measure of the quantity of air that is wound into the roll with the film, *i.e.*, the roll density. Formulating, for winding in vacuum, consists of adding particles to the surface of the film to maintain film-to-film spacing, and therefore the roll density, in the absence of air.

It is common to want to assess the roll hardness, and operators often can be seen knocking rolls with a knuckle to listen to the sound produced as a method of testing the roll hardness variations. In fact, this method has been improved upon by use of a stick or Billy club to do the knocking (*Figure 1.9-1*). These subjective methods can be variable from operator to operator and dependent on operator experience. However, these simple methods can clearly show the location of hard and soft regions in the roll and can provide important information to winding and film line operators.

Figure 1.9-1 Testing the Roll Hardness Using Billy Club Method (*tapping stick designed and made by operator*)



It is possible to purchase a simple test meter that standardizes the same process. There are several products on the market that can be calibrated to give a reproducible measure of roll hardness. One method uses a spring-loaded plunger, that for a given force, will press into the roll and a measure of the deflection will correlate to the hardness. Another test is to use a projectile, such as a ball bearing, that is dropped from a standard height and the height of the

bounce is measured. The harder the roll the higher the bounce. This same principle can have the projectile mechanically fired so that the test can be carried out at any angle and not just vertically downward.

In the book by Good & Roisum¹⁷ the following methods or instruments are described: the Billy Club and its variants; RhoMeter and RhoHammer (previously the Beloit RhoMeter); Backtender's Friend; Schmidt (Concrete) Hammer; Parotester; and TAPIO RQP¹⁸⁻²¹.

It is best to measure the roll several times across the whole width so that any variations are identified. Some of the above meters now can store many readings and transfer the data to computers to simplify recordkeeping.

A word of caution: Measuring the hardness, whether by the knuckle, Billy stick or automatic machines that bounce a projectile or press a probe against the surface can damage the film. The indentation can penetrate more than 25 layers into the roll. The harder the roll, the greater the depth the damage can penetrate. This damage may only be of consequence to some of the more demanding products, such as transparent conducting oxides or similar ceramic coatings as used in optical or electronic applications. Decorative metal coatings may be completely unaffected by this type of testing.

1.10 Roll Density Measurement

The density of a roll of film is a composite measurement of the relative amounts of air and film in the volume of the roll, representing a fundamental roll structure measurement²². It is one of the most useful measures of the combined impact of winding conditions, winder construction, and film formulation available and has been used to control roll quality²³. In all winding operations, the desired or optimum density of the roll must be determined by use of the film in subsequent processes as a means of maintaining the film without damage from shrinkage, compressive forces or roll structure failure (telescoping, corrugations). Roll density control can increase process productivity several percentage points due to more uniform unwinding conditions and uniformity in web handling. Optimum roll density values for a given film and process using the rolls must be determined experimentally.

Many discussions of roll formation only take into account winding conditions and machine construction and ignore the impact of film formulation. However, it is best to realize that film composition and formulation, especially surface formulation, will interact with the winding conditions to generate the roll density as a function of wound-in tension²². The need to form a firm base to prevent telescoping must be balanced with the need to not stretch the film to minimize compressive forces in a roll. Therefore, the winding tension and nip pressures generally do not remain uniform during the formation of a roll. Generally the tension is tapered, or programmed, from the start of a roll to the end. Controlling winding conditions will control the air thickness between film layers, which will generate the final roll density.

Film formulation can play an important role in controlling roll density as film formulation can be used to control film-to-film separation. This can be accomplished by the use of particulate materials, called antiblocking agents, in the film surface. This is especially important for metallizing operations because the film will be wound in the absence of air (*see Section 8.3*). Films formulated for winding in vacuum will also improve the winding in air as the film-to-film spacing can be designed to exceed that of the wound-in air layer. This uncouples, or greatly reduces, the dependence of the roll density on winding conditions.

In addition to the winding stresses imposed on the film, the film may not be dimensionally stable for several days after manufacture (caused by secondary crystallization effects in oriented polypropylene) or in the presence of environmental changes such as variations in humidity (polyester films). Also, film tensile properties for the same material will vary from different manufacturing processes (cast and oriented films) and between materials. Because of this, the proper amount of air to be wound into the roll varies between films of different polymers, surface formulations, and film manufacturing processes, as well as at different stages of a film's manufacture, converting, and storage history.

Roll density is defined as the weight of the film in the roll per unit volume. It is calculated by measuring the combined volume of air and film and the film weight of the roll. Roll film weight is measured directly by weighing and subtracting the weight of the core from the total. This net film weight measurement is commonly available for all rolls in a manufacturing process. Film volume is calculated by assuming the roll is a cylinder, measuring the roll width (W), and outside diameter (D_o), calculating the volume from the roll's outer dimensions, then measuring the core diameter (D_c) and subtracting the empty volume of the core from the volume of the roll's outer dimensions. This then yields the roll density as:

$$\text{Roll Density} = \frac{\text{Film Weight}}{\left(\frac{\pi W}{4}\right) * (D_o^2 - D_c^2)}$$

Measuring the roll's average outside diameter is done several ways, either as a calculation, at the winder, from the known thickness and length of film^{23, 22} or by direct measurement of the circumference or diameter at the roll end, or by distance from a fixed point, using an ultrasonic distance measurement²².

In general, the direct circumference measurements using a flat tape may be preferred as it can be applied to all rolls in all situations where automatic systems do not exist, and relative to calculated diameters, it is a direct measurement devoid of assumptions. Flat steel tapes, which read in diameter directly (PITAPE®), are designed specifically for this purpose and commercially available²⁴. However, standard steel measuring tapes can be used to measure the circumference.

1.11 Section 1 References Cited

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Section 2: Electrical, Optical, and Metal Thickness Relationships

Introduction

It is always useful to be able to compare any measurements taken from metallized films. Within this section there are graphs of various parameters plotted against each other.

What it is important to understand is that this data is only a snapshot of what was produced on one machine in a particular climate on one day and should be used as an example only. It is not necessarily the best or the worst that can be produced.

Some machines will be dirtier or working in a higher humidity environment and with less stable power supplies or with different substrates or with a different pre-treatment. Any of these factors will certainly result in differences in coating structure and quality. Having said this, these graphs provide a useful guide to what can be achieved.

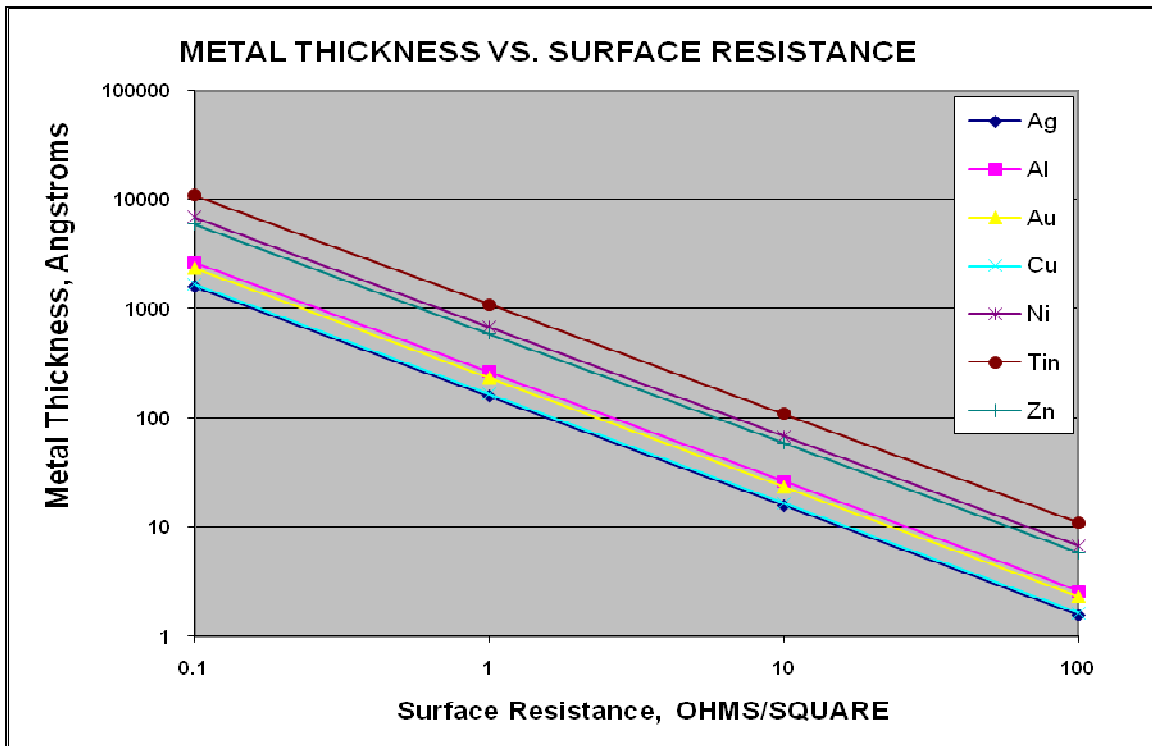
The optical density and surface resistance may be measured as described in the AIMCAL Test Procedures TP-101-78 and TP-102-78, respectively (*see Sections 1.1 and 1.2*).

Optical density is a measure of the percent light transmission of the film (*Figure 2.4-1*) and serves as a key measure of the light barrier of the product. However, as discussed in the Introduction of Section 3 for barrier properties, the optical density obtained for a given aluminum deposition is determined by the surface free energy of the polymer film surface. From this it would be expected that the aluminum thickness vs. optical density curve would be different for PET and oriented polypropylene or any other polymer film used for metallization. In addition, the optical density vs. resistivity curves should be determined experimentally for each film metallized if one measurement is to be used in place of another. Further correlations of resistivity or optical density with barrier properties and actual aluminum thickness should also be measured for each new base sheet to obtain the most accurate results.

The results presented here represent one set of values obtained for the films and conditions listed and should only serve as a guide to the properties obtained from a particular set of films and metallizing chamber conditions.

2.1 Metal Thickness vs. Surface Resistance

Figure 2.1-1 Metal Thickness vs. Surface Resistance (Courtesy of Advanced Ceramics Corp.)



The lines are determined using resistivity values from Pollock¹ from Equation 2.1-2².

Equation 2.1-2

Thickness = Resistivity x 100 / (Surface Resistance)**

The values of the bulk resistivities in micro ohm cm (moc) used in the calculation of Figure 2.1-1 are given below:

Ag	Al	Gold	Cu	Ni	Tin	Zn
1.59	2.65	2.35	1.67	6.84	11	5.91

The surface resistance is measured in ohm/square.

**According to Powell and Rossnagel², the surface resistance R_s of a thin film of thickness, t , should be referred to as “sheet resistance” and not, as is common practice, as the sheet resistivity or sheet ρ .

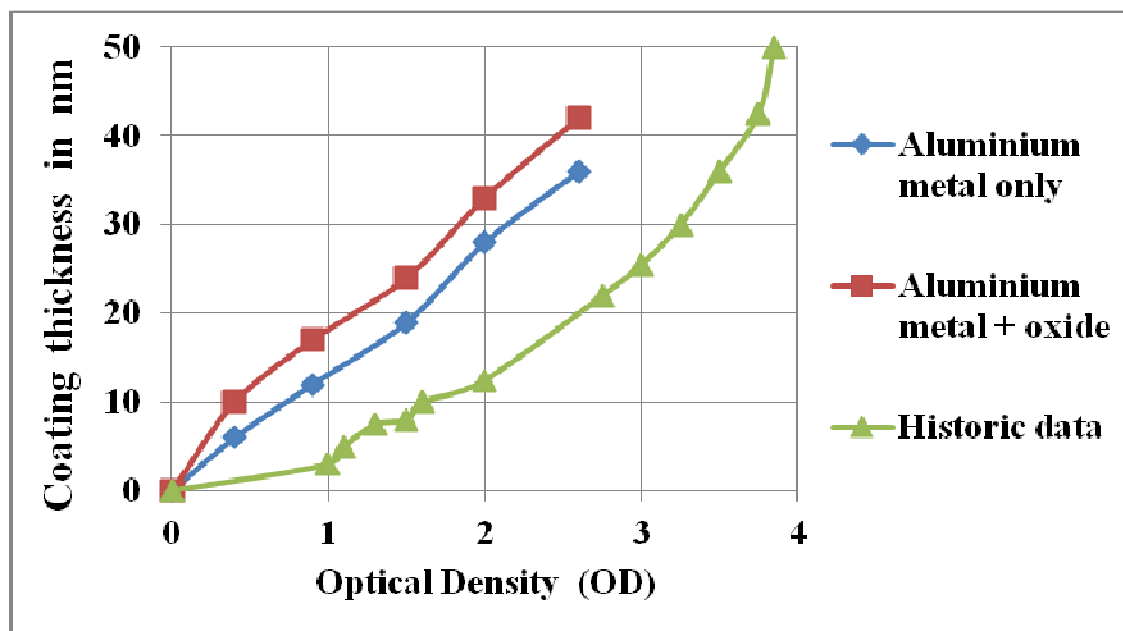
Additional resistivity data³ for ideally pure elements are listed as a function of temperature.

Electrical Resistivity of Ideally Pure Elements ($\mu\Omega$ cm) Micro-ohm/cm (adapted from Donnelly³)

T (K)	Ag	Al	Au	Cu	Ni
295	1.61	2.74	2.2	1.70	7.0
273	1.47	2.50	2.01	1.55	6.2
250	1.34	2.24	1.83	1.40	5.4

2.2 Relationship of Aluminum Thickness to Optical Density

Figure 2.2-1 Relationship of Aluminum Thickness to Optical Density (Polyester Film)

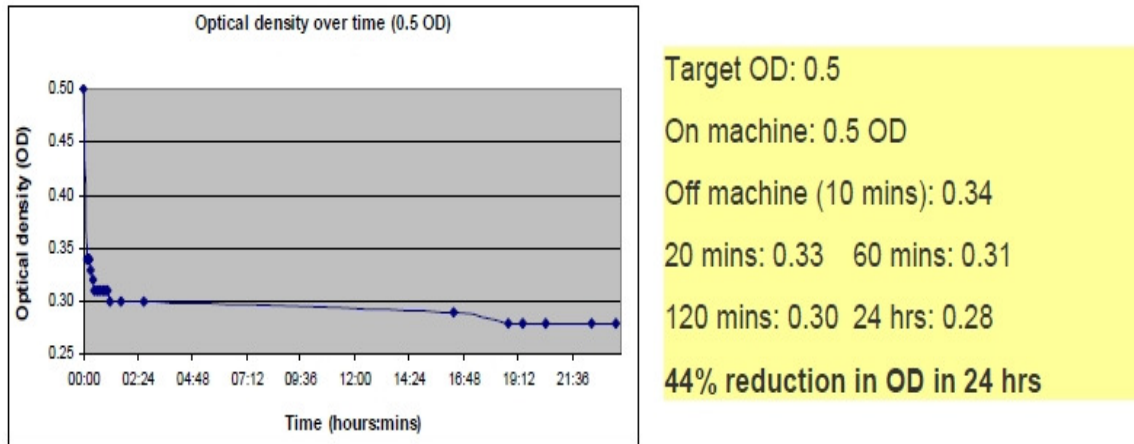


There was a recent piece of work done to revisit the relationship between the coating thickness and optical density⁴⁻⁸. It had been noticed that many coatings varied significantly from the graph⁹ published in earlier editions of this reference book.

It is known that the aluminum coatings oxidise (*see also Section 6.3 Oxidation of Aluminum*) and the properties such as resistivity and optical density changed with time. As such, the point in time when the coatings are measured adds an error to the measurement, and it was thought that a revised graph using materials produced on a more modern vacuum metallizing system and with a more detailed understanding of the metal thickness would be helpful.

Aluminum coatings were deposited in a standard metallizer with the in-vacuum optical monitor being used to control the optical performance of the coatings. As shown in Figure 2.3-1, a series of coatings was produced onto a corona-treated 12-micron PET film from Optical Density (OD) 0.5 through to OD 2.5 in 0.5 OD steps. The OD of the coatings was measured again once outside the vacuum system and further OD measurements were taken after one, five and 221 days. Typically the largest amount of oxidation takes place during the first 24 hours. Figure 2.2-2 shows the oxidation for a coating of initial OD 0.5 for the first 24 hours.

Figure 2.2-2 Aging of OD 0.5 Aluminum Coating Over Time Due to Oxidation⁴⁻⁸.



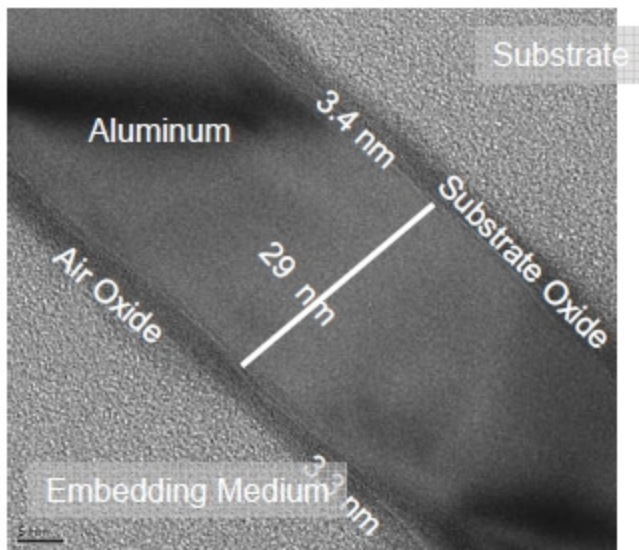
There can be a difference between different measurement systems for in-vacuum OD measurement and that used for the off-line OD measurement, but once the material is outside the vacuum, the repeat measurements are all done on the same measurement equipment. As expected the greatest change was seen after venting the system, with the next greatest change taking place during the following day.

The difference between the on-line measurement and the first off-line measurement is ~30% loss in OD. The oxidation and loss of OD continues, and there is a 44% loss in OD during the first 24 hours. Across an extended period, there would be a further small increase in the oxide thickness. So, it could be expected that there would be a 50% loss in OD for a roll of material stored for a long time. The loss of OD as a percentage is less for thicker coatings with a coating of OD 2.0 being reduced by ~20% in 24 hours.

This corresponds to the aluminum being oxidized with the oxide coating providing protection against further oxidation. A key part of this work was to verify the oxide thickness grown on the aluminum. Scanning transmission electron microscope (STEM) images were obtained, and then using X-ray fluorescence images to confirm the chemical differences, it was confirmed that there was an aluminum oxide layer on both sides of the aluminum metal coating. An STEM image of an oxidized aluminum coating is shown in Figure 2-2.3.

Although it was expected that there would be some aluminum oxide at the polymer/metal interface it was not expected that it would be of the same thickness as on the metal/air side of the coating.

Figure 2.2-3 STEM Image of a Nominal OD 2.0 Aluminum Coating on 12-micron PET Film⁵



As the PET substrate and aluminum oxide are substantially optically transparent, the main contribution to providing the OD is from the metal. Hence on our graph of coating thickness versus OD, there is a plot of the total coating thickness labeled “Aluminum Metal + Oxide.” The growth of the aluminum oxide is similar irrespective of the metal coating thickness, and so for the thinner coatings (around OD 0.5), the variation between the OD measured inside the vacuum system and OD measured later outside the vacuum system can be more noticeable as a higher proportion of the available metal is consumed in forming the oxide. The STEM images show that an oxide thickness of the order of 3.5nm is formed on each side of the aluminum metal. This means that if the deposited metal thickness is only 5-7nm it may all be consumed in forming the oxide, and the coating will “disappear.”

This property is used to some advantage in metallizing systems that are designed to produce transparent barrier alumina coatings where oxygen is deliberately added to the growing coating to encourage the oxidation to take place within the vacuum system. Where the aim is to produce aluminum metal coatings with a specific OD, the time when the OD is measured becomes very important. Relying on a measurement of OD during the metallizing process may result in disputes from customers measuring the OD on receiving the coated material some time later when the aluminum has undergone further oxidation. The coatings were produced in the UK and the samples measured in the UK and USA. It is also important to note that the oxidation can be accelerated if the material is heated and if the humidity is high. This means that for samples produced elsewhere in the world, the values may change from those shown.

Also shown in Figure 2.2-1, labeled “Historic data,” is the original graph produced more than 25 years ago. As can be seen, there is a significant difference between the current and historic data. It is impossible to go back and look at what might have contributed to the differences. Possibilities include process factors such as substrate, pretreatment, system pressure, and deposition rate, as well as calibration and stability of the measurement systems. In this current work the aim was minimize the source of errors and use alternative techniques to confirm measurements where possible. Hence there is a high level of confidence that the current graphs of coating performance against thickness are more accurate than those previously available⁹.

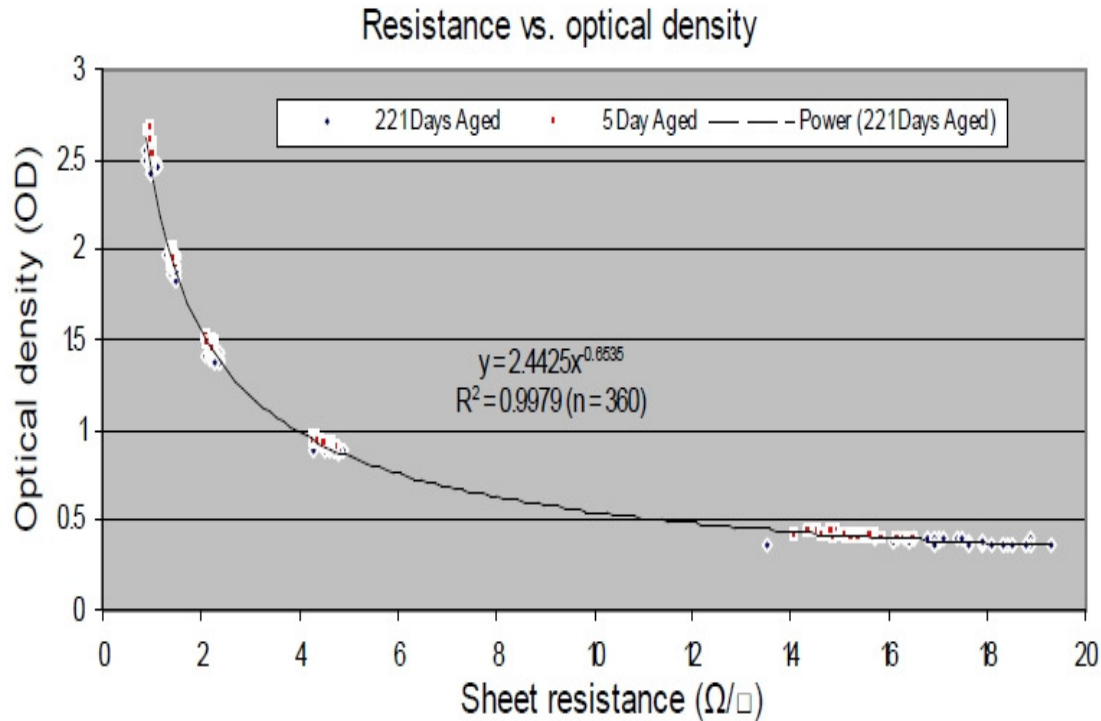
However it is worth reiterating the following warning.

It is important to understand that this data is only a snapshot of what was produced on one machine in a particular climate on one day and should be used as an example only. It is not necessarily the best or the worst that can be produced.

2.3 Relationship of Resistivity to Optical Density and Oxygen and Water Vapor Barrier and Optical Density for Aluminum Metallized Polyester Film

The resistivity also changes with time, and when investigated, it was shown that this too relates to the available metal left in the coating. As the oxide reduces the metal thickness the optical density falls and the electrical resistivity rises. This is shown in Figure 2.3-1 as a plot of OD and sheet resistance. This graph would suggest that a coating of 1.5 - 2.0 OD is required to have a reasonable resistivity.

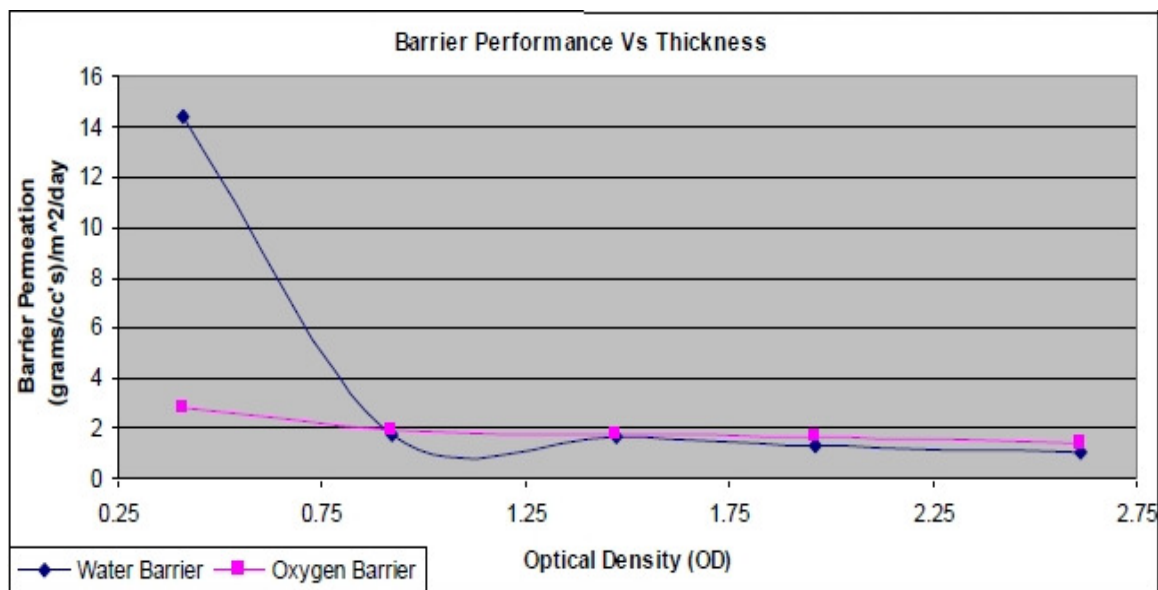
Figure 2.3-1 A Plot Showing the Relationship Between OD and Resistivity⁴⁻⁸



The conductivity or resistivity of a coating will depend on the deposition process. For the same thickness coating, the conductivity will be higher if the crystal size of the material is large. A significant cause of loss of conductivity is the disruption of electron flow at each grain boundary. So, for a large crystal size there are fewer grain boundaries per unit length and fewer losses. For a coating with a small crystal size there will be many more high resistance grain boundaries and a higher loss of conductivity. The nucleation and growth of the material can be affected by the substrate surface energy and roughness as well as the substrate temperature, energy of the depositing flux, and system pressure. This means that it would be easy to produce coatings that vary significantly from the graph above.

However for any system and process it should be possible to produce a similar graph of the coating performance that can be expected from that particular system.

Figure 2.3-2 A Plot Showing the Relationship Between Oxygen and Water Vapor Barrier and OD⁴⁻⁸.



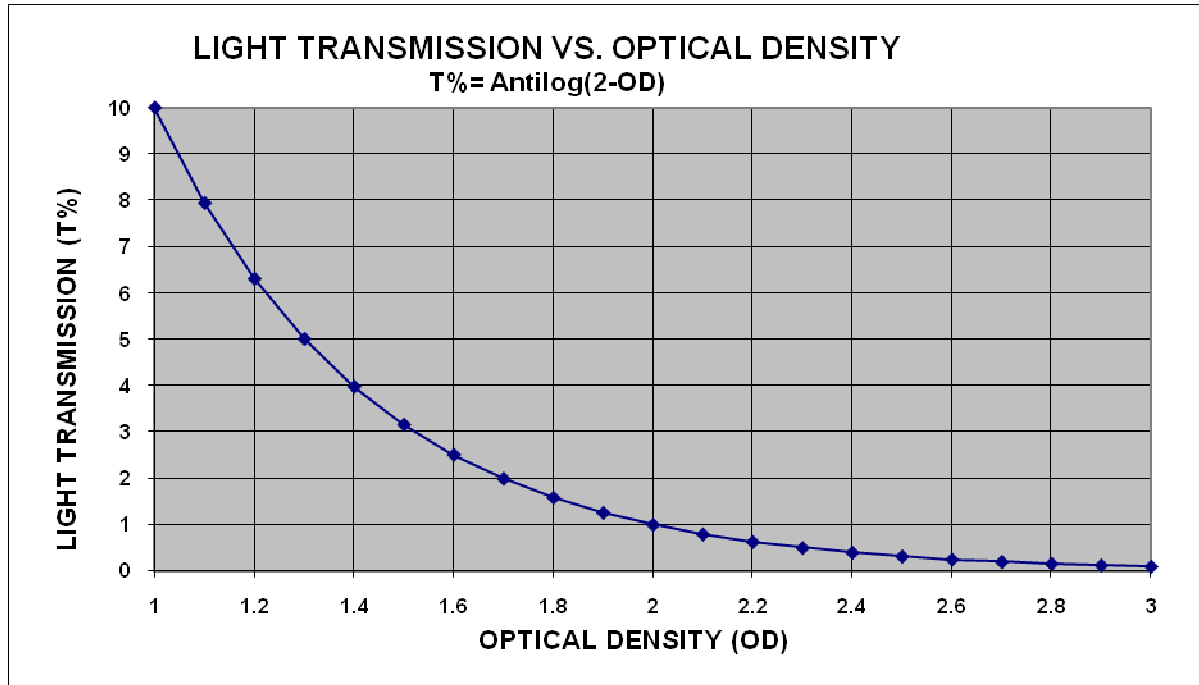
OTR = Oxygen barrier = cc/m²/day (27°C @ 50 % RH)

WVTR = Water barrier = g/m²/day (37.8°C @ 90 % RH)

This plot of barrier performance also gives an indication of what can be produced. It will be seen in Section 3 (Barrier Properties of Metallized Films) that the performance can be hugely affected by the surface quality of the substrate. The above plot indicates that for this grade and quality of 12-micron PET, the best combination of water vapor and oxygen barrier would require a minimum coating of OD 1.5 for consistent performance.

2.4 Light Transmissions vs. Optical Density

Figure 2.4-1 Light Transmissions vs. Optical Density (Courtesy of Advanced Ceramics Corp.)



Optical density is a term used to indicate the ability of a material such as a metallized film to block or absorb the radiant energy in an incident light beam. It is correctly termed the absorbance⁴ when used for measuring the attenuation of a light beam passing through a material.

Optical density is calculated as the $\text{Log}_{10}(1/T)$

Where:

- T is the transmittance defined as P/P_0
- P is the unabsorbed energy remaining in the light beam passing through the sample
- P_0 is the energy of the incident light beam

The optical density is used to indicate the level of light barrier supplied by a metallized film. Thus an optical density (OD) of 2 represents a film with 1% light transmittance or 99% of the light is reflected and absorbed by the film.

2.5 Electrical, Optical, and Barrier Properties of Various Metallized Oriented Polypropylene (OPP) Films

Figure 2.5-1 Electrical, Optical, and Barrier Properties of Various Metallized OPP Films (Courtesy of Quantum Performance Films)

Gauge	Optical Density	Light Transmission %	Resistivity (ohms/square)	WVTR*	OTR*
70	2.1	0.8	2.2	0.025	3.0-4.0
70	2.4	0.4	1.8	0.025	2.0
80	2.1	0.8	2.2	0.07	8.0-10.0

* Flat laminated samples

WVTR = gm/100 in²/24 hrs. at 100°F (37.7°C), 90% RH

OTR = cm³/100 in²/24 hrs. at 73°F (22.7°C), 0% RH

2.6 Emissivity of Polyester and Kapton H and HN Films

Figure 2.6-1 EMISSIVITY OF POLYESTER FILMS
(Courtesy of Metallized Products)

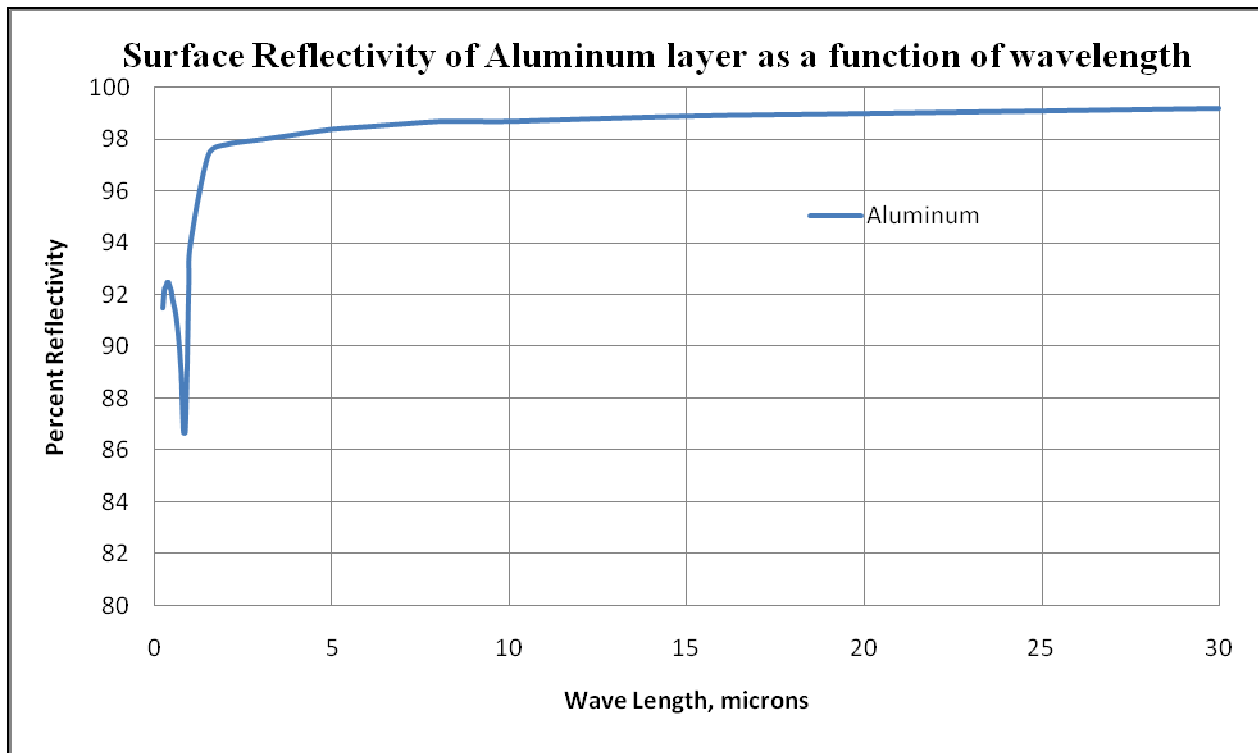
	Ohms/Sq	Emissivity Range
25 Gauge	0.4	.03 - .04
	0.5	.03 - .045
	0.6	.025 - .04
	0.7	.025 - .04
	0.8	.025 - .04
	0.9	.025 - .04
	1.0	.025 - .045
	1.1	.03 - .04
	1.2	.035 - .05
	1.3	.03 - .04
	1.4	.025 - .06
	1.5	.03 - .04
	1.6	.04 - .045
	1.7	.035 - .04
	1.8	.03 - .05
	1.9	0.05
	2.1	0.05
	2.4	.04 - .06
92 Gauge	2.6	0.04
	3.0	0.05
	3.3	0.05
	0.6	0.03
	0.9	0.04
	1.0	.03 - .035
	1.1	.035 - .04
	1.2	0.04
2 mil	1.3	.025 - .04
	1.4	0.04
	1.5	0.04
	1.6	0.04
	1.7	0.025
	0.4	0.04
	0.5	.03 - .04
	0.6	.025 - .04
	0.7	.025 - .04
	0.8	.025 - .04
3 mil	0.9	.03 - .04
	1.0	.035 - .04
	1.1	.02 - .04
	1.2	.025 - .04
	1.3	.03 - .04
	1.4	.035 - .04
	1.7	0.04
	1.8	0.04
	0.6	0.03
	0.7	0.03
5 mil	0.8	.02 - .03
	1.0	0.025
	1.1	.025 - .03
	1.3	0.025
	1.4	.035 - .04
	1.6	0.03
	1.7	0.03
	0.4	0.03
	0.5	.025 - .035
	0.6	0.035
	0.7	0.04

Figure 2.6-2 EMISSIVITY OF KAPTON H AND HN FILMS
(Courtesy of Metallized Products)

	Ohms/Sq	Emissivity Range
30H	0.3	0.025
	0.4	.02 - .035
	0.5	.025 - .04
	0.6	.025 - .04
	0.7	.03 - .04
	0.8	.025 - .04
	0.9	.025 - .04
	1.0	.03 - .04
	1.1	.03 - .04
	1.2	.03 - .04
	1.3	.04 - .045
	1.4	0.04
	1.5	0.045
	1.7	.045 - .05
	1.8	0.05
50H	0.6	.035 - .04
	0.7	.025 - .03
	0.8	.02 - .04
	0.9	.02 - .04
	1.0	0.03
	1.1	.03 - .04
	1.2	.03 - .04
	1.4	0.04
100HN	1.7	.035 - .04
	0.4	.03 - .04
	0.5	.03 - .035
	0.6	.03 - .04
	0.7	.03 - .04
	0.8	.03 - .04
	0.9	.025 - .045
	1.0	.03 - .04
	1.1	.03 - .04
	1.2	0.04
200HN	1.3	.03 - .04
	1.4	.035 - .045
	1.5	0.04
	0.4	0.03
	0.5	.02 - .04
	0.6	.025 - .04
	0.7	.015 - .04
300HN	0.8	.025 - .04
	0.9	.02 - .045
	1.0	.025 - .04
	1.1	.03 - .04
	1.2	0.04
	1.3	.03 - .05
	1.4	.03 - .04
	1.5	0.04
500HN	1.6	.03 - .035
	0.6	.02 - .035
	0.7	.03 - .035
	0.8	.03 - .04
	0.9	.03 - .04
	1.0	.025 - .035
	1.1	0.035
	1.3	0.04
	0.4	.02 - .035
	0.5	.02 - .04
	0.6	.02 - .04
	0.7	.02 - .04
	0.8	.02 - .04
	0.9	.025 - .04
	1.0	.03 - .035
	1.1	0.04

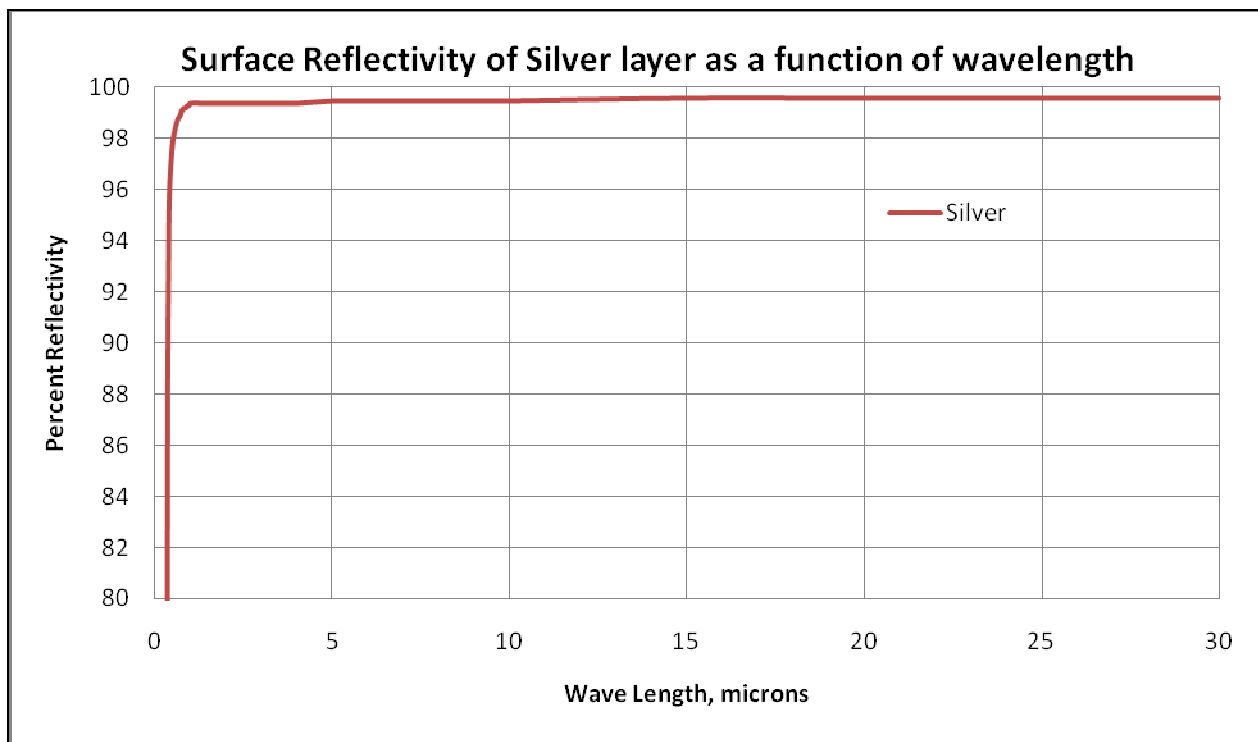
2.7 Surface Reflectance of a Vacuum-deposited Aluminum Layer

Figure 2.7-1 Surface Reflectance of a Vacuum-deposited Aluminum Layer⁵



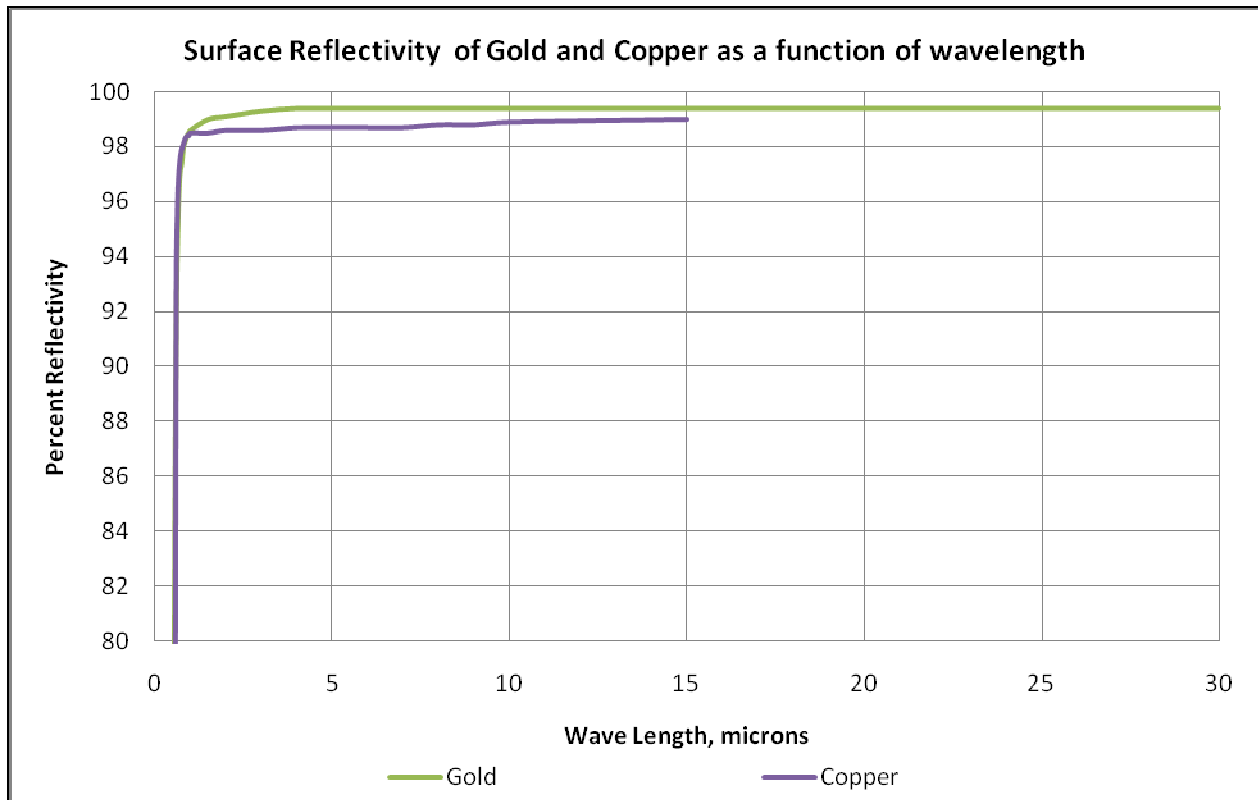
2.8 Surface Reflectance of a Vacuum-deposited Silver Layer

Figure 2.8-1 Surface Reflectance of a Vacuum-deposited Silver Layer⁶



2.9 Surface Reflectance of a Vacuum-deposited Gold and Copper Layer

Figure 2.9-1 Surface Reflectance of a Vacuum-deposited Gold and Copper Layer⁶



Data in Figures 2.7-1, 2.8-1, and 2.9-1 show the reflectance of several metals prepared without over-coatings for use as mirrors in the ultraviolet (UV) to the far infrared region of the spectra. Only aluminum has sufficient reflectivity at extremely short wavelengths, (in the Vacuum UV region) to function as a mirror.

2.10 Section 2 References Cited

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 - ⁸ McClure D.J., Copeland N.J., "Evaporated Aluminum on Polyester: Optical, Electrical & Barrier Properties as a Function of Thickness and Time," Proc. Society of Vacuum Coaters 54th Annual Technical Conference 2011, pp 634-640.
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Section 3: Barrier Properties of Metallized Films

Introduction

Dedication

This chapter is dedicated to Dr. Bernard M. Henry, who passed away unexpectedly in February of 2007 at the age of 42. Dr. Henry dedicated a major part of his research work to explore the function of thin film barrier coatings. He proposed different models, particularly of water vapor permeation, that much better describe the difference in permeation behavior between oxygen and water vapor. Dr. Henry hypothesized and demonstrated experimentally that water dissociates within the barrier layer, allowing the separated hydrogen ions to permeate through the barrier layer at a much greater rate, re-associating with oxygen at the opposite side of the layer.

Dr. Henry's contributions to the field of vacuum-coated barrier films will be greatly missed. It is for this reason that AIMCAL together with the Society of Vacuum Coaters Foundation has established a Scholarship Fund in his memory. The Bernard Henry Scholarship will be given to students who do research in the field of vacuum-deposited barrier coatings. For contributions to this fund please contact the AIMCAL office.

Sincerely,
Charlie Bishop
Eldridge M. Mount III
Wolfgang Decker

The barrier properties of metallized films presented below represent a special set of values, determined on particular base sheets and need to be understood in a larger context of barrier property development. During the last 15 years, the understanding of metallized film barrier development has improved and broad film barrier properties can be shown in terms of aroma and moisture and oxygen barrier for a wide range of metallized films¹. In addition, the chemical (aroma) barrier properties have been shown to be dependent upon the base sheet to a large degree².

This is especially true for the barrier properties of oriented polypropylene (OPP), where it has been shown that the barrier properties are dependent upon the film surface polymer³, the metallizing vacuum level^{4,5}, an optimum optical density⁶, and the chemical composition of particular metallizing surfaces⁷. Consequently, the data presented here should serve as a guide. For specific applications, requiring a particular barrier level, the product should be checked by barrier testing the film produced from a particular chamber and the process capability of the chamber determined.

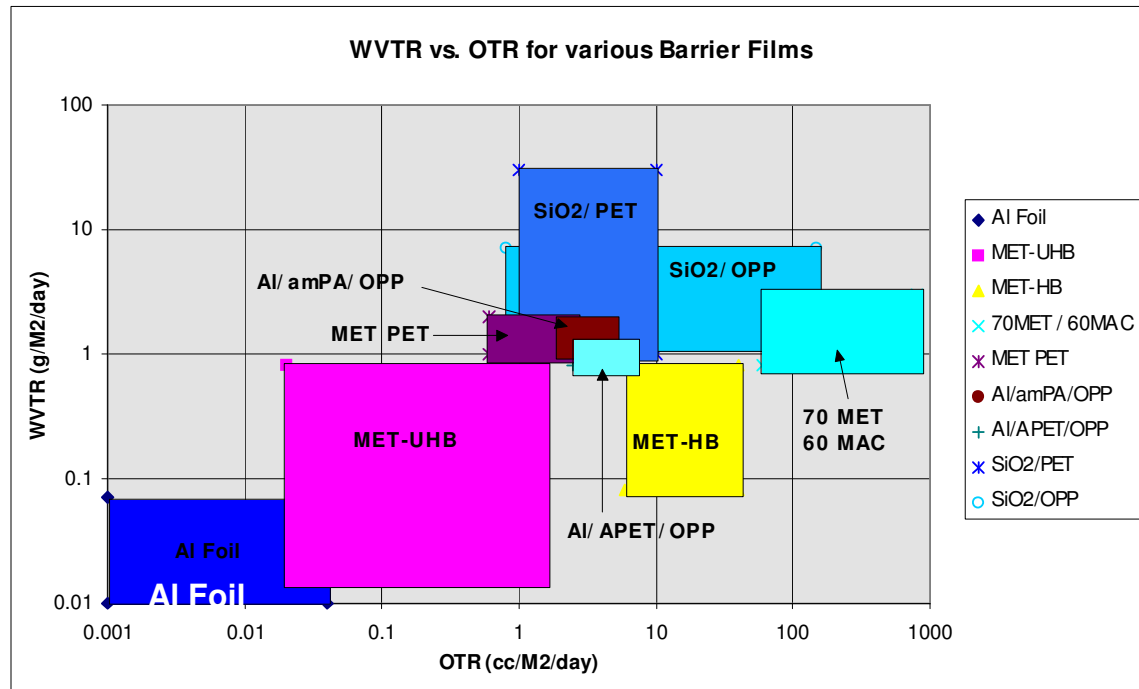
New technologies for the treatment of film in the metallizing chamber are also showing that the film optical density, at a fixed aluminum deposition rate, depends on the surface treatment level of the film surface⁸. Thus for a particular metallized film product, the process used to make it will determine the ultimate properties of the aluminum layer deposited, based upon the thickness and density of the aluminum and any other interfacial effects which might be important (*see Section 3.14*). From this it can be seen that it is unlikely that a unique relationship will exist for the optical density vs. barrier properties of a film, especially for OPP where the metallized surface may have many compositions.

Polyester films should be more uniform in this regard, as the surface polymer is fairly uniform in composition from film to film. However, some PET films are surface modified to improve adhesion⁹ by coating and care should be taken to better understand the impact of surface polymer modifications in PET films on metallized film barriers. But in general, the metallized PET films seem to be less likely to have surface polymer modifications by coextrusion or extrusion coating as found with metallized OPP films.

Consequently, special care must be taken when specifying OPP films for applications requiring gas and moisture barrier versus metallized films for light barrier and decorative applications as described in Reference 2 (*see references at end of section*). The first figure of this section shows a broad mapping of film properties on the WVTR-OTR space. It compares many film types vs. aluminum foil. Many of the films shown are OPP with high surface energy polymers such as, amorphous nylon¹⁰ (amPA), amorphous PET¹¹ (APET), HDPE¹² (MET-HB), EVOH¹³ (MET-UHB) and EP copolymer¹⁴.

3.1 Water Vapor Barrier (WVTR) vs. Oxygen Barrier (OTR) for Various Barrier Films

Figure 3.1-1 Plot of WVTR vs. OTR for Various Barrier Films²



WVTR: ASTM test method F1249-90, 37.8°C (100°F) and 90% relative humidity (RH).

OTR: ASTM test method D 3985-81, 23°C (73.4°F) and 0% RH.

This graph demonstrates the wide range of barrier levels possible with a wide range of OPP-based films and reinforces the comments above about the wide range of properties as a function of optical density, while, also showing the relative barrier property data range for metallized PET and silica-coated films¹⁵.

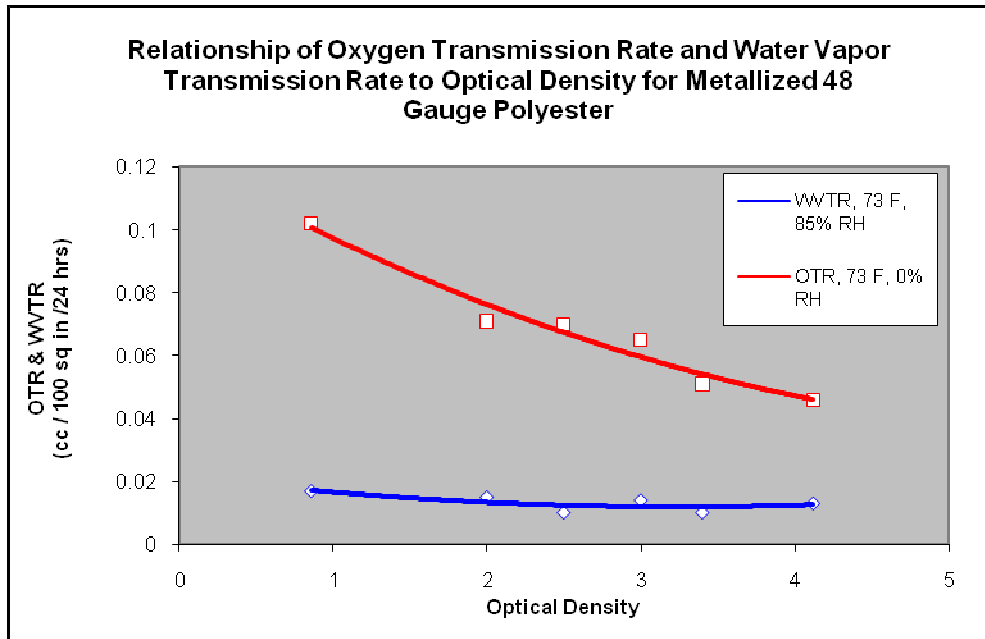
Care must be exercised in handling of metallized films to insure that the barrier properties are not compromised during the many times the film is wound and unwound as it is processed into final products. All rolls must be clean and turning at film speed to minimize any scratching of the metal surface.

Also, film tension must be at a level to prevent stretching such that the metal layer is cracked (less than 1% elongation). For example, film tension load levels (in kg/cm or lb/in of film) will need to be lower for OPP than for PET films of equal thickness to maintain the same range of film elongation due to differences in the films tensile strengths (kg/cm² or lb/in²).

Also as film thickness decreases, tension loads on the film (kg/cm² or lb/in²) have to be reduced to keep tensile stresses (kg/cm² or lb/in²) low enough to maintain low elongations and prevent metal damage. Film suppliers should be able to specify appropriate tension ranges for the products they supply. This is also important in each of the subsequent lamination processes to which the metallized film is subjected. Most films, used for moisture and gas barrier properties, are laminated to place the metal layer in the middle of the structure to protect it from further scratching. While laminated films generally show some barrier improvement over the flat film barrier properties, the metal layer is still subject to cracking if film tension levels and lamination conditions are not controlled.

3.2 Relationship of Oxygen and Water Vapor Transmission Rate to Optical Density for Metallized 48-gauge Polyester

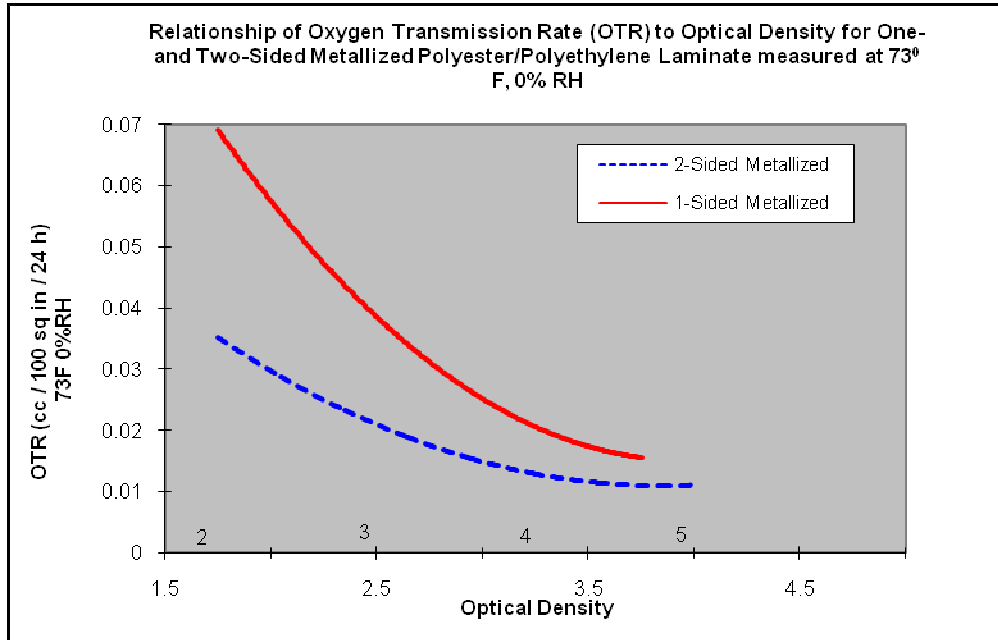
Figure 3.2-1 Relationship of OTR and WVTR to Optical Density for Metallized 48-gauge Polyester
(Courtesy of Camvac International, Inc.)



Note: Graph represents measured results

3.3 Relationship of Oxygen Transmission Rate (OTR) to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate

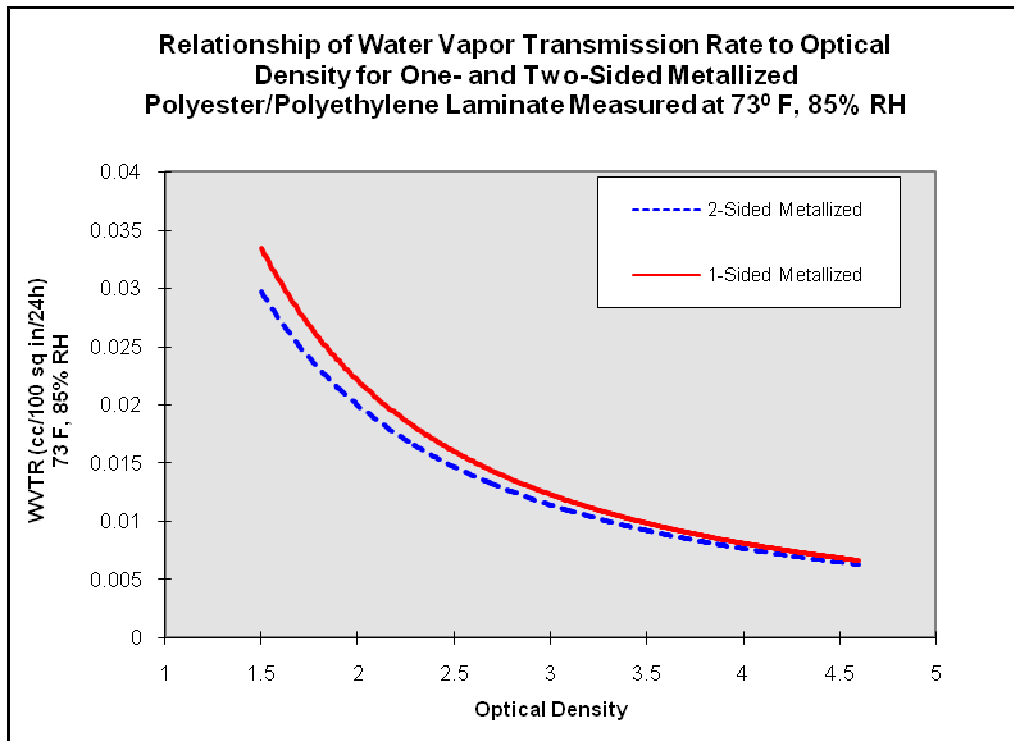
Figure 3.3-1 Relationship of OTR to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate (Courtesy of Camvac Ltd.)



Note: Two-side-metallized film has an approximate equal thickness of aluminum on each side.

3.4 Relationship of Water Vapor Transmission Rate to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate

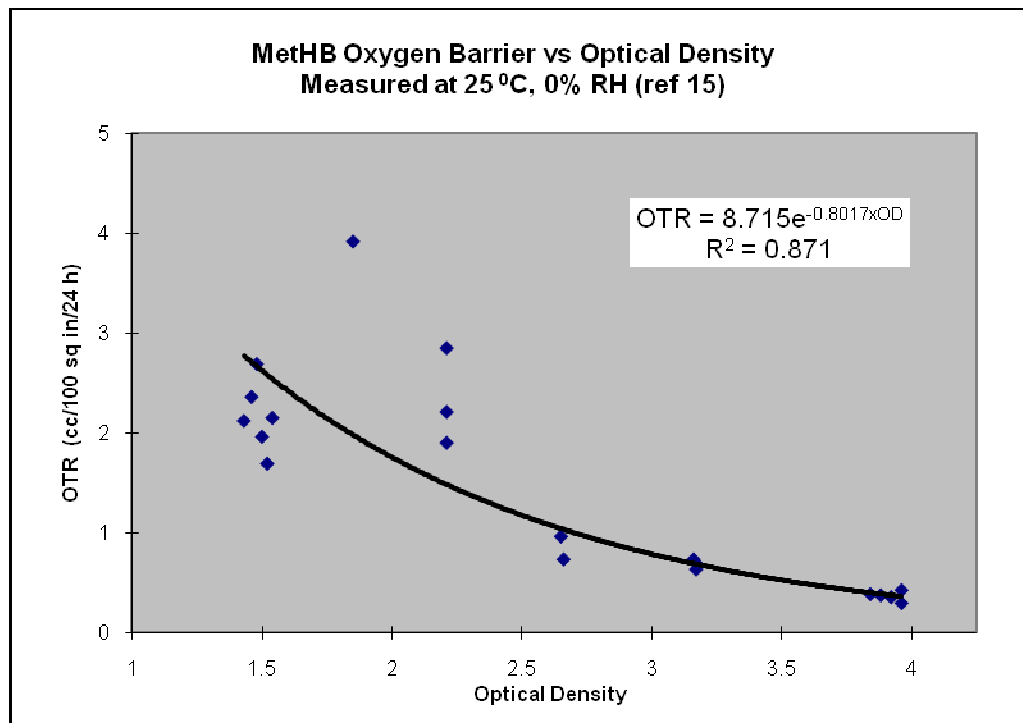
Figure 3.4-1 Relationship of WVTR to Optical Density for One- and Two-Side-Metallized Polyester/Polyethylene Laminate (Courtesy of Camvac Ltd.)



Note: Two-side-metallized film has an approximate equal thickness of aluminum on each side.

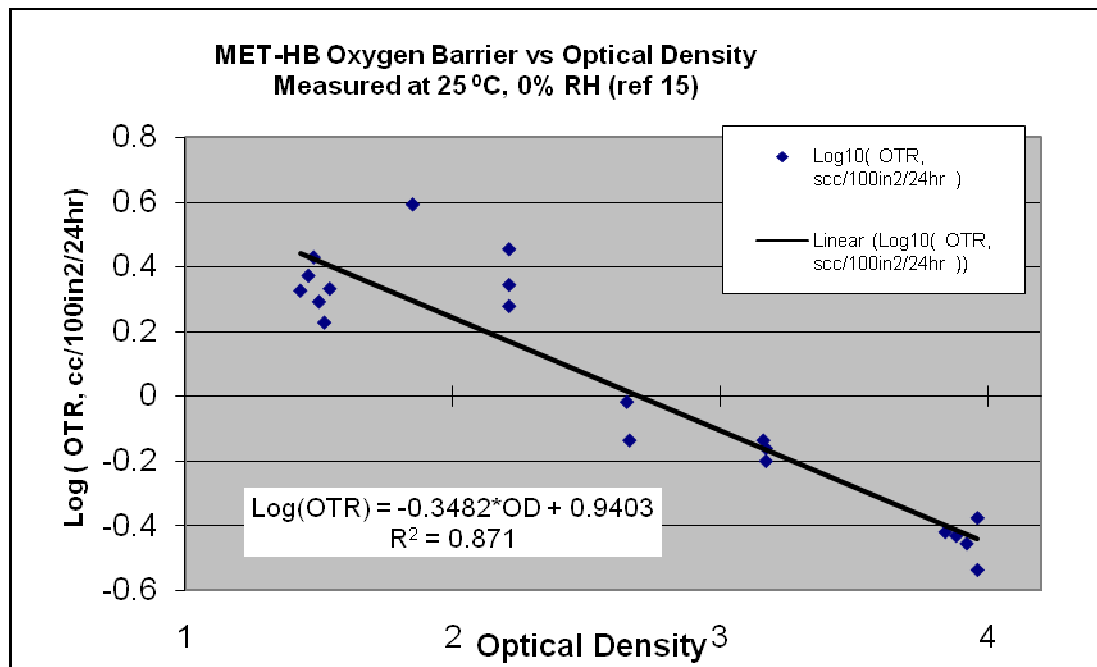
3.5 Oxygen Barrier vs. Optical Density for Metallized OPP, MET-HB

Figure 3.5-1 Oxygen Barrier vs. Optical Density for Metallized OPP, MET-HB¹⁶



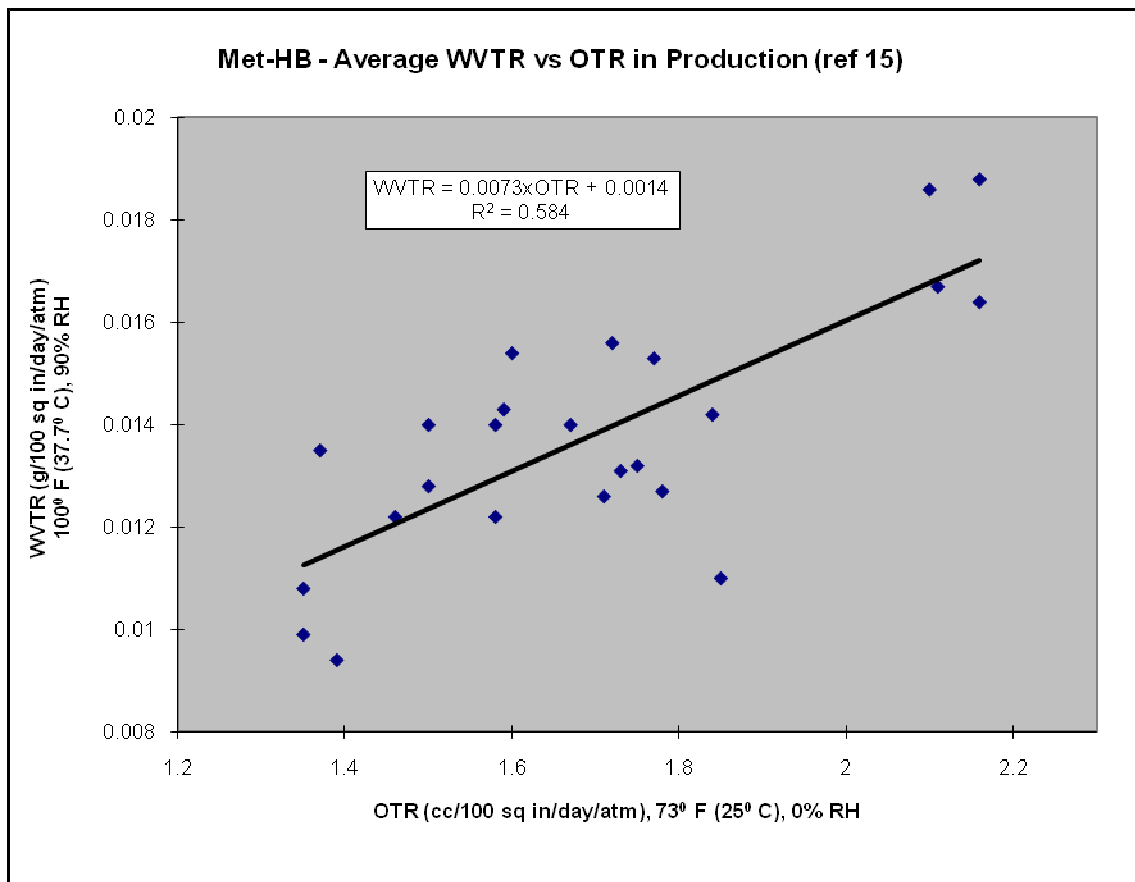
3.6 Log₁₀ (Oxygen Barrier) vs. Optical Density for Metallized OPP, MET-HB

Figure 3.6-1 Log₁₀ (Oxygen Barrier) vs. Optical Density for Metallized OPP, MET-HB¹⁵



3.7 Average Water Vapor Transmission Rate (WVTR) vs. Oxygen Transmission Rate (OTR) for Metallized Production, MET-HB

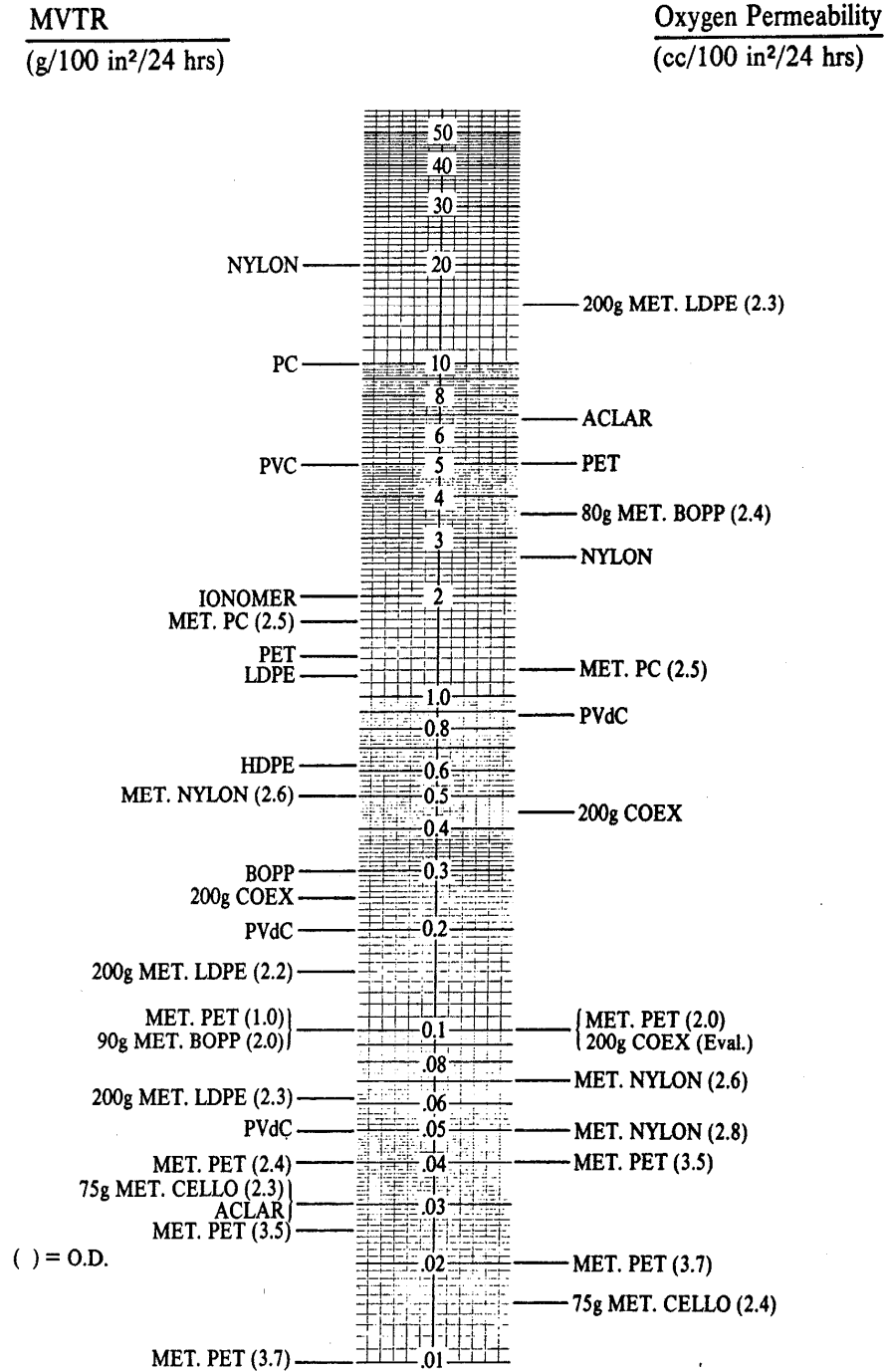
Figure 3.7-1 Average WVTR vs. OTR for Metallized Production, MET-HB¹⁵



3.8 Barrier Properties for Metallized Films- 1

Figure 3.8-1 Barrier Properties for Metallized Films- 1 (Courtesy of ICI Americas Inc.)¹⁷

Barrier Comparison

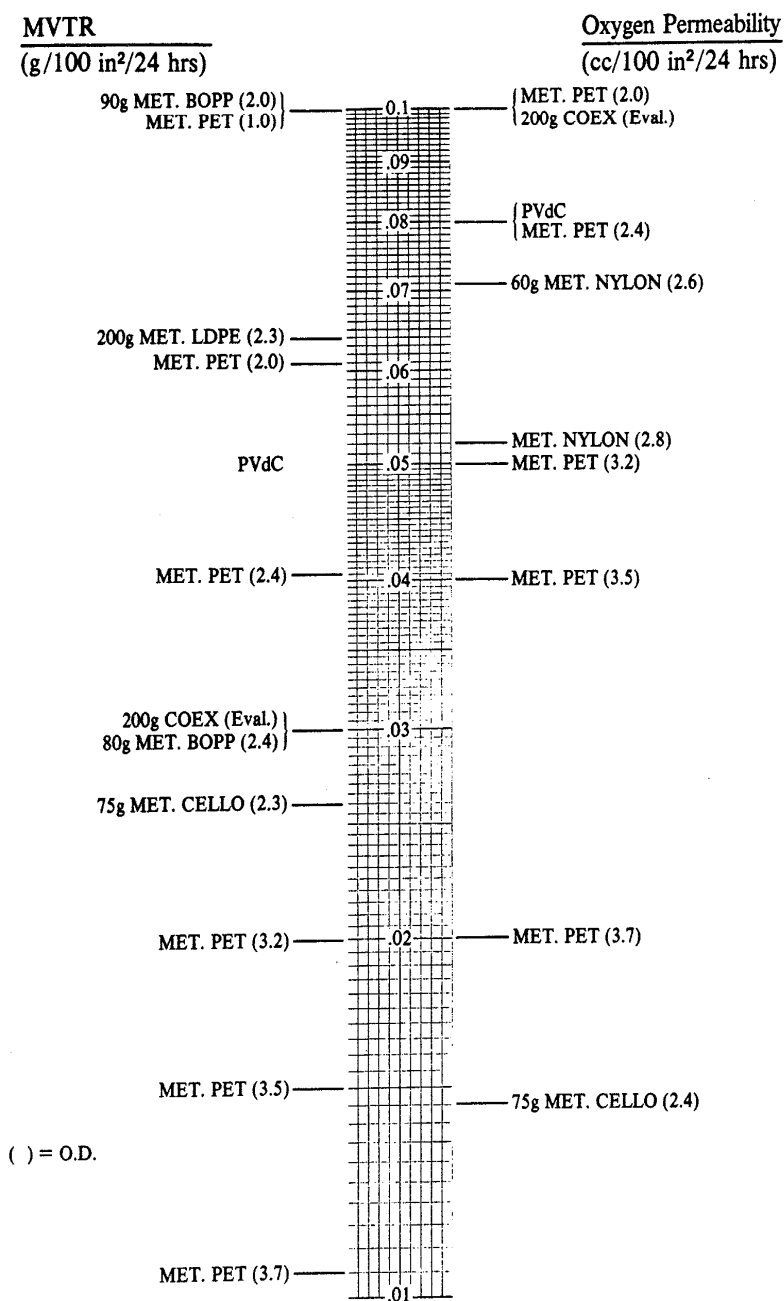


*Base substrates 1 mil/MET. substrates 48g unless noted otherwise

3.9 Barrier Properties for Metallized Films-2

Figure 3.9-1 Barrier Properties for Metallized Films- 2 (Courtesy of ICI Americas Inc.) ¹⁷

Barrier Comparison

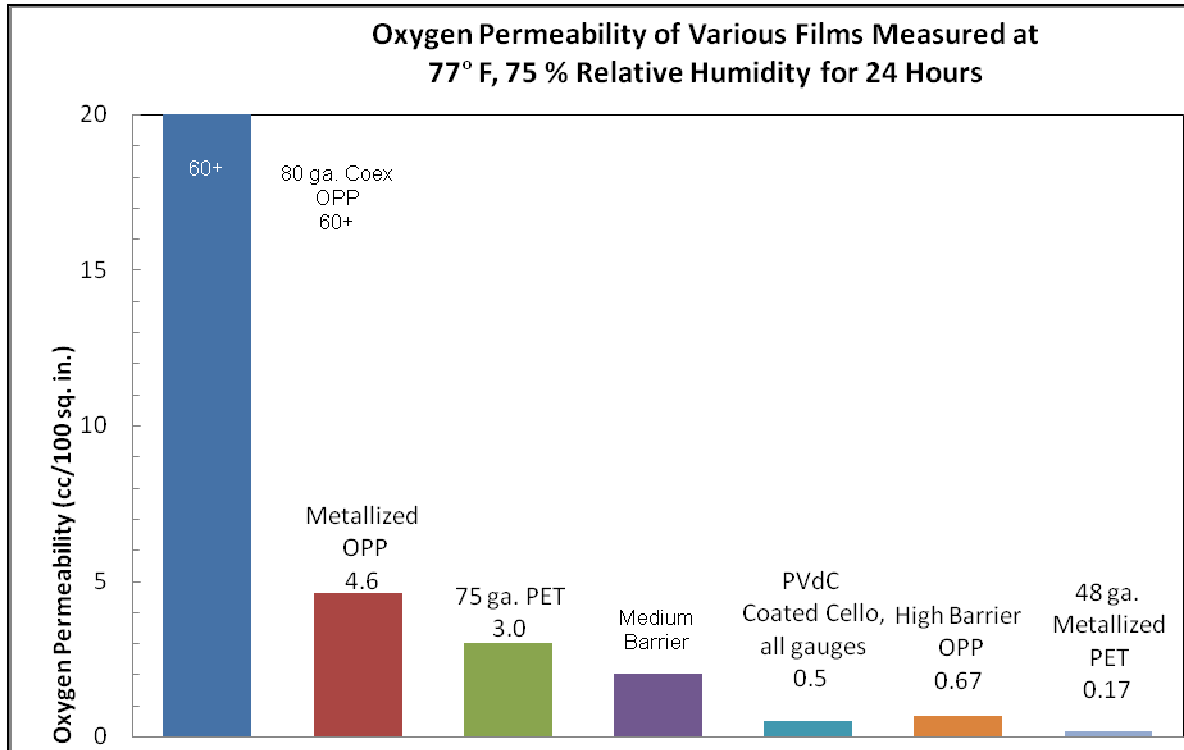


() = O.D.

*All substrates 48g unless noted otherwise

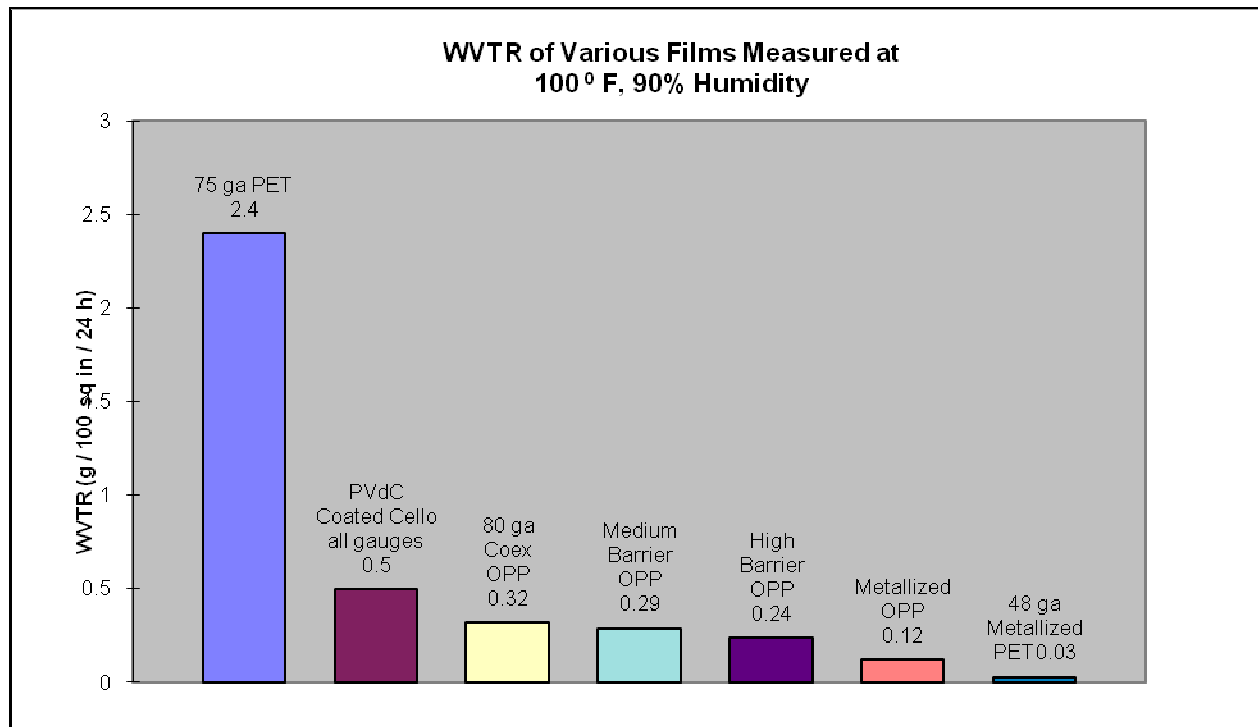
3.10 Oxygen Permeability of Various Films

Figure 3.10-1 Oxygen Permeability of Various Films (Courtesy of ICI Americas Inc.)



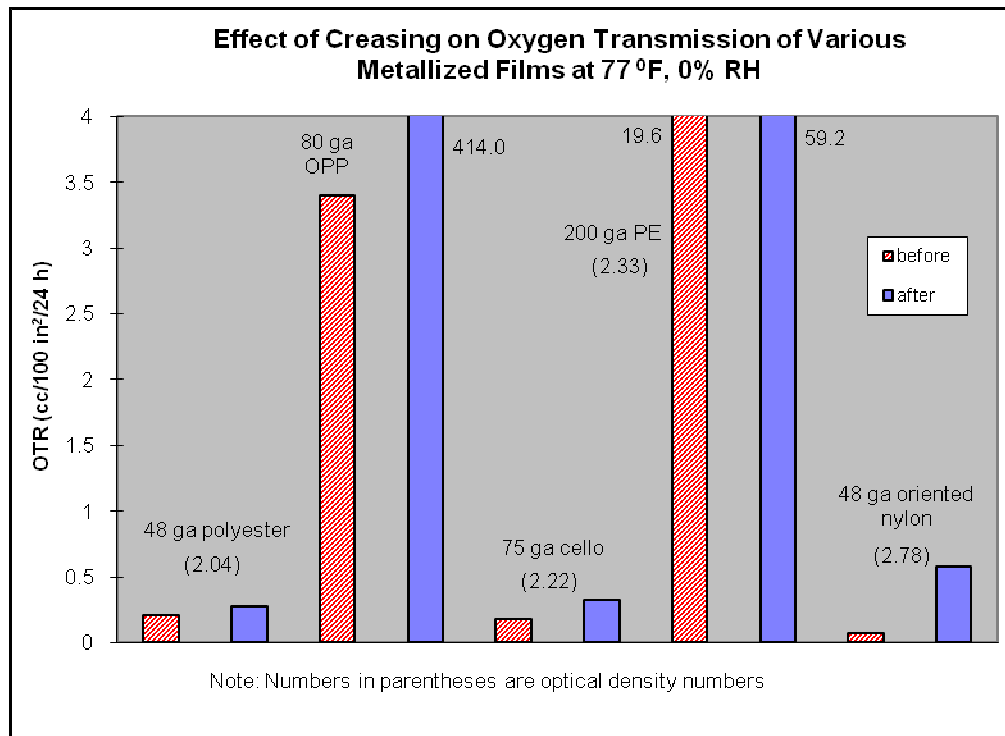
3.11 Water Vapor Transmission of Various Films

Figure 3.11-1 Water Vapor Transmission of Various Films (Courtesy of ICI Americas Inc.)



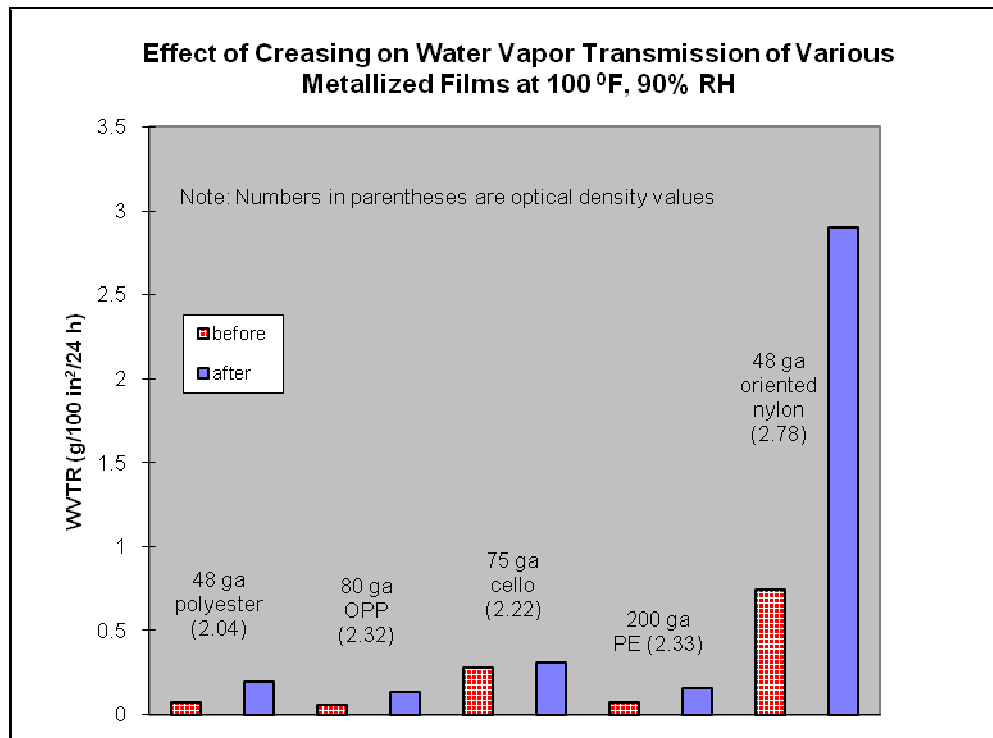
3.12 Effect of Creasing on Oxygen Transmission of Various Metallized Films

Figure 3.12-1 Effect of Creasing on OTR of Various Metallized Films
, values listed in Section 3.14 (Courtesy of ICI Americas Inc.)



3.13 Effect of Creasing on Water Vapor Transmission of Various Metallized Films

Figure 3.13-1 Effect of Creasing on WVTR of Various Metallized Films
Values listed in Section 3.14 (*Courtesy of ICI Americas Inc.*)



3.14 Comparison of Barrier, Electrical, and Optical Properties of Various Metallized Films

Figure 3.14-1 Comparison of Barrier, Electrical, and Optical Properties of Various Metallized Films
(Courtesy of ICI Americas Inc.)

	<u>OPTICAL DENSITY</u>				RESISTIVITY (ohms/□)	<u>WVTR</u> (gms./100sq.in/24hrs)		<u>OXYGEN PERMEABILITY</u> (cc./100sq.in/24 hrs)	
	Metallized Film Type	Optical Density	%VLT	%UV		Flat	Creased	Flat	Creased
PET	48 ga.	2.04	0.91	1.71	1.8	0.061	0.190	0.105	0.258
	48 ga.	2.38	0.42	0.83	1.6	0.045	0.118	0.080	0.144
OPP	80 ga.	2.32	0.48	1.00	2.3	0.029	0.056	3.419	414.000
	90 ga.	1.91	1.22	2.52	2.5	0.103	0.120	16.850	37.300
Cello	75 ga.	2.22	0.61	0.99	1.6	0.266	0.271	0.143	0.304
PE	200 ga.	2.33	0.46	0.94	1.8	0.062	0.132	19.600	59.240
	125 ga.	2.01	0.98	1.92	2.6	0.279	0.429	16.720	152.200
Cast Nylon	100 ga.	2.22	0.60	1.08	1.9	8.335	8.335	0.129	0.387
Oriented Nylon	60 ga.	2.88	0.13	0.29	1.8	0.14	0.24	0.071	1.245
	48 ga.	2.78	0.16	0.38	1.2	0.15	0.61	0.052	0.561

WVTR conditions: 100°F, 90% RH

Oxygen permeability conditions: 77°F, 0% RH

3.15 Section 3 References Cited

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Section 3: Addendum

The following is a paper given by Dr. Wolfgang Decker and Dr. Bernard Henry. This paper is presented in its entirety as an excellent plenary on barrier.

Basic Principles of Thin Film Barrier Coatings

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ABSTRACT

There is hardly any application used in Thin Films Technology, which is used as widespread as the Barrier Coating on Packaging Films. But despite this fact, there is – still – not a comprehensive understanding as to why and how Barrier Coatings actually work. In fact even the biggest producers of barrier-coated materials seem to struggle when it comes to explanations why certain coatings and procedures work or fail. It has become apparent though, that sometimes even the most fundamental aspects of the function of barrier coatings are not understood. This report will try to put together some of the basic principles of Thin Film Barrier Coatings. In addition it will point to some of the aspects of the function of barrier coatings, which require more attention in the scientific field.

THE FUNCTION OF A BARRIER COATING

A Barrier Coating in the simplest terms is a membrane that creates a partition between two areas, for example, preventing certain materials – gases, liquids or solids – on one side of a separation from reaching the other side. Packaging film functions in a similar manner: It contains a packaged product and locks out environmental impact to the degree desired. Because of the effect of diffusion, there is always a certain amount of gases passing through the barrier materials.

Diffusion is the movement of matter through a medium driven by the kinetic motions of atoms and molecules. Directional diffusion – that is the preferred flux of matter in a specific direction – is caused by nature's need for equilibrium and occurs in the opposite direction to the gradient of concentration of the particular species.

The rate of diffusion (j) of matter (i) is described by Fick's 1st law, which states:

$$j_i = D_{ij} \frac{dc_i}{dx} \quad [\text{Eq. 1}]$$

Where D_{ij} is the diffusion coefficient of the matter i in a medium j , and $\frac{dc_i}{dx}$ describes the concentration gradient in the diffusion direction. Values of D_{ij} are typically in the order of $10^{-1} \frac{\text{cm}^2}{\text{s}}$.

Diffusion in gases is quite well understood, and the diffusion coefficient for most gases within gases can be calculated using gas kinetic theories as shown in Equation 2:

$$D_{ij} = 0.002662 \frac{\sqrt{T^3}}{p \sigma_{ij}^2 \sqrt{M_{ij}} \Omega_D} \quad [\text{Eq. 2}]$$

Where σ_{ij} represents the gas kinetic parameters of the Lennard Jones length [$\text{\AA}$] and Ω_D the diffusion collision integral, which is a correction factor for the interaction between the gas molecules. M_{ij} is the reduced molar mass which recognizes the molar distribution of the gas mixture. The derivation of these gas kinetic parameters can be found in Reference 1.

The diffusion of gases in solids however is not as simple to calculate. This is dependent on many more factors such as the degree of crystallinity, porosity, chemical interaction with the gas atoms, polarity of the gas, etc.

Diffusion of gases, such as oxygen, through totally crystalline materials is typically very slow due to the very constricted interstitial spaces of the lattice. The diffusion mechanism, when no chemical reaction takes place, is considered to occur by a “random walk” of atoms or molecules through the interstices of the lattice. In this case, the diffusivity is given by:

$$D = \gamma \bar{\lambda}^2 \Gamma \quad [\text{Eq. 3}]$$

where γ is a geometric factor, $\bar{\lambda}^2$ is the average of the squares of the jump distances and Γ is the jump frequency. For gas diffusion in single crystals, lattice defects and to a lesser extent crystallographic orientation often control the diffusivity. For partially crystalline solids, diffusion is largely dictated by the amorphous content as migration through the amorphous phase occurs considerably faster than through the crystalline lattice. This is also the case for the grain boundaries, which act as short circuit pathways for diffusion. It should be noted that if the solid is porous then a number of additional mechanisms exist for the flow of gases through the structure; these include viscous flow where the gas migrates unhindered through a pore of around 100nm in diameter, Knudsen flow, which involves the occasional collision between the diffusing gas and the pore wall (pore diameter between around 4nm and 100nm), and surface diffusion (involves the gas migrating along the pore wall). The extent to which each mechanism contributes to the overall gas flow depends on the size of the permeating molecule (or atom) and the number and size of each class of pore present in the solid. The diffusion coefficient of these solids is considerably lower than those for the cases previously outlined.

Table 1: Permeation Values of Oxygen in Basic Polymers

Polymer	Permeability $\left[\frac{cm^3 O_2}{m \cdot d \cdot atm} \right]$	Permeation $\left[\frac{cm^3 O_2}{m^2 \cdot d \cdot atm} \right]$	Film Thickness [μm]
LDPE	$3.61e^{-1}$	6000	60
HDPE	$6.70e^{-3}$	100	60
PP	$1.51e^{-1}$	2500	60
PET	$1.20e^{-3}$	100	12

The diffusion behavior of water vapor in solids differs significantly to that of oxygen, due to its reactive nature and bi-polar structure. A non-interactive model based on a Fickian mechanism of molecular water vapor diffusion in silicon oxide has been proposed but to date has not been supported by experimental evidence. The data suggests that the migration of water vapor through some solids involves attractive interaction and/or reaction. In some cases, hydrolysis has been shown to be a partial pathway in a non-Fickian mechanism of water diffusion through oxides.

Generally the diffusion coefficient of gases in solids is many orders lower than in gas. The diffusion coefficient for Oxygen (O_2), which is considered to be non-reactive, in air is about $0.15 \frac{cm^2}{s}$ and in Polyester, depending upon crystallinity, is in the region of $10^{-9} \frac{cm^2}{s}$. Typical barrier materials like metal oxides or aluminum however have a diffusion coefficient for oxygen several magnitudes lower than this, for example, silicon oxide has a reported value of $10^{-13} \frac{cm^2}{s}$. Oxygen diffusion in aluminum is a special case, since oxygen chemically reacts with aluminum forming aluminum oxide, which in turn alters the diffusion properties of the solid. In the absence of defect space within the matrix, diffusivities as low as $10^{-18} \frac{cm^2}{s}$ have been measured for water vapor migration through silicon oxide at standard temperature and pressure.

Knowing the diffusion coefficients of unwanted gases through the packaging material still does not really tell us how much gas actually penetrates the package and reaches the product. Knowing the permeation would provide us with this information. If we substitute Henry’s law [Eq. 2], which describes the solubility S of gas in solid materials (c_x and p_x are the concentration and the pressure respectively),

$$c_x = S \cdot p_x \quad [\text{Eq. 4}]$$

into [Eq. 1] we yield the molar flux

$$j_i = \frac{D_{ij} \cdot S \cdot (p_1 - p_2)}{d} = \frac{P_m^*}{d} \cdot \Delta p \quad [\text{Eq. 5}]$$

Where: $P_m^* = D S$ is the **molar Permeability of the film** normalized to a unit thickness. Dividing P_m^* by the film thickness, t delivers the **Permeation P**:

$$P = \frac{P_m^* \cdot M}{\rho \cdot d} \left[\frac{\text{cm}^3}{\text{m}^2 \text{ day atm}} \right] \quad [\text{Eq. 6}]$$

Typical values of oxygen permeation through polymer packaging materials are listed in Table 1, and one can see that indeed a significant amount of oxygen reaches the contents of the package.

PERMEATION THROUGH BARRIER COATINGS

Typical polymers used for packaging have permeation values for oxygen and water vapor that would either make them unsuitable for barrier packaging or require extremely thick films. Today thin film barrier coatings such as metals and metal-oxides are used and provide much lower diffusion coefficients than the polymer materials. Below is a list of such barrier constructions. (See Table 2.)

Table 2: Oxygen and Water Vapor Barrier Performance of Different Barrier Materials (/15/, unless stated otherwise)

Barrier Material	Thickness of Substrate or Coating	Oxygen Transmission [ccm/m ² /day/atm]	Water Vapor Transmission [g/m ² /day/atm]	Deposition Process	Film Composition	Source
PET / Blank	12.00 µm	100	64.64			
PVDC	24.00 µm	8	0.3			/2 /
EVOH	24.00 µm	0.16 - 1.86 *	N/A			/2/
m-OPA	15.00 µm	30				
Aluminized PET ⁺ (single)	~ 30 nm	0.31 - 1.55	0.31 - 1.55	Evaporation	Al	/3, 4/
Aluminized PET ⁺⁺ (double)	~ 30 nm each	0.03	N/A	Evaporation	Al	/3, 4/
Aluminum on PE	7 µm Al	0.001	N/A	Laminated	Al	
SiO _x on PET	10 - 80 nm	0.35 - 10**	0.46 - 1.24	Evaporation	SiO _x	/5, 6, 7, 8/
SiO _x on PET	10 - 80 nm	0.08 - 1.55	0.5 - 5.0	PECVD	SiO _x	/9, 10, 11/
Al ₂ O ₃ on PET	20 nm	1.5	5.0	Reactive Evaporation	Al ₂ O ₃	/2, 3/
Al ₂ O ₃ /SiO _x on PET	50 nm	2.0 - 3.0	1.0	Evaporation	Al ₂ O ₃ / SiO _x	/13
Diamond-like Carbon on PET	20 nm	2	1.50	PECVD	Diamond	/14/
* depending on relative humidity and ethylene content ** depending on used process +: Al 30 nm / PET 12 µm ++: Al 30 nm / PET 12 µm / AL 30 nm						

Surprisingly the barrier values listed in Table 2 are actually a lot worse than one would expect from the diffusion values of oxygen or water in these coating materials (*see above!*). Therefore there must be other mechanisms or effects that reduce the barrier property of the thin film coatings.

To calculate the success of a barrier coating, a factor is introduced at this point, which will be called the **Barrier Improvement Factor**, or **BIF**, which is defined by the expression:

$$BIF = \frac{P_{\text{Substrate}}}{P_{\text{(Substrate+Barrier)}}} \quad [\text{Eq. 7}]$$

and describes the improvement in barrier property achieved by applying a specific barrier layer. The impact, which the substrate's properties (thickness and permeability) as well as the barrier's properties (defect ratio and permeation) have on the BIF, can be shown using different models (*Figure 1*). None of the models shown below is valid alone, but supplement each other. Furthermore, they don't cover all aspects of permeation through thin defective or defect-free layers, but they do give some insight into what needs to be achieved in order to obtain a barrier improvement of the film structure. The three models covered here are the

- Electrical Analogy Model
- Coverage Model
- Pinhole Model

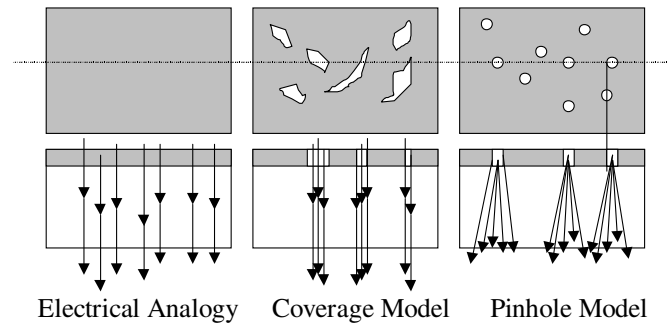


Figure 1: Schematic of Permeation Models

Electrical Analogy Model

The electrical analogy model assumes that permeation through different layers is like electrical current passing through a series of resistors (*Figure 2*).

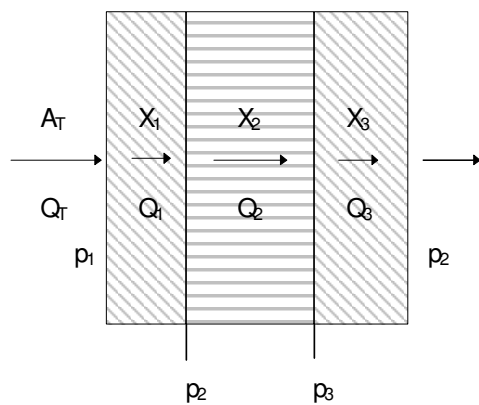


Figure 2: Electrical Analogy Model

The gas volume Q_T permeating through an area $A_T=A_1=A_2=A_3$ of a multilayer structure with the total thickness $d_T = d_1 + d_2 + d_3$ is constant in each layer, so that

$$Q_T = Q_1 = Q_2 = Q_3 \quad [\text{Eq. 8}]$$

Substituting in [Eq. 5] leads to

$$Q = \frac{P^*}{d} \cdot A \cdot (\Delta p) \quad [\text{Eq. 9}]$$

Or:

$$Q_T = \frac{P_1^*}{d_1} \cdot A_T \cdot (p_1 - p_4) = \frac{P_1^*}{d_1} \cdot A_1 \cdot (p_1 - p_2) = \quad [\text{Eq. 10}]$$
$$\frac{P_2^*}{d_2} \cdot A_2 \cdot (p_2 - p_3) = \frac{P_3^*}{d_3} \cdot A_3 \cdot (p_3 - p_4)$$

Rearranging eq. [Eq. 10] and with $(p_1-p_4)=(p_1-p_2)+(p_2-p_3)+(p_3-p_4)$ yields

$$\frac{Q_T d_T}{A_T P_T^*} = \frac{Q_T}{A_T} \left[\frac{d_1}{P_1^*} + \frac{d_2}{P_2^*} + \frac{d_3}{P_3^*} \right] \quad [\text{Eq. 11}]$$

Therefore;

$$\frac{d_T}{P_T^*} = \frac{d_1}{P_1^*} + \frac{d_2}{P_2^*} + \frac{d_3}{P_3^*} \quad \text{or} \quad \frac{1}{P_T} = \frac{1}{P_1} + \frac{1}{P_2} + \frac{1}{P_3} \quad [\text{Eq. 12}]$$

Since the diffusion through a multilayer structure can be calculated like resistors in series (when $\frac{1}{P}$ reflects the resistance of the particular layer), this model is called the “Electrical Analogy Model” and is commonly referred to as the ideal laminate. Assuming the thin film barrier layer is free of defects the BIF can be expressed by:

$$\text{BIF} = 1 + \frac{P_{\text{Substrate}}}{P_{\text{Barrier}}} \quad [\text{Eq. 13}]$$

and therefore:

$$\text{BIF} = 1 + \frac{P_{\text{Substrate}}^*}{d_{\text{Substrate}} P_{\text{Barrier}}} \quad [\text{Eq. 14}]$$

Coverage Model

If the thin film barrier material can be considered to have a perfect barrier, that is, it is impermeable, then any permeation through the barrier layer must take place through defects. The Coverage Model assumes that the permeation is directly proportional to the defect area ratio (Θ) exposing the area fraction of the uncovered polymer. It is furthermore assumed that the diffusion through the substrate polymer is unidirectional or totally attributed to the bare polymer surface. Thus, the permeation P through the structure is

$$P = \Theta \frac{P_{\text{Substrate}}^*}{d_{\text{Substrate}}} \quad [\text{Eq. 15}]$$

[Eq. 15] in [Eq. 5] shows, that the BIF for a polymer is independent of the thickness d and inverse proportional to the defect area ratio:

$$\text{BIF} = \frac{1}{\Theta} \quad [\text{Eq. 16}]$$

Although the unidirectional diffusion seems to be unlikely, the coverage model is valid for the extreme cases that either the thickness of the substrate is very small (thus the spreading can be neglected) or the coverage is very small $[(1 - \Theta) \ll 1]$. The Metallized Pinhole Model, outlined below, highlights the restricted application of the coverage model in an analytical way.

Metallized Pinhole Model (Pinhole Model)

The unidirectional diffusion assumed in the coverage model is an idealization that does not match reality. It is more realistic to assume that the gas diffusing through a hole will spread out in all directions in the polymer. In 1958, PRINS and HERMANS /16/ proposed a model for the gas diffusion through pinholes in metallized polymers.

For this model the following assumptions are made:

- No permeation through the thin film barrier material
- Permeation is through the defects only
- All the defects are circular and can be described by their mean radius r_0 and their number per unit area n
- All defects are sufficiently far apart from each other to assume additivity

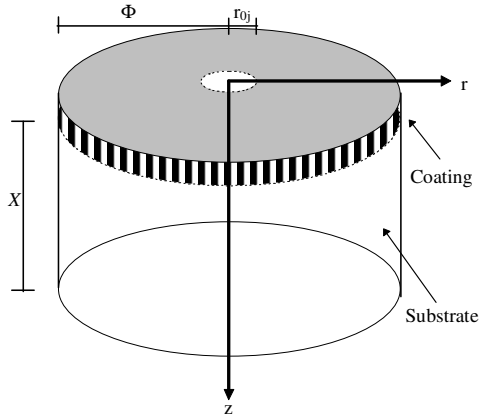


Figure 3: Pinhole Permeation Model

The defect area ratio is then defined by:

$$\Theta = \frac{r_0^2}{\Phi} \quad [\text{Eq. 17}]$$

In a cylindrical coordinate system (see Figure 3), the steady state diffusion equation to solve is the following:

$$\frac{\partial^2 c}{\partial r^2} + \frac{1}{r} \frac{\partial c}{\partial r} + \frac{\partial^2 c}{\partial z^2} = 0 \quad [\text{Eq. 18}]$$

Applying the following boundary conditions:

$$\begin{aligned}
 c(r,0) &= c_2, \quad r \leq r_0 && \text{Oxygen concentration at the surface inside the pinhole is equal to the solubility of oxygen in the polymer at 1 atm} \\
 \frac{\partial c}{\partial z} \Big|_{z=0} &= 0, \quad r > r_0 && \text{No permeation directly under the coating} \\
 c(r, X) &= c_1, \quad \forall r && \text{Zero oxygen on the opposite surface,}
 \end{aligned}$$

the following approximation for the oxygen flux has been found¹

$$J = D \cdot \frac{(c_1 - c_2)}{d} \cdot \Theta \cdot (1 + 1.18\lambda) = D \cdot S \cdot \Theta \cdot (1 + 1.18\lambda) \quad [\text{Eq. 19}]$$

When $\Theta \ll 1$ and $d > 0.3 r_0$, λ is the ratio between substrate thickness and pinhole diameter, thus:

$$\lambda = \frac{d}{r_0} \quad [\text{Eq. 20}]$$

and hence [Eq. 19] is valid for $\lambda > 0.3$. The Permeation P then is given by:

$$P = \frac{J}{\Delta p} = \frac{P_{\text{Substrate}}^*}{d} \cdot \Theta \cdot (1 + 1.18\lambda) \quad [\text{Eq. 21}]$$

and the BIF is:

$$\text{BIF} = \frac{1}{\Theta(1 + 1.18\lambda)} \quad [\text{Eq. 22}]$$

When the film thickness is small compared to the pinhole radius ($\lambda \ll 1$), [Eq. 22] goes over into the trivial solution

$$P = \frac{P_{\text{Substrate}}^*}{d} \cdot \Theta \Rightarrow \text{BIF} = \frac{1}{\Theta} \quad [\text{Eq. 23}]$$

proving the validity of the coverage model for small substrate thickness.

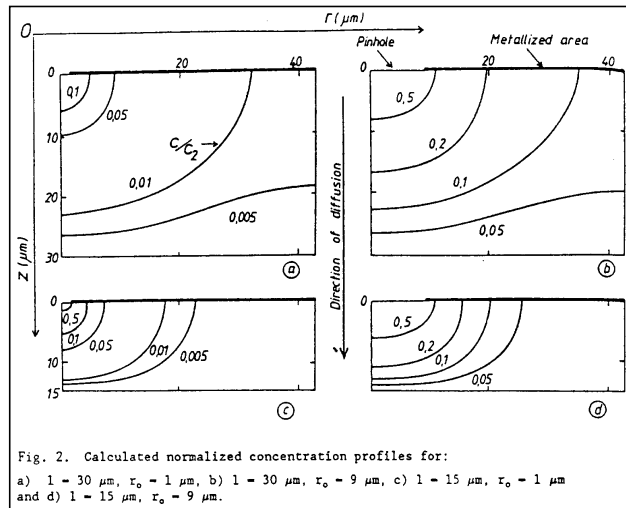
It is interesting to also look at the inverse case, where only small islands of coating cover the surface $(1 - \Theta) \ll 1$. In this case the boundary condition changes to:

$$\begin{aligned}
 c(r,0) &= c_2, \quad r > r_0 && \text{Oxygen concentration at the surface outside the island is equal to the solubility of oxygen in the polymer at 1 atm} \\
 \frac{\partial c}{\partial z} \Big|_{z=0} &= 0, \quad r > r_0 && \text{No permeation directly under the coating} \\
 c(r, X) &= c_1, \quad \forall r && \text{Zero oxygen on the opposite surface}
 \end{aligned}$$

Here again for small substrate thickness there is the same trivial solution given in [Eq. 23]. It is interesting that the same result is obtained for $\lambda \gg 1$, indicating that small amounts of coverage don't have an influence on the permeation at all.

¹ The exact mathematical derivation can be found in Reference 16

Figure 4: FEM Simulation for Diffusion through Pinholes



BEU and MERCEA¹⁷ performed FEM analysis (see Figure 4) on metallized polymers, using the pinhole model. The solutions show that with increasing polymer thickness, the gas concentrations at the outer radial boundaries ($r=\Phi$) increase as well, causing higher diffusion over these boundaries. If the density of holes increases so the pinhole areas touch each other and the concentration gradient over the radial border becomes equal to zero and diffusion does not take place. At small thickness the radial diffusion becomes more and more neglectable.

Besides these limitations of the Pinhole Model it is interesting to have a look on the practicable outcome, *i.e.*, what coverage actually is necessary for a specific polymer to even see the barrier effects from the coating. Figure 5 shows the influence of the pinhole diameter on the BIF with changing defect areas on a 12 μm PET substrate. In addition a curve for the Coverage Model is also drawn. It is seen that the smaller the hole diameter becomes, the smaller the BIF will be at a given defect ratio. This is explainable through the larger number of holes required to get a certain defect ratio and by that again the better spreading of gas in the polymer. But once again, one has to obey the limitations of the model.

Within the limitations of the model it appears that coverage higher than 95% is necessary to begin to see some improvement on the barrier function of the system. This high value explains the humongous impact small pinhole defects in barrier coatings have on the overall performance of the barrier coating. If, in addition, the barrier material does allow permeation, as assumed in the Electrical Analogy Model, then the required coverage will increase even higher!

INTERFACE PHENOMENA

Although the equations of the Pinhole Model have been widely accepted and repeated²⁶, they do not represent the whole wisdom about barrier functions. In fact, there have been reports about interesting phenomena, which contradict parts of these equations. Examples of these disagreements are:

- Metallization of Polymer Films with varying degrees of pre-treatment shows different Barrier Properties, even though optical density and defect count appear to be the same.
- Metallization of Polymer Films with Electron Beam Evaporation vs. Resistive Evaporation or Free Span vs. Supported Web yields different Barrier Properties, even though the material properties such as the optical density and defect density of the coating are the same.
- Metallized Polymer Films laminated with either adhesive coatings or extrusion lamination show different Barrier Properties, to the extent that sometimes the barrier after lamination is higher than the barrier properties of the single layers would suggest.

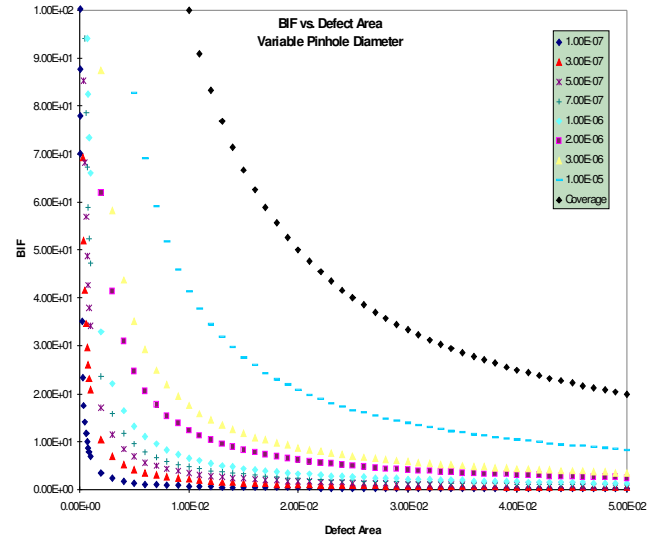


Figure 5: Influence of Pinhole Diameter and Coverage on the BIF for 12 μm Substrate

In all three cases there is a relation to the surface of the substrate and the coating, which raises questions about the impact of the substrate before, during, and after the coating. It is therefore the mating plane, or the interface, between substrate and coating, which has substantial impact on the barrier performance of the coating. It has to be mentioned though, that the physical and chemical effects taking place at the interface are far too complex to describe completely within the limited scope of this paper, so only the very basic aspects will be covered. Much more information can be found within the referenced literature. Some of the aspects described here have not been completely analyzed or examined and should be investigated to complete our understanding on how to make the perfect barrier.

Pre-Treatment: How the Surface Prior to Coating Impacts the Outcome

In a quote attributed to the great physicist Linus Pauling, it was said: “God created Matter, but it was the Devil who created the surface!” There is a great deal of truth in this statement, as that in thin film technology the properties of the surface immediately before the deposition of any coating has tremendous impact on the properties of this coating. Coating polymer films presents a particular challenge; since the chemistry of polymers is so complex that even minor process modification can significantly alter the surface chemistry and topography of the substrate. For Barrier Coatings the goal must be to create a surface, which allows the formation of a perfect coating; the problem is, however, that most polymers have intrinsic physical and chemical properties that go contraire to the requirements.

Polymers like polypropylene, possibly the most common plastic film metallized for barrier applications, exhibit a surface energy as low as 32 dyne. Polypropylene is chemically rather inert, and as a homopolymer is not metallizable. The saturated Carbon-Hydrogen chain does not give the aluminum or any other metal any locus to chemically connect with the surface. Metallization in such a case only happens based upon loose physical connection. For this reason most polymers are pre-treated. Treatment of polymer surfaces has been described in numerous publications^{18–22}. The most common methods are:

- Corona Treatment
- Flame Treatment
- Inline Plasma Treatment

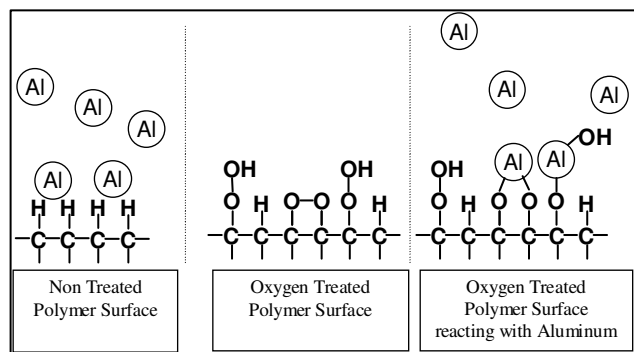
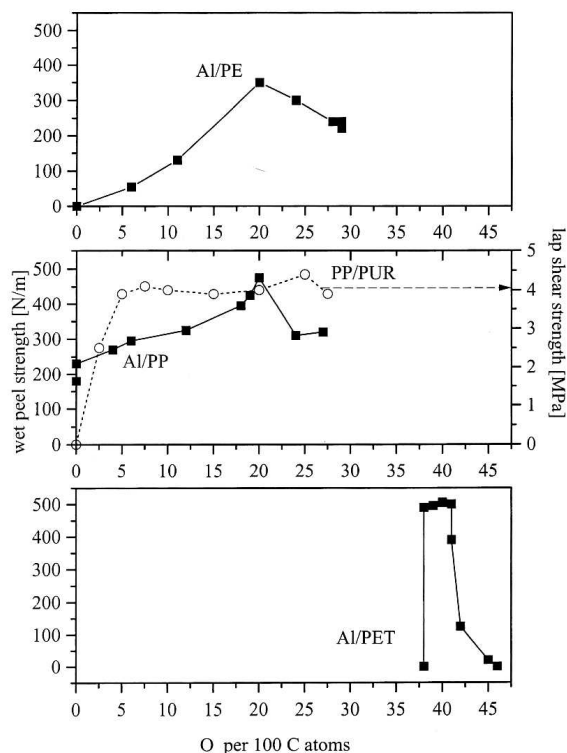


Figure 6: Non-Treated and Treated Polymer Surface at Metallization

All these methods expose the polymer surface to activated, ionized gases. Pretreatment causes three reactions on the polymer: Chain Scissioning, Crosslinking, and Grafting. The latter is the most important, since it inserts reactive chemical groups onto the polymer. Typical is the oxidation of the polymer surface, which creates C=O, C-O-O-C, C-O-O-H or C-O-H groups on the surface.

Figure 6 shows schematically how such a surface looks without and with treatment. The obvious advantage of pretreatment is that now the polymer exhibits a reactive surface with which the metal can react and form chemical bonds. Friedrichs *et al.*²⁴ analyzed plasma-modified polymer surfaces after metallization and did find that the deposited metal indeed reacts with the activated surface creating metal-organic bonds. This is also shown through the higher adhesion between the metal layer and the polymer surface after treatment with oxygen. (See Figure 7. The investigation was aimed at improvement of adhesion of metal layers on the polymer surface, so what is the impact on the barrier property?

Figure 7: Correlation Between O₂ Plasma-induced Oxygen Uptake of PE, PP and PET Surfaces as Measured



by XPS and Peel Strengths of Al-PE, Al-PP, and Al-PET²³.

The deposited metal layer starts initially growing at condensation points, also known as nuclei. These sites are loci with surface energy high enough to allow the metal species arriving in the vapor cloud to discharge its energy and to connect with the surface. A treated surface, however, offers enough nucleation sites to start the condensation of the metal. The left half of Figure 8 depicts the nucleation phase of silver deposition on non-treated (*outmost left*) and treated (second from left) polymer surfaces. Obviously the treated surface (*right half of Figure 8*) shows a much higher density of nucleation points. During the coating process, the metal layer grows into a dense structure at a lower thickness (*outer right*), while at the same average thickness the non-treated material still exhibits voids.

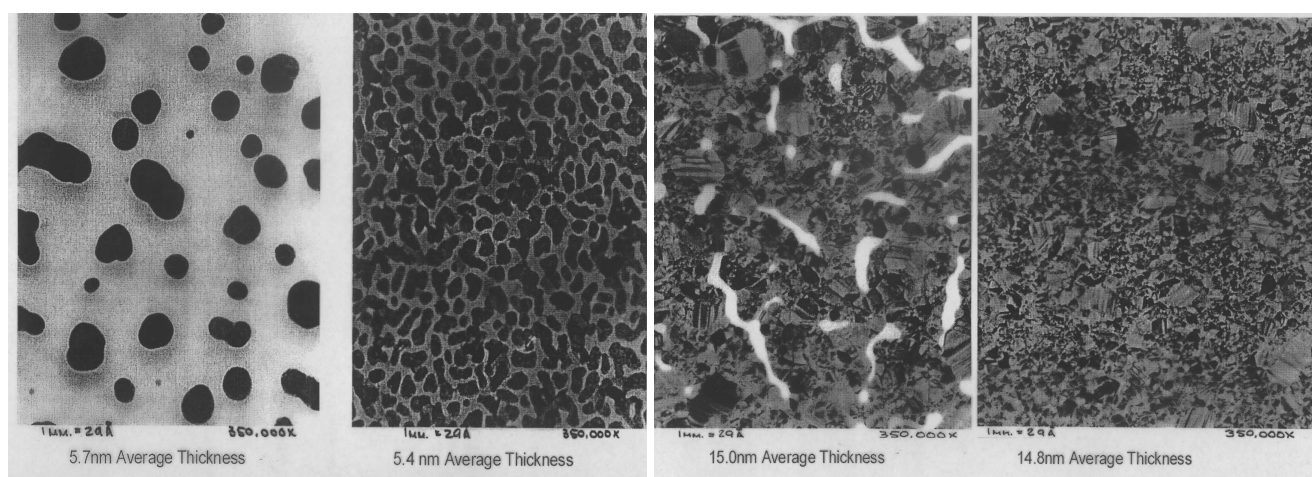


Figure 8 : Nucleation and Layer Growth on Non-Treated and Treated Polymer Surfaces /25/

Figure 9: Impact of Nucleation Density on Barrier Property

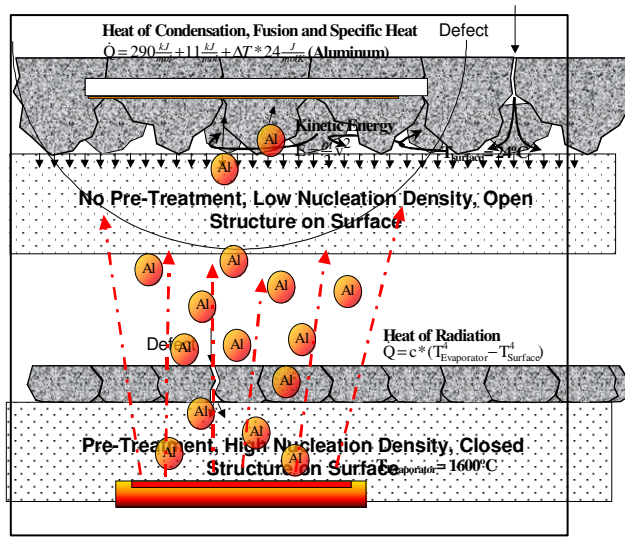


Figure 9 schematically depicts, why the higher nucleation density creates the better barrier. With low density there are voids left on the immediate surface, which allows gas permeating through a crack to have a relatively broad area where it can diffuse into the polymer surface. With high nucleation density, a better surface coverage is achieved; gases permeating through cracks basically have only the narrow area between neighboring metal grains in which to diffuse into the polymer.

Another aspect of the surface that one has to be aware of is its morphology. A metallizable surface should be as smooth as possible, and that is for two reasons:

1. The higher the surface roughness the larger the effective surface area becomes. The effective surface area is the area exposed to the environment, and that means for rough surfaces, gases have a larger area available to penetrate into the polymer.
2. With the exception of the Sputter and PA-CVD processes, the coating processes are “line-of-sight” coating. This means that high surface roughness increases the probability of having non-coated areas, specifically in grooves and crevices.

Mode of Operation: How the Different Coating Methods Affect the Coating

The coating techniques employed to apply the barrier layer can have an impact on the barrier property of the final coating as well. Commonly used processes for the deposition of barrier metals or oxides are:

- Resistive Evaporation of Pure Metals
- Resistive Evaporation of Metals with Plasma-Enhanced Oxidation
- Electron-Beam Evaporation of Metals and Oxides
- Sputter-Deposition of Metals and Oxides, Reactive and Non-Reactive
- Plasma-Enhanced Chemical Vapor Deposition

Depending on the coating process used there are different sources of energy that work on the substrate surface:

- Radiation:
 - Heat Radiation from the Evaporator Boats or the Crucible
 - Electron Radiation from Reflected Electrons During E-Beam Evaporation
 - Electron-, Ion-, and UV-Radiation from Plasma During Sputtering or Plasma-Enhanced Processes
- Heat of Condensation and Fusion as Well as Specific Heat from Material Deposited (*See Figure 10.*)
- Kinetic Energy of Material Impacting the Surface, Specifically During Sputtering.

Figure 10: Sources of Heat Impacting the Substrate Surface During the Deposition Process in Evaporative Coating

The energy input into the web surface during metallizing a 50nm layer of aluminum from heat of evaporation, fusion and cooling alone is about $1.70 \frac{kJ}{m^2}$, which is about the same energy used for heavy plasma pretreatment. This heat does not include the heat transferred by radiation. It is therefore to expect that the coating process itself does have an impact on the substrate material. In another case it has been reported /23/, that PET being coated in an E-Beam evaporator exhibited oxygen transmission rates which were about 10 times higher than the base material would allow. So what exactly can happen on the surface:

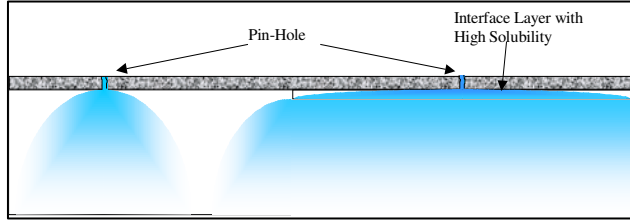


Figure 11: Change of Barrier Function Through Degradation of Interface Layer

The impact of the deposition energy can actually change or alter the chemical and physical composition of the interface layer. The alteration can create a layer with either much higher solubility for the specific gas or with lower solubility. Since the solubility becomes higher the lower the degree of crystallization in the polymer is, chances are, that the energy impact will degrade the crystalline structure, thus leading to a higher solubility in the interface layer. This of course leads to a faster radial diffusion under the pinholes, and with that, the barrier value decreases.

The higher kinetic energy of the depositing material can have a positive impact on the barrier property. Erlat *et al.*²⁷ indicate in their analysis of materials deposited by different processes, that sputter coating creates better barrier layers. They found that the higher kinetic energy in the deposited particles creates a denser layer, reducing the number of micro- and meso-defects, which contribute to the malfunction of the barrier coating. This thesis is supported by the work of Schiller *et al.*²⁸

Another important aspect, specifically in conjunction with plasma, plasma treatment and electron-beam evaporation is the creation of surface charges. Especially with biased plasmas and the electron beam, it is possible to create surface charges that exceed the breakdown voltage of the polymer. If this happens, a discharge through the polymer occurs and holes – sometimes microscopic in size – in the polymer result.

Barriers after Lamination

According to the Pinhole Theory the barrier properties of the barrier-coated film should increase by the same function as the single-layer sheet:

$$J^{-1} = \left(D_1 \cdot S_1 \cdot \Theta \cdot \left(1 + 1.18 \frac{d_1}{r_0} \right) \right)^{-1} + \left(D_2 \cdot S_2 \cdot \Theta \cdot \left(1 + 1.18 \frac{d_2}{r_0} \right) \right)^{-1} \quad [\text{Eq. 24}]$$

where Index 1 defines the properties of the base substrate and Index 2 the properties of the sheet laminated on top of the barrier layer. If base substrate and laminate sheet are identical, the diffusion through the laminate would therefore be cut in half.

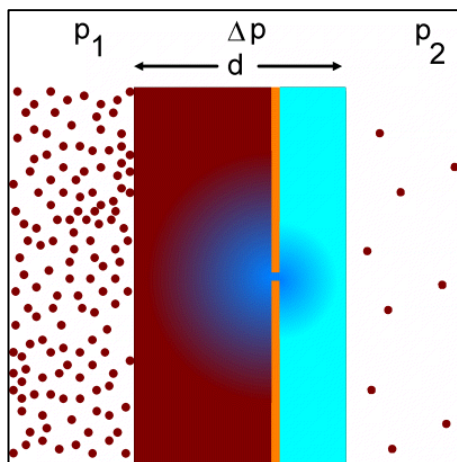


Figure 12: Diffusion Through a Laminated Barrier Construction /28/

Why, however, are the results varying: Sometimes the barrier property does not improve at all, in other cases, the barrier again is better than the sum of the single layers would suggest. In the first case, the answer can sometimes be straightforward: Handling damage caused by stretching, scratching, touching with dirty rollers in the converting equipment can damage the thin barrier layer to a degree that the initial barrier on the base substrate has decreased dramatically. In such cases the laminate does not improve the overall barrier, it rather recovers – sometimes barely – the damage inflicted on the base barrier layer.

In other cases however, that surface chemistry can take effect again. Hot lamination or lamination with chemically very reactive adhesives can create metal-organic interfaces, which can positively impact the overall barrier property. Research in this area has not progressed very far at this point and more investigation should be promoted in this area.

CONCLUSION

The function of Barrier Coating is a very complex matter, witnessed by the enormous amounts of publications and the ongoing research in this area. The basic principles however, as described in this paper, should enable the converter to develop the required understanding of the impact of his procedures on the properties of his product. At the same time, it is obvious that there is still a big piece missing, specifically, when it comes to the change in barrier behavior during subsequent converting procedures. For this, additional research, particularly in the area of the interface phenomena, is required and seems to have drawn the interest of several scientists working in this field.

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Section 4: Useful Relationships for Metallized Films

Introduction

This section provides the tables and equations needed to carry out various practical conversions that are often required within the metallizing industry. This includes converting the density of materials into film area, the roll diameter into roll length or weight, as well as providing a method of costing the film per unit area from a cost per unit weight.

4.1 Typical Area Yield for Various Films of Various Thicknesses

Figure 4.1-1 Typical Area Yield for Various Films of Various Thicknesses
(Courtesy of Camvac International Inc.)

GAUGE	TYPE	YIELD	GAUGE	TYPE	YIELD
		in ² /lb			in ² /lb
24	PP	128,800	120	PVC	16,985
28	PP	103,100	125	PE	24,000
32	PP	96,600	140	PVC	14,811
36	PP	85,700	150	PE	20,000
40	PP	77,300	170	PE	17,645
48	PP	64,400	175	PE	17,140
48	NYLON	51,000	200	PVC	10,560
55	PVC	40,000	200	PE	15,000
60	PVC	35,000	200	Tedlar	9,100
60	NYLON	40,800	280	PVC	7,400
70	ACRYLIC	33,000	500	PVC	4,100
70	PP	43,600	1000	Polystyrene	2,630
76	PP	40,700	40lb	Paper	10,800
80	PP	38,200	43lb	Paper	10,050
80	PVC	25,400	58lb	Paper	7,450
90	PP	34,000	60lb	Paper	7,200
100	PE	30,000	61lb	Paper	7,080
100	PVC	23,625			
100	PP	31,000			
100	NYLON	24,500			
100	PP Cap	30,400			

4.2 Typical Area Yield for Polyester Film of Various Thicknesses

Figure 4.2-1 Typical Area Yield for Polyester Film of Various Thicknesses
(Courtesy of Hoechst Celanese Corp.)

THICKNESS		YIELD			
Gauge	Microns	in ² /lb	ft ² /lb	yd ² /lb	m ² /kg
25	6.3	83,000	576.4	64.0	118.0
36	9.1	55,000	382.0	42.4	78.2
47	11.9	42,132	292.6	32.5	59.9
48	12.2	41,254	286.5	31.8	58.7
50	12.7	39,604	275.0	30.6	56.3
60	15.2	33,003	229.2	25.5	46.9
75	19.1	26,403	183.4	20.4	37.6
85	21.6	23,296	161.8	18.0	33.1
88	22.4	22,502	156.3	17.4	32.0
92	23.4	21,524	149.5	16.6	30.6
142	36.1	13,945	96.8	10.8	19.8
200	50.8	9,901	68.8	7.6	14.1
250	63.5	7,921	55.0	6.1	11.3
300	76.2	6,601	45.8	5.1	9.4
400	101.6	4,901	34.4	3.8	7.0
500	127.0	3,960	27.5	3.1	5.6
700	177.8	2,829	19.6	2.2	4.0
750	190.5	2,640	18.3	2.1	3.7
1000	254.0	1,980	13.7	1.5	2.8
1400	355.6	1,414	9.8	1.1	2.0

4.3 Formula for Calculating Length or Outer Diameter of a Roll of Polyester Film

Figure 4.3-1 Formula for Calculating Length or Outer Diameter of a Roll of Polyester Film
(Courtesy of ICI Americas Inc.)

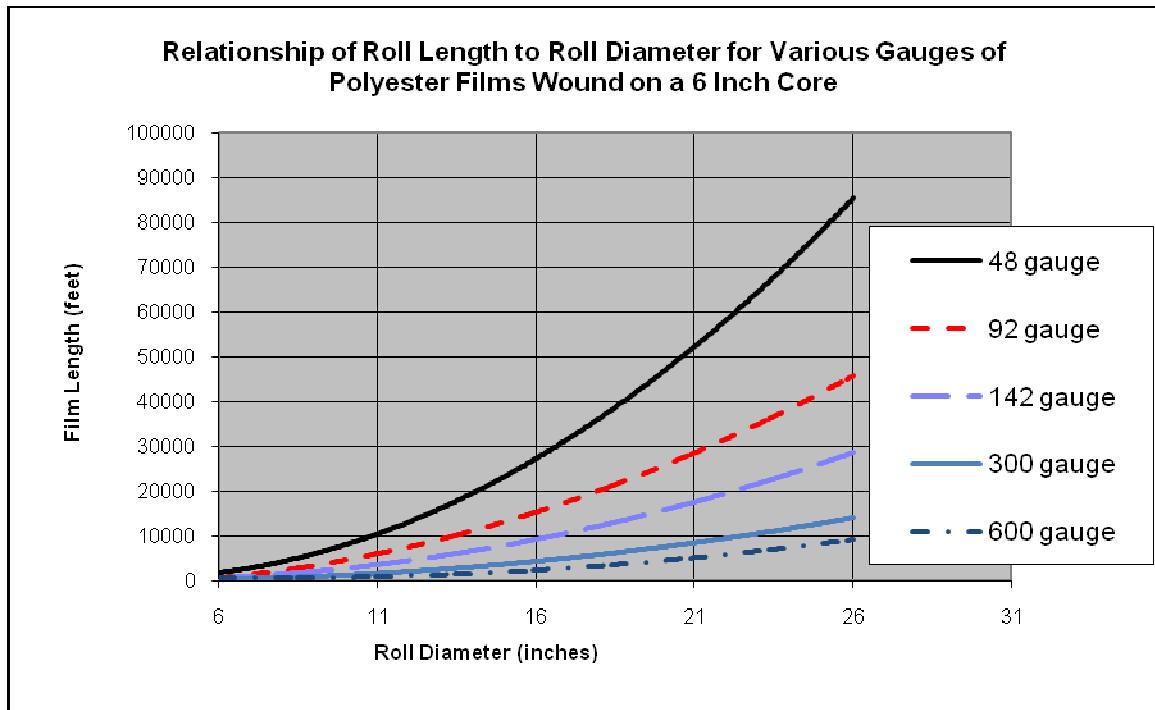
$$Roll, OD(in) = \sqrt{\left[\frac{(thickness(ga) + 2.58) * length(ft)}{6545} + [coreOD^2(in)^2] \right]}$$

$$Length(ft) = \frac{\left[rollOD^2(in)^2 - coreOD^2(in)^2 \right] x 6545}{thickness(ga) + 2.58}$$

If knurled film, substitute knurl height for 2.58. Use 2.58 for slit film without knurl.

4.4 Relationship of Roll Length to Roll Diameter for Various Thickness Polyester Films Wound on a 6-Inch Core

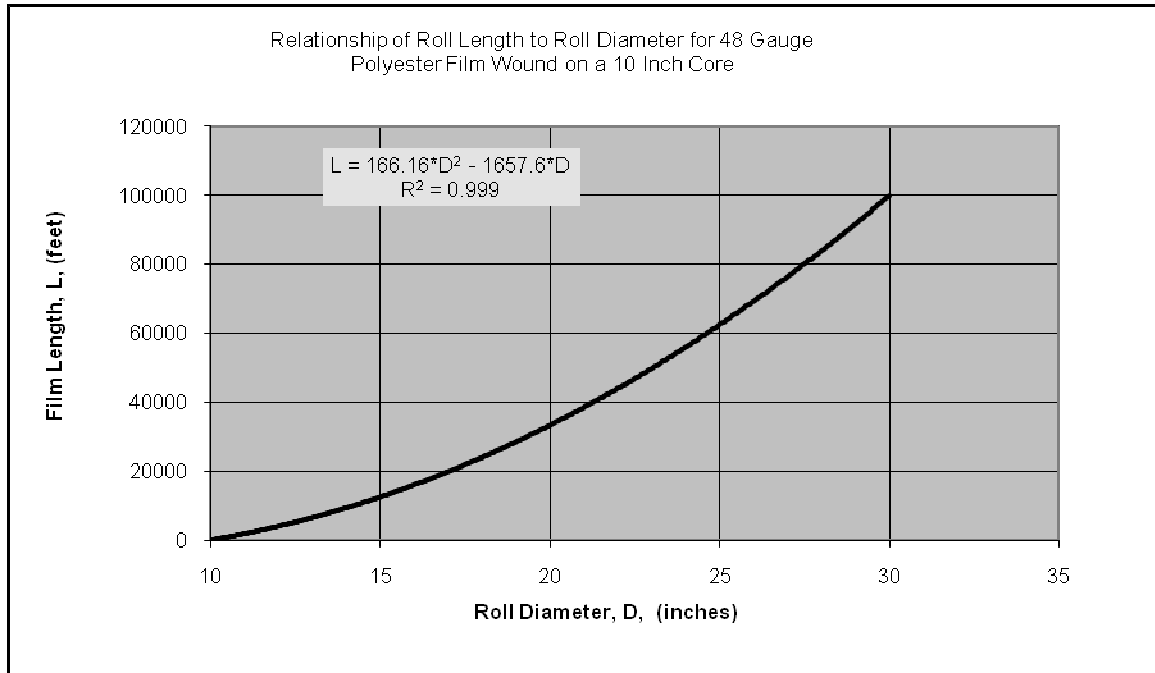
Figure 4.4-1 Relationship of Roll Length to Roll Diameter for Various Thickness Polyester Films Wound on a 6-Inch Core (Courtesy of ICI Americas, Inc)



Note: Graph represents measured results.

4.5 Relationship of Roll Length to Roll Diameter for 48-Gauge Polyester Films Wound on a 10-Inch Core

Figure 4.5-1 Relationship of Roll Length to Roll Diameter for 48-Gauge Polyester Films Wound on a 10-Inch Core (Courtesy of Camvac International, Inc.)



Note: Graph represents measured results.

4.6 Formula for Calculating Roll Weight of Polyester, Nylon, and Propylene Film

Figure 4.6-1 Formula for Calculating Roll Weight of Polyester, Nylon, and Propylene Film
(Courtesy of Hoechst Celanese Corp.)

$$\text{Roll Weight (pounds)} = \text{Th} \times \text{L} \times \text{W} \times \text{K}_1$$

Th = Thickness in inches

L = Length in feet

W = Width in inches

K₁ = 0.606 for polyester

0.489 for nylon

0.394r polypropylene

4.7 Formula for Calculating MSI (1,000 Square Inches) Cost of Polyester, Nylon, and Propylene Film

Figure 4.7-1 Formula for Calculating MSI (1,000 Square Inches) Cost of Polyester, Nylon, and Propylene Film (Courtesy of Hoechst Celanese Corp.)

$$\text{Cost (\$ / 1,000 in}^2\text{)} = \text{Price} \times \text{Th} \times \text{K}_2$$

Cost in \$/msi

Price in dollars per pound

Thickness in mils

$\text{K}_2 =$ 0.0504 for polyester
 0.0406 for nylon
 0.0327 for polypropylene

4.8 Factors for Converting Various Basis Weights of Metallized Paper

Figure 4.8-1 Factors for Converting Various Basis Weights of Metallized Paper
(Courtesy of Camvac International, Inc.)

To change from one basis weight to another, use the following conversion factors:

<u>Convert from</u>	<u>Convert to</u>	<u>Do</u>
Grams per square meter (gsm)	lb/25x38	Multiply by 0.675
lb/25x38	lb/24x36	Divide by 1.100
lb/25x38	msi/lb	Divide 475 by lb/25x38
lb/24x36	msi/lb	Divide 432 by lb/24x36

4.9 Formulas for Calculating Wound Roll Density Characteristics

Figure 4.9-1 Formulas for Calculating Wound Roll Density Characteristics

$$\text{Roll Density} = \frac{4}{\pi * W} \left(\frac{G_{total} - G_{core}}{D_{outside}^2 - D_{core}^2} \right)$$

where :

G = weight in lbs or Kgms

D = diameter in feet or meters

W = width of roll face

Example: A roll of polypropylene film is measured and weighed to calculate its density:

Roll width	= 410cm
Roll outside diameter	= {(79.15 + 78.46 + 78.42 + 78.42 + 78.57)/5} = 78.60cm
Core outside diameter	= 55cm
Weight of roll + core	= 915kg
Weight of core	= 100kg

$$\text{Roll Density} = \frac{4}{\pi * 4.1m} \left(\frac{915Kgms - 100Kgms}{(0.786m)_{roll}^2 - (0.55m)_{core}^2} \right)$$

$$\text{Roll Density} = 792.6Kgms / m^3$$

$$\text{Roll Density} = 0.793gms / cm^3$$

Because the wound roll will contain air, its density will be less than that of the polymer from which it is made. Therefore, the percentage air in the roll can be estimated as:

$$\% \text{ Air in Roll} = \left(\frac{\rho_{material} - \rho_{roll}}{\rho_{material}} \right) \times 100$$

$$\% \text{ Air in Roll} = \left(1 - \frac{\rho_{roll}}{\rho_{material}} \right) \times 100$$

where :

$\rho_{material}$ = density of polymer or film (lbs/ft³ or kgms/M³)

ρ_{roll} = density of roll (lbs/ft³ or kgms/M³)

$\frac{\rho_{roll}}{\rho_{material}} \equiv \text{Density index}$

For polypropylene film roll with Density of 792.6 Kgms/M³

$$\% \text{ Air in Roll} = \left(\frac{905 - 792.6}{905} \right) \times 100 = 12.4\% \text{ air}$$

$$\text{Density Index} = \left(\frac{792.6}{905} \right) = 0.876$$

Where :

905 Kgms/m³ is the density of polypropylene (Table 4.10)

4.10 Density for Various Polymer Films

Figure 4.10-1 Density for Various Polymer Films¹:

Polymer	Density (gm/cm ³)*	Density (lb/in ³)**
Polypropylene	0.90 - 0.91	0.03245 - 0.03285
Polyester (PET)	1.23 - 1.395	0.0444 - 0.0504
Nylon 6,6	1.1 - 1.14	0.0397 - 0.04115
Nylon 6	1.129	0.04076
High-density polyethylene	0.94 - 0.965	0.0339 - 0.0348
Linear-low-density Polyethylene ²	0.915 - 0.94	0.0330 - 0.0339
Low-density polyethylene	0.91 - 0.93	0.0328 - 0.0336
Polystyrene	1.05	0.0379
Polycarbonate	1.20	0.0433
Cellophane	1.40 - 1.55	0.0505 - 0.0560
Cellulose acetate	1.28 - 1.31	0.0462 - 0.0473
Ionomer	0.94	0.0339
Polyvinylidene chloride (Saran®)	1.64 - 1.71	0.0592 - 0.0617
Polyimide	1.42	0.0513
Flexible Polyvinyl Chloride	1.20 - 1.45	0.0433 - 0.0523
EVOH ³	1.14 - 1.20	0.0412 - 0.0433

* To convert from gm/cm³ to kg/M³, multiply by 1,000

** To convert gm/cm³ to lb/in³, divide by 27.7

** To convert gm/cm³ to lb/ft³, divide by 62.4298

4.11 Conversion Table for Commonly Used English/Metric Units of Measure

Figure 4.11-1 Conversion Table for Commonly Used English/Metric Units of Measure
(Courtesy of E.I. DuPont de Nemours & Co., Inc.)

<u>Abbreviation</u>	<u>START WITH:</u> <u>Description</u>	<u>MULTIPLY BY:</u>	<u>Abbreviation</u>	<u>TO OBTAIN:</u> <u>Description</u>
<u>AREA</u>				
in ²	square inch	6.452x10 ⁻⁴	m ²	meter ²
ft ²	square foot	9.290x10 ⁻²	m ²	meter ²
<u>ENERGY</u>				
Btu	British Thermal Unit	1.056x10 ³	J	joule
cal	calorie	4.187	J	joule
kw-hr	kilowatt-hour	3.600x10 ⁶	J	joule
<u>FLOW</u>				
gal/min	gallons/minute	6.309x10 ⁻⁵	m ³ /s	meter ³ /second
<u>HEAT CAPACITY</u>				
c (heat capacity)	Btu/lb • °F	4.187x10 ³	J/kg•K	joule/kilogram-Kelvin
<u>LENGTH</u>				
Å	angstrom	1.000x10 ⁻¹⁰	m	meter
mil	thousandths of inch	2.540x10 ⁻⁵	m	meter
in	inch	2.540x10 ⁻²	m	meter
ft	foot	3.048x10 ⁻¹	m	meter
yd	yard	9.144x10 ⁻¹	m	meter
<u>MASS</u>				
oz(mass)	ounce(mass)	2.835x10 ⁻²	kg	kilogram
lb(mass)	pound	4.536x10 ⁻¹	kg	kilogram
T	ton(short)	9.072x10 ²	kg	kilogram
<u>PERMEABILITY-GAS</u>				
cc/m ² • 24 hrs • atm	centimeter ³ / meter ² • 24 hrs • atmosphere	9.87	ml/m ² • Mpa • d	milliliter/meter ² mega pascal day
cc/100in ² • 24 hrs	cc/100in ² • 24 hrs	15.50	cc/m ² • d	cc/meter ² • day
<u>PERMEABILITY-WVTR</u>				
g/100in ² • 24 hrs	gram/100 inch ² • 24 hrs	15.50	g/m ² • d	gram/meter ² • day
<u>POWER</u>				
Btu/hr	Btu/hr	2.930x10 ⁻¹	W	watt
hp	horsepower	7.460x10 ²	W	watt
<u>PRESSURE</u>				
m•bar	millibar	0.760	mm Hg	millimeters mercury(Torr)
mm Hg	millimeters mercury (Torr)	1.316	m•bar	millibar
atm	atmosphere	1.013x10 ⁵	Pa	pascal
cm Hg	centimeter	1.333x10 ³	Pa	pascal
lbs/in ² (psi)	pounds per square inch	6.895x10 ³	Pa	pascal
<u>TEMPERATURE</u>				
°F	degrees Fahrenheit	.555 (°F – 32)	°C	degrees Celsius
°C	degrees Celsius	(1.8x°C) + 32	°F	degrees Fahrenheit
°C	degrees Celsius	°C + 273.15	K	Kelvin
°F	degrees Fahrenheit	<u>°F + 459.67</u> 1.8	K	Kelvin
<u>THERMAL CONDUCTIVITY</u>				
k (thermal conductivity)	Btu • in/sec • ft ² • °F	5.192x10 ²	W/m • K	watt/meter Kelvin
k (thermal conductivity)	Btu • in/hr • ft ² • °F	1.442x10 ⁻¹	W/m • K	watt/meter Kelvin
<u>VOLUME</u>				
in ³	cubic inch	1.639x10 ⁻⁵	m ³	meter ³
ft ³	cubic foot	2.832x10 ⁻²	m ³	meter ³
gal	gallon (U.S. liquid)	3.785x10 ⁻³	m ³	meter ³
<u>YIELD</u>				
in ² /lb	inches ² /pound	1.442x10 ⁻³	m ² /kg	meter ² /kilogram

Note: These are basic units, i.e., meter. Make proper adjustments (kilo, milli, centi, etc.) so that the final answer is in the form you need from .01 to 1,000.

4.12 Section 4 References Cited

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Section 5: Metal Evaporation and Heated Boat Data

Introduction

This section provides information about metal evaporation and evaporation boats. This includes influences on boat life and suggestions for maximizing boat life expectancy, a discussion of feed rates, and calculations related to evaporation rates.

5.1 Some Factors Influencing Boat Life

ELECTRICAL CONTACT
INITIAL HEATING CYCLE
INITIAL WETTING OF THE CAVITY WITH ALUMINUM
OPERATING TEMPERATURE
UTILIZATION OF THE CAVITY AREA/CLEANING
VACUUM LEVEL
NUMBER OF THERMAL CYCLES
CORRECT RESISTIVITY
COOLING OF BOATS
BOAT DEFECTS
EVAPORATION RATE
BOAT COMPOSITION

5.2 Recommendations for the Use of Evaporation Boats

To assure optimum boat performance and to maximize boat life, the following procedures should be adhered to:

1. In order to assure adequate electrical contact during thermal cycling, both ends of each boat should be covered with flexible graphite foil and firmly pressed into place in clean copper clamps. A prerequisite for uniform electrical contact is parallel surfaces between the clamps and the boats.
2. The aluminum feed wire should be positioned so that it meets the boat approximately 30-50% from the end of the cavity.
3. For the initial heat-up cycle, the temperature of the boats should be slowly raised to operating temperature during a period of 10 to 15 minutes. This procedure assures maximum boat life and performance. Subsequent heat-up cycles can be performed at whatever rate provides the best machine productivity.
4. When the desired operating temperature has been reached (approximately 1450C) and the boat cavity is uniformly wetted with molten aluminum, the coating process can begin. For optimum performance, care should be taken to operate each boat at the temperature at which a thin layer of molten aluminum covers the entire cavity during operation.
5. Upon completion of the coating cycle and after the wire feeder has been stopped, the boats should be kept under power until all residual aluminum on the cavity surface has evaporated.

Adherence to these five key procedures can significantly extend boat life and improve the quality of the metallized film.

5.3 Boat Temperature vs. Approximate Boat Life

Figure 5.3-1 Boat Temperature vs. Approximate Boat Life

Operating Temperature (° C)	Boat Life (Hours)
1500	X
1600	3/4 X
1700	1/2 X

5.4 Boat Resistivity (HT) and Approximate Service Life at Different Temperatures for Aluminum Evaporation

Figure 5.4-1 Boat Resistivity (HT) and Approximate Service Life at Different Temperatures for Aluminum Evaporation

Service Life Hours	Resistivity (moc)		
	1500C	1600C	1700C
0	3500	3700	3900
1	2500	2700	2900
2	2400	2600	2800
3	2350	2550	2750
4	2300	2500	2700
5	2300	2500	2700
6	2300	2500	2000
7	2300	2500	1000
8	2300	2500	0
9	2300	2500	
10	2300	2500	
11	2300	2000	
12	2300	1000	
13	2300	0	
14	2300		
15	1800		
16	1000		
17	0		

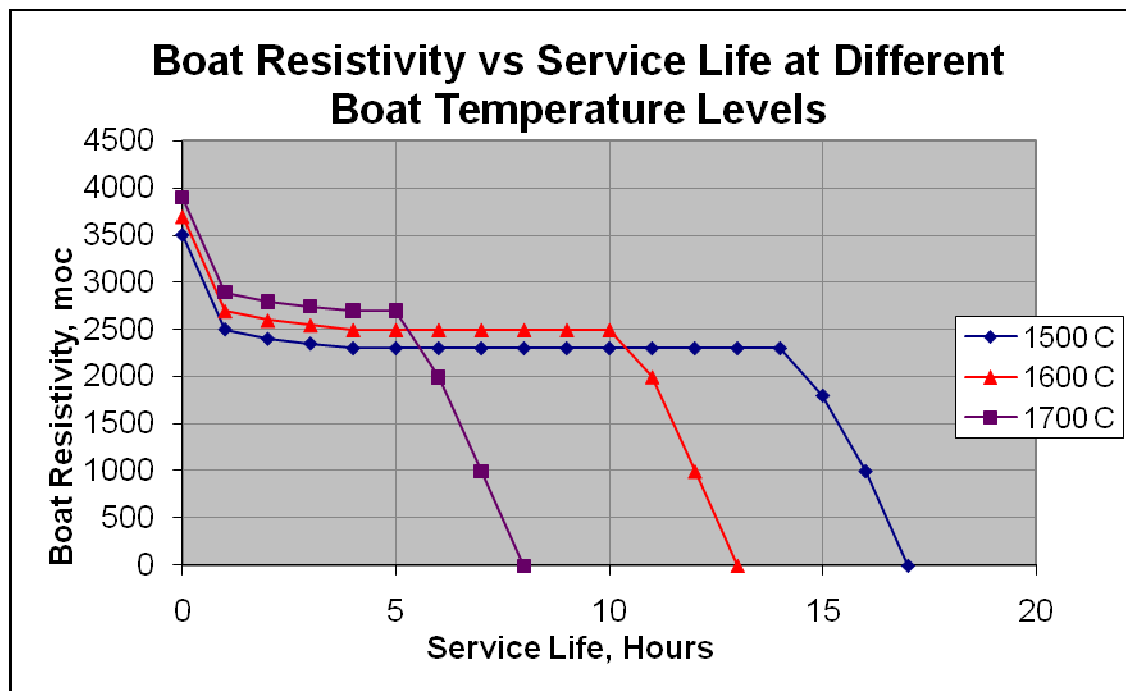
Boat size: 10 x 30 x 150mm

Cavity: 1.5 x 26 x 90mm

Resistivity at room temperature: 1000 moc

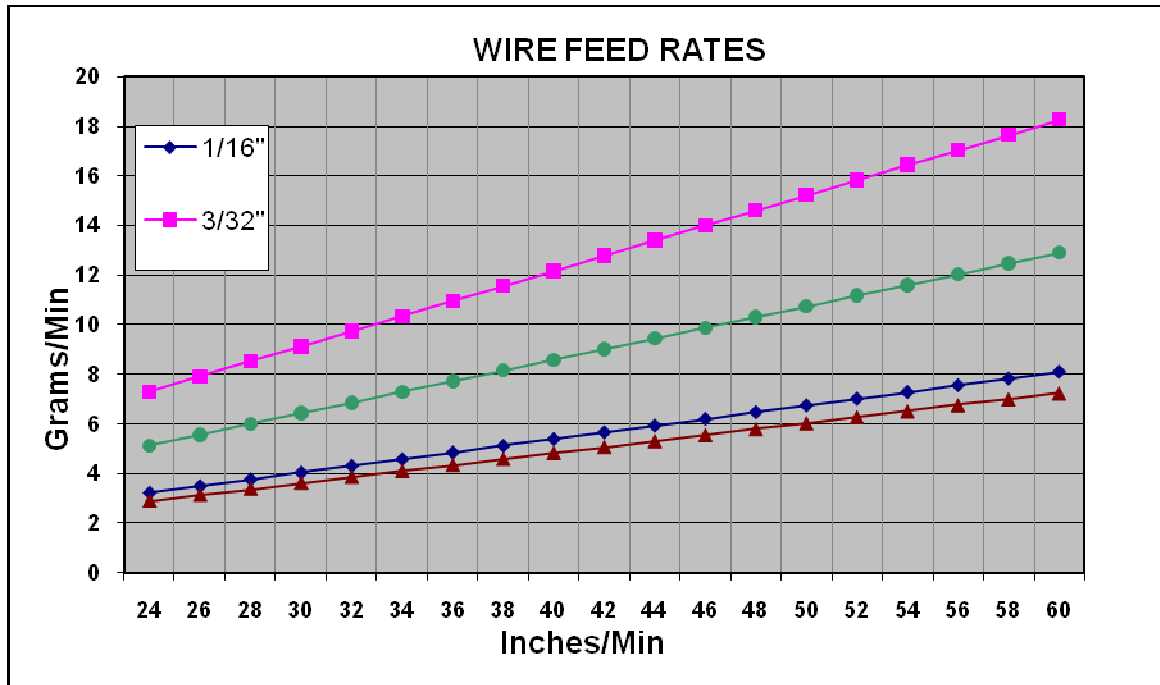
Wire- feed rate: 8 g/min of aluminum

Figure 5.4-2 Plot of Data in Table 5.4-1



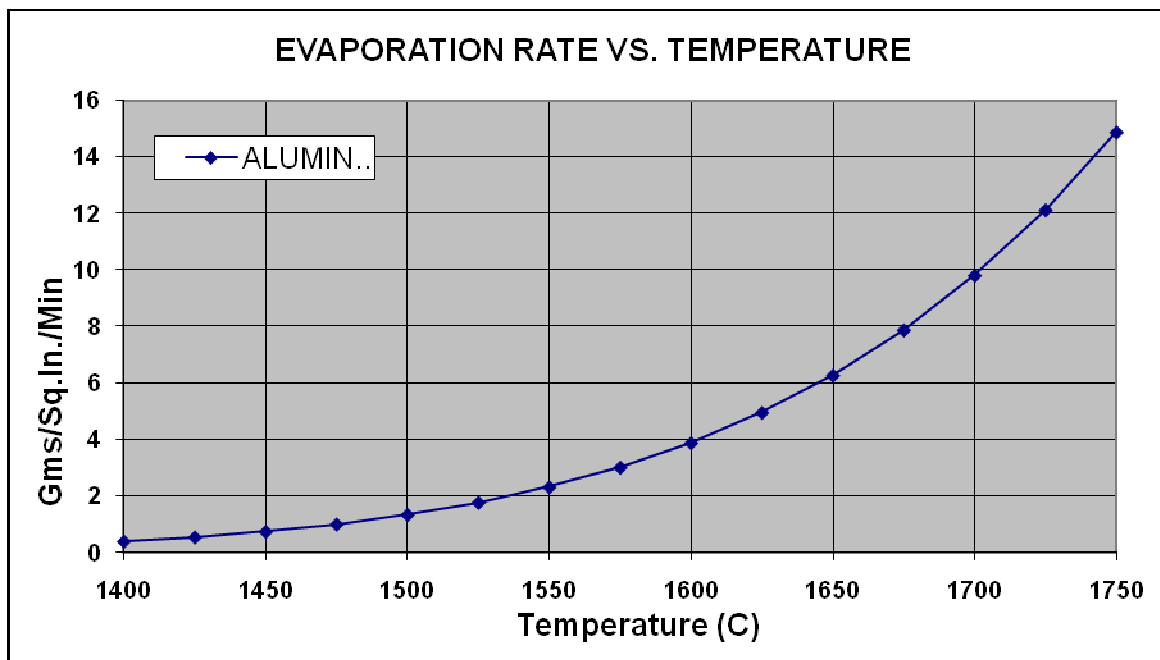
5.5 Aluminum Wire Feed Rates

Figure 5.5-1 Aluminum Wire Feed Rates (Courtesy of Advanced Ceramics Corp.)



5.6 Evaporation Rate vs. Temperature for Aluminum Metal

Figure 5.6-1 Evaporation Rate vs. Temperature for Aluminum Metal (Courtesy of Advanced Ceramics Corp.)



$$\text{Rate of Evaporation, gm/in}^2\text{/min} = 22.586 \times \text{Vapor Pressure} \times [\text{Molecular Weight/Temperature, } ^\circ\text{K}]^{1/2}$$

Note: Aluminum vapor pressure from *RCA Review*, June 1969, p 295, and molecular weight from *Principles of Engineering Materials*, Bartlett, Nix, Tietleman, pp 540-541.

5.7 Calculation of Evaporation Rates

A discussion of the evaporation rate of various materials can be found in texts dealing with the ideal gas laws^{1,2}. These discussions show that evaporation rates can be estimated from a consideration of molecular dynamics of saturated vapors. Several analyses of this sort exists and there are several sources of evaporation rate equations^{2,3} and the vapor pressure data needed for the calculation^{2,4}.

Theoretically, evaporation rates may be estimated by using the fact that for a saturated vapor, evaporation and condensation are occurring at the same rate. While it is difficult to calculate the evaporation rate, the rate of condensation may be calculated from the kinetic gas theory as the rate of molecules striking the surface of a solid or liquid and is given as Equation 5.7-1:

Equation 5.7-1

$$\alpha v = \frac{1}{4} n v_a = p / [2\pi RT]^{1/2}$$

Where:

- α = The condensation coefficient
- v = Rate of gas molecule impacts per unit area per unit time
- n = Number density of gas molecules
- v_a = Average molecular velocity
- p = Pressure
- R = Ideal gas constant
- T = Absolute temperature

For a vapor at equilibrium with a liquid, this is just equal to μ , the rate at which molecules evaporate from the surface, giving Equation 5.7-2

Equation 5.7-2

$$\mu = \alpha v = \frac{1}{4} n v_a = p / [2\pi RT]^{1/2}$$

The ideal gas considerations allow the calculation of the maximum possible rate of evaporation, which would occur if all the vapor is removed as it is formed such that there is no condensation¹. Dushman² makes use of Langmuir's⁵ observation that the rate of evaporation of Tungsten in high vacuum is the same as the rate of evaporation in the presence of a saturated vapor. Dushman² also makes use of Langmuir⁵, who showed that for metal atoms condensing on a metal, the value of α is equal to 1, to calculate the mass of gas or vapor incident on a unit area per unit time as:

Equation 5.7-3

$$G = mv = \frac{1}{4} \rho v_a$$

Where:

- ρ = ρ , the gas density
- m = Mass of molecules

This can be written in terms of vapor pressure and temperature as:

Equation 5.7-4

$$G = 5.833 \times 10^{-2} P_{\text{mm}} [M/T]^{1/2}$$

Where:

- G = gm/cm² sec
- P_{mm} = Vapor pressure in mm Hg or Torr
- T = Absolute temperature in degrees Kelvin
- M = Molecular weight

Equation 5.7-4 may be modified through its constant to allow use of different units for the pressure or to obtain different units for the rate of evaporation as used in the calculations of Figure 5.6-1 in terms of gm/in²/min.

For much of the tabulated vapor pressure data of metals, at pressures below 1 mm Hg, it has been observed that the vapor pressure may be represented as a function of temperature by Equation 5.7-5²:

Equation 5.7-5

$$\text{Log } P = A - B/T$$

Where, A and B are constants.

The equation for evaporation rate data G, may be rewritten and expressed in terms of the constants A and B through the vapor pressure as follows:

Equation 5.7-6

$$\text{Log } G = 5.7660 + 0.5 \log M + \log P_{\mu} - 0.5 \log T$$

Where:

P_{μ} is measured in microns (1 micron = 10^{-3} Torr)

Substituting in Equation 5.7-6 for $\log P_{\mu}$ and rearranging yields Equation 5.7-7:

Equation 5.7-7

$$\text{Log } G = C - 0.5 \log T - B/T$$

Noting that:

Equation 5.7-8

$$C = A + 0.5 \log M + 5.7660$$

Where:

P_{μ} measured in microns, (1 micron = 10^{-3} Torr)

Dushman² tabulates values of the constants for a wide range of metals for the pressure in microns (1 micron = 10^{-3} Torr). Figure 5.8-1 lists the constants for metals important in commercial metallization of films and papers. Much of the data in Dushman² is based upon the data compilation of Honig⁶, in the *RCA Review*, which has been widely cited, but difficult to locate. An additional source of vapor pressure data is that of Nesmeyanov⁴. Nesmeyanov reviews the vapor pressure literature data for a wide range of vapor pressure determinations for the elements. Nesmeyanov⁴ fits and tabulates the constants for vapor pressure data according to Equation 5.7-9:

Equation 5.7-9

$$\text{Log } P_{\text{mm}} = A - B/T + C \cdot T + D \cdot \log T$$

as opposed to Equation 5.7-5 of Dushman². The constants A, B, C and D are listed for solid and liquid condensed vapors. Nesmeyanov⁴ also tabulates boiling point as a function of pressure, heats of fusion and heats of vaporization for the elements.

5.8 Evaporation Rate G , gm/cm² sec, Constants for Various Metals

Figure 5.8-1 Evaporation Rate G , gm/cm² sec, Constants for Various Metals for Use in Equation 5.7-7 Taken from Dushman²

Metal	A, for P_μ in microns 1 micron = 10^{-3} Torr	B	C
Aluminum	11.79	15940	8.27
Copper	11.96	16980	8.63
Zinc	11.63	6560	8.30
Indium	11.23	12480	8.03
Tin	10.88	14870	7.69
Chromium	12.94	20000	9.56
Nickel	12.75	20960	9.4
Silver	11.85	14270	8.63
Gold	11.89	17580	8.80

5.9 Vapor Pressure Constants of Nesmeyanov for Various Elements Where the Condensed Phase of the Element Is a Solid

Figure 5.9-1 Vapor Pressure Constants of Nesmeyanov⁴ for Use in Equation 5.7-9 for Various Elements Where the Condensed Phase of the Element Is a Solid, Pressure in mm Hg

Metal	A	B	C	D
Aluminum	15.41687	16710.43	0.00198518	-1.81636
Copper	-0.70729	16869.44	-0.00091963	3.50147
Zinc	13.50846	7011.863	0.00040185	-1.58311
Indium	470.88627	26891.71	0.10315698	-179.48306
Tin	-150.16773	10870.18	-0.03682439	62.12498
Chromium	4.36192	19723.49	-0.00055955	2.03941
Nickel	-10.94190	16193.89	-0.00176813	7.29936
Silver	21.68194	15662.85	0.00069459	-4.01891
Gold	-24.40936	16439.69	-0.00239291	11.25366

5.10 Vapor Pressure Constants of Nesmeyanov for Various Elements Where the Condensed Phase of the Element Is a Liquid

Figure 5.10-1 Vapor Pressure Constants of Nesmeyanov⁴ for Use in Equation 5.7-9 for Various Elements Where the Condensed Phase of the Element Is a Liquid, Pressure in mm Hg

Metal	A	B	C	D
Aluminum	12.40857	15999.58	-0.00000352	-0.99926
Copper	-285.06133	-14786	-0.01263166	91.90151
Zinc	-7.91900	5330.735	-0.00161644	5.59972
Indium	10.82068	12298.11	0.00007147	-0.82159
Tin	-9.32188	14023.92	-0.00088122	5.62012
Chromium	165.22012	44507.12	0.00366345	-45.49926
Nickel	30.68264	19678.72	0.00038550	-6.08393
Silver	10.80805	14323.13	-0.00009019	-0.53927
Gold	-6.59202	16370.99	-0.00064886	4.79166

5.11 Section 5 References Cited

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Section 6: Aluminum Wire Chemistry, Temper, and Oxidation

Introduction

In the same way that the boat is a core technology that can significantly affect the process, so too is the quality and metallurgy of the aluminum used in the process. Impurities can increase the corrosion of the boats, shortening the boat lifetime, as well as being a source of spits from the boats that can damage the polymer substrate and pinhole the aluminum coating. The first part of this section covers the aluminum purity and temper details and follows with the chemical and physical properties of aluminum. The final part of the section covers the problematic issue of aluminum oxidation. The kinetics of oxidation gives rise to what is believed to be the mechanism of oxidation, which is reproduced here. Also included is a brief description covering how the aluminum oxidation affects the metallization process and why care is needed while cleaning any deposition shields because of the exothermic oxidation reaction.

6.1 Aluminum Wire Alloys

The primary concern for aluminum wire is the purity of aluminum versus the other alloying elements. All aluminum alloys with a first digit of 1 possess a minimum aluminum purity of 99%. The second digit of the series indicates any specific control used over the individual impurities in the particular alloy. The third and fourth digits are the same as the two digits to the right of the decimal point indicating minimum aluminum content required.

Alloy/Impurity	1100	1350	1188	1199
Silicon (Si)	0.95 Si+Fe	0.10	0.06	0.006
Iron (Fe)		0.4	0.06	0.006
Copper (Cu)	0.05-0.20	0.05	0.005	0.006
Manganese (Mn)	0.05	0.01	0.01	0.002
Magnesium (Mg)			0.01	0.006
Chromium (Cr)		0.01		
Zinc (Zn)	0.10	0.05	0.02	0.006
Titanium (Ti)			0.01	0.002
Others	0.15	0.10	0.01	0.003
Aluminum (AL)	99.00	99.50	99.88	99.99

All element values for aluminum wire are maximum values except the aluminum, which is a minimum. The above analysis is in conformance as specified by the Aluminum Association Standards and Data¹³.

Aluminum wire used in metallizing can be produced by cold rolling or drawing and, thus, supplied at different strengths and tempers. A soft temper or “O” temper is produced by annealing the wire after cold working it to its final dimensions. A hard temper of H18 is a result from strain hardening of about 75% (full hard). Intermediate tempers are produced by strain hardening annealed wire by an amount that increases tensile strength by one-quarter, one-half, or three-quarters the difference in strength between O and H18. These tempers are called H12, H14, and H16 respectively.

The wire used for feeding the evaporation boats can vary in composition, diameter, and hardness. There is a trade-off to all of these features. The wire needs to be thick to have a good volume-to-surface area ratio. All aluminum is easily oxidized in atmosphere, so to minimize the amount of oxide, it is preferable to maximize the volume-to-surface ratio. The limit to this is that the wire has to be flexible enough to be curved down to the boats. If the wire is thick it needs to have been softened more to enable it to be bent round and to enter the boat easily without causing too much erosion. The amount of processing can affect the thickness of the surface oxide present on the wire with the oxide thickness being thicker for wires with the greater amount of processing³. The higher purity grades of aluminum will produce less slag in the molten pool that can be a source of spitting (*Section 6.4*). The higher the purity, the greater the cost of the wire; similarly if there needs to be additional processing to soften the wire after drawing to the final diameter, this will result in a higher cost also⁴. However, higher purity can give a better boat life. These costs need to be balanced against the quality of the product required, the amount of spitting and hence pinholes that are acceptable vs. the boat life that can be achieved.

6.2 Chemical and Physical Properties of Aluminum¹

“A hard, strong, silvery white metal, protected by oxide film from reacting with air and water. Soluble in hot concentrated HCL and NaOH solution. Hundreds of uses as metal and alloys.”¹

Abundance: Earth's crust 83,000 ppm; seawater 0.01 ppm

Atomic number 13
Relative atomic mass 26.981

Atomic radii, pico meters 143.1

Melting point, °K 933.52
Boiling point, °K 2740
Heat of fusion, kJ/mol 10.67
Heat of vaporization, kJ/mol 290.8

Solid density, kgm/m³ 2698 [298K]
Liquid density, kgm/m³ 2390 [liquid at m.p.]
Thermal conductivity, W/mK 237 [300K]
Electrical resistivity, ohm-m 2.6548 x 10⁻⁸

Vapor pressure and maximum evaporation rate as a function of temperature²

Pressure, microns	10 ⁻²	10 ⁻¹	1	10	100	1000
Temperature °C	882	972	1082	1207	1347	1547
Evaporation rate gm/cm ² sec	8.92x10 ⁻⁸	8.59x10 ⁻⁷	8.23x10 ⁻⁶	7.88x10 ⁻⁵	7.53x10 ⁻⁴	7.10x10 ⁻³

6.3 Oxidation of Aluminum

Aluminum owes much of its use to its corrosion resistance, a property that is derived from the oxidation of the surface. Any freshly formed aluminum surface will immediately oxidize. If the aluminum surface is scratched, breaking the oxide layer, it will self heal as the newly exposed aluminum is oxidized.

The oxide that forms has a molecular volume that is stoichiometrically 1.5 times that of the aluminum used up in oxidation. This makes an oxide film that is in compression that not only covers up the aluminum but can also tolerate an amount of flexing without the oxide breaking. It has been assumed that this oxide layer is amorphous with the outer surface of the oxide being a hydrated aluminum oxide⁵.

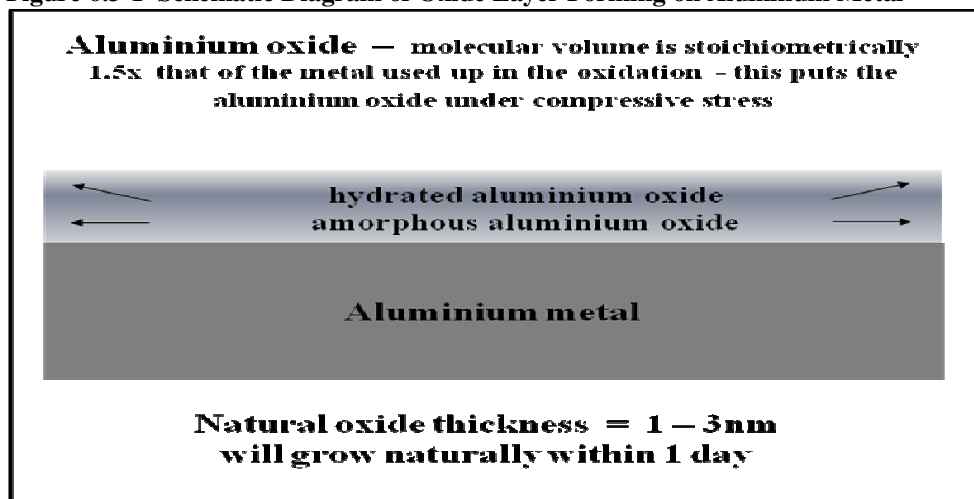
The kinetics of the oxide growth is complex⁶. On a freshly produced aluminum surface, an oxide layer of 2-3 nm will be formed within one day. The formation of the oxide on the surface is a logarithmic process, and so it ought to be regarded as a dynamic rather than self-terminating process. The reason for the process being logarithmic is that once the oxide forms a monolayer over the surface, it provides a thin barrier coating, slowing down diffusion, through the compressed oxide, of more oxygen and/or moisture to the underlying metal.

Aluminum, as indicated by its position in the electromotive force series, is a highly reactive metal. In fact among structural metals, only beryllium and magnesium are more reactive. Vacuum-deposited rolls of aluminum-coated film have a slightly different dynamic of oxidation because of the film being rewound into a roll in a vacuum system, limiting the arrival rate of oxygen to the aluminum surface. Thus the oxide thickness growth in a roll may be slower than that of a freely exposed aluminum surface. The following approximation has been suggested as a likely progression of oxidation.

1 monolayer	immediately
2nm	1 day
2-3nm	1 week
3+nm	3 months

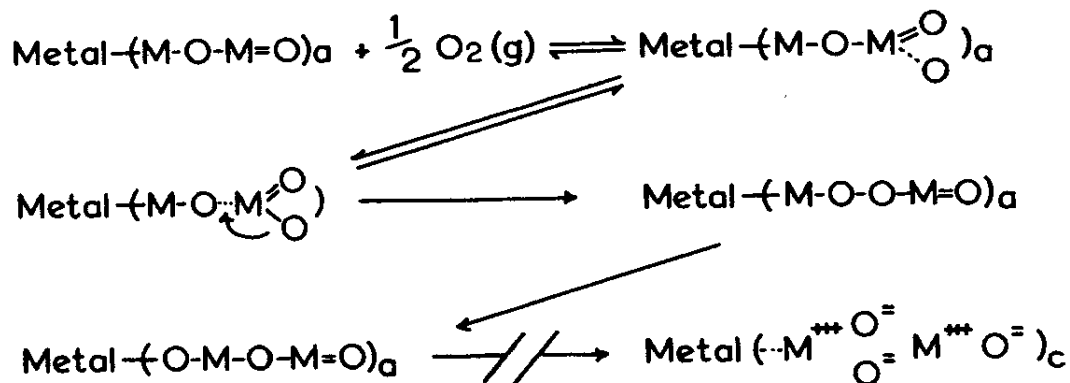
This is only suggested as a working rule-of-thumb and may differ if the film has been exposed to high levels of humidity or temperature that can increase the oxidation rate.

Figure 6.3-1 Schematic Diagram of Oxide Layer Forming on Aluminum Metal



A study of the kinetics of aluminum oxidation¹¹ has shown that the likely mechanism for the growth of the film is due to transfer of the oxygen from the surface of the oxide layer to the metal interface as shown in Figure 6.3-2.

Figure 6.3-2 Mechanism for the Uptake of a Monolayer of Oxygen by Aluminum (from Reference 11)



The summary from Reference 11 describes the experimental results of Reference 12, which found an inverse logarithmic rate law for the growth of the oxide layer on an aluminum single crystal, at 20C and 1 atmosphere of dry oxygen. The oxide layer grew to 35 angstroms (3.5 nm) over the course of seven days and then stopped. It was also found¹² that in humid oxygen, the initial rate of oxide growth at 25C was initially faster (logarithmic) than that of dry oxygen for approximately 10 hours and then followed an inverse logarithmic rate law after that. The authors proposed that the included water in the oxide film assisted in the migration of the oxygen ions through the oxide layer. This hypothesis appears to be supported by the observation that oxide layers in humid oxygen at 25C reached a limiting thickness of 50 angstroms (5.0nm).

The recent work¹⁴⁻¹⁹ that has looked at the thickness of aluminum oxide at both the surface and film interface of aluminum metallized thin films of optical density 0.5 – 2.5 has shown that there is an oxide present that is consistent with the 3.5nm found experimentally in References 11 and 12. For more details see the earlier Sections 2.2 and 2.3 on the relationship of aluminum thickness to optical density and other properties.

This work also found that the oxidation of the metal coating was also present at the polymer/metal interface and not just at the metal/air interface. It was also found that the thickness of the oxide was equal on both sides of the aluminum coating. It had been thought that the oxide thickness at the metal/polymer interface might be thinner than the surface oxide thickness, but this was found not to be the case. There is obviously sufficient oxygen present at the surface of the polymer or within the bulk of the polymer or available in the water vapor present in the polymer that the aggressive oxidation of the aluminum is not restricted in any way, resulting in a similar thickness of oxidation at the metal/polymer interface.

The oxide layer (alumina) is a much harder material than the aluminum metal. This is a noticeable factor in metallized film. If a metallized film is processed too soon after metallization, it can be highly prone to scratching with the production of a high level of pinholes. If the same film is left for longer before processing, the level of pinholes or scratching is considerably reduced. Thus, films, which are to be double-side metallized, benefit from having the second side metallized one or more days after the first side is metallized.

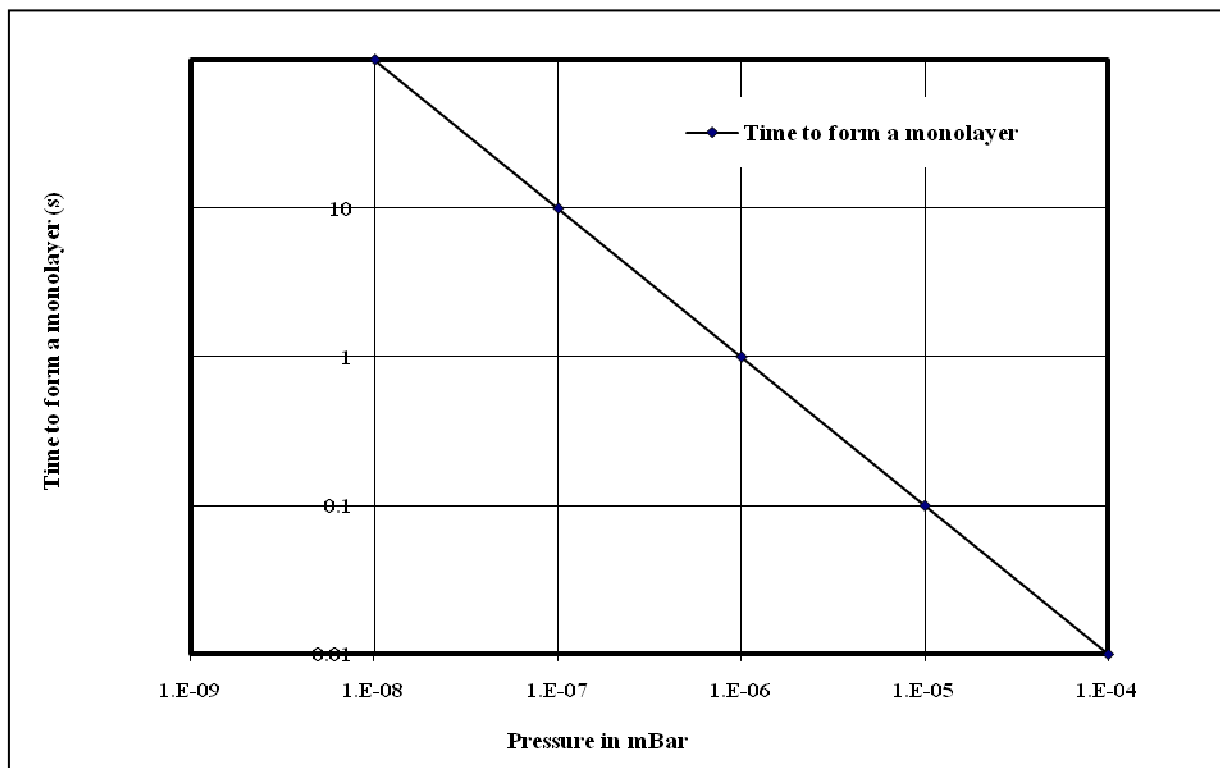
Although a vacuum system is used as a method of removing oxygen from the atmosphere to allow the aluminum to be deposited without oxidation, it is not a perfect process. If we look at the time to form a monolayer of gas on a surface, we find it is related to pressure or molecular density. If we assume that the sticking coefficient, the probability that a particle hitting a surface will stick to it, is one,

At 1×10^{-3} mbar (~1 mT), a monolayer will form in 1 millisecond.
 At 1×10^{-6} mbar, a monolayer will form in 1s.

The aluminum does not grow as a fully dense coating. There will be areas of low-density material at the grain boundaries of the crystals and even some trapped gas within the crystals. The same information about the arrival rate can be used to indicate the potential contamination rate of the growing coating due to residual gases within the system.

Looking at Figure 6.3-3, a graph of the time it takes to form a monolayer of atoms on a surface, it can be seen that at a pressure of 1×10^{-6} mBar it takes 1s to form a monolayer.

Figure 6.3-3 The Time It Takes to Form a Monolayer of Gas in Vacuum



The aluminum metallization of polymer webs typically requires the system to be pumped out into the 10^{-6} mBar pressure range, and when the deposition starts, the pressure typically rises into the 10^{-4} mBar region. Taking the pressure to be 1×10^{-4} mBar, the time to form a monolayer is only 0.01s. Typically the polymer passes through the aluminum deposition zone in 0.1s (or less) and so there could be 10 monolayers of contaminants arriving at the same time as the aluminum arrives. At the typical metallizing thickness, this represents a contamination of approximately 1% - 2%. For slower deposition processes, this contamination level will be higher.

Most vacuum systems at the deposition pressures will have water vapor as the major proportion of the remaining contaminant gas (*see Section 11*). Aluminum will react with the oxygen from the water vapor to form aluminum oxide, but there will be a proportion of aluminum hydroxide formed which is not as tough as pure aluminum oxide.

6.4 Wire Oxidation, Spitting, and Pinholes

The aim is to have the aluminum evaporating at a consistent rate, however, there are several factors that can prevent this, including aluminum wire quality. Ideally the aluminum would be 100% pure, and the surface would be free from oxide. The reality is that neither is true.

The choice of wire can affect the process. A larger diameter wire will have a better volume-to-surface ratio, but it will be stiffer and more difficult to handle, and it may cause more severe erosion to the boat, leading to early boat failure. Similarly, lower purity aluminum wire can have impurities that can attack the grain boundaries of the boat material, also increasing the boat erosion rate. A finer diameter wire is easier to handle, but there will be a much higher surface area, and the volume-surface area will be lower. The wire surface area is a factor because the surface of the aluminum oxidizes, and the boat will not evaporate the oxide.

Similarly, impurities may add to the slag on the surface of the molten pool. This oxide or slag tends to collect around the edges of the molten pool. This mixture of metal and oxide can be moved about as the pool size is changed, and the wetting characteristics will be different from pure, clean metal. This can mean that some material gets flash evaporated off the surface. This very rapid flash evaporation can carry some debris from the pool edges, and this will be incorporated in the thin film coating. This debris can appear as coating defects. If adhesion is poor, these defects can be knocked loose or abraded during winding leaving pinholes of uncoated web¹⁰. The oxide is likely to be very hard and can induce scratches into the freshly coated surface.

If the molten aluminum pool is expanding in size and the boat is hot enough, the molten pool may not wet the new surface of the boat that it covers. In this case, the aluminum will bead and may spit, causing quite large spots of aluminum to be deposited⁷. An analogy would be adding water to a very hot surface where the water may be seen to bead up and bounce around on the surface and spit without wetting the surface. As the surface cools, the water will then wet out the surface, and the spitting stops.

6.5 Wire Feeding

Modern metallizers can be expected to coat 40km of web of up to 4.5m widths at a speed of 1000m/min at an optical density of 2.0 or more. Allowing for a collection efficiency of 50% this could require approximately 7.0kg of aluminum per boat. It would be quite a slow process to heat the whole quantity of aluminum from the start of the process. The technique adopted is to heat a boat and to feed the aluminum gradually into the boat for evaporation. There are various methods of feeding material for metallization. Common methods include vibrating powders or pellets into a crucible and rod feeding through a water-cooled crucible into a molten pool from underneath. The method that has been adopted for aluminum is a wire feed where a spool of aluminum wire is gradually uncoiled and pushed toward the molten pool on the surface of the boat. Ideally the aluminum wire is pre-heated as it approaches the molten pool and just as it is about to touch the pool it reaches the melting point where it will drip into the pool.

The control of wire feeding has improved over time. Initially, all the spools of wire were on the same shaft, so if the rate of wire feed increased for one boat it increased for the others too. The first step forward was to use slipping electromagnetic clutches so each spool feed could be independently controlled. The electromagnetic clutches were controlled by a series of pulses. Varying the duration of the pulse controls the feed rate. This system works well up to winding speeds of around 800m/min, above which the speed of response of the clutches is a limiting factor. The slow nature of the response can cause the feed to oscillate between slightly too fast and slightly too slow. This causes the molten pool size to increase and decrease slightly with a subsequent increase in the rate of spitting from the wetting and de-wetting of the boat surface around the perimeter of the pool.

The latest change has been to use a stepper motor drive on each wire feed that achieves much higher precision and faster response⁷. The stepper motors have a set of windings that receive a series of high voltage pulses that cause the shaft to rotate by a fixed angular amount per pulse. The stepper motors have a high torque output and high precision, so that given a signal, the shaft will rotate a fixed amount very quickly without any need for any feedback to check the position. The high voltage signal also is unaffected by electromagnetic interference, so the quality of the control is good too. Typically the stepper motors have 400 pulses per revolution of the shaft resulting in an angular motion of less than one degree per pulse. It is possible to achieve a speed control to the wire feed of +/- 0.02%. This precision control keeps the pool size constant and minimizes spitting.

6.6 Oxidation Hazard

The oxidation of aluminum is an exothermic reaction. Where the aluminum is adhered to a surface, there is usually sufficient mass that the heat liberated by the exothermic reaction is only able to raise the temperature of the mass a limited amount. If, however, the aluminum were to flake off the surface, the thermal mass becomes very much smaller and the temperature rise can become much higher during oxidation. If the aluminum flake is thin and the dimensions small, the temperature can rise enough to exceed the auto-ignition temperature, in which case, the flake becomes incandescent. There are two occasions where this can become a problem in the metallizing process:

The first is when venting the system. It is possible during venting, for the stray material deposited onto protection shields to flake from the surface generating aluminum dust. The surface of the deposited coatings can be powdery, particularly if the depositing material has been scattered and is depositing at many angles including low angles. If the system is not regularly cleaned, this aluminum dust can accumulate. Some of this may already be partially oxidized but there may be un-oxidized aluminum. If some of this reaches the auto-ignition temperature and becomes incandescent, it can trigger the ignition of the rest of the dust. This can generate enough heat that some of the porous aluminum deposited on the shields may ignite. This can go unnoticed; it may be seen as a flash with an accompanying noise and may not be a problem, or it can cause a serious fire.

The second occasion when oxidation of aluminum can be a hazard is when cleaning the shields of the stray aluminum deposits, which can be accomplished by hitting the shields with a hammer, scraping with a sharp blade, wire brushing or other mechanical actions. This will produce much more dust generated by the multiple fracturing of the aluminum. Consequently, there will be plenty of new aluminum surfaces to be oxidized. A proportion of this dust can be airborne and this can increase the possibility of an explosion^{8,9}. This may not in itself be dangerous; usually the quantity of aluminum is limited. What may be more of a hazard is the reaction of any operator in the vicinity or the presence of a flammable solvent. In the operator's case, there is a tendency to recoil rapidly from the flash fire, and this becomes a hazard as an injury can be caused if the operator falls and/or bumps into nearby objects. If there are flammable liquids or solids in the vicinity of the operation, the flash can ignite them, causing a more serious fire.

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Section 7: Surface Treatment of Webs for Aluminum Vapor Deposition

INTRODUCTION

Web-based materials for aluminum vapor deposition often require surface preparation or chemical modification to insure adequate adhesion of the aluminum layer to the web surface. With today's improved understanding of the impact of film surface chemistry, there may also be good reasons to chemically modify web surfaces to enhance moisture and gas barrier¹ properties as well as adhesion or improved chemical resistance of the aluminum.

7.1 Discussion of Surface Treatment Technologies

There are many web materials available for metallization with a wide range of surface chemistries and topographies in paper, nonwoven polymers, cellophane, and synthetic polymer films. These webs have different chemical functionalities, which will influence what surface treatments, if any, will be necessary to prepare them for metallization. In preparation for metallization, the web is surface modified, or functionalized, prior to placement in the vacuum chamber and at its point of manufacture. The surface modification could take place or be enhanced in the vacuum chamber during, but just prior to, the deposition of the aluminum. The two treatment locations will be broken down into two broad categories of atmospheric pretreatments and in-chamber pretreatments for this manual.

Atmospheric pretreatment consists of coating or chemically priming, corona treatment, flame treatment, chemical treatment, and atmospheric plasma treatment of the web surface outside the vacuum chamber. Most atmospheric pretreatments result in the direct oxidation of the web surface. In-chamber pretreatment is performed by plasma treatment using various treater designs and gas plasmas to optimize film properties². The choice of the treatment method can be a key decision in determining the final web properties. The greatest care must be exercised in the definition of the final properties required for the metallized product to determine which method of surface modification is best suited for the product.

In general, the web surface to be modified is relatively flat and nonporous, composed of a natural or synthetic polymer. Even with paper, the metallizing surface generally is coated with a polymer layer to smooth the surface so it will accept the metal deposit. The fiber structure of the paper is not important in the metallization. The discussion of treatment here is restricted to the methods used and their impact on natural and synthetic polymer materials, focusing on the treatment of polyester (PET), polyethylene (LDPE, HDPE) and polypropylene (OPP) films with reference to other polymers such as polystyrene, polyimides and polyamides (nylon) when appropriate.

Most surface pretreatments are designed to chemically modify the top-most layers of a web and not to penetrate into the bulk material. Coating of films serves the purpose of placing a new polymer onto the carrier film and represents a surface modification, by changing the near surface bulk composition, of the film or paper for metallization. Consequently, coating will not be discussed in detail, as we will focus on the interaction of various polymers with the treatment method and not simple changes in surface polymer. Many of the films used for metallization are coextruded multilayer polymer structures with the outermost polymer layers being selected for particular purposes such as sealability, printability or for metallization. This section will focus on the methods and mechanisms of surface oxidation of PE, PP and PET films, paper and foil by corona, flame, and plasma treatment. Means of measuring the surface treatment and surface oxidation will also be covered.

Oxidation of a material in this context will mean the addition of an oxygen atom into the structure of the polymer. PE and PP have no oxygen in their chemical structure as compared to PET, which contains oxidized carbon atoms in the form of carboxylic acid (HO-C=O) and alcohol (C-OH) functions built into the molecular structure of the polymer in the ester linkage of the polyester. Carbon atoms are capable of forming four chemical bonds, and oxygen atoms can form two chemical bonds. Thus the state of oxidation of a carbon atom may be determined by counting the number of carbon-oxygen bonds. Therefore, a carbon atom in PE molecules with two C-C and two C-H bonds is unoxidized. An alcohol (-C-OH) with one carbon-oxygen single bond is the least oxidized carbon, a carbonyl carbon (-C=O) with a carbon-oxygen double bond is more oxidized than the alcohol, a carboxylic acid, (HO-C=O) with three carbon-oxygen bonds is even more oxidized and carbon dioxide (O=C=O) with two carbon-oxygen double bonds is completely oxidized. These chemical group names and oxidation levels will be used throughout this section to explain the impact of the various surface treatment methods on the polymers.

SURFACE TREATMENT MEASUREMENT

Measurement of the treatment level as surface energy can be conducted in different ways by obtaining simple wettability measurements, detailed information about the nature of the surface forces, the level of surface oxidation, and the composition of the surface chemistry. Each method is more complex, time consuming, informative, and expensive than the preceding method. Each will be briefly described and its use outlined. Those wishing to study this topic further are directed to the recent book by Chan³, which reviews the many methods of polymer surface modification and characterization.

For the simple determination of the existence of treatment, pre-made treatment pens may be purchased to wipe on a film surface to check for minimum surface treatment levels. However, due to contamination of the pen tip by actual testing of the film by surface wiping, the pens should not be routinely used to determine the treatment level but as a go/no-go test for the presence of treatment. If no treatment pens are available, it is possible to determine which commercial markers will show treated and untreated surfaces. As some markers are solvent-based, and others are water-based, it should be quite possible to test some to determine which show proper film treatment levels for your process by the formation of a uniform mark, or a mark which breaks up and does not wet your substrate.

Surface Treatment Level: This is a straightforward and fast measurement to determine the surface tension or wettability of the surface. The test is detailed in the ASTM procedure D2578 for the measurement of surface tension of polyolefin films from 30 to 56 dynes/cm. While simple to do, reading the results is somewhat subjective and therefore prone to operator variability. The test consists of wetting a cotton-tipped swab with a series of solutions of formamide and ethylene glycol monoethylether (ethyl cellosolveTM) blends (*Table 7.1*) and wiping the wet swab lightly over the surface. Alternatively, several drops of solution may be spread or drawn down by a metering rod. Approximately one square inch of surface is covered and then the breakup of the liquid film is noted. The surface tension is defined as the solution composition where the liquid film breaks up at two seconds, determined by testing the surface with a series of solutions. Great care is needed at each step, which is why the operator's experience is important. A fresh swab needs to be used for each attempt. Too much fluid on the cotton will impact the reading by varying the liquid film thickness; too much pressure on the swab tip can disturb the film surface changing the result. The cotton itself must be free of sizing chemicals and the solutions need to be fresh to insure proper composition. In general, the test yields a value, which can vary 1 to 2 dynes for a single tester and between testers.

Contact Angle: This is a more time-consuming test to determine the treatment level of treated films. It may be determined in several ways, and it is described in great detail in many texts and publications. In the *equilibrium sessile drop method*, a controlled-volume, stationary drop of liquid is placed on a planar surface and the angle between the solid, the saturated vapor or air and the liquid is related to the height, h , and base diameter, d , of the drop, which are measured.

Equation 7.1-1

$$\tan \theta/2 = 2 h/d$$

Contact angle results may be reported as the contact angle in degrees, θ or $\cos \theta$, the cosine of the contact angle θ . The apparent contact angle may be affected by the chemical composition variation and the roughness of the surface. These effects may be checked by measuring the advancing and receding contact angle. These tests are performed by making the drop edge move across the surface and slowly increasing or decreasing the volume of the liquid drop. Then the drop is held at constant volume and the contact angle is measured after the drop edge has come to rest. The difference between the two measurements, the hysteresis, can give information about the relative surface roughness, because a rough surface will change the contact angle as the liquid retracts along the surface, when the drop is decreased in volume.

The *Wilhelmy plate or tensiometric methods* also give the equilibrium, advancing and receding contact angles by calculating the force required to lower or raise a sample into a fluid using a micro-balance. This test is gaining more use as it is less subjective in its measurement due to the instrumentation used, and it gives several measurements with a single sample setup.

Solid Surface Free Energy: There is no direct, reliable method for determining the surface tension of a solid⁴. However, there are several indirect empirical and semi-empirical methods based on contact-angle measurements. The solid surface tension is important to know, because of its potential for determining the adhesion between materials. It has been suggested that the surface free energy of a solid surface is composed of several intermolecular

forces comprising dispersive, polar and hydrogen bonding. It is postulated that two interacting materials which share these intermolecular forces adhere together and that the work of adhesion is comprised of three intermolecular forces. Measuring the contact angle of the solid surface with various liquids allows the estimation of the solid surface energy, γ_s , which can be shown to be composed of two parts; the polar part, γ_s^p , and the dispersive part, γ_s^d . These parameters are thought to be important in that they can be used to estimate the work of adhesion between a solid and another material. The polar portion is made up of the combination of the dipolar interactions and the hydrogen bonding attractions between two surfaces. Hydrogen bonding is the intermolecular attraction of an alcohol hydrogen with a molecular dipole such that a weak bond-like attraction is formed between two molecules keeping them in close association. In treated PE surfaces the auto adhesion between two films has been shown to be due to hydrogen bonds between –C-OH groups on one surface and C=O functions on the other surface⁵. For polyester, the auto adhesion is between terminal phenols in one surface and carbonyls in the other⁶. Equation 7.1-2 is used to determine the dispersive and polar solid surface energies by measuring the contact angle with two different fluids and solving the set of two simultaneous equations for γ_s^p and γ_s^d :

Equation 7.1-2

$$\gamma_{LV} (1 + \cos\theta) = 2(\gamma_s^d \gamma_{LV}^d)^{1/2} + 2(\gamma_s^p \gamma_{LV}^p)^{1/2}$$

Values for γ_{LV} are tabulated in Table 7.3 for well-characterized liquids, and a combination of any two sets can be used for the determination⁷. Often, water/methylene iodide or water/formamide are the liquid pairs used.

Surface Oxygen Concentration: Total surface oxygen concentration may be determined by X-ray photoelectron spectroscopy (XPS) or, as it is often termed, electron spectroscopy for chemical analysis (ESCA). In this method, a sample is placed in a high-vacuum chamber and bombarded with monochromatic X-rays. The X-rays interact with the uppermost atoms of the surface and eject electrons from the innermost electron shells surrounding the nucleus. The chemical environment of the atoms, *i.e.*, the nature of the chemical bonds comprising the molecules, determine the binding energy required to remove the electron from the surface atom. By carefully measuring the kinetic energy of the ejected electrons, the atoms of the surface may be determined from the binding energy, and the relative concentrations of the atoms also may be determined. Electrons can only escape from the uppermost layer of the sample, making this a very surface-sensitive method (upper 50nm of the sample). In its simplest form, the spectra show peaks representing the binding energy of the atoms, but without further detailed analysis, no chemical functionality. For instance, the various carbon-oxygen binding energies are very close together and in low resolution spectra, appear as a single peak. For polymers, the C 1s electron spectrum is used, and for treated surfaces it is ratioed with the O 1s electron to determine the molecular concentrations of oxygen in the treated surface.

Surface Chemical Composition: As described above, the XPS spectrum contains a single peak for carbon-oxygen species on the surface, which is not resolved in terms of the contributions of individual surface chemical species. However, it is possible to obtain a calculated value of the surface composition by estimating the relative contributions of each chemical group and performing a curve-fit to reproduce the actual peak shape. This approach, while not exact, represents a practical approach to characterizing the treated surface and makes use of model compound studies. It has been widely used in literature studies of polymer-metal surface interactions. It is not possible to differentiate between alcohol and ether groups with this method, which is why the results can be considered speculative⁸.

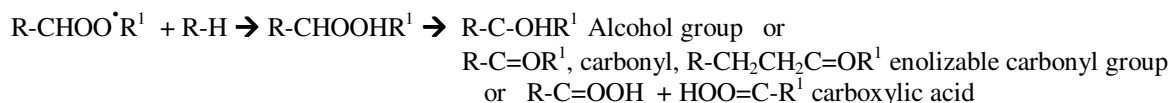
The chemical functionality of a treated surface may be determined from direct chemical derivitization and analysis of the surface. This is a complex but classical chemical analysis technique where specific reagents are reacted with a surface to form unique chemical groups, which may then be analyzed by XPS and other spectroscopic methods. While making its own set of assumptions about relative reactivity, etc., in combination with XPS, it allows the determination of alcohol, acid, ketone, and aldehyde groups directly. By varying the reactant chemicals, it is also possible to determine the amount of enolizable carbonyl groups and, by a difference, the amount of ether linkages. (See *free radical mechanism in Treatment Mechanisms section*.)

TREATMENT MECHANISMS

A general observation of the effect of treatment mechanisms is that they all generate oxidized polymer surfaces. Detailed studies of the various treatment methods do indicate that there are variations of the mechanism or variations in the ratio of functional groups produced, but these differences seem larger between polymers than between the different treatment methods. All the methods can be applied aggressively, resulting in highly oxidized, low molecular weight materials on the substrate surface, which can be termed “over treatment.” While it can be shown that the low molecular weight material might not be harmful for some inks or for aged or washed PET surfaces^{6,9}, in

general, its impact on aluminum adhesion is the same as for low molecular weight additives, which can form on the film surface and destroy the aluminum adhesion¹⁰. The low molecular weight molecules on the surface lower adhesion of metallized layers by producing a weak interface, which is not desired, therefore, the film surface should have been treated less for optimum metal adhesion.

Mechanism of free radical oxidation of PE:



The basic mechanism of treatment starts with the generation of a free radical on the polymer chain. A free radical is a very reactive chemical species consisting of an unpaired electron. The polymer radical is produced by a high-energy impact of a free electron, ultraviolet radiation or a disassociated molecule produced by the high energy or high temperature in the corona, flame or plasma treater gap. All of the treatment methods generate free electrons and/or free radicals, which interact with the substrate to create a radical on the surface. The part of the polymer molecule that the radical is produced on controls the various differences in the oxidation between various polymers. Thus on PE, radicals are produced on the carbon backbone of the polymer molecule; on PP, the radical is produced on the backbone carbons or on the pendant methyl group; and for PET, the radical is produced on the aromatic ring or by chain scission. The radical formed rapidly reacts with an oxygen molecule, O₂, which forms a peroxide. The peroxide then decomposes by various routes to form ethers, alcohols, aldehydes, ketones, or carboxylic acid groups on PE and PP and phenols in PET. Almost all of the studies involving self adhesion, ink adhesion, and aluminum interactions show that the carbonyl groups play a key role in adhesion to the treated surface through formation of Al-O-C bonds or hydrogen bonding to alcohol groups in the inks or opposing film surface. Enolizable carbonyl groups play an important role in adhesion because they may rearrange from the carbonyl into an unsaturated alcohol by tautomerism and thus, may participate in adhesion improvement groups as a carbonyl or as an alcohol.

The carbonyl and alcohol groups are formed first and can be further oxidized to carboxylic acid or carbonyl groups, respectively, which would be the impact of high or aggressive over treatment. Thus a definition of over treatment could be described as the treatment intensity, which forms carboxylic groups at the expense of carbonyl and alcohol groups.

TREATMENT AGING

Surface treatment values are found to decrease with time from the moment treatment is complete. This has been found in all of the treatment methods employed to date. The amount and cause of the aging is dependent on the level of treatment and the formulation of the film. In general, aging results from the migration of highly oxidized, low molecular weight material on the film surface, rotation of the highly polar components attached to the polymer backbone chain into the film surface to minimize surface free energy when stored in air, the bloom* of film additives to the surface, covering the surface with a low free energy layer, and the transfer of film additives across a film-to-film contact plane in wound film rolls. Typical additives, which show this latter behavior, are fatty acid amides (slip agents) and silicone oil, both of which have low wetting tensions. One notable exception to aging has been reported, and that is the atmospheric plasma treatment with a helium/acetylene blend¹¹. These samples have shown high, stable surface treatment values, indicating that a different oxidized surface structure is being produced.

*The term "bloom" has its roots in the rubber vulcanization process where excess sulfur was added to crosslink rubber latex. The extra sulfur would come to the surface of the rubber¹² creating a surface pattern, which was called bloom (reported by Dr. Billmeyer as due to the pattern looking like a flower).

EFFECT OF FILM ADDITIVES

Additives are added to most film products to modify the surface properties of the film. In general, there are two classes of film additives; blooming and non-blooming. Blooming additives are designed to migrate to the surface of a film and change its surface properties. Non-blooming, or non-migratory, additives are designed to stay in a fixed location in the film, generally on the film surface, to modify properties. In general, the blooming additives are called "slip" additives because they lower the films coefficient of friction (COF), while the non-blooming additives are called "anti-block" because they hold the film surfaces apart and prevent the auto-adhesion or "blocking" of the film, which leads to film tearing during unwinding. Migrating additives are more likely to be added to polyolefin films such as PE, polyvinyl chloride (PVC) and PP, than to polyester-based films. PVC films are also highly modified, or plasticized to allow processing of the material into flexible PVC films.

When a slip-containing film has been surface treated, the surface is covered by the slip additives, and the treated surface is hidden under the new surface layer. At first the treatment seems to have gone away, while in fact, it is under the new surface layer. Washing the surface will remove the slip, and the treatment should reappear. Depending upon the chemistry of the additive, it might be possible to raise the apparent treatment of the film. However, most migratory slip agents are relatively non-polar, which gives a lowered treatment measurement. The principal detriment of slip agents for metallization is that this surface coating lowers the COF because it is a weak adhering surface layer. If metal is deposited over this slip layer, the adhesion of the aluminum to the underlying film surface will not develop as described in the patent to All and Duncan¹⁰. Typical slip additives are fatty acid amides such as erucamide, or stearamide, and high-viscosity silicone oils. In addition, polymer stabilizers may migrate to the film surface. These blooming additives can also interfere with adhesion of ink or polymer to aluminized surfaces since they will form a weak layer by transfer from one film surface to another while the film is wound in roll form. Storage time and temperature affect the amount of film additive that migrates and transfers to the surface of the film.

Antiblock particles are often added to reduce the amount of slip additive required to lower the COF of the film. In addition for vacuum winding of film they are used to fix the separation distance between film layers (replace the air layer) and improve winding and tracking over the rolls in the metallizer. Because they are in the surface of the film and stick above the surface they might cause film scratching if film layers move across each other. Some particles are embedded in the film surface, while others are more loosely bound falling out of the film and causing buildup on equipment.

The additives are required in film because the final products would not work in the chain of use required for the film. Film suppliers balance the relative amounts and types of additives to produce the many different properties needed for successful winding, metallization, lamination, and machinability at the end use of the film and film structures. The proper choice of film type and treatment method and treatment location are important in determining how well the film performs in the metallizer and how well the metallizer can handle a wide range of products such as by surface cleaning by treatment.

ATMOSPHERIC TREATMENT TECHNOLOGIES

Surface treatment of polymers has been studied extensively in terms of the chemical modifications to PET, OPP, and HDPE films by corona, flame, and ultraviolet (UV) methods. While the newer atmospheric plasma treaters have not been studied extensively, the mechanism of treatment should be essentially the same, and similar results are to be expected for the straightforward treatment of the films. Each method will be discussed to show the equipment and methods used.

Corona Treatment

In corona treatment, a high frequency electric circuit is established that contains an air gap capacitor. The electric voltage is raised to the point where electric breakdown of the air occurs, and a current begins to flow across the gap. The electrons in the gap are accelerated by the high potential across the gap and impact gas molecules in the air, creating ions and additional electrons, which are accelerated toward the grounded surface. This causes an avalanche of high-energy electrons, which impact the grounded surface in the circuit. During the breakdown of the air, ozone is formed, and an intense blue light is emitted from the air gap. On close inspection, many small sparks can be seen in the air gap. The electrons, ozone, and the UV light all participate in the chemical modification of the polymer surface. The high frequency AC circuit constantly raises and lowers the voltage in the gap to keep the ionization process going. The high frequency is used to prevent treatment skips as the voltage cycles.

In commercial practice, the air gap capacitor consists of an electrode held at high voltage and a grounded metal roll covered by a dielectric material to increase the voltage drop in the air and prevent arcing to the grounded roll. This dielectric surface may be formed from an epoxy, a HAPOLON or silicone rubber sleeve, or a glass or a ceramic coating. All of this equipment is placed inside a box, which is vented to remove ozone by high airflow.

In order to treat the film, it is passed through the air gap during breakdown. High-energy electrons impact the polymer creating free radicals, which rapidly react with oxygen or water molecules in the air and in the film, forming hyper-peroxides in the upper layer of the film. These hyper-peroxides then decompose into a range of chemical species such as alcohol, epoxy, ether, ketones, aldehydes, and carboxylic acids. The level of treatment obtained is determined by the energy density, kw/m^2 dissipated in the treater gap.

While the corona treatment process may be used successfully for many products, due to the high frequency and high voltage used, it is possible to create oxidized sites on the film surface against the roll (the back side of the film), as in all atmospheric installations there is always a thin layer of air between the film and the roll surface. While the backside treatment can be managed, it can never be eliminated. For metallizable film this is problem, which must be

controlled, being a principal source of metallized product defects. Many manufacturing process variables and equipment condition factors are interrelated in determining the baseline level and the distribution of backside treatment. Factors effecting the level of backside treatment include treater roll surface finish, nip roll condition and alignment, film line speed, film web tension, film gauge uniformity, and treater power level.

Flame Treatment

In the flame treatment process, air and combustible gas are mixed and sent to a burner where a high-velocity flame is started on a burner grid in close proximity to the film surface. There are two basic flame configurations: a short compact flame or a longer “softer” flame. In the past, only the long flame was used and the process was primarily used for treating complex three dimensional articles. Currently, the largest commercial applications for film treatment use the short flame with a well-defined flame front. The treatment process is dependent upon the air/gas ratio, the air/gas flow rate, position of the film surface relative to the flame (distance from the burner), and the gas composition, all of which combine to determine the flame temperature and the oxidizing or reducing nature of the flame. Treatment level, as determined by surface oxygen concentration, is found to pass through an optimum air/gas ratio, to increase with air/gas flow rate and to decrease with increasing distance from the burner. Air/gas ratio controls the flame temperature and the oxidizing condition of the flame. The air/gas flow rate represents total thermal power, while the position in the flame front affects both the flame temperature and the oxidizing environment at the flame/film interface.

Process control has been greatly improved over the years with a focus on the air/gas composition control, either through oxygen composition of the combustion blend, or by measuring the flame temperature of a side stream, which is burned, heating a thermocouple. Control is through feedback to control the gas input at a venturi mixing valve in the process air supply to the burner. The safety and reliability of these new systems have been greatly improved during the last 20 years. Due to the high temperature of the flame, the film must be cooled on a water-cooled roll to protect it from the thermal load of the flame. Generally, the burner is placed below the water-cooled roll and the film is passed above the burner against the roll surface. This is to allow the formation of a stable flame front and a stable treatment zone, but this also results in the highest thermal load on the film. A key advantage of flame treatment for metallizable base films is that it prevents back side treatment as there is no mechanism for having the flame front behind the film.

The chemical species produced and the treatment mechanisms for PE surfaces are similar to those found with corona treatment. However, due to the complex chemistry of the flame, additional mechanisms may be present causing the balance in chemical species to be somewhat different. Also, the treater may be set to yield potentially different film product performances for products treated by both methods. It has been shown that the air/gas ratio controls the surface chemistry developed^{13,14}. In the case of PP, the mechanism of oxidation does appear to be different than that of corona treatment. In flame treatment, the XPS evidence of the oxidation groups formed shows that the pendent methyl group undergoes a sequential oxidation from alcohol to carbonyl to carboxylic acid group and the polymer chain is not attacked¹⁵.

Atmospheric Plasma

This is a new treatment technology just recently described and patented¹⁶. It is being commercialized and has the potential to permit more flexibility for chemical surface modification of polymer films by control of treatment gas composition. It is a low-voltage (5000 volts) discharge from a hollow cathode electrode using helium as the discharge medium. The use of helium allows a plasma discharge to form at much lower voltages than in generating a corona in air. The use of helium for generating the discharge and due to the control of helium in the treater bar, the concentration of helium behind the film should be close to zero. This eliminates the potential for backside treatment, since the voltage is too low to create a plasma with air.

Conceptually, this new process falls between a corona and a flame treater. As in flame treatment, the helium gas is delivered to the treatment gap by passing it through an electrode faced with a sintered metal surface. But like a corona treater, the plasma in the treatment gap is excited by a high-voltage, high-frequency alternating current, as the pores of the treater function as hollow cathodes. In many ways, it may be characterized as a “cold flame.” Treatment of the film surface is effected with 100% helium or with blends of helium/oxygen or helium/nitrogen. Initial reports of the treatment result show an oxidized surface, just as with flame and corona treatment. The treatment level is found to age, just as with other forms of surface oxidation, and the extent of treatment is in the same range. It is not yet known how well the relative concentrations of carbon-oxygen species may be controlled at the surface. Another exciting aspect of this technology is the use of hydrocarbon gas molecules blended with the helium carrier gas. This has the potential to dramatically change the surface treatment chemistry and hence treatment results, relative to flame and corona treaters. The use of acetylene has shown some dramatic surface treatment results with both Teflon and HDPE surfaces in that a high treatment level of 60 dynes may be developed and the

dyne level does not age with time¹⁵. This indicates the formation of a new surface structure than for flame, corona or for simple helium plasma treatment.

Metallization

Another method of surface treatment which is not widely discussed as such, is metallization itself. Harold Schonhorn, *et al.*, at Bell Labs generated an extensive body of knowledge on the effects of metal evaporation with various metals on polymer surface properties. For instance, they showed that using aluminum deposition on fluoropolymers is a means of improving adhesive bond strength of gold-metallized Teflon™ after stripping the aluminum from the surface¹⁷. It was shown by XPS studies that the Teflon surface had reduced fluorine abundance and increased oxygen attached to the carbon of the polymer. Thus, the aluminization process of the Teflon causes a surface oxidation of the fluoropolymer. This surface oxidation effect of metallization can explain the development of excellent adhesion of aluminum layers to untreated polymers, sometimes observed when the wrong, *i.e.* untreated, surface is metallized. There are several studies of these effects in the literature today^{18,19}.

In-Chamber Plasma Treatment

There are several principal plasma generation technologies in use: the original DC and AC magnetron sources with reactive gas blends, and a new hollow cathode AC plasma generator²⁰ with reactive gas blends. The essential features of all the plasma generators are that they supply high-energy electrons in sufficient quantities to generate heavy ions and additional electrons from the gas blend for bombardment of a film surface. In traditional magnetrons, the ions and electrons are constrained in a ring-shaped magnetic field above the generator, and the film is passed through or near the upper regions of the plasma (*Figure 7.10-1*). These systems may also be magnetically enhanced to open the field lines toward the film. In the hollow cathode design, the magnetic field above the hollow cathodes are modified by permanent magnets placed behind the film as it passes through the treater. This has the effect of inserting the ions and electrons in the open magnetic field and directing them toward the film surface (*Figure 7.11-1*) as the polarity of the generator cycles from positive to negative. These various electrical systems, in conjunction with the gases fed into the various plasma generators, clean and activate the surface for chemical modification just prior to the deposition of the aluminum. The results published by the various plasma treater manufacturers all show improvements in the metallized film properties of barrier, metal adhesion, and metal quality with various gases, gas blends, and flow rates. The future of the various processes lies in determining which one allows the most flexibility and operating reliability in generating polymer surface modifications for property enhancement in addition to improved adhesion.

While the treatment occurs just prior to the deposition of the aluminum, it is likely that only oxygen species are generated on the film surface under the metal layer, even for plasmas generated without added oxygen. This can be inferred from the results described above where the deposition of aluminum results in oxidized surfaces. In general, the plasma systems used commercially today are based upon argon or blends of argon and oxygen. Where does the oxygen come from for treatment using argon plasma in the vacuum? One source would be dissolved oxygen in the film itself followed by oxidation of any intermediate species formed after venting of the film to air and on rewinding for conversion. This is likely due to the high oxygen transport of polymer films. Also, there are very few reports of any other form of stable chemical species other than oxygen groups on treated films.

It is apparent from discussing the plasma pretreatment process with suppliers of metallization equipment and plasma treaters, that the process is not new, especially for those suppliers that market sputtering equipment or machines for capacitor applications. Indeed, according to leading metallizing equipment manufacturers, technologies for the pretreatment of films in the metallizing chamber have been in use for many years. However, the principal uses were to pre-clean the web and for chemical modification of the film surface to improve bonding. Generally, this was accomplished with a planar magnetron-based technology, with or without magnetic enhancement, and modified to use argon gas to create plasma to clean or etch expensive films to accept sputtered layers. In these cases, the process speed was not a limit, and the cost of extra equipment and increased process complexity were warranted by the high price and quality demands of the products.

However, today the use of in-chamber pretreatments is expanding from the low-speed sputtering applications into the high-speed process typical of aluminum metallization of packaging films. This change has been sparked by the realization, with existing film designs, of improved barrier properties²¹, metal layer improvements^{22, 23, 16}, and metal adhesion²¹ in the aluminized films produced. These improvements are brought about by the chemical modification of the film surfaces with the evolving pretreatment technologies.

Summary of Surface Treatment for Aluminum Deposition

All of the treatment processes practiced by the film manufacturers suffer in some regard, relative to treatment at the time of metallization, because of the passage of time from the point of film manufacture to the metallization chamber. As the rolls are stored in wound form, the surface to be metallized may be contaminated from chemical additives in the film and contact with the opposite film surface. In addition, most surface oxidation techniques give surfaces which decrease in surface energy with time due to the diffusion of chemical groups into the surface of the film. Consequently, plasma treatment at the point of metallization would seem to hold a great deal of promise because it can supply higher treatment surfaces at deposition due to no ageing, provide better metal layers due to the surface cleaning, and may supply better surface chemistry due to new gas blends for the desired surface functionalization. There is, however, a potential disadvantage to the technology, especially when used with existing films that are treated by the film manufacturer. As known by the film industry, it is possible to over treat the film and cause a reduction in metal bonding. Thus, when installing and starting up a new pretreatment process it is worthwhile working with your film supplier to obtain film with reduced surface treatment to permit the plasma treatment process to be optimized for film properties.

7.2 Dyne Level Testing Solution Compositions

Table 7.2-1 Dyne Level Testing Solution Compositions²⁴

% Formamide By volume	% Cellosolve By volume	Wetting Tension Dyne/cm
0	100	30
2.5	97.5	31
10.5	89.5	32
19.0	81.0	33
26.5	73.5	34
35.0	65.0	35
42.5	57.5	36
48.5	51.5	37
59.0	41.0	39
63.5	36.5	40
67.5	32.5	41
71.5	28.5	42
74.7	25.3	43
78.0	22.0	44
80.3	19.7	45
83.0	17.0	46
87.0	13.0	48
90.7	9.3	50
93.7	6.3	52
96.5	3.5	54
99.0	1.0	56

7.3 High Surface Energy Test Mixtures

Table 7.3-1 High Surface Energy Test Mixtures²⁵

% Formamide By volume	% Water By volume	Wetting Tension Dyne/cm
100.0	0	56
81.2	18.81	58
65.0	35.0	60
47.0	53.0	62
30.6	69.4	64
18.2	81.8	66
8.6	91.4	68
3.6	96.5	70

7.4 Surface Free Energy of Various Liquids at 20C

Table 7.4-1 Surface Free Energy of Various Liquids at 20C
(From Table 3.1 of Rance²⁶)

Liquid	γ_{LV} mJ m ⁻²	γ_L^d mJ m ⁻²	$\gamma_{LV} - \gamma_L^d$ mJ m ⁻²
Water	72.8	21.8	51.0
Glycerol	63.4	37.0	26.4
Formamide	58.2	39.5	18.7
Methylene Iodide	50.8 50.76	49.5 (50.76)	1.3 -
Ethane-1, 2-diol	48.3	29.3	19.0
Dimethyl sulphoxide	43.54	34.86	8.68
Tri-cresyl phosphate	40.70	36.24	4.46
Pyridine	38.00	37.16	0.84
Dimethyl formamide	37.30	32.42	4.88
2-ethoxyethanol	28.6	23.6	5.0
n-hexadecane	27.6	(27.6)	-

7.5 Components of Surface Energies for Various Polymers

Table 7.5-1 Components of Surface Energies for Various Polymers
(From Table 4 of Brewis and Briggs²⁷)

1 dyne/cm = 1 mNm⁻¹

Polymer	γ_s mNm ⁻¹	γ_s^d mNm ⁻¹	γ_s^p mNm ⁻¹
LDPE	33.1	32.0	1.1
Poly (vinyl fluoride)	36.7	31.3	5.4
Polypropylene	30.2	30.2	0
Polypropylene Flame treated	37.6	33.5	4.1
Poly (methylene oxide)	33.3	21.8	11.5
Poly (ethylene terephthalate)	41.3	37.8	3.5
Polytetrafluoroethylene	18.0	16.9	1.1
Polytetrafluoroethylene (etched)	50.6	36.1	14.5

7.6 Inherent Surface Energy Levels of Polymer Substrates

Table 7.6-1 Inherent Surface Energy Levels of Polymer Substrates (From Markgraf²⁸)

Polymer, Untreated	Dyne/cm Level
Nylon	46
Polyester, Mylar	43
Cellulosics	42
PVDC, Saran	40
Vinyl, acrylic	39
Ionomer (high)	37
Ionomer (low), polystyrene	33
Polyethylene, (LD & HD)	31
Polypropylene	29

7.7 Critical Surface Tensions of Polymers

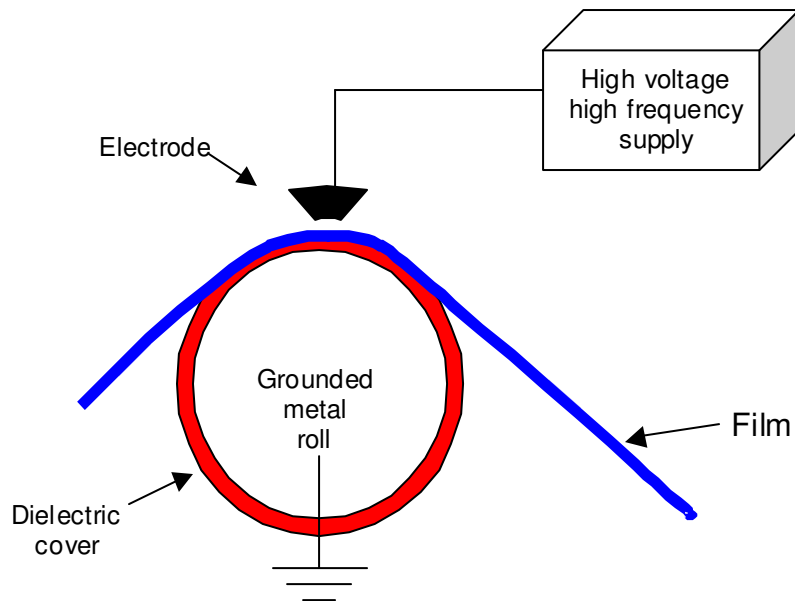
Table 7.7-1 Critical Surface Tensions of Polymers²⁹ (From Table 3 of Brewis and Briggs³⁰)

1 dyne/cm = 1 mNm⁻¹

Polymer	Critical Surface Tension mNm ⁻¹
Polyethylene, PE	31
Polypropylene, PP	31
Poly (vinyl chloride), PVC	39
Polystyrene, PS	43
Poly (ethylene terephthalate), PET	43
Poly (hexamethylene adipamide), Nylon 6,6	46

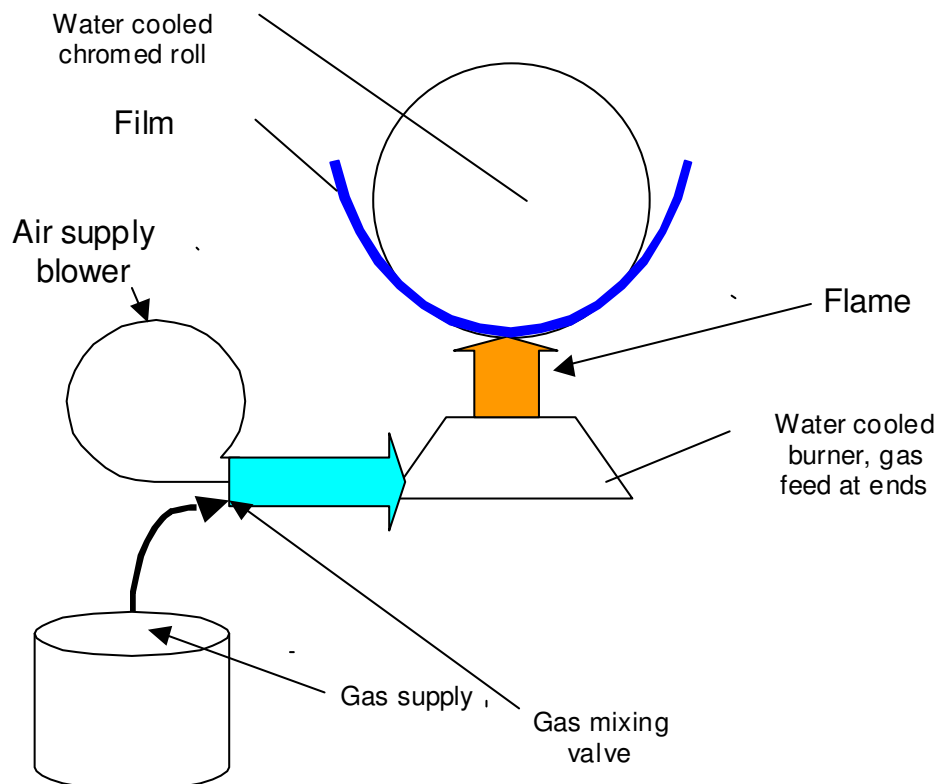
7.8 Schematic of Corona Treatment

Figure 7.8 -1 Schematic of Corona Treatment



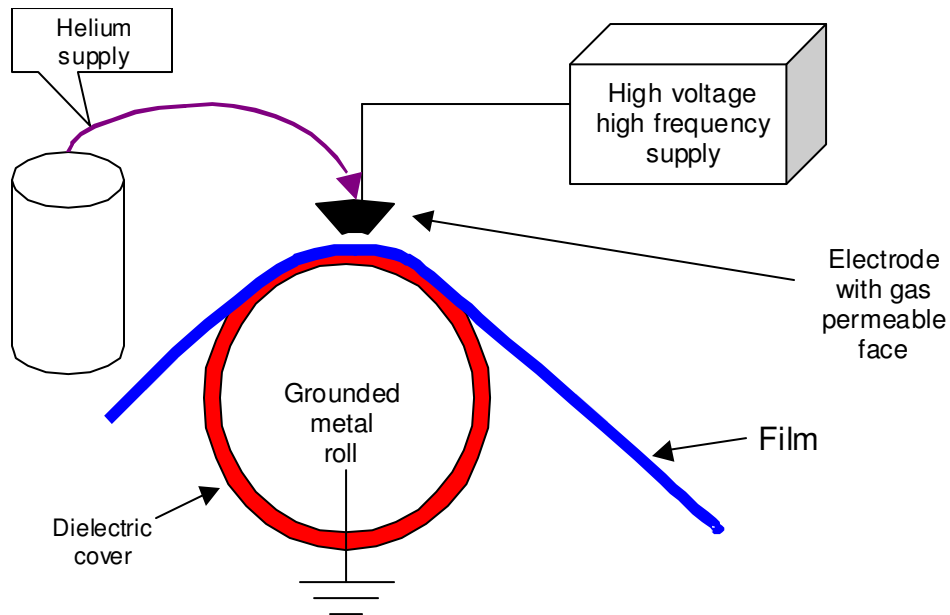
7.9 Schematic of Flame Treatment System

Figure 7.9-1 Schematic of Flame Treatment System



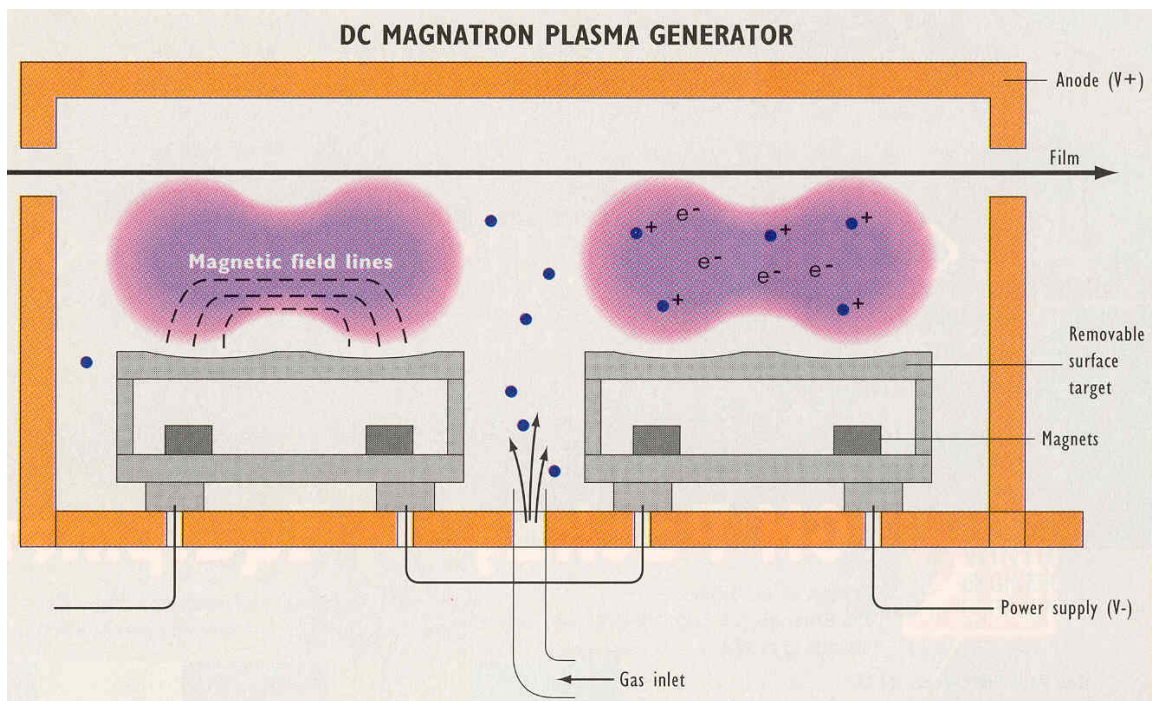
7.10 Schematic of Atmospheric Plasma System

Figure 7.10-1 Schematic of Atmospheric Plasma System



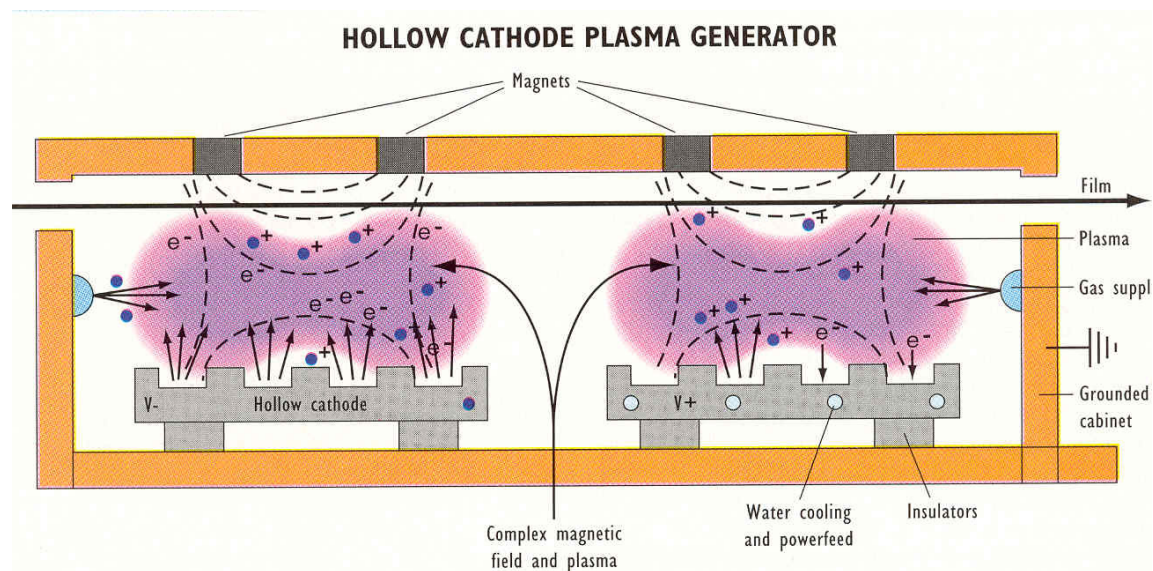
7.11 Schematic of Magnatron Plasma Treater

Figure 7.11-1 Schematic of Magnatron Plasma Treater



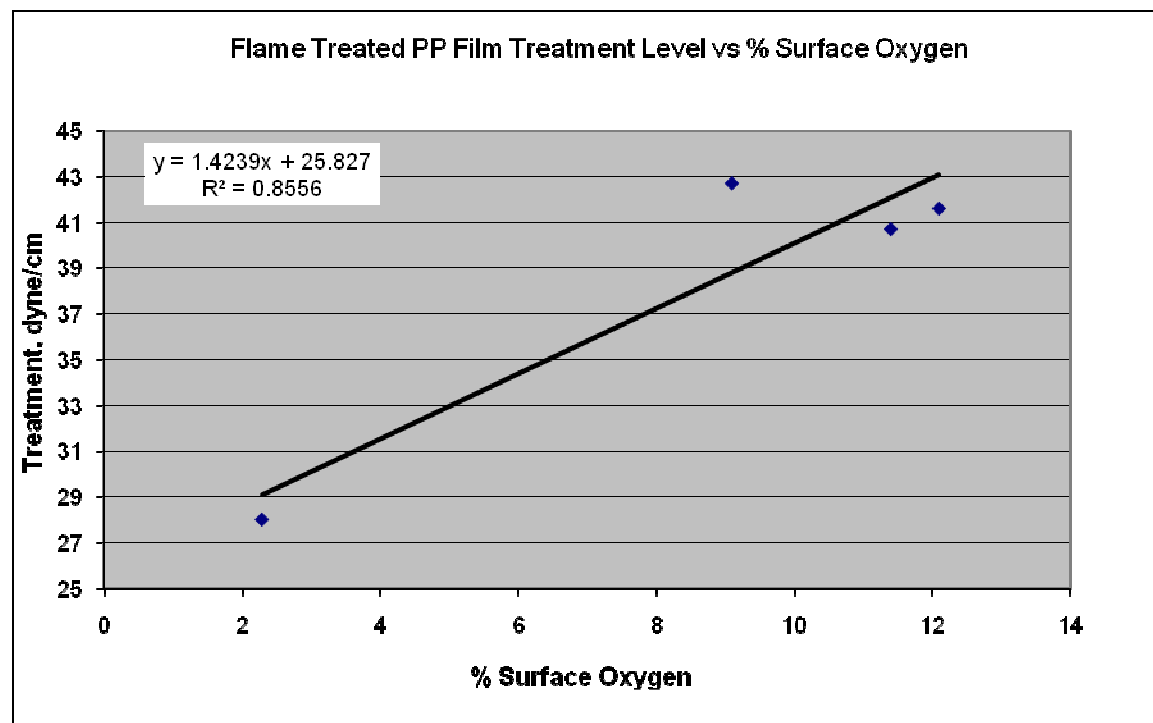
7.12 Schematic of Hollow-Cathode Plasma Treater

Figure 7.12-1 Schematic of Hollow-Cathode Plasma Treater



7.13 Flame-treated PP Film Surface Oxygen Level as Function of Treatment Level for One-pass Treatment

Figure 7.13-1 Flame-treated PP Film Surface Oxygen Level as Function of Treatment Level for One-pass Treatment (From Garbassi et al.³¹)



7.14 Section 7 References Cited

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²⁸ Ibid., Markgraf, Table II, p. 76.

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Section 8: Substrate Materials for Metallization

Introduction

Metallization substrates form supports on which the aluminum layer is condensed during the metallization process. The support's principal purpose is to supply the mechanical and thermal properties required for the film application in which the metallized film will be used. In addition (*see Section 3*), the substrate design and composition can play a key role in determining the final gas, moisture, and chemical barrier properties of the metallized film. In this section we will not consider metal foils as a substrate for metallization although they can be used.

What makes a substrate acceptable for metallization is a complex question primarily related to the required end-use properties. For example, the difference in expected behavior between metal transfer products, where 100% metal lift is required, and a metallized laminating film where 0% metal lift is desired and 100% metal lift is considered a failure. Aside from these property requirements, the apparent suitability of a substrate also may be determined by the metallizer itself. This is understood in terms of the thicknesses and film strength a particular machine is able (or designed) to handle without mechanical damage from the web-tension system, potential thermal damage from the deposition, the web-cooling systems capacity, or for paper metallization, water pumping capabilities.

These aspects of the suitability of a substrate have to be determined in conjunction with the operation and setup of the metallizer and cannot be separated. In general, we have noticed that if the substrate can be unwound, it can be metallized and the only question is, if the final metallized product is suitable (*i.e.*, properties and manufacturing costs) for its intended application. If not, then how can THIS particular metallizer be made to successfully produce the product economically? For the remainder of this section, we will assume that the substrates described are inherently suitable for metallization, but recognize that particular metallizers may have difficulty with any given one, based upon its design and operation.

8.1 Polymer-based Substrates

The principal materials to be considered as metallizing substrates here are natural and manmade polymer films and webs. Useful substrates for vacuum metallization are materials that are in a form that can be wound and unwound into cylindrical rolls without difficulty. The size of the roll that can be metallized is primarily controlled by the size of the vacuum chamber into which it will be placed, and to some degree, the properties and manufacturing methods used to produce the substrate. As discussed in Section 11, the formation of the roll and its density can have some impact on the metallization process productivity and general operating success.

Following the production of an acceptable roll form for the substrate, the primary consideration of a substrate is two-fold: the polymer, that comprises the bulk of the substrate and supplies the primary mechanical and thermal properties; and the surface polymers of the substrate onto which the metal will be deposited. To some degree, the opposite surface that will contact the metallized surface when the metallized film is wound and stored also should be considered. Since film and web manufacturing processes are very versatile, today there is an endless array of polymer combinations available for metallization. Consequently, when specifying a substrate, it is important to understand the product design and how it can impact the success of the metallization process. While most substrates are multilayer, first focus should be on the primary polymers used to form the substrates in addition to the manufacturing methods used to produce them, since this impacts the physical and thermal stability of the materials. For example, in the case of biaxially oriented polypropylene (BOPP) and cast polypropylene (CPP), both are made from PP, but the mechanical and barrier properties of each substrate are very different due to the methods used to produce them as film. Paper- and polymer-fiber-based substrates will be considered separately.

At the end of this section are graphical comparisons of various substrate properties for 13 representative polymers widely used as films (*Figures 8.1-4 to 8.1-12*). The properties, shown with a minimum and a maximum for each polymer, can vary widely depending on the film thickness and the manufacturing process used. The graphs are not to be considered universal and should be used only as a broad comparison between the materials. When actually contemplating the use of a substrate, the reader should obtain information about it directly from the manufacturer.

This will insure that the user has the best information available when setting up the metallization process, or in determining the likelihood of the success of that substrate when metallized on particular equipment. Property information about substrates is also useful in evaluating the converting and end-use applications being contemplated.

Substrate Property Development

As described below, polymer properties are dependent on the structure in the polymer, the environmental conditions in which they are used (primarily on the temperature, but perhaps on moisture levels), and the rate, or time interval, at which they are subjected to forces. Therefore, properties should be tested at conditions approaching the intended environment in which they will be used or processed. Also, some knowledge of the substrate's physical structure, continuous film or fiber matt, polymer crystallinity, molecular orientation levels, and how they were formed is valuable in interpreting the potential and real performance of the substrate.

To gain a better understanding of the impact of polymer selection and the substrate manufacturing process on the property ranges developed in various substrates, a discussion of variables controlling basic mechanical properties of polymers and the various manufacturing processes used to produce films are presented. Basic polymer mechanical, barrier, and chemical properties are given for molded polymer samples, representing the polymer in its simplest form under standard testing conditions. Polymer processing is designed to develop specific structure/property combinations, *e.g.*, polymer properties based upon the development of specific polymer morphologies. Final mechanical, barrier, and chemical properties of the substrates manufactured may vary greatly. Therefore, it is important to obtain information about the properties of the polymer in the form it is actually to be used, perhaps from the supplier from which it is purchased.

Basic Polymer Principles

The strength of a polymer-based substrate web is its most important property in the consideration of its use as a metallizing substrate. The web strength will depend on many variables. Figure 8.1-1 shows the five principal solid-state behaviors of a polymer.

Figure 8.1-1 The Five General Behaviors of Polymeric Materials

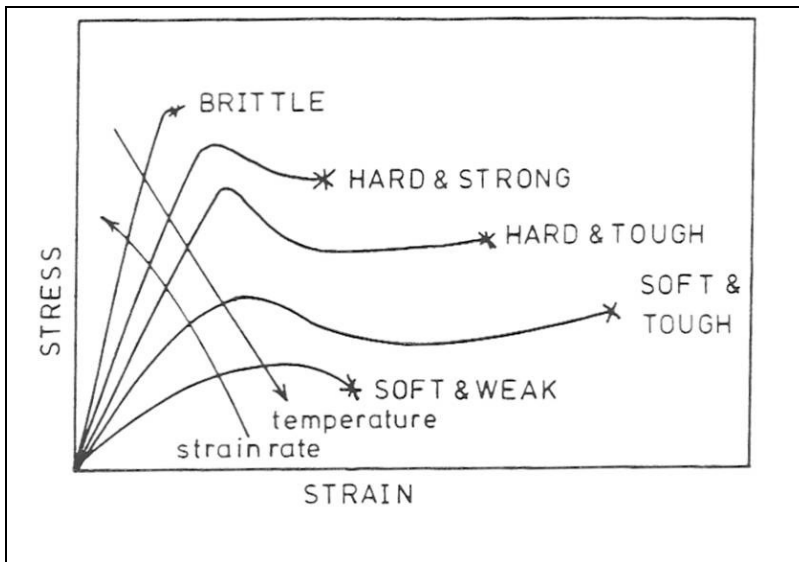
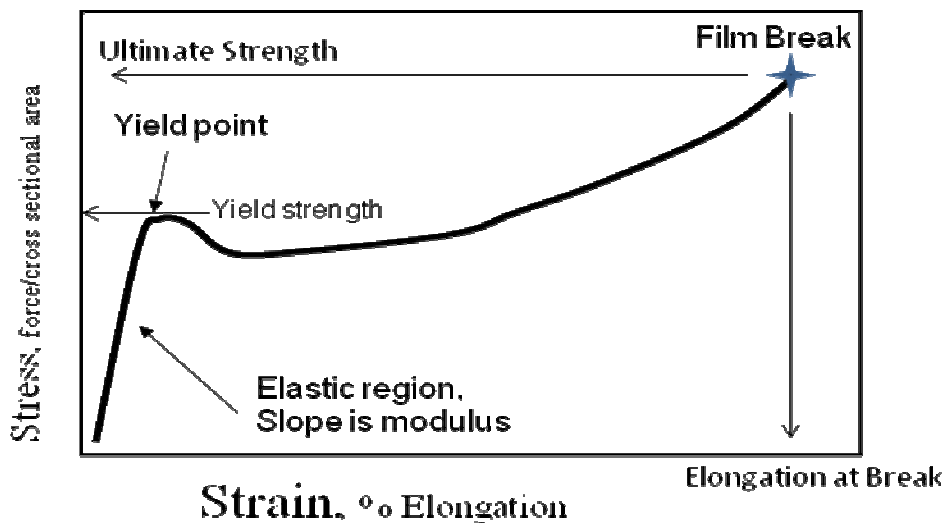


Figure 8.1-1 is the stress (force/cross sectional area) vs. strain (increase in length) graph, showing brittle-to-ductile changes and strong, tough, and weak behaviors of the polymer. In Figure 8.1-2, the chart is redrawn with a single stress-strain curve to help better define the portions and meaning of the curve.

“Brittle” is defined as a material that when stretched, breaks without yielding (permanently deforming), giving no warning. “Ductile” describes a material that stretches and elongates beyond the yield point, before breaking, giving a clear indication that the polymer web is failing in tension. “Strong” is represented as the initial slope of the stress vs. strain plot, and it is defined as the “modulus.” The higher the modulus (steeper the slope), the harder it is to stretch and elongate the polymer. If the slope, or modulus, is decreased, the polymer is weaker, easy to stretch, and not as “strong.” The definition of “toughness” is the area under the stress-strain curve. This represents a measure of the work (force times distance) required to break the sample, and the tougher the material, the more work required to break it.

Figure 8.1-2 Plot of Tensile Curve for Polymer Materials



It is important to realize that all polymers will show all five behaviors, depending on the temperature and the length of the test time. Which curve the polymer falls on at room temperature is generally the characteristic we assign to it. For example a piece of Silly Putty® is weak and easy to stretch at room temperature while a polystyrene cup is hard, strong, and brittle. A ball of Silly Putty® placed on a table will slowly flow under the force of gravity and form a shapeless puddle, like a liquid. But, if the Silly Putty® is put in liquid nitrogen, the molecules of the Silly Putty® are no longer able to move past each other and the polymer becomes hard, strong, and brittle, shattering like glass if hit with a hammer. By the same token, if the polystyrene cup is put in the oven at 150C (302F), it would soften and shrink into a shapeless puddle like the Silly Putty®.

The temperature at which the polymer changes from a hard, strong solid to a weak, soft liquid is called the glass transition temperature (T_g). All polymers have a unique T_g . The T_g for different polymers vary from -200C to +400C. Polymers that have only a T_g are called amorphous polymers because they exist only as a liquid or frozen glass with no long-range molecular order. Other polymers are able to form crystals with long-range order, and they will also have a crystalline melting point (T_m). Crystallizable polymers are never 100% crystalline. They are generally described as semicrystalline, and the degree of crystallinity will impact their mechanical, barrier, and thermal properties such as seen with low- and high-density polyethylene (LDPE and HDPE). Some polymers such as polyethylene terephthalate (PET) with a relatively high T_g (77-80C), can be produced in an amorphous or semicrystalline state, depending on the conditions to which they are subjected. In comparison, polymers such as HDPE cannot be produced in a 100% amorphous state, because they have a rapid crystallization rate and a T_g (-83°C)¹, far below room temperature. The amount of crystals formed will affect the stiffness and the density of the polymer while the form of the crystal will affect its melting point.

The wide range of physical properties a polymer can display is related to the size, and in particular, the length, of the polymer molecule. Polymer molecules are in the form of a long chain, being so long that they entangle, much like long pieces of string or spaghetti. It is primarily this chain entanglement that gives the polymer its mechanical properties, for, in effect, if we pull on a polymer molecule, we are really pulling on many polymer molecules because of the entanglements. For instance, the chemical composition of lighter fluid and paraffin wax is identical to that of PE. Lighter fluid is a liquid, paraffin wax is a highly crystalline and brittle solid, while PE is semicrystalline, stronger, more ductile, and a tougher solid than the wax. A single chain of PE may contain 100,000 or more separate building block molecules (called monomer units), while the lighter fluid contains three and the paraffin may contain as many as 50. Most non-polymeric materials, (*e.g.*, sugar, salt etc), only contain one molecular unit.

In addition to the length of the polymer molecule, its orientation can impact the strength of the polymer. The strengths of the chemical bonds along the chain are much stronger than the molecular interactions between the polymer chains. Therefore, if the chains are ordered (oriented) in one direction more than another direction during manufacture, the polymer properties will be dependent on the direction (anisotropic) in which they are measured in the polymer sample, relative to the principal orientation direction. With the possible exception of solvent casting, each film manufacturing process imparts some level of molecular orientation to the substrate, which is then a source of dimensional instability since the polymer will relax (shrink) and dissipate some portion of the orientation. How quickly this shrinkage occurs depends on the temperature at which the polymer substrate is heated relative to the manufacturing temperatures and the T_g and T_m of the polymer.

Aside from the balance in the orientation, the level of orientation will profoundly affect the substrate's mechanical properties. In general, during substrate manufacture, the polymers are melted in an extruder and then formed from the melt into its final form (blown or cast film or nonwoven fiber mat), and they will retain some level of melt orientation in the solid polymer. Higher levels of polymer orientation can be obtained by drawing the polymer from the solid state, as in uniaxial (one-direction) orientation or biaxial (two-direction) orientation (*see below*). In amorphous polymers the drawing is performed slightly above the T_g in the rubber state, and in semicrystalline polymers, the drawing is performed above the T_g , but below the T_m of the polymer. How dimensionally stable the drawn films are depends on the manufacturing conditions and any subsequent heating and/or relaxation of the film.

Because of the wide range of physical and thermal properties potentially available with substrates produced from different materials, different manufacturing processes, and between different manufacturers, it is important to understand the properties of the films purchased and their intended use.

Substrate surface properties will be dependent primarily on the polymer at the substrate surface, as well as any chemical modifications made to the substrate surface. These may include surface treatment (*see Section 7*), surface coating, or priming, performed prior to metallization by the substrate manufacturer, or as with in-chamber processes such as plasma treatment or polymer overcoating by the metallizer. In today's substrate manufacturing environment, it is very unlikely that a commercially available substrate will be a single-layer material. In general the substrate will be at least two layers with three to seven layers being the norm. Improvements in polymer technology, polymer availability, and multilayer extrusion equipment capabilities are leading to the manufacture of substrates with ever more layers. Currently, commercial substrates of 25 to 125 microns (1 to 5 mils) with as many as 132 to 1000 layers are available.

Table 8.1-3 Polymers Commonly Used for Metallizing Substrates

Polymer	Acronym(s)	T _g , °C	T _m , °C	Key characteristic as film
Polypropylene	PP	-20	166	Low density, moisture barrier
Polyethylene Terephthalate	PET	79	265	High strength, temperature resistance
Polyethylene Naphthalate	PEN	125	268-270	High strength, temperature resistance
Polyamide	Nylon 6	45-50	218	Tough, oxygen barrier
	Nylon 66	45-50	265	Tough, oxygen barrier
Low Density Polyethylene	LDPE	< -80	110	Low cost, many uses
Linear Low Density Polyethylene	LLDPE	< -80	125	Toughness, many uses
High Density Polyethylene	HDPE	< -80	135	Moisture barrier, dead fold
Polystyrene	PS	100	na	Clarity,
Polyimide	PI		na	High temperature resistance
Polyvinyl Chloride	PVC	80-90	175	Stretch and shrink films
Polyvinylidene chloride	PVDC, Saran ®	-4	160	Good gas and moisture barrier properties
Polytetrafluoroethylene	PTFE, Teflon ®	125	326	High temperature usage
Polymonochlorotrifluoro ethylene	PCTFE, ACLAR®	55	210	Lowest moisture barrier
Polycarbonate	PC	150	na	High temperature resistance
Polyvinylidene Fluoride	PVDF	-40	170	UV resistance, high temperature
Cellulose	Cellophane		burns	Easily modified by coating
	Paper			Porous, dead fold
Cellulose Acetate			232	Clarity and gloss
Polyvinyl Alcohol	PVOH	85	burns	Water soluble, highest oxygen barrier
Polyethylene Vinyl Alcohol	EVOH		157	Excellent oxygen barrier
Polyurethane	PU	-51	Varies	Toughness, flexibility & durability

Figure 8.1-4 Comparative Yield of Various Polymers as Films of 25-Micron (0.001in) Thickness

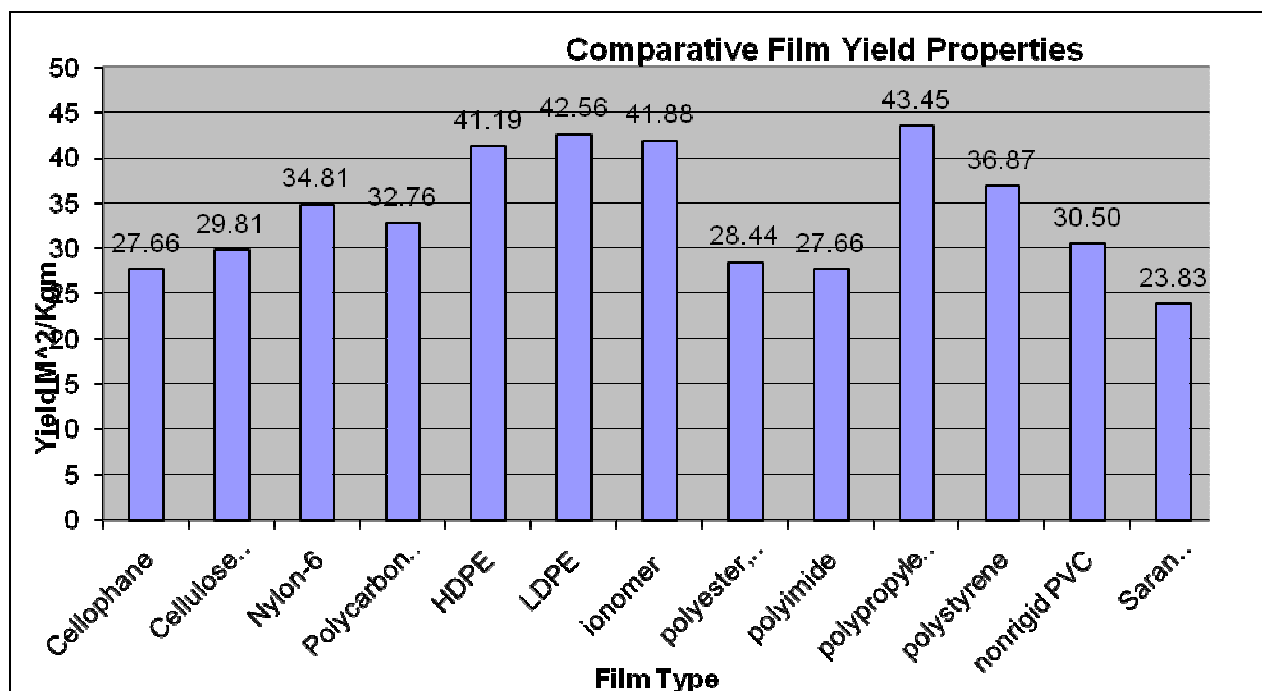


Figure 8.1-5 Comparative Specific Gravity Values for Polymers

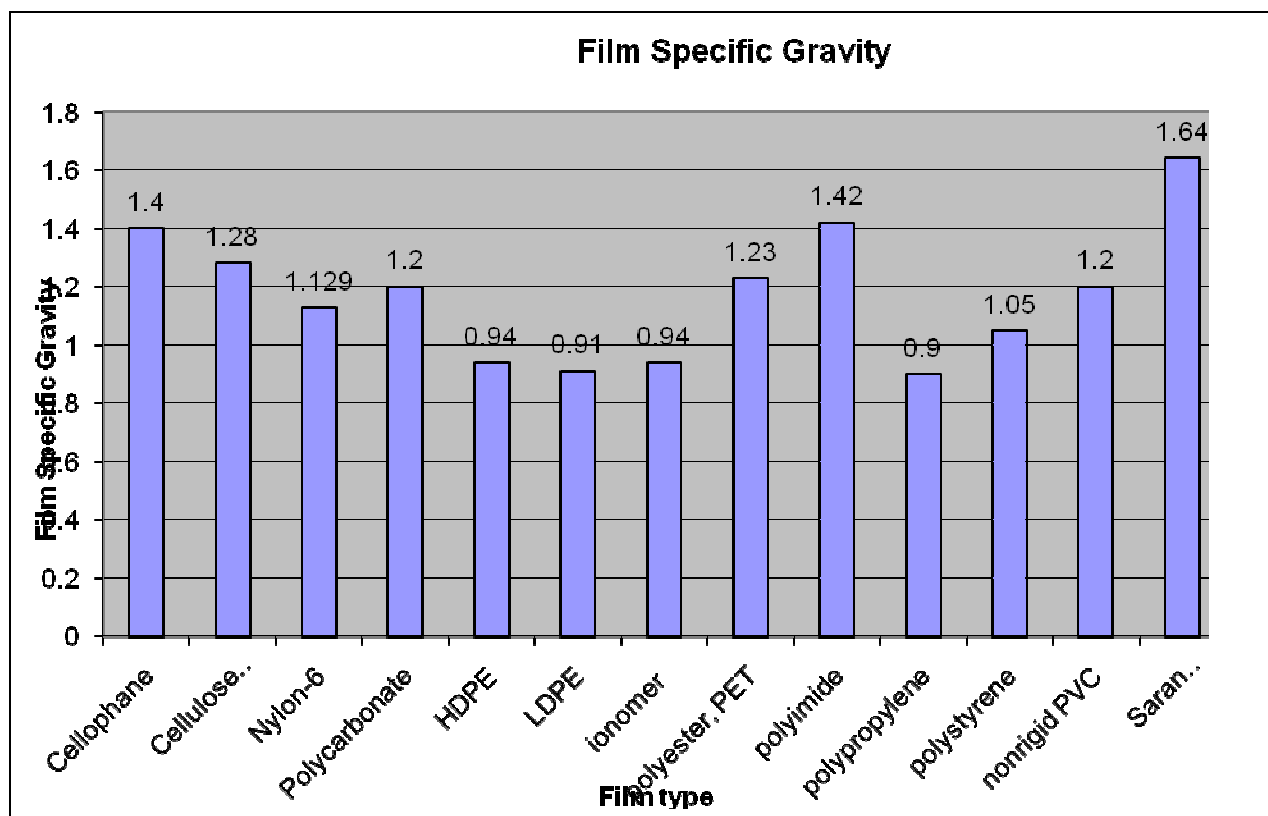


Figure 8.1-6 Comparative Film Modulus

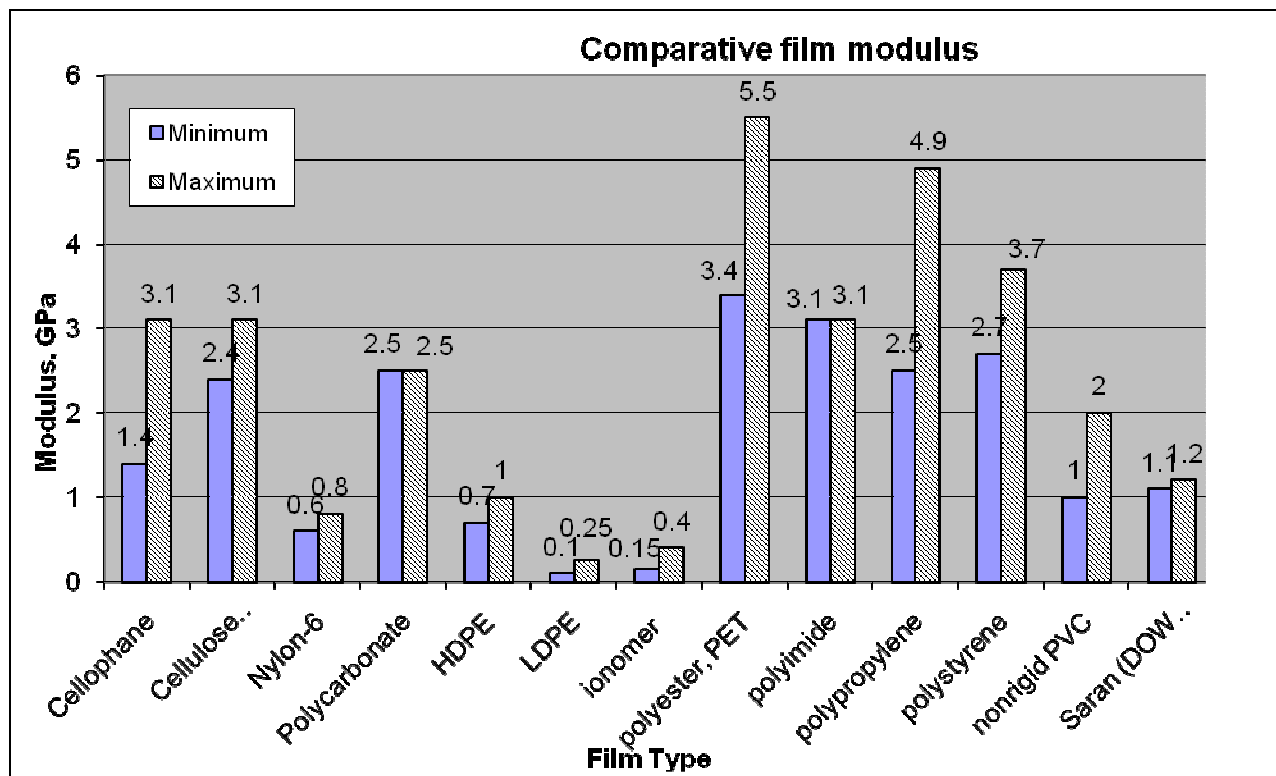


Figure 8.1-7 Comparative Film Ultimate Strength

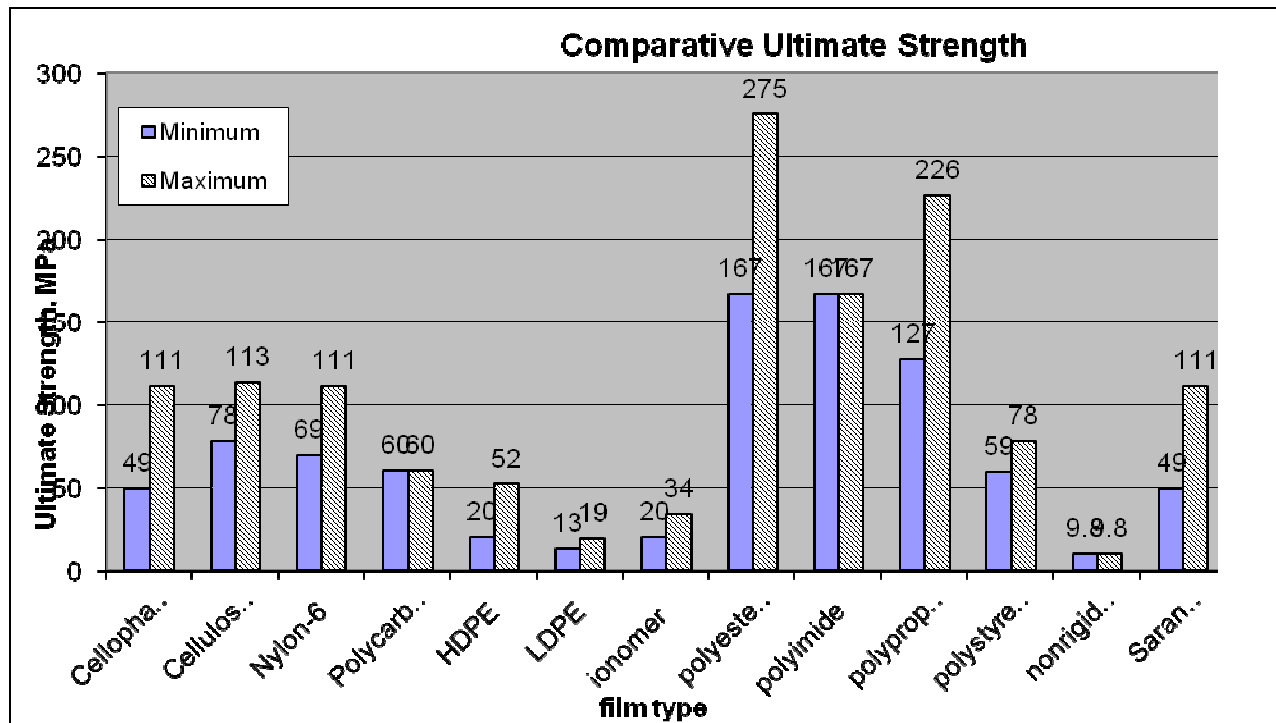


Figure 8.1-8 Comparative Elongation to Break

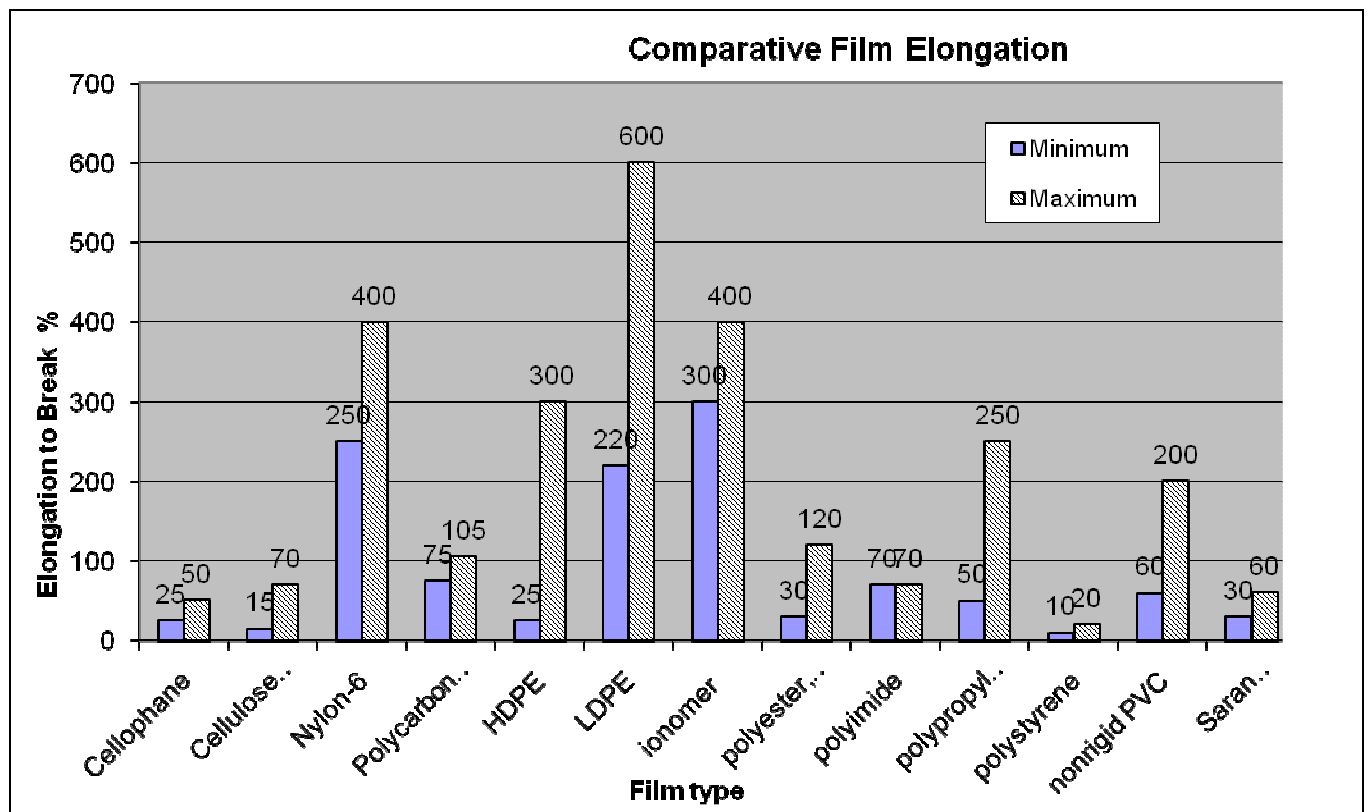


Figure 8.1-9 Comparative Tear Strength

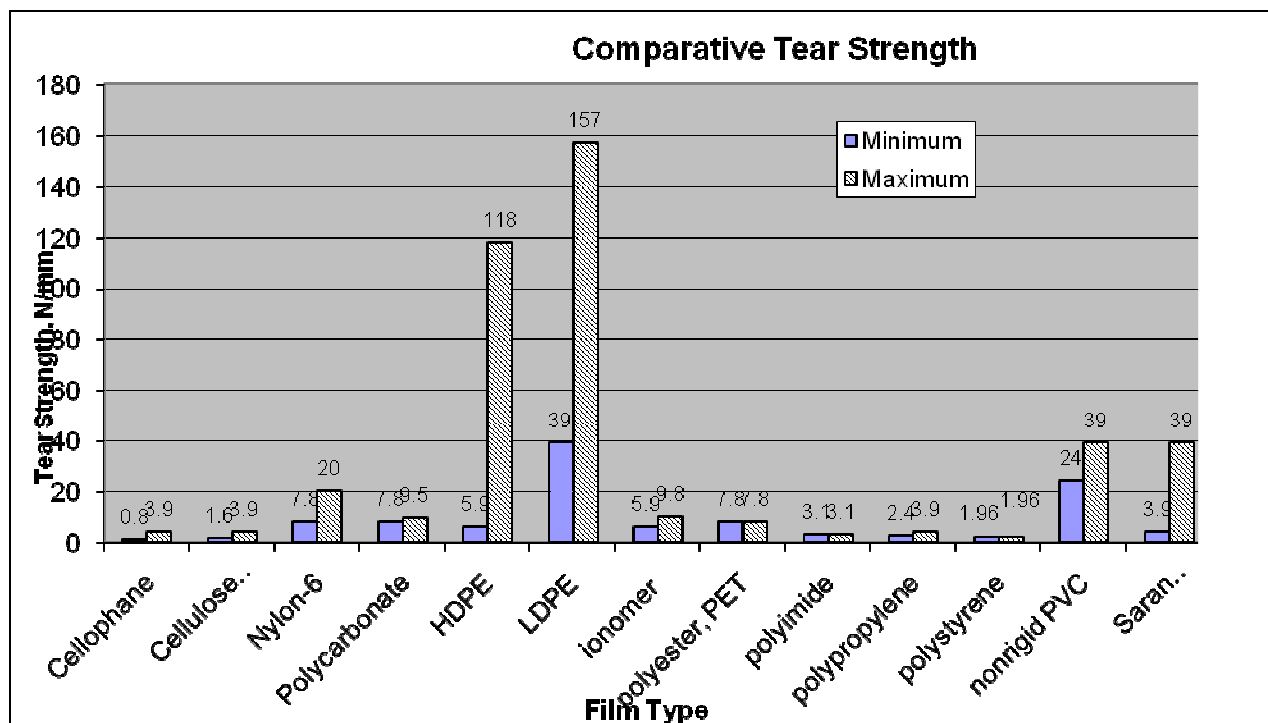


Figure 8.1-10 Comparative Moisture Barrier (WVTR)

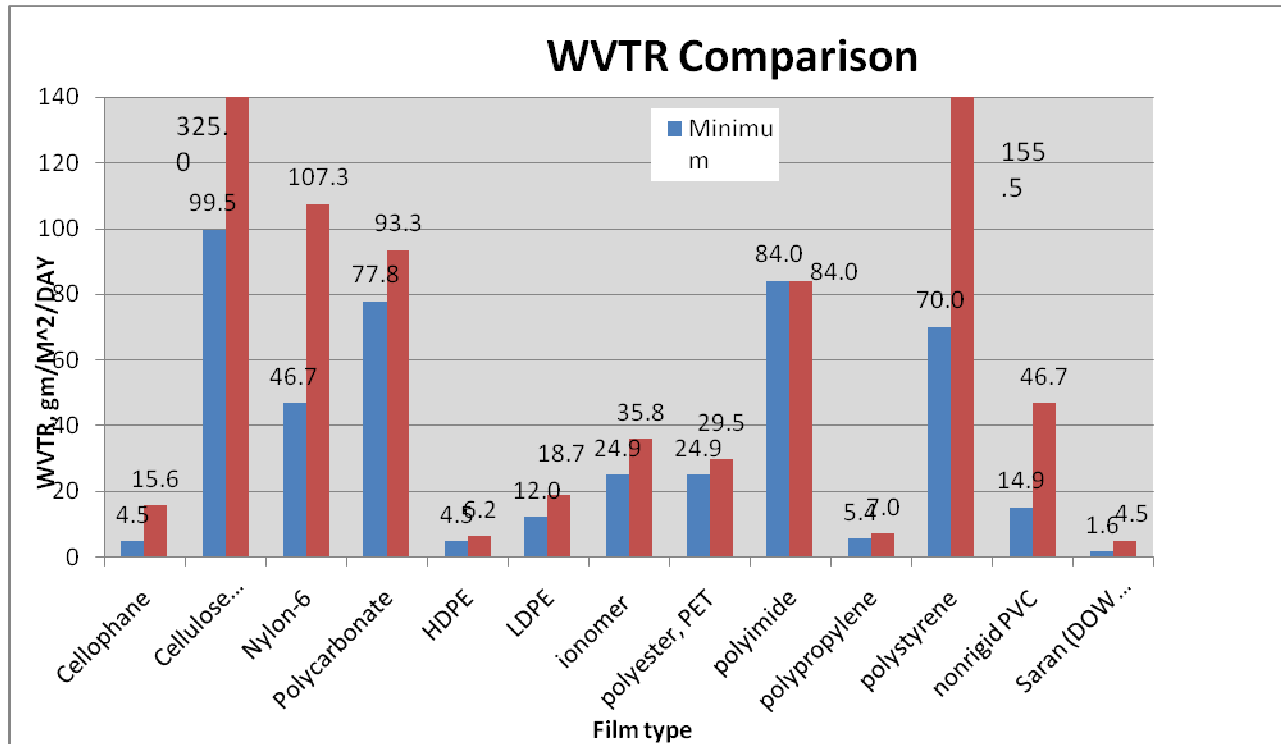


Figure 8.1-11 Comparative Oxygen Permeability (TO2)

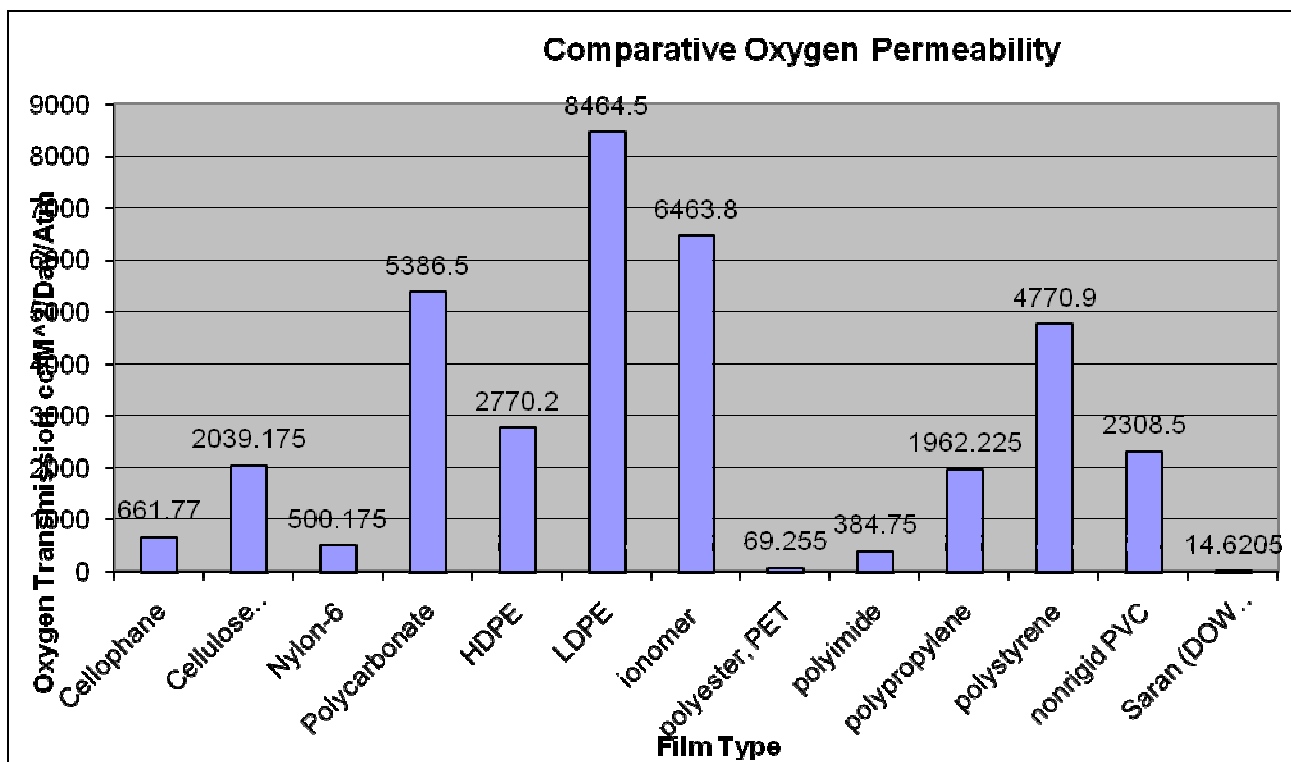
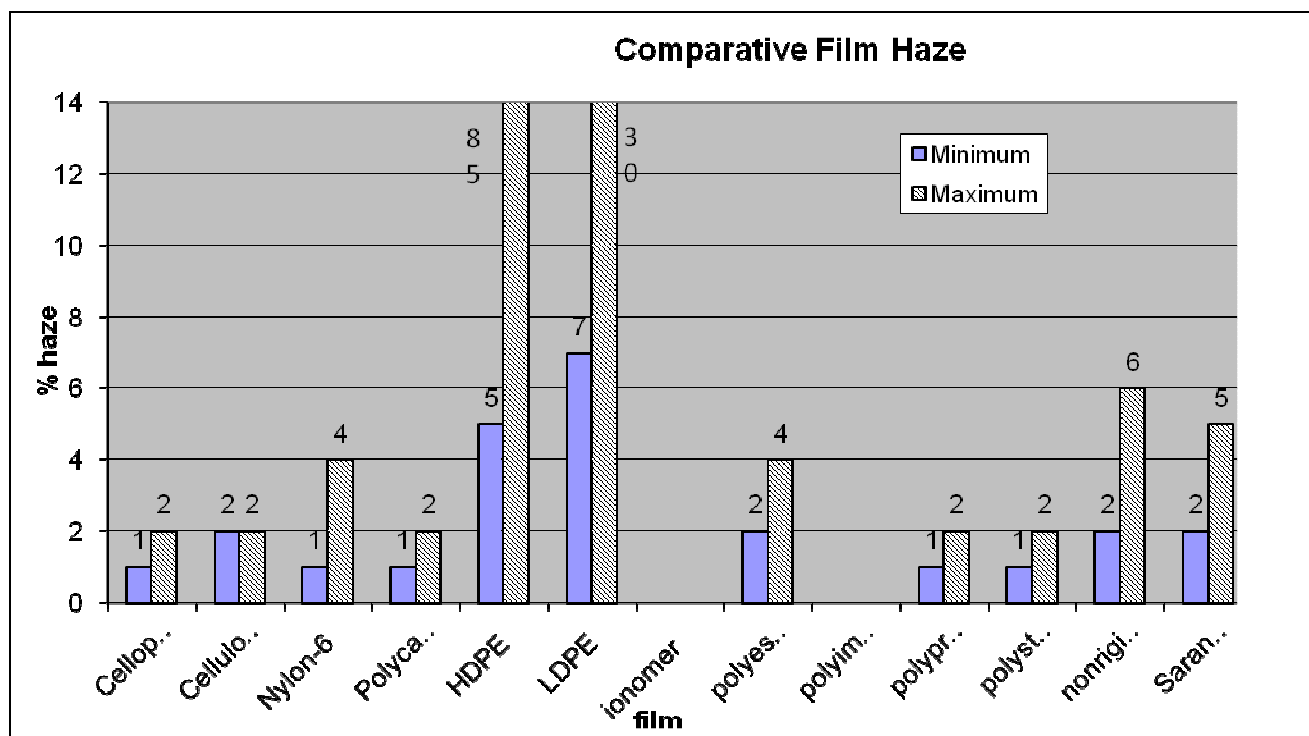


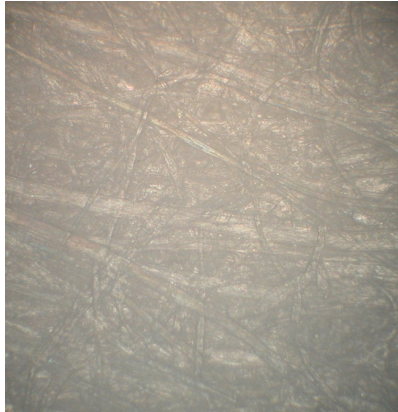
Figure 8.1-12 Comparative Film Haze



8.2 Paper and Other Substrates

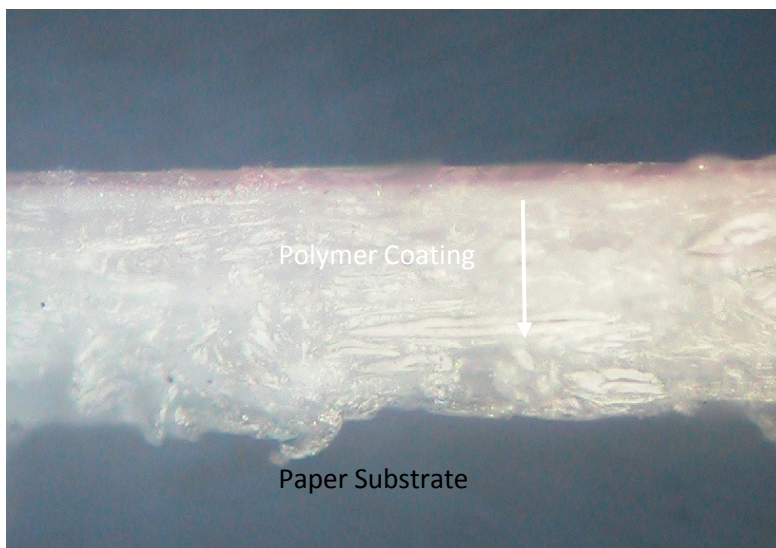
Metallized papers are generally used in decorative applications. They depend upon a highly reflective aluminum layer as the starting point in conjunction with embossing for the production of holographic and refractive images and effects. Figure 8.2-1 shows the magnified surface of a typical paper surface at 250X magnification and clearly shows the fibrous nature of the native paper surface.

Figure 8.2-1 250X Magnification of Uncoated Metallizing Paper Surface



This fiber matt is not suitable for direct metallization as it will not produce a highly reflective surface. Therefore papers for metallization are routinely coated with a polymer layer to give a smooth high gloss surface for the deposition of aluminum much like a smooth glass surface is prepared to produce a mirror. The polymer surfaces are also necessary for the production of refractive surfaces, since the surface coatings are heated and pressed into a textured embossing roll to transfer the refractive image to the film surface. The polymer coatings often are based upon acrylic polymers, which are readily formed into solutions for coating and have thermal properties that allow them to be readily embossed. Figure 8.2-2 shows a cross section of a coated paper at 400X magnification.

Figure 8.2-2 Cross Section of a 0.118mm- (0.0465 in-) thick Metallized Paper Showing Polymer Coating on Paper Surface (magnification 400X)



The use of paper-based substrates for metallization requires special care in the selection of the paper support. This is due to the sensitivity of the aluminum to chemical attack and subsequent corrosion (*see Section 5*). The paper manufacturing process requires the use of aggressive, caustic chemicals to aid in the separation of individual cellulose fibers in the pulping process. If these caustic materials are not removed, they will chemically attack the metallized aluminum layer and convert it to aluminum hydroxide or aluminum oxide when the metallized paper is wound into a roll. For this reason, special clean grades of paper are necessary for the successful production of metallized paper.

The moisture content of the paper needs to be controlled as well, to minimize the impact on vacuum chamber pressures and oxidation of the aluminum in the vapor state, which can result in poor reflectivity or discolored aluminum films². Typically, the moisture level of the paper entering the metallizer should be 2% or lower to maintain a controllable metallization process. Paper metallizers are usually equipped with additional water pumping capacity in the form of polycold units designed to capture water vapor and hold it as ice during the metallization process. Once the film has been metallized, it should be humidified to improve the base properties of the paper for subsequent converting and use.

Woven and Nonwoven Polymer Substrates

Aside from paper, other polymer-fiber-based substrates exist, which can be coated and metallized as paper. Some can be metallized directly without coating. Typical nonwoven polymer substrates commercially available are:

Table 8.2-3 Several Polymers and the Corresponding Commercial Nonwoven Products Trade Name

Polymer	Trade Name
HDPE	Tyvek®
PP	TYPAR®
Polyimide	Kevlar®

In general these applications are decorative and used primarily for novelty items. For instance metallized TYVEK® has been used as an outer cover for novelty jackets used for company marketing and they have a dull metallic luster.

8.3 Section 8 References Cited

¹ Billmeyer Jr., F.W, *Textbook of Polymer Science*, second edition, John Wiley & Sons, New York (1962, 1971), p. 230.

² Bishop, C.A., “*Vacuum Deposition onto Webs, Films, and Foils*,” William Andrew, Inc., Norwich, NY, (2007), pp. 164-165.

Section 9: Substrate Manufacturing Technologies

Introduction

There are a number of technologies used to manufacture the substrates for metallization. Section 9 focuses on polymer substrates and the various processes used to create these films.

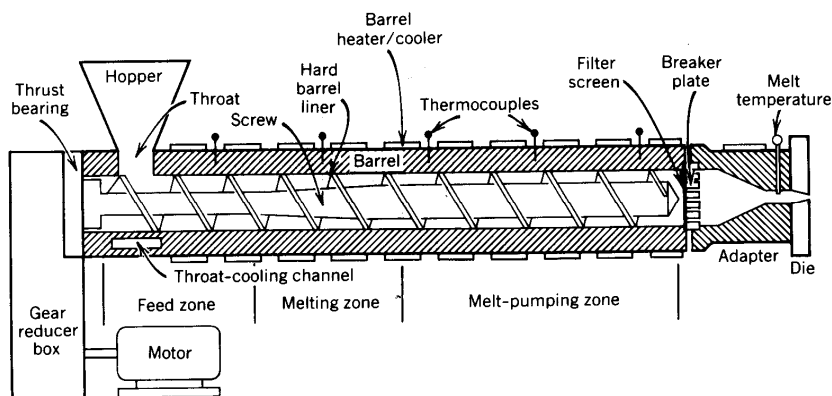
Table 9.0-1 Comparison of Polymers and Possible Film Manufacturing Methods

Polymer	Manufacturing Methods						
	Solvent cast	Melt cast	Blown	Uniaxially oriented	Biaxially oriented (tenter)	Biaxially oriented (double bubble)	Nonwoven (or fiber)
PP	No	Yes	Yes	Yes	Yes	Yes	Yes
PET		Yes		Yes	Yes		Yes
PEN		Yes		Yes	Yes		
Nylon 6		Yes	Coextruded	Yes	Yes	Coextruded	Yes
Nylon 66		Yes	Coextruded	Yes	Yes	Coextruded	Yes
LDPE	No	Yes	Yes	Yes		Yes	
LLDPE	No	Yes	Yes	Yes	Yes	Yes	
HDPE	No	Yes	Yes	Yes	Yes	Yes	Yes
PS	Yes	Yes	Yes	Yes	Yes		
PI	Yes	Yes		Yes	Mandrel		Yes
PVC	Yes	Yes	Yes	Yes	Yes	Yes	
PVDC, Saran ®	Yes	Yes		Yes		Yes	
PTFE, Teflon ®	No						
PCTFE Polymonochloro trifluoroethylene		Yes		Yes	Yes		
PVF	No						
PVDF	No	Yes		Yes	Yes		
Cellophane	Yes	No	No	No	No	No	Yes
Paper	No	No	No	No	No	No	Yes
PVOH	Yes	No					
EVOH	Yes	Yes		Yes	Yes	Yes	
PVA	Yes						Yes
PU	Yes	Yes	Yes				

9.1 Single-screw Extruder

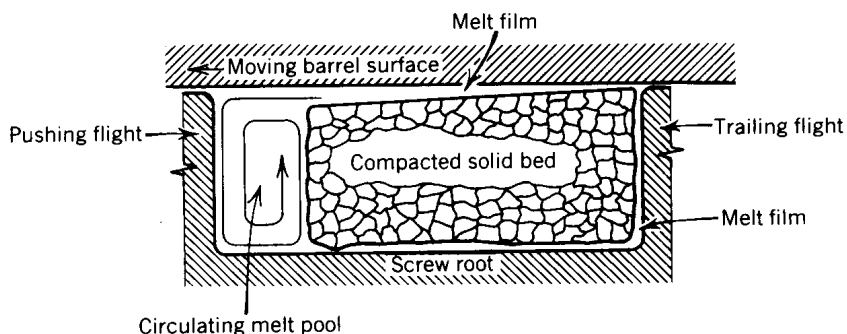
With the exception of solvent casting, where the polymer is first dissolved in a solvent, the majority of polymers are formed into substrates by first melting the polymer in an extruder and then forming the film¹. A single-screw extruder, shown in Figure 9.1-1, is a device consisting of a framework with a motor, gear box, heated barrel, and helical screw turned by the motor².

Figure 9.1-1 Schematic of a Single-screw Extruder



Polymer is fed to the screw from a hopper through an opening in the barrel. As the screw rotates, it picks up and pushes the polymer into the screw where it is compacted and rubbed against the hot barrel surface (*Figure 9.1-2*) forming a thin, highly sheared, viscous film.

Figure 9.1-2 Schematic of the Melting Mechanism of Single-screw Extruders



This process converts the mechanical power of the motor into heat by viscous dissipation, creating the energy needed to melt the polymer at the high rates required to produce film. For modern, high-capacity film orienting lines running 2200 to 5500kg/hr, the use of a tandem extrusion system (two extruders: one to melt and feed; the second to pump) or a twin-screw extruder (two screws that intermesh and rotate) is becoming more common.

In most instances, films are melt cast, melt blown, melt cast and stretched or solvent cast. Solvent casting is done in cases where the polymer does not melt (cellulose or PVOH) or where very uniform physical or optical properties are desired. Oriented films are made by first melt casting a flat or tubular polymer form and then reheating and stretching the flat sheet in a tenter or the tubular sheet in a double process. Therefore, the following will describe casting, followed by melt blowing film and orientation by tenter and double bubble processes. Each manufacturing method is important to understand since it determines the level of molecular orientation, and therefore, the physical properties, film surface, and optical characteristics of the polymer film.

9.2 Cast Films

Cast films are made by laying down polymers from a solution or as a melt onto a moving surface with a minimum amount of orientation. This gives the films the basic thermal and mechanical properties of the polymer, and usually with good film optics. The quality of cast films can cover a large range with films that are suitable for optical and photographic end uses as well as films for pallet wrapping for industrial transportation and bundling, food wrapping and storage, and mulching films for ground cover. The films are generally uniform in properties, but may have small amounts of directional residual orientation from the fabrication method, which can permit some film shrinkage on heating. Typically, the films are noted for their overall toughness and elongation, but this is dependent on the material and fabrication method. Overall, casting of films is an important area of commerce, and many cast films find their way into many converting operations including metallization and coating.

Films may be cast directly from molten polymer or from polymer solutions pushed through a slot die to produce thin films. Cast films are characterized by the general low levels of orientation in the finished films, and the casting method chosen will control the level of residual orientation in the films. The low orientation levels give them a range of toughness, elongation, and tear resistance which is not possible with oriented films, but they are also not as strong and are much more sensitive to tension levels and stretching than oriented films of the same polymer. Of course polymer selection will control the resistance to tension and stretching through the modulus of the polymer. This low residual orientation gives cast films excellent sealing performance in terms of water- and air-tight package formation as well as permitting thermoforming of the material into blister packages and other forms. In many ways cast films are like blown films in that the blown films are also essentially unoriented (*see Blown Film Production*). For melt casting, the use of cooled casting systems permits higher production rates compared to blown films, but for some end uses, this is offset by higher capital costs for the equipment.

There are two principal methods of producing cast film: melt extrusion and solvent or solution casting. The method chosen to cast the film is dependent on the processing characteristics of the polymer and the optical and mechanical uniformity of the film required in the final application. For instance, cellulose is not melt processable (It burns before it melts.), due to its molecular structure, and consequently must be dissolved, precipitated, and dried to give cellophane films. In films for photographic applications where excellent dimensional stability as well as uniform optical properties are required, solution cast films are used, rather than melt cast, where residual and directional stress will exist in the films.

Cast film surface properties also vary depending on the polymer, the rate of production, surface quality of the casting surface, film layer structure, and film additives. In general, there are two surfaces to the film: one from the contact of the material with a metal surface and the second surface, which does not contact any solid surface, and is against air. In some instances, the film can be formed between two metal rolls (polishing), but this is typically used with films thicker than 0.12mm (5 mils).

Usually, the air side has the surface texture inherent to the polymer as it is cooled. This can range from smooth and glossy for amorphous polymers to rough and cloudy, giving a matte surface, for semi-crystalline polymers, the relative degree of surface roughness depending on the material. The film surface against the metal surface tends to take on the finish of the metal surface, which can be highly polished or matte depending on the film material and thickness. Also, the much higher quenching rate achievable against the metal surface can result in less crystalline surfaces, enhancing the film surface smoothness and the gloss of the semi-crystalline polymers.

Very thin cast films often are produced with a matte roll to lessen surface contact with the melt, in order to permit removal of the quenched film from the roll surface with minimal distortion of the film. The matte surface traps a layer of air between the film and the roll surface while still permitting high heat-transfer rates for high productivity. Matte surfaces -also can be used when the surface optics are not critical or to insure that uniform surface textures are achieved. For instance, the intermittent contact of a melt with a polished roll surface can give a mottled film appearance, which may be more objectionable than a lower gloss, but uniform surface. The melt also can be cast directly into a water bath without contacting a metal surface to create films without pinning. This has been used

when thin films of exceptional clarity are required. Surface defects in this operation are related to air entrainment in the water bath or water boiling at the film surface, both of which cause craters in the film due to uneven cooling.

A principal difficulty with casting against a metal surface is the removal of the air behind the molten polymer film to give a uniform surface contact with the roll, which may be desired for uniform cooling or surface appearance. Uniform surface contact is accomplished by a process called “pinning” where a force is applied to the melt surface to push it against the metal roll. If the pinning force is not high enough, air will be trapped behind the sheet in small areas preventing the polymer from contacting the metal surface, resulting in a variation in surface smoothness, which is visible when metallized or viewed at low angles. This surface mottling is often called “water marks” because the variation of surface gloss looks much like a waxed surface on which water has dried and left a residue as shown in the following picture.

Figure 9.2-1 Picture of Water Marks (Darker Areas and Some Circled) on Metallized Films Approximately 60X Magnification



Pinning of the melt to the roll may be accomplished with an electrostatic charge (10,000 to 30,000 volts at 5 to 20 microamps), a polished nip roll, an air knife, a vacuum box, or a combination of the air knife and vacuum box. In most cases, the air knife and vacuum box are the method used in combination with a polished or matte roll surface. The pinning method chosen depends on many factors but ultimately is determined by film quality requirements. Most pinning methods apply the force without touching the melt with a solid surface, only nip polishing pins the melt by actually touching the melt surface. However, variations in the pinning force across the film width can cause defects such as MD or TD lines. Air knives may also cause particles to be embedded in the film surface.

In addition to surface pinning, the film edges often are pinned to control film necking at the casting point. Again, this is most often done with air jets, and occasionally, with a static charge placed on the film edges. Once the film is removed from the casting section, it is cooled, the edges are removed and recycled by slitting, and the film is surface treated, if required, and wound into a roll.

9.3 Solution Casting

Solution casting is a means of producing films from polymers which are not melt-processable, for substrates which require low levels of residual strain in the film, or substrates that require films that are optically isotropic. Films may be cast from solutions using two basic methods. One is solvent casting, where a film is formed by evaporation of a solvent; the other is by precipitating a film from a solution, followed by drying. Both methods have been used to produce commercial products, and they have some common aspects, such as the formation of a polymer solution and the formation of a polymer gel film followed by drying to produce the final film. The method chosen depends on the ability of the polymer to form a solution and the nature of the solvent required for the polymer. This section explains the differences and similarities of the methods and the nature of the films that are formed.

In general there are two potential rate-limiting steps, the dissolution and the final drying of the film. Polymers, due to their high molecular weight, require a significant amount of time to completely dissolve. In general, this will be the rate-limiting step in any film formation process. Polymer must be added to a stirred tank of solvent and uniformly dispersed into the solvent in such a manner as to prevent coagulation and the formation of large polymer aggregates. The solution may be heated while the polymer is stirred and dissolution proceeds. Table 9.3-1 lists several polymer solutions, which are prepared for solution casting of polymers.

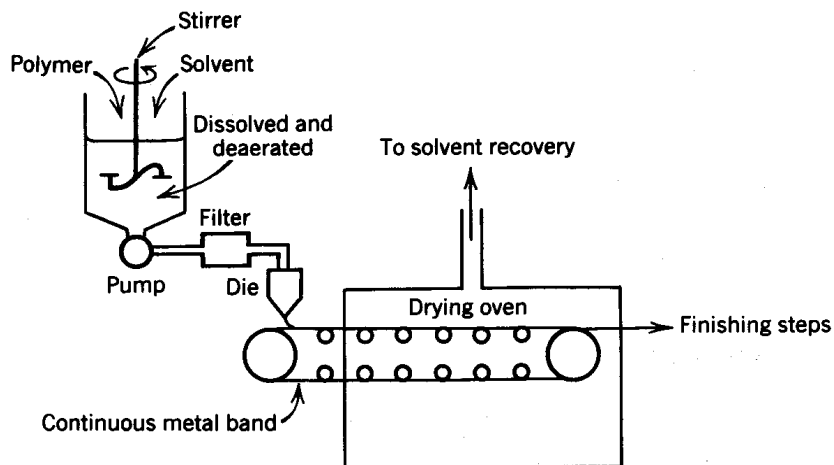
Table 9.3-1 Polymer Solutions for Solvent Casting Films

Polymer		Solvent	Film Forming Method
Nitro cellulose		Various	Drying
Cellulose triacetate		Methylene chloride	Drying
Cellulose		Carbon disulfide/ sodium hydroxide	Precipitation
Polyvinyl alcohol		Hot water	Drying
Polyvinyl chloride		Various	Drying
Polyimide		Aqueous sulfuric acid	Precipitation

Figure 9.3-2 shows the overall solvent casting and evaporation film-forming process. In this process the polymer is dissolved in a solvent and aged. It is then pumped and filtered on its way to a slot die. The solution is metered from the die onto a highly polished continuous metal belt. It is a low- viscosity solution as it travels onto the belt, and it will, therefore, self-level and be stress-free and uniform in all directions through the solution. The coated belt is then taken into an evaporation oven where the solution is heated to drive off the solvent. As the solvent evaporates, the polymer solution reaches a saturation concentration and begins to form a gel of polymer, containing solvent. During drying, the solvent is recovered for reuse. Continued drying of the solvent from the gel layer forms a solid polymer film with little or no residual stresses. The dried film is then stripped from the belt and transported to subsequent steps where it might be further dried or conditioned to avoid curling or blocking. Finally it is trimmed and slit to desired width and wound into rolls.

Films made from solvent casting and evaporation are noted for their lack of molecular orientation and are used when dimensional stability and optical uniformity are critical properties. This is generally the case for photographic films used for projection or for negatives.

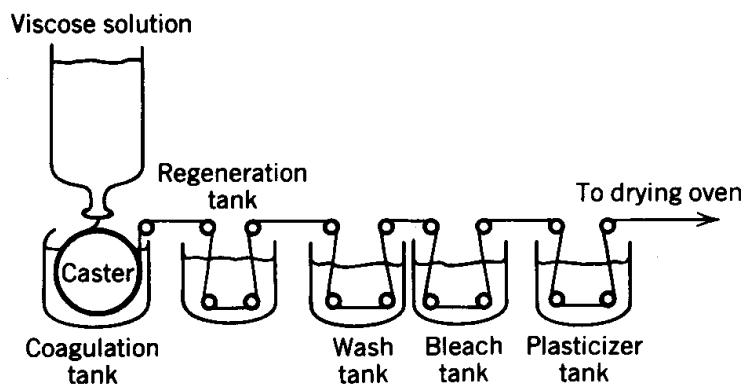
Figure 9.3-2 Schematic of Solvent Casting of Film



Precipitation

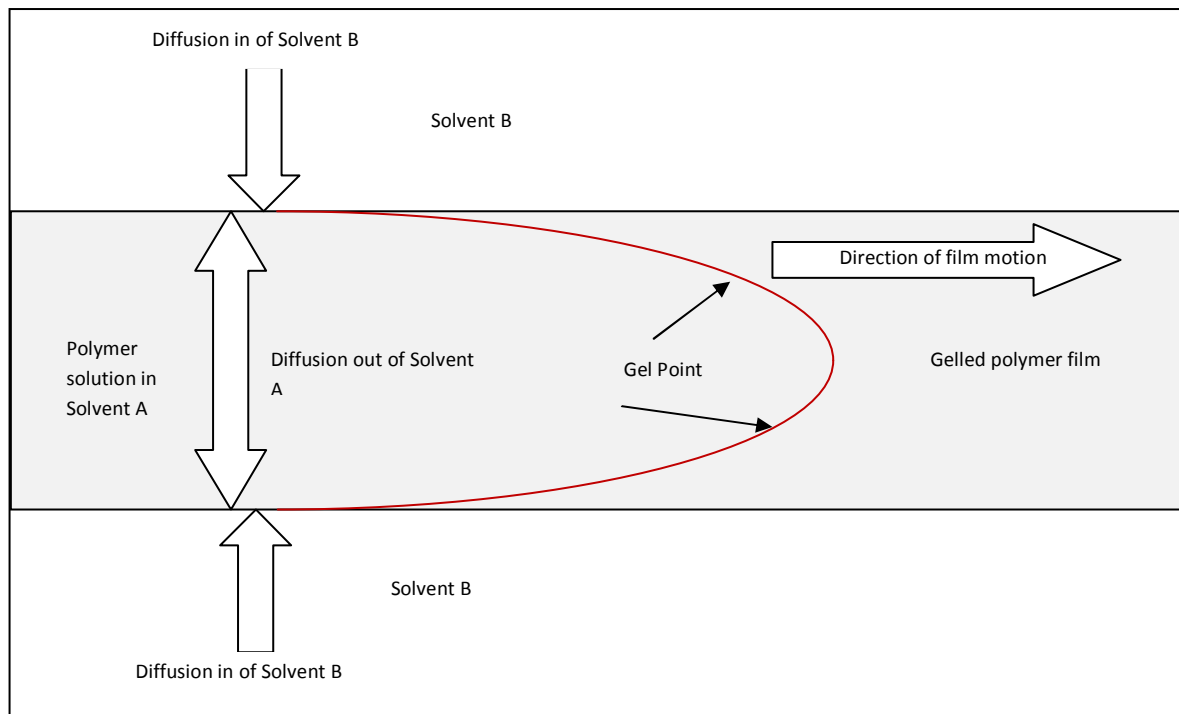
Figure 9.3-3 shows the overall process for precipitating a gel from a solution as would be done with cellophane production³.

Figure 9.3-3 Schematic of Solvent Casting and Film Precipitation as Applied to Cellophane Production



In this process, as in solvent casting, the polymer is dissolved in a solvent, aged, and the solution is pumped and filtered to remove impurities. The polymer solution is then extruded from a slot die into a liquid bath. The bath is used to form a polymer gel film from the polymer solution, which is entering by coagulation or by precipitation. The gel forms at the interface between the solution and the bath liquid and thickens as the film passes through the coagulation bath (*Figure 9.3-4*).

Figure 9.3-4 Schematic of Gel Film Production During Precipitation



The mechanism by which the gel is formed by coagulation depends upon the differential solubility of polymers in a blend of solvent and non-solvent and solubility changes resulting from concentration changes of added chemicals. In general, the polymer is in solution with a good solvent (A) for the polymer. The coagulation bath is generally composed of a solvent, which is a poor solvent for the polymer (B or B+A). As the polymer solution enters the bath, the good solvent (A) diffuses out of the solution and the poor solvent (B) from the bath diffuses into the moving polymer solution creating a solvent blend (A+B), which will not dissolve the polymer, causing it to precipitate out of solution and form the gel film. As diffusion of each species (A & B) continues, the gel layer thickens until a complete gel film is formed. Subsequent steps wash the film to remove various solvents from the gel, which is then dried and wound into rolls.

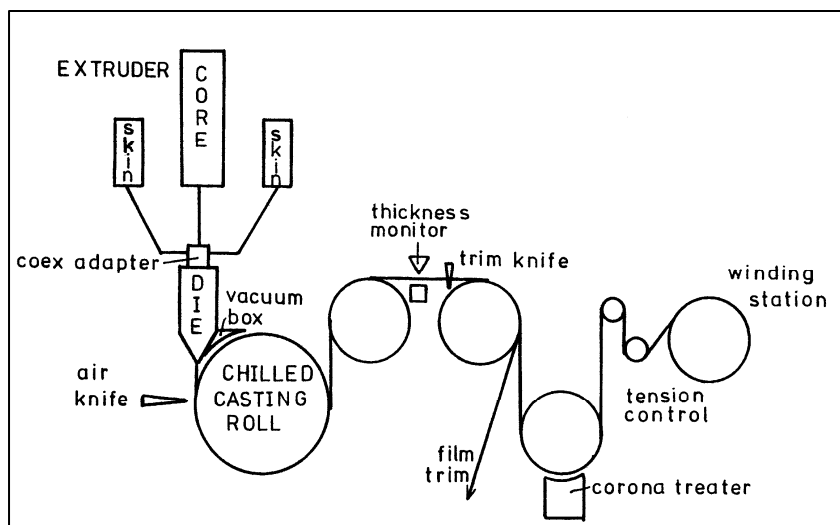
For example, solvents A and B are a good solvent and poor or non-solvent, respectively, for polymer Y. A solution of Y is prepared in A and extruded into a bath of solvent mixture AB. The concentration difference causes diffusion of A out of the solution into the bath and the diffusion of B into the polymer solution. As the concentration of B increases and that of A decreases, the polymer becomes less soluble and eventually precipitates as insoluble gel. Complete evaporation of A and B leaves the finished film.

An alternative for solvated polymers is coagulation caused by chemical additives in the bath, which may reverse the solubility or change the chemical nature of a polymer in solution, transforming it into an insoluble, stable gel. A good example of this technology is the casting of cellophane where a carbon disulfide/sodium hydroxide solution is used to dissolve the cellulose. The solution is then pumped through a slot die to extrude a sheet of solution, which when passed into an aqueous solution of sulfuric acid, neutralizes the sodium hydroxide, driving the cellulose out of solution forming an insoluble gel. The gel is then washed and bleached in water and various aqueous solutions to remove all traces of the carbon disulfide as well as the acid. Plasticizer and softening agents such as ethylene glycol can be added in subsequent baths before the film is dried to remove water before being wound into rolls.

9.4 Melt Casting

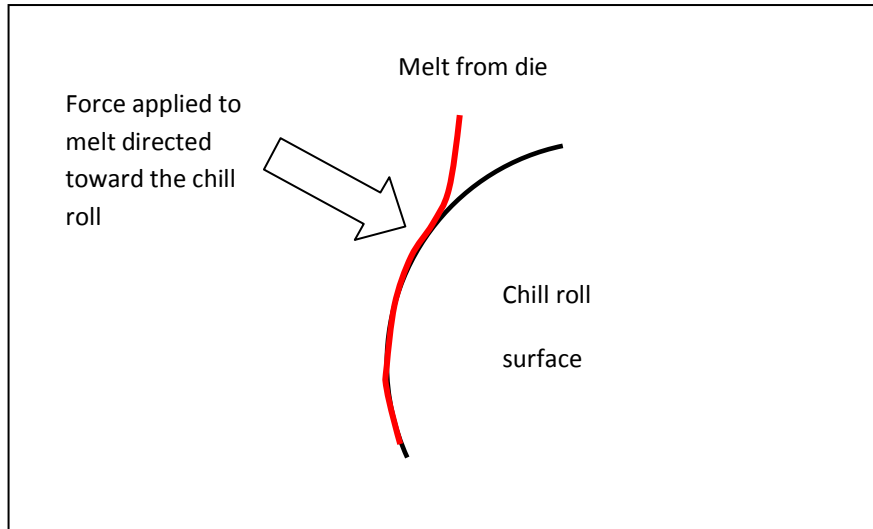
In many cases, it is possible to melt a polymer and to extrude a film directly from the melt⁴. In this process (*Figure 9.4-1*), a solid polymer is fed to one or more extruders where it is melted by the action of the extruder screw. This creates a high-temperature liquid, which is passed through a filter media and pumped to a die. The uniformity of the melt is important in determining the ultimate uniformity of the film properties and thickness so it is important to have an extrusion system optimized for the polymer to be extruded. This includes the extruder screw, filtration system, melt transfer piping, and die. The die shapes the polymer into a wide, thin shape, which is extruded as a melt curtain. The melt curtain is contacted with a chilled metal roll traveling with a surface speed at or slightly in excess of the melt velocity exiting the die. This pulls the melt from the die, thinning and narrowing the melt curtain. The relative position of the die and the casting system is important in obtaining a uniform flow of polymer to the chill roll with minimum interaction of the melt curtain and the die lips.

Figure 9.4-1 Schematic of the Melt Casting Process



To improve melt cooling and obtain high production rates, the melt is forced against the metal chill roll by a process called pinning (*Figure 9.4-2*). The pinning process can use an electrostatic charge placed on the film by a charged wire, an air jet from an air knife to push the film against the chill roll, or a vacuum box behind the die and sealed to the cast roll to pull the melt against the chill roll (really to push the film against the chill roll using atmospheric pressure difference across the melt curtain). In all of these pinning steps, the goal is to obtain intimate contact of the melt with the chill roll. In addition, it is often the case that separate edge pinners of compressed air streams, electrostatic charge, or cooled rollers are used to minimize the melt curtain width reduction or “neck-down” between the die and the cast roll.

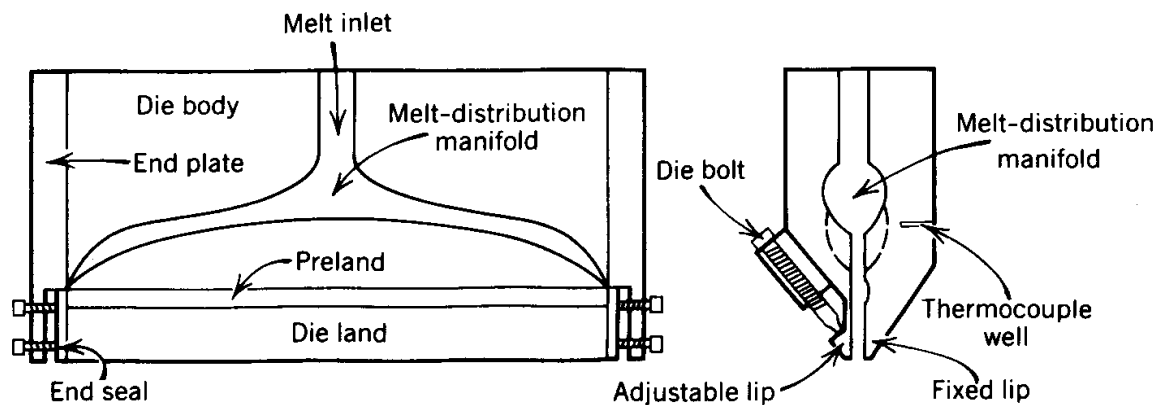
Figure 9.4-2 Schematic of Pinning Force Applied to a Polymer Melt Against a Chill Roll



Proper location of the chill roll, die, and pinning source are important in obtaining the best quality film without pinning-induced “flutter” or other defects to the melt from the pinning force. The vacuum box changes the angle at which the melt exits the die and contacts the roll surface. A vacuum box can sometimes be used to increase line speed by changing the total wrap of the melt on the roll thereby increasing the residence time the melt is on the drum, or to minimize melt surface defects caused by motion of the melt relative to the die lips. Ideally the vacuum level is adjusted to cause the melt to fall directly away from the die. However, too high a vacuum level can cause the melt to drag against the back die lip causing melt defects.

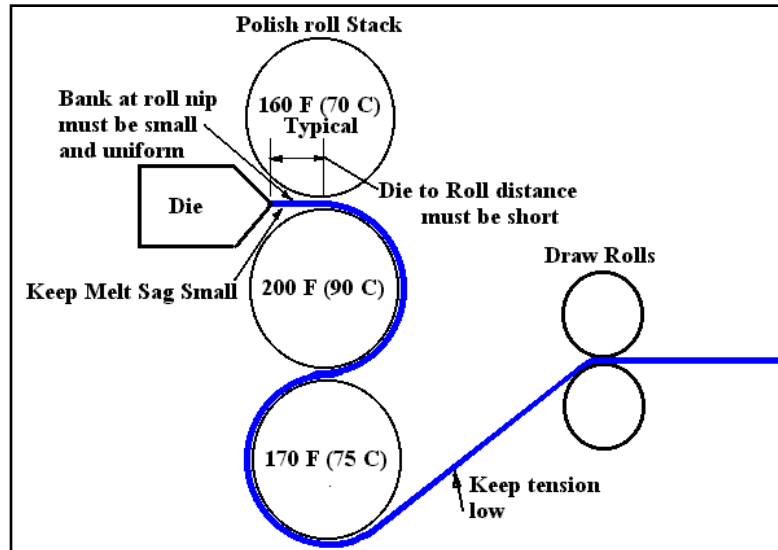
Once the film is quenched, it is removed from the chill roll and gauged to measure its thickness uniformity. The gauge profile is displayed, and the die is adjusted either manually or by an automated system by changing the die gap in local regions controlled by bolts (*Figure 9.4-3*). The die bolts are generally on 1in centers and can be turned to open or close the die or heated and cooled to change the length of the bolts and thus the die gap. After gauging, the edges of the sheet are removed by slitting because they will be thicker than the majority of the film due to the neck-in of the melt in the gap between the die and the casting roll. The film may then be surface treated and wound into rolls for shipment or subsequent converting steps.

Figure 9.4-3 Schematic of Single-cavity Die with Lip-adjusting Bolts



The casting drum may be supplied with a high-gloss or a matte surface depending on the film type and thickness. Matte rolls are often used to impart a desired matte finish to the film. Matte rolls also insure a uniform appearance with some resins, which poorly wet the roll surface, due to controlled contact of the melt with the roll surface from an air layer trapped between the film and roll. For a high-gloss film surface, the use of a highly polished casting drum is required. Generally, there is a side-to-side difference in film gloss due to the existence of a free surface opposite the casting drum surface. For thicker films it is possible to extrude the melt between two polished rolls (Figure 9.4-4) to obtain high gloss surface on both sides of the film.

Figure 9.4-4 Schematic of Nip Polishing System



This nip polishing can be used to make films ranging in thickness of approximately 0.100 to 6.35mm (0.004 to 0.25in) in thickness. Thinner films ranging from 0.006 to 0.127mm (0.00025 to 0.005in) in thickness are generally made using the process of Figure 9.4-1 where the extruded melt curtain is drawn from the die and forced against the rotating casting drum to rapidly quench the melt into a film. Since the casting process is based upon the rapid cooling of a melt onto a metal drum, it can be operated at much higher line speeds and output rates than blown film where the film is cooled by air. Also the high cooling rates can improve film optical properties.

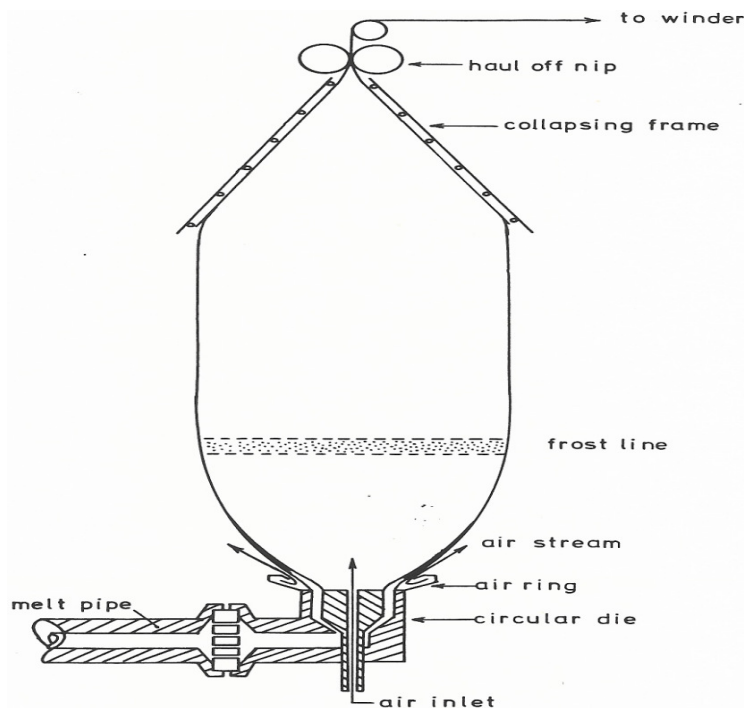
Because cast film is made in a straight, continuous in-line process, gauge uniformity across the width of the film is important to insure that the wound film is not stretched between portions of the roll that differ in thickness. If the gauge is found to vary significantly, the roll will be different diameters across the width, which implies that the film becoming the next layer on the roll has to fit around different lengths. This will cause the film to stretch in the high spots of the roll. When it is unwound, portions of the film will be loose compared to adjacent unstretched areas, and the film will not lay flat as it goes through various processes. This poor layflat will cause wrinkles and creases to form as the film travels over rolls, or it can tear if the film must be slit in a loose region. Poor web flatness is called baggy film, and because it is generally in lanes, it is referred to as baggy lanes. Baggy film requires higher tension levels to stretch the short portions of the film to keep it flat in the converting process but the higher tensions can further damage the film for subsequent converting steps. Baggy lanes are controlled by improving the film flatness through gauge control, good winding conditions, and in some instances, by oscillating the film back and forth sideways to the running direction as it passes through the process to minimize the diameter buildup in the thicker regions of the film. Almost all film processes are subject to this phenomena and much time and effort goes into controlling gauge quality and film flatness.

Gauge control in cast films can also be superior to that found in blown films because flat dies can be built to permit localized adjustments of the polymer flow by die gap thickness changes as compared to circular dies, which have rigid walls that can be hard to center. As the film is removed from the cast roll and transported, it can be gauged by several means, *e.g.*, a direct thickness measurement by touching the film surface or by noncontact methods such as optical absorption using infrared radiation. Generally, film thickness is measured using the absorption of beta particles (using Beta gauges) released from a radioactive source by the attenuation of the particle beam. The radioactive source is held in a moving head with the detector and is scanned across the film as it is running between the source and detector. The change in the particle beam intensity is dependent on the mass of polymer it passes through. Film absorption is measured and compared to that for no film and the mass density of the film is plotted as a function of position. Changes can then be made to the die gap manually by an operator adjusting the bolts by turning them in or out or automatically by a computer. Computer control is most frequently performed by heating or cooling the bolts with compressed air to change their length, thereby changing the die lip opening. In some cases, a robot has been designed to automatically adjust the die by turning the bolts, but this is infrequently done today. The adjustment sensitivity depends on the number of threads per length of bolt. Typically, cast film thickness can be controlled to 1-2% as long as the extrusion system delivers a uniform temperature and pressure melt. Thickness adjustment in solvent cast or precipitated films, while more complex, can be performed in a similar manner by measuring final film thickness and adjusting the rate of flow from the die.

9.5 Blown Films

The blown film process is melt-based. After the polymer melts, it is delivered under pressure to a circular die. As the melt exits the die, it is drawn away from the die face faster than it is flowing out, by two nip rolls which pinch the film together, pulling on the film. At the same time that the melt tube is pulled, it is also expanded by trapping air in the bubble formed by the nip rolls (*Figure 9.5-1*).

Figure 9.5-1 Schematic of Blown Film Process



The speed ratio of the film's motion to the melt speed at the die is called the draw-down ratio (DDR), and it imparts machine direction (MD) orientation into the molten polymer. The diameter expansion is called the "blow up," which imparts transverse (TD) orientation to the melt. Blow-up ratio (BUR) is the ratio of the final film diameter to the circular die diameter. As the molten polymer is drawn and expanded, it is thinned and cooled as it freezes into a

solid at the point called the frost line. At the frost line, the existing melt orientation is frozen into the film as the film becomes a solid. Because the blown film process orients the melt in both the MD and TD direction, it is a biaxial orientation process. The relative balance between the DDR and the BUR determines how uniform, or balanced, the MD and TD film properties are.

Much of the blown film process centers around the die and the cooling of the molten polymer as it is extruded. Special structures called air rings have been developed to improve the cooling of the melt with high velocity air. This air is only applied to the outside surface of the film. The inside surface of the film is cooled at a much slower rate. Because of this, the inside surface has a longer time to relax stresses in the film surface from the die exit, so it is generally smoother and glossier than the outside surface of the bubble. Improved interior bubble cooling is possible by the use of a cooling stalk located inside the tube on the die face through which cool air is blown in and warm air is removed.

Blown film mechanical properties are similar to those of cast films, but due to the slower quenching rates, optical properties are generally not as good. However, by modification of the DDR and the BUR, it is possible to create films with widely varying directionally biased mechanical properties in the MD and TD directions. Films can be made ranging from low DDR and high BUR (preferentially TD oriented), to high DDR and low BUR (preferentially MD oriented), as well as films with balanced MD and TD properties.

Thickness uniformity of blown films is generally poorer than that for cast films. The poorer thickness uniformity is due to the construction of the circular blown film dies, as it is generally not possible to directly adjust the thickness of the film profile by variations in the die gap. Thickness uniformity for blown films ranges from 3% to 20%, and more typically, from 5% to 10%. Some automatic thickness control is possible by selectively varying the heating or cooling rate of the melt as it exits the die. This can be accomplished by blowing heated air against the molten polymer to modify the cooling effect of the air ring, thereby changing the melt stretching in the bubble, permitting thick regions to thin more. Cooling of thinner regions would give less thinning and would increase the thickness in these regions.

Due to the relatively poor gauge control of blown films, it is common practice to rotate the film to randomize the location of thickness variation. This results in films with poorer tracking characteristics than cast films. The rotation is generally carried out by rotating the inner or outer die lip but may also be accomplished by rotating the winders around the center line of the bubble.

Highly Oriented Films

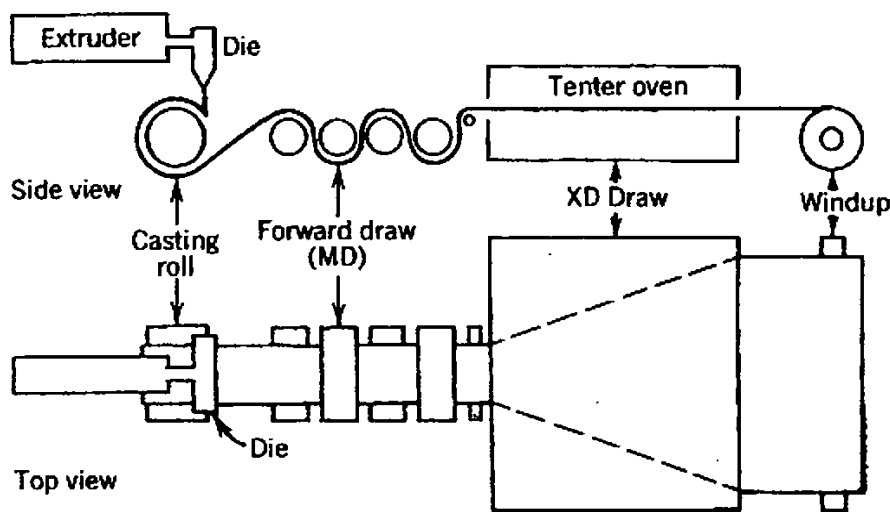
Films with high levels of molecular orientation may be produced by solid-state stretching of polymers. In this case, the polymer is generally stretched above its glass transition temperature (T_g) and below the melting point (T_m). Solid-state stretching can be performed from a flat sheet using sequential or simultaneous MD and TD stretching, generally called the tenter (or sometimes stenter) process and from cast tubes by the double bubble process.

In both the tenter and double bubble processes, the stretching forces are high, leading to very high levels of polymer orientation, and therefore, film mechanical properties. Film properties are controlled by the stretching temperature, the stretching rate, and the extent of stretching used in the stretching process. Thickness uniformity of the films can be excellent, with the flat tenter process having better overall control than the double bubble process. However, in this instance the double bubble process yields films with better overall thickness variations than that of melt blown films, providing commercial film with thickness variations of 3% to 4%.

Tenter Process

The tenter process is shown schematically in Figure 9.5-2. It is comprised of an extrusion system, a flat die, a pinning process (which depends on the polymer being used), and a casting system. The cast film is transported to a machine direction orienter where it is reheated to its orientation temperature and then MD-oriented with a roll speed change. The MD-oriented sheet is transported to an oven containing a continuous chain with film-gripping clips. The film is heated at constant width in the first section of the oven and then the chain path diverges sideways to stretch it to induce TD orientation. After the TD stretching section, the chain travels at constant width (or a slight convergence of from 1% to 4%) through an annealing and cooling zone where it is stress relieved and/or cooled. After exiting the oven, the film is released from the clips and the thick film edges are slit and ground for recycling. The film surface may be treated with an oxidizing process such as corona or flame treatment, then wound into rolls for aging or further processing to meet customer requirements.

Figure 9.5-2 Schematic of the Tenter Frame Process



Tenter frame process.

In contrast to the double bubble or blown film process, the tenter frame process permits the direct manufacture of films with improved dimensional stability. This improvement is possible because of the heat setting and the film stress relaxation that can be performed in the tenter oven after the TD stretching has been completed. This improved dimensional stability is very important for products requiring stable film dimensions such as printing in registration or photographic and graphic arts applications. This improved stability is possible due to controlled heating and cooling of the film while it is restrained in the tenter clips and/or with the controlled TD relaxation at a fixed temperature.

A wide range of polymers may be MD or TD uniaxially oriented and MD x TD biaxially oriented. Commercial tenter operations produce biaxially oriented films comprised of PE, PP, polystyrene, polyethylene-co-vinyl alcohol (EVOH), Nylon 6 and Nylon 66 and PET.

In some cases, the separate MD orientation step may be omitted with the film simultaneously oriented in the tenter oven using specially designed chain elements. These chains can be comprised of a pantographic stretching mechanism which advances the clips in the MD direction while the chains diverge in the TD direction. This

mechanical process is well known, but it limits the speed of the process. Alternatively, the clips can be passed to a lead screw with a fixed rate of MD clip advancement, as the lead screw is angled away from the direction of motion, again causing a TD stretching to take place simultaneously with the MD stretching. This simultaneous stretching process has not been widely used relative to the pantographic chain, and it also limits the operating speed of the film process. This simultaneous pantographic process has been used primarily for the orientation of HDPE and nylon (polyamide) films.

A more modern development in simultaneous stretching uses LISIM® technology, which is a rail and clip system comprised of linear motors and a magnetically levitating rail system to drive the clips. This eliminates the mechanical connection between clips of the pantographic or standard tenter system, and it has been used commercially for simultaneously biaxial orientation of polyester and PP film production. These lines are not limited in speed or output as are the pantographic lines, but remain expensive due to the technology of the linear motors and the clip drive control system. The LISIM® process also has the potential of permitting an improvement in film dimensional stability by an MD relaxation in addition to the TD relaxation of the tenter frame process.

The high level of orientation achievable in the tenter frame process significantly increases film mechanical properties, improves optical properties, and decreases the permeability of the film to gases and moisture.

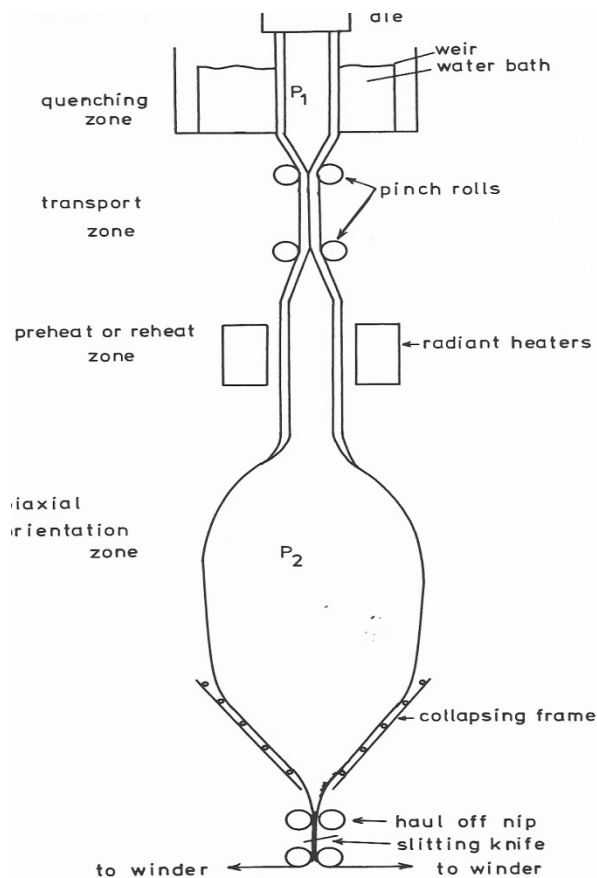
Double Bubble Process

Figure 9.5-3 is a schematic of the double bubble process. In this process a cylindrical tube is extruded from a circular die and quenched with a water bath. Once formed, the tube is pinched shut forming the first bubble. The pinch rollers control the speed at which the cast tube is formed and may impart some melt orientation to the tube. The cooling rate of the tube to set the initial molecular structure is critical since it can impact the ease of orientation in subsequent steps. Additional cooling of the tube may be obtained by the use of an internally cooled mandrel or with a specially designed tubular casting section⁵. Once formed and collapsed, the tube is maintained at a uniform temperature and transported to the reheating and stretching sections. In some instances, such as the production of heat-shrinkable films, the collapsed tube may be irradiated with high-energy electrons or ionizing radiation to crosslink the polymer⁶.

As the collapsed tube enters the reheating and stretching section, it is isolated by a second and third set of pinch (haul-off nip) rolls. The tube is inflated between the second and third pinch rolls by the insertion of compressed air. The relatively thick and cooled tube does not initially stretch once it is inflated. It is passed into a preheating section where it is heated to the stretching temperature. On reaching the stretching temperature, the film is simultaneously drawn in the MD and TD under the combined air pressure and MD drawing force of the third nip rolls. The third set of pinch rolls are moving at a speed greater than that of the second pinch rolls and will stretch the heated film in the MD. The speed ratio of the second to the third nip roll determines the MD stretch ratio. The film stretches in the TD direction by the inflation caused by the compressed air inserted inside the tube. The air is captured between the second and third nips. The air is captured and is stationary acting like a rigid body over which the film must stretch⁷. The amount of air trapped in the tube determines the TD stretching ratio. The pressure of the trapped air is determined by the stretching force of the polymer, which is polymer dependent, the temperature of the tube, and the quenching conditions used.

After stretching the film, the third nip rolls collapse the film tube, transporting it away from the nip toward the winder. The film is slit into two independent sheets, which are separated from each other. The individual sheets are surface treated (*see Section 7*), if desired, and wound into rolls for subsequent aging and processing to customer requirements.

Figure 9.5-3 Schematic of the Double Bubble Process



Films produced by the double bubble process are less dimensionally stable than tenter-based oriented films. This is because the process does not permit a high temperature stress relaxation of the film. In some instances a third bubble⁸ has been added specifically to permit stress relaxation to be performed to improve the dimensional stability of double bubble films.

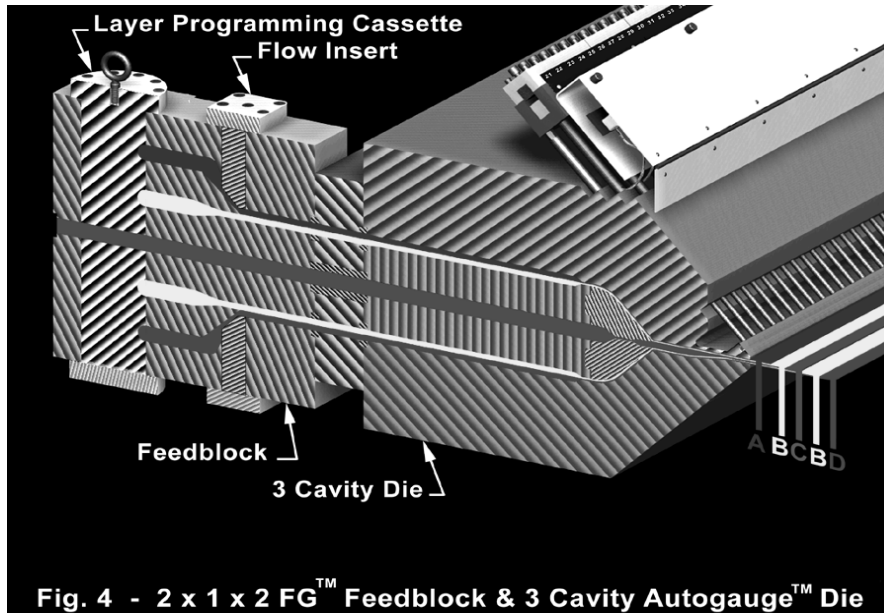
9.6 Coextrusion

Most modern metallizing substrate films are produced from multiple materials with the purpose of modifying the film properties in a way not possible by the use of a single material. While it is possible, and sometimes useful, to produce single layer, multiple polymer films by blending, this is not the most widely used or most flexible method for optimizing substrate film properties. The most widely used method is to layer the polymers together by a process called coextrusion, which segregates the individual polymers into distinct layers. Modern films containing from two to several hundred layers are now common.

Coextrusion technologies have been available since the 1940s and have grown dramatically in films since the mid-1970s. There are many reasons to coextrude, such as for control of metallized barrier properties (*see Section 3*) and to control surface and winding properties of the films in vacuum (*see Section 11*).

Coextrusion is possible with all of the melt-based film manufacturing operations as well as some solution-based films, however, coextrusion is most widely associated with melt-based processes. When coextruding flat tenter-oriented or melt-cast films, the flows from two or more extruders are collected in an adapter and delivered to a feedblock and single-cavity die combination, a feedblock and multicavity die combination (*Figure 9.6-1*) or to a multicavity die.

Figure 9.6-1 Schematic of Feedblock and a Three-cavity Die (Courtesy of Cloeren Inc.)



In feedblock coextrusion devices, the individual polymers are collected, shaped, and then stacked together into rectangular⁹ or circular¹⁰ cross sections before being spread in the die. In a multicavity die the individual polymers are gathered and routed to each die cavity where they are spread into thin sheets prior to joining together. Once the die has spread the polymer to form the film, it is extruded from the die and into the film-quenching system of the various film making processes described above.

In circular extrusion for blown film and the double bubble processes, the individual polymers generally are distributed around the circumference of the die prior to joining together. The circular die's individual distribution structures can be arranged radially (Figure 9.6-2) or stacked (Figure 9.6-3). The stacked dies are more compact, easily machined, and are used for films of three and more layers, while the radially arranged dies are less frequently used above five layers due to their increased complexity, increased diameter, and difficulty in manufacturing assembly and maintenance.

Figure 9.6-2 Schematic of Three-layer Radially Arranged Blown Film Die

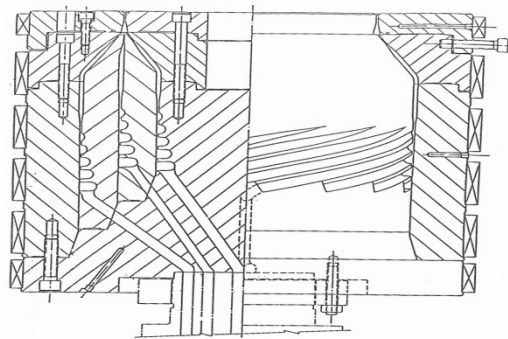
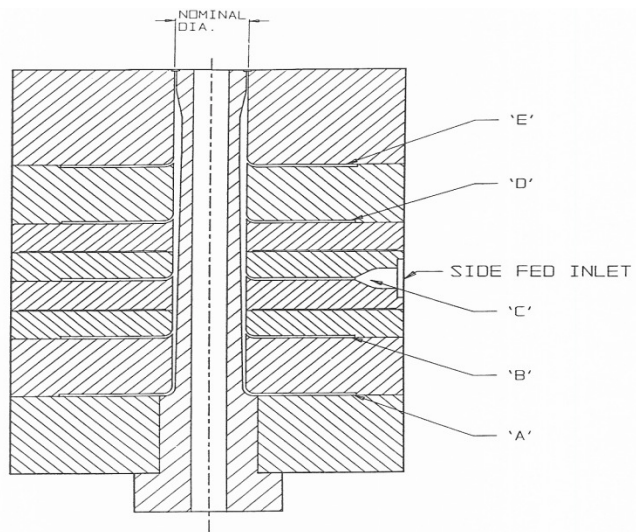


Figure 9.6-3 Schematic of Stacked Five-layer Blown Film Die



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⁹ Cloeren, P., U.S. Patent No. 4,152,387, "Method for Forming Multi-layer Laminates," U.S. Patent and Trademark Office, Washington, DC, (1979).

¹⁰ Nissel, F.R., U.S. Patent No. 3,833,704, U.S. Patent and Trademark Office, Washington, D.C., (1974).

Section 10: Contamination, Web Cleaning, Planarization, and Heat Stabilization

Introduction

Contamination is important because a contaminant on the surface of the web can result in the loss of adhesion, loss of barrier performance or reduction in reflectivity. Polymer webs can be contaminated from the instant they are produced and can become progressively worse with successive processing and rewinds. The source of the contamination can be from both internal and external sources.

10.1 Contamination

The external contamination source is generally airborne particles. This can include everything, from condensed monomer powder or slitting debris generated during the manufacturing process to natural airborne debris such as pollen, brought in from outside the factory, or even skin debris from the process operators. Anything that can be carried in the air can be transported to the web surface and become a contaminant¹. Contaminants can be a major, but not the only, source of pinholes in any vacuum-deposited coating, as the coating covers the debris and then if the debris gets moved, it leaves behind an uncoated area with no light, gas or moisture barrier performance at all^{2,3}.

Debris can be attracted to the surface of the web by two principal mechanisms: gravity and electrostatic charge. The larger, or more precisely, the heavier the debris, the faster it falls to earth and the film surface. Polymer webs winding over rollers will generate a triboelectric static charge, which can actively attract debris onto the surface. The amount and magnitude of the charge is dependent on the materials involved and their relative position in the triboelectric series⁴⁻⁷. Not only can this static charge attract debris to the surface, it is a key mechanism for keeping the debris on the surface, since the physical contact point can attract moisture that can act as glue further holding the debris in place.

Any debris landing on the surface can be wound into the roll. How much and how deeply the debris is pressed into the substrate surface depends on particle size and how hard the roll is wound (*see Section 1.9 and Section 11*). Large debris will be pressed hard and may dig deep into the surface and thus become difficult to remove. Smaller debris will not be pressed as hard into the surface and so can be more easily removed. The dents and film surface distortions produced in the surface in this manner are another potential source of coating defects. The edges of the pits or dents cause a change in angle of the coating and this point in the coating can be vulnerable to stress cracking. Any cracking of the coating will be a further source of reduction in the barrier performance.

Debris starts to contaminate the web from the point where the film is first produced. For example within the stenter ovens (*see Section 9*), some of the degraded or low-molecular-weight polymer will be given off into the hot air. In the case of PET, this vapor may condense onto cold stenter internal surfaces producing a white powder or “snow,” while for OPP low-molecular-weight oil will condense and drip onto the film surface. The white powder accumulated on internal stenter surfaces may be disturbed by the circulating air and can fall onto the polymer web.

In some cases the air is filtered as it recirculates, but if the filters are too fine, air flow is limited. Therefore, filtration particle size limits are a compromise, and web surface contamination at some level is to be expected. An alternative to filtration would be to discard the process air after passing once through the stenter. However, the volume of process air needed to be heated for a stenter, which uses the air only once, is a high-cost option as discarding heated air will significantly increase the manufacturing cost.

Manufactured films are slit, and slitting produces debris. This becomes worse with blunt blades where more debris is generated. As the film passes over rollers, it builds up a surface charge, which attracts slitting debris to the film surface. There is also evidence that every time the film is wound, the level of contamination increases⁸.

Within the vacuum system the web can continue to build up an electrostatic charge, attracting any airborne particulates to the surface. In fact the same electrostatic charging process can be used to reduce the free debris in a vacuum system⁹. For example, a rod that is charged will attract dust to the surface in exactly the same way that television screens collect dust, and these are known as electrophoretic collectors. An electrophoretic collector collects the dust as it is moved around during the time when the gas flow in the system is turbulent, such as in the initial pump down and when venting the system.

Figure 10.1-1 Schematic of Vulnerable Cracking Points Where Coating Covers Dents or Bumps

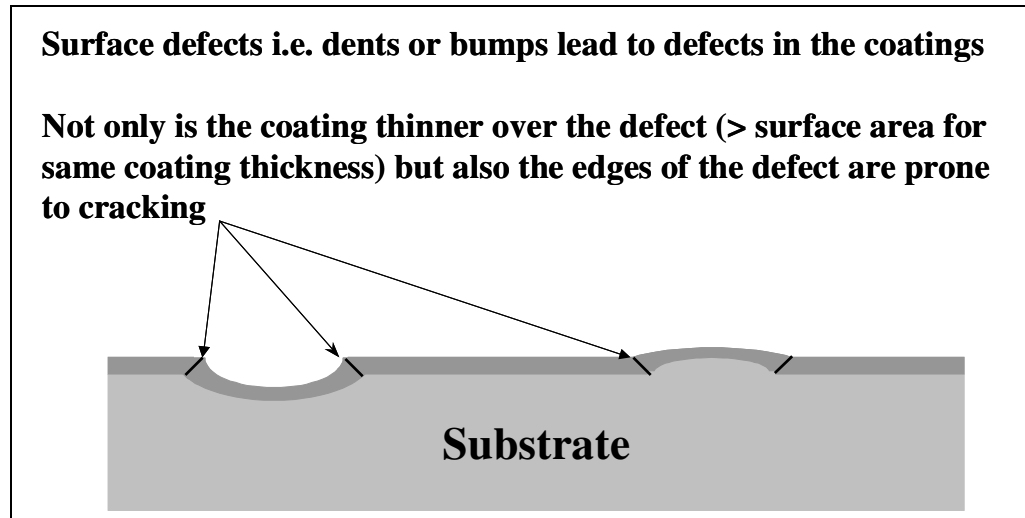
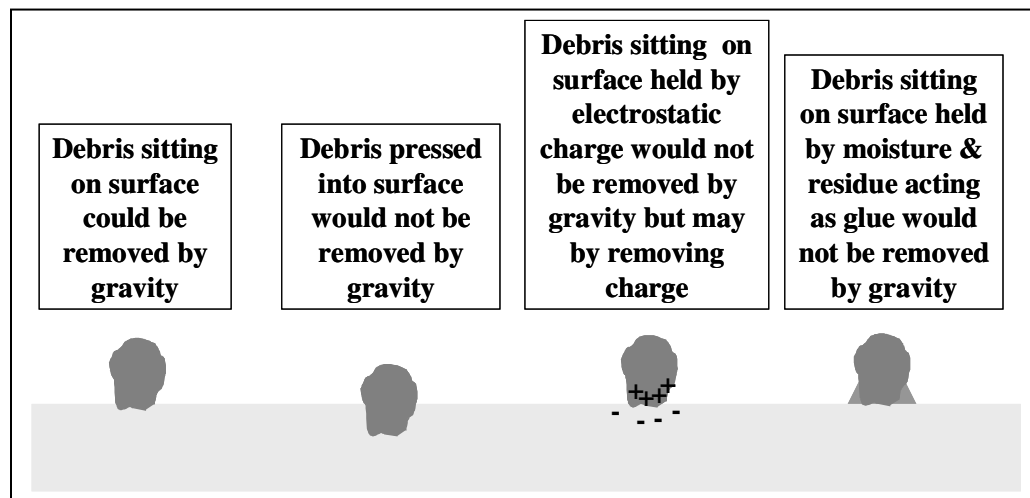


Figure 10.1-2 Schematic Showing Different Ways Debris May Sit on a Substrate Surface



A second possible source of external contamination to the web within the vacuum system is oil. Given the right circumstances, oil may backstream onto the vacuum chamber. Most often this happens when using oil diffusion pumps and when pumping down to a low base pressure, particularly if the system is left pumping for a long time. The oil mist may directly contaminate the web or may contaminate rolls that transfer the contamination to the web.

The other type of contamination is from internal sources of the substrate. This contamination can derive from the polymer chemistry such as the presence of oligomers or from the huge variety of possible additives such as nucleation promoters, antioxidants, radical scavengers, ultraviolet absorbers, crosslinking agents, gloss improvers, surface active agents, antistatic, anti-mist, slip agents or lubricants, anti-blocking fillers, color (pigments or dyes) and/or whitening agents (optical brighteners), voiding agents, viscosity modifiers, dispersion stabilizers, flame retardants, foaming agents or some fillers.

A number of these additives are not locked into the polymer structure and will migrate to the film surfaces. Oligomers and slip agents are low-molecular-weight materials¹⁰. These materials, once present on the polymer surface, can lower the surface energy and form a weak boundary layer that reduces adhesion. These materials can be removed from the surface, but as they are present within the bulk of the polymer, more will always migrate out and re-contaminate the surface.

As with all contaminants, the very nature of the web being wound into a roll means that there is always the potential for any contaminant from one surface being transferred to the other surface. Therefore, if cleaning the web is proposed then cleaning both surfaces of the web ought to be considered.

10.2 Web Cleaning

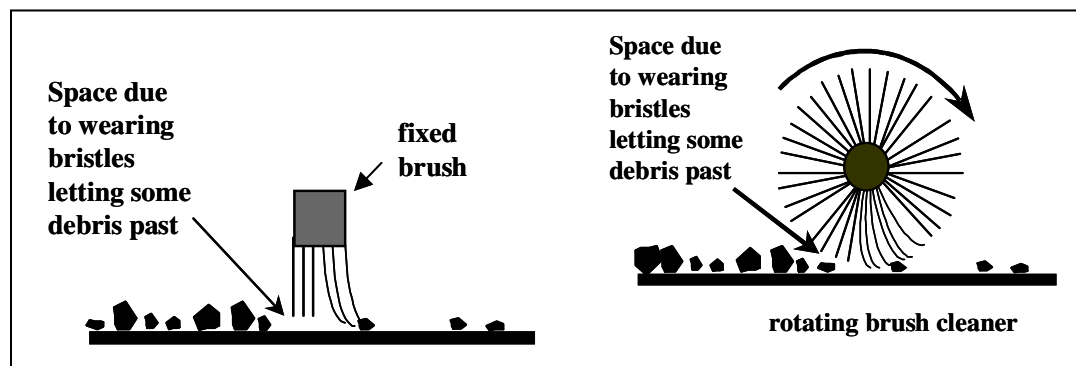
The requirement for cleaning is an indication of a failure to keep the web clean. Cleaning is a fix not a cure. The cure is to find a way to prevent contamination.

However, as mentioned above, it may be both difficult and expensive to keep the whole atmosphere around a film line clean and to maintain a neutralized film surface, thus alternative strategies are used. This may range from doing nothing at all to prevent or clean the web, to adding clean air hoods, vacuum extraction around the slitting process and some electrostatic neutralisation and cleaning process before the wind-up roll. However, this only gets the roll as far as the end of the film manufacturing process. The roll may be re-wound as it gets slit into smaller rolls and other coatings or pre-treatments are used. Every time the roll is re-wound, there will be another opportunity for it to become contaminated.

As with the different types of contamination, there are different methods of cleaning, although generally, only a limited number are used in practice. One way of classing the techniques is to separate them into mechanical and chemical. Mechanical techniques include wiping, brushing, vibration, air jets, carbon dioxide “snow” jets, vacuum extraction, and adhesive, or tacky, rolls. Chemical techniques cover everything from liquid baths to all the plasma techniques and include overcoating.

The danger with some of the mechanical techniques such as wiping or brushing is that the debris is not immediately removed but may be pushed around the surface. Brushes can be fixed or rotating. Any debris that the brushes move either accumulates at the brushing point or with time may escape past the brushes. Because the brushes have to be in contact with the web surface they wear and gradually allow material past. With brushing some debris can also pass through depending on the number, diameter, and density of the bristles. The debris may damage the polymer surface depending on the type of the debris. If the debris is harder than the polymer, it may easily scratch the surface.

Figure 10.2-1 Schematic of Fixed and Rotating Brush Cleaners



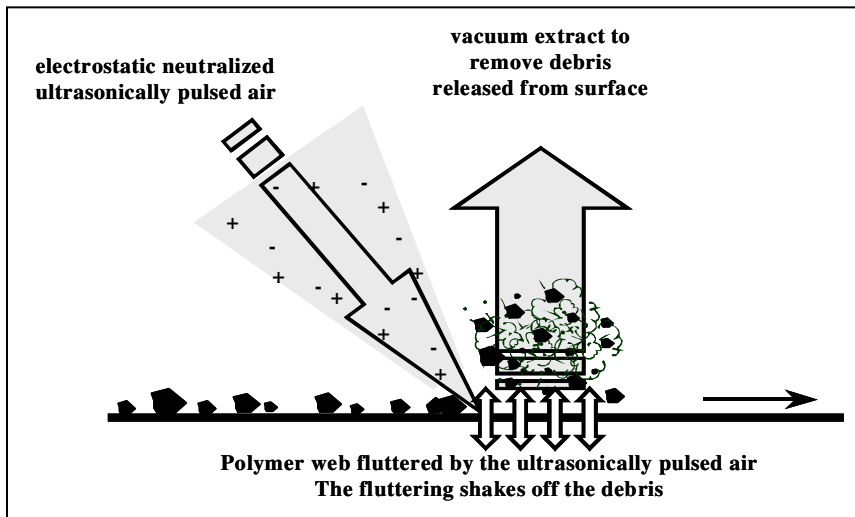
Because of the concern about the possibility of scratching the web, brushing and wiping techniques are not favored.

When winding, the film surface carries a boundary layer of air, which is dragged along with the web. The faster the film is moving, the thicker this layer of boundary air. This can become a problem with cleaning techniques such as an air knife, as the air stream from the air knife can be deflected from the film surface by the boundary layer and thereby fail to remove all, or even any, of the debris. Similarly, with the vacuum technique, the vacuum can end up only removing material that is loose and above the air boundary layer.

To overcome the boundary layer problem, one solution is to use a combination of an air knife and a vacuum (*Figure 10.2-2*). In order to help the air knife break down the entrained boundary layer of air, the air is ultrasonically pulsed.

The pulsing air jet makes the film flutter, and the rapid change in direction shakes the debris from the surface where it can be vacuumed away. To minimize the effect of any electrostatic charge pulling any debris back to the surface, the air knife air jet is also electrostatically neutralized. This combination of techniques effectively removes all debris greater than 0.3 micron in size. Without the ultrasonic vibration, a simple air knife or simple vacuum cannot break through the boundary layer and debris of sizes up to the boundary layer thickness can escape past the cleaning zone. This can be typically as large as 25 to 50 microns.

Figure 10.2-2 Schematic of the Ultrasonic, Neutralized Air Jet with Vacuum Cleaning Process

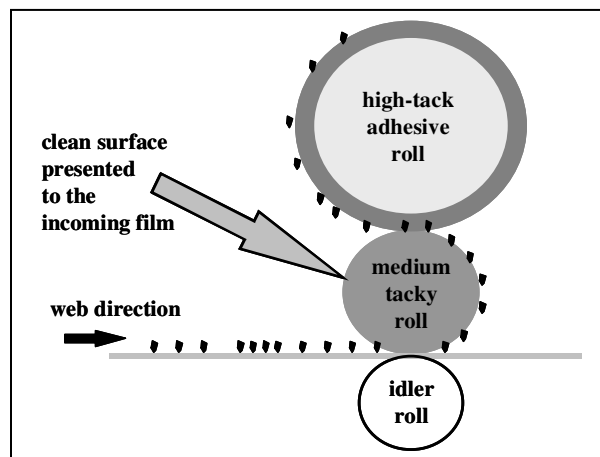


A different approach to removing debris is to bring a moderately sticky surface into contact with the contaminated surface¹¹⁻¹³. The tacky surface is usually in the form of a roll, and the level of adhesion has to be carefully chosen. The level of adhesion makes the roll surface sticky enough to pick up as much debris as possible but not sufficiently sticky to permanently adhere to the film it is meant to clean. As this surface does not adhere to the film but does allow easy release from the surface, it is usually referred to as a “tacky” roll. The debris adheres to the tacky roll in preference to the contaminated surface; so when it is removed, it takes the debris away.

To be able to clean a whole roll of film requires a tacky surface with enough capacity relative to the length of the roll to be cleaned. As contaminants are removed by the tacky roll, its surface becomes clogged with debris, reducing surface adhesion and effectiveness until it stops working. The arrangement that is more often adopted is to have a tacky roll in contact with the web and then a second high-tack roll that is in contact with the intermediate roll (*Figure 10.2-3*). The debris is transferred from the web to the tacky roll and then to the high-tack roll. Since the high-tack roll also would become clogged, it is usually made up of layers of high-tack material. As each layer becomes clogged, a layer can be stripped off, revealing a fresh high-tack surface.

Tack rolls have typically been used on film production lines immediately prior to the rewind. What potentially makes this cleaning process of greater interest to vacuum coaters is the possibility of using it inside a vacuum system. This is not without difficulties but by transferring the cleaning within the vacuum, it can mean the roll is as clean as possible before coating. The materials that make up the composition of the tacky rolls may not be stable in a vacuum and may harden and lose some tack properties. This does not prevent them being used, but it may limit their effective lifetime. The second difficulty is the surface area of the high-tack roll relative to the length of the surface to be cleaned. Normally an operator strips off the high-tack layers before they become clogged, but in a vacuum system this needs an alternative arrangement. As this technology is in its infancy it has, so far, been left for individual users to make their own arrangements.

Figure 10.2-3: Schematic of the Use of a Single-side Tacky Roll Cleaning System



The tacky roll techniques remove all debris down to a level of 0.3 micron. It can have the advantage of pulling out some of the debris that is pressed into the surface that other techniques may not remove. The reason that these techniques tend to be limited to removing debris down to this 0.3-micron level is that as the debris size gets smaller the Van der Waals force holding the debris to the surface becomes more significant. Below 0.3 micron this force is so high that it is virtually impossible to pull the debris from the surface.

The tacky roll can become clogged, therefore, if the film is to be consistently cleaned it is essential that there is always sufficient tacky surface to remove everything that can be lifted from the surface. Otherwise the size of debris remaining on the surface will be greater than the 0.3 micron that could be achieved.

Where leaving debris of up to 0.3 micron on the surface is unacceptable, an alternative strategy is to add a coating to the surface to cover the debris. This can be done before the vacuum deposition process, but not only does the coating have to be filtered to a high degree, it also needs to be protected from re-contamination. This coating process has been developed as an option that can be carried out in a vacuum system. The polymer coating is metered into the vacuum system where it is injected as a mist onto a flash evaporator and condensed on the film where it is cured, usually by an electron beam¹⁴. This polymer coating would have to be quite thick to cover all the debris, so pre-cleaning by one of the above techniques to minimize the debris and then a coating layer of less than 1 micron could be acceptable.

All of these cleaning processes are designed to remove the loose debris but are not effective on other types of contamination such as low-molecular-weight material that has exuded to the surface or oil contamination. Organic contamination is better treated using some form of plasma treater (*see Section 7; In-Chamber Plasma Treatment*). Plasma treaters vary in what they can achieve depending on the gases and power used.

Plasma treatment is a molecular process whereby the plasma bombardment can break bonds allowing some crosslinking to take place or for the fragments to sputter or combine to make a compound that can be volatilized and removed. It is often a misunderstanding of the plasma cleaning process that it will completely remove any and all organic contamination. Given time it is possible to remove virtually anything from the substrate surface but, as the plasma will not only be bombarding the organic contamination but also the polymer substrate, care must be exercised. For example, if there is a large quantity of organic contamination, such as an oil spot, by the time the whole of the oil spot has been removed, the surrounding polymer, where the oil is not present, will have been plasma etched, roughening the surface.

Where there is gross contamination, 1 micron or more, plasma treatments are of limited benefit. Where the contamination is only a few molecules deep, the plasma treatment is very effective. Thus residual oligomer, slip agents and similar low-molecular-weight materials can be either locked into the bulk polymer by crosslinking or by combining any carbon atoms with oxygen they can be volatilized and removed.

Inert gas (argon) plasmas cannot remove organics. The organics can be sputtered, but the sputtered material is likely to fall back onto the surface and can become a contaminant. The sputtered material re-depositing onto the surface will be carbon powder and not highly bonded, making the surface powdery leading to poor adhesion. Inert gas plasmas have heavy atoms to break bonds easily and so mixed gas plasmas of inert and reactive gases such as argon and oxygen are preferred to enable carbon that is produced to be combined with oxygen making a gaseous product than can be pumped away. Critical to this type of cleaning is a suitable gas flow. Once the contaminants have been volatilized, the gaseous product needs to be swept away, otherwise the active plasma gas will become dominated by the exhaust products rather than the inert and reactive gases and the quality of the cleaning will be degraded.

One of the less frequently used cleaning processes for hydrocarbons is to use ultraviolet (UV) lamps to activate the surface¹⁵. UV light produces ozone from the water or oxygen present, and the energy from the UV excitation coupled to the ozone or atomic oxygen efficiently converts the hydrocarbons to volatiles that can be pumped away. In reality this is part of what is happening within the plasma. The plasma generates UV light through to the vacuum-UV and it has been estimated that ~50% of the organic surface reactions are as a result of vacuum-UV photochemistry.

Summary

Contamination generally falls into one of two types, debris that falls or is attracted onto the substrate surface and molecular contamination that may be splashed or dripped onto the surface or exuded to the surface from within the bulk polymer.

The cleaning required is different for each type of contamination. Debris requires a physical technique such as an ultrasonic neutralized air jet with vacuum extraction or a tacky roll system. Molecular contamination requires a chemical cleaning process such as one of the plasma techniques.

Cleaning has limitations. Generally debris can only be removed down to a level of approximately 0.3 micron because of the strong Van der Waals forces present below this size. Chemical contamination can only be removed if it is present in limited quantities. Microns-thick layers of chemical/organic contamination have to be removed by other means to leave only a few molecular layers to be removed by the plasma. If the contamination is too thick, not all of it will be removed or if it is removed, the un-contaminated polymer surface may be roughened during the aggressive plasma treatment.

10.3 Planarization

Planarization is the process whereby the substrate is coated in order to provide a smoother, flatter surface than the original surface.

Although polymer films appear smooth and flat to the eye, they are not smooth or flat when looked at in fine detail such as by atomic force microscopy (AFM). The polymer surface will have debris that has fallen onto the surface during manufacture. Depending on the polymer composition, it also may have exudates such as oligomer forming on the surface and surface defects caused by fillers protruding from or pressing into the surface. All of these defects are potential sources of flaws in vacuum-deposited coatings. In barrier coatings debris can cause pinholes, which can result in a large loss of barrier performance. In transparent conducting coatings, flaws can initiate cracks and reduce conductivity.

The idea of the planarizing layer is to fill surface pits or dents and cover any exudates or debris on the surface producing a defect-free surface. This may require the polymer web to be cleaned before the planarizing layer is applied to minimize the thickness (and cost) of the planarizing layer.

The planarizing layer may be applied as part of the film manufacturing process before it is delivered to the vacuum coating system^{16, 17}. If the coating is applied to the substrate at this stage, the coated surface needs to be protected from recontamination. This means between the coating being applied and the roll being rewound, the surface needs to be in a controlled atmosphere where the air is filtered to prevent any further debris from falling onto the surface.

It is also possible for the planarizing layer to be applied inside the vacuum system^{18, 19}, printing polymer onto the web surface between the unwind roll and the deposition process. Typically the printed polymer is cured using UV light or an electron beam. To get the best out of the coating, the surface must be smooth. The speed of the process needs to be matched to the polymer coating viscosity, and space available as the coating, once applied, needs a certain amount of time to allow the surface to level^{20, 21}. If the coating is cured too quickly, artifacts from the coating process may remain in the surface, defeating the point of applying the coating. As the coating is required to be smooth, flat, and defect-free, curing needs to be controlled so the surface remains free of such curing defects as shrinkage cracks. As the coating is to be of high quality, the monomer needs to be of similar high quality and filtered to minimize gels or particulate contamination.

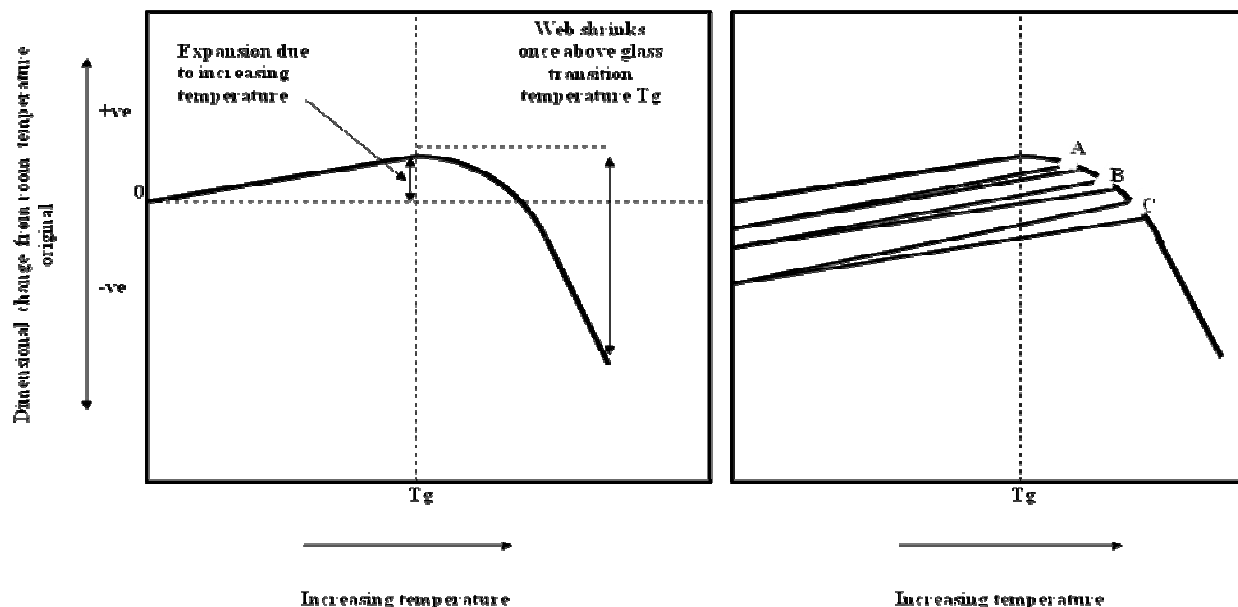
Planarizing layers applied in atmosphere before the roll enters the vacuum system may be subject to contamination from oligomers. It is possible that the planarizing layer can be resistant to the migration of oligomer^{16, 17} but as the planarizing layer is applied to one side it is still possible for oligomer that is present on the back surface to be transferred to the planarized front surface as the roll of film is stored.

10.4 Polymer Substrate Heat Stabilization

Another source of problems for a number of electronic applications is the instability of the polymer substrate during downstream processing. We will see in Section 11.4 on heat load during vacuum deposition that the substrate temperature can rise considerably during the deposition process. The polymer films that have been biaxially stretched will have some residual stress frozen into the film so that when the film is next heated above the glass transition (T_g) temperature, the film will relax and undergo a dimensional change.

The left hand side of Figure 10.4 shows what happens to the dimension of a polymer film as it is heated. Initially the film expands with normal thermal expansion, but once the T_g temperature is exceeded, relaxation takes place, and the film shrinks. In the right hand plot, the film temperature is increased beyond the T_g by a small amount, to Point A and then cooled. It can be seen that there is the normal thermal contraction but the shrinkage that occurred above the T_g remains as a constant dimensional change. If the film is heated again beyond the T_g , it does not shrink but returns back up the cooling line until it reaches Point A. Once the temperature exceeds Point A, additional shrinkage occurs. If the film is taken to the temperature at Point B and cooled, the film will again contract and be left with the total shrinkage seen at Point B. If this is repeated and the film taken back to Point B, it will again expand but have no additional shrinkage unless the temperature is taken beyond Point B. The same characteristic is seen of the film is taken to Point C; then if heated beyond Point C, further shrinkage can be seen.

Figure 10.4 Effects of Heat on Dimensional Stability of the Polymer Film



The learning from this is that for the likes of polyester films such as PET with a T_g of 78°C or PEN with a T_g of 120°C, it is possible to stabilize the films to enable them to have vacuum-deposited coating applied with minimal dimensional changes¹⁶. This is achieved either during the film manufacturing or on a separate heat stabilization process. Both rely on the film being heated to the high stabilizing temperature but with the winding tension minimized to enable the residual stresses to be relaxed. Once the stabilization process has been done, the film can be heated and cooled any number of times (providing the temperature always remains below the stabilization temperature), and the dimensional change will remain minimal.

It can be difficult to match this relaxation process to a film manufacturing line and so in many cases this has been done as an off-line process. To heat the film and wind it slowly with minimal tension is not a low-cost process. So the add-on cost to the basic film can be significant. The heating of the film for a period of time will also enable more residual oligomer or other exudates to reach the surface faster than for the untreated film.

For the most critical applications such as ultra-barrier or electronic displays, the most stable and clean substrates are required, and films that have been both heat stabilized and planarized are preferred.

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Section 11: Winding in Vacuum

Introduction

There is one key difference between winding webs at atmospheric pressure and winding webs in vacuum and that is the lack of air in the vacuum. At atmospheric pressure, air is carried along with the web and becomes entrained and forms a lubricating layer between successive layers of film. This can be a problem if too much air is allowed to be included as the wound roll will be very soft and prone to telescoping.

The amount of air in the roll is controlled by the winding tension and the film design. Rolls such as lay-on rolls are used in contact with the roll being re-wound to squeeze out most of the air and act as a substitute for web tension in controlling the air content (density) of the roll. Without the air layer, which is dragged along with the web, the web is in much harder contact with all roll surfaces including the deposition drum. This lack of air will always produce a hard roll, which will mean that once rewound in vacuum, any web profile problems will become more clearly apparent and can lead to gauge bands forming baggy lanes in the film on unwinding or lead to blocking. Therefore, it is a common practice to rewind metallized rolls soon after production to readmit air to the roll and to minimize the damage done by the tightness of the wound roll.

In order to improve the wound roll quality and prevent damage to the film, when wound in the absence of air, there is a need to control the film-to-film spacing when the film is wound in vacuum. This may be done with film formulations and specifically by “formulating for winding”. Formulating for winding is used to control the film-to-film spacing by the use of surface roughening additives located at the film surface. By controlling the film-to-film spacing we are in essence controlling the roll density.

11.1 Vacuum Winding Systems

It has often been said that a good winding system can help minimize the problems caused by poor quality film whereas a poor winding system is likely to add additional winding problems. It is a common battle that if a roll handles poorly the supplier will blame the winding system, and the owner of the winder will blame the quality of the roll. If all rolls wind poorly and in the same way, then there will most likely be a problem relating to the winding system. But if each roll is different, and some wind badly but in different ways, it is more likely that the problem is in the incoming rolls.

Most roll-to-roll vacuum systems separate the metallizer internal volume into two vacuum zones: a high vacuum zone for the deposition process; and a higher pressure (lower vacuum) zone where the unwind and rewind rolls are positioned. This higher pressure zone is designed to manage the entrained air that is released as the roll unwinds. Controlled roll density in the incoming rolls can aid in improving the reproducibility of the vacuum process control.

To keep the system volume as small as possible, there are often a number of rolls included that redirect the web path enabling the winding system to be compacted. One key component that is probably the starting point to any winding system design is the deposition drum diameter. The deposition drum diameter can be used to change the deposition collection efficiency¹. The larger the diameter of the deposition drum, the greater the possible collection efficiency for any given distance of the sources from the deposition drum. If a double row of evaporation sources is used, it is certain that a larger diameter drum will be needed to maintain the material deposition efficiency. There is a choice of source-to-substrate distance and also acceptable deposition angle. As the angle that the depositing material arrives at the substrate increases away from the normal to the drum surface, the deposit can become more porous. This material represents the initial nucleating material and also the last of the material deposited as the material enters and leaves the deposition zone respectively. This can affect both the adhesion of the coating as well as the reflectivity from the top surface. Thus it is common to restrict the deposition angle using shields. The deposition drum needs to have a diameter substantially larger than the length of the deposition zone. Otherwise the rapid curve of the deposition drum will also make the angle of deposition too far away from the normal, lowering the coating performance.

Working from the deposition drum outward, there are two more decisions to make. The first is to decide if the film is to have any plasma pre-treatment, and if so, what type and where it is to be sited. The plasma treater can be sited against the deposition drum or it can be located as a completely separate unit before the web reaches the deposition drum.

The second decision is what monitoring system is required following deposition. Optical monitoring may require some space around the web between two sets of rollers to allow for a light source on one side of the web and the detectors on the other. An eddy current monitoring system may require a non-conducting roller for the eddy current unit to be sited next to it.

After this, the other decisions will be to decide on the core size, maximum roll size, range of film thickness and substrate type to determine the range of web tensions to be accommodated in the design.

Control of tension is another critically important factor. The minimum number of drives would be three, where the unwind, deposition drum, and rewind are all driven. This can allow some, but limited, difference in tension between the unwind and rewind drives with the deposition drum acting as a tension isolation roll. This can lead to the re-wound roll being wound with a higher tension than would be preferred. Winding rolls with too much tension can result in the polymer web blocking². Blocking is where layers of polymer web do not slip over each other easily, and under certain conditions, may stick together making it difficult, or even impossible, to unwind. Conversely, reducing the tension on the rewind will also result in a reduced tension around the deposition drum, which can reduce the efficiency of removing the heat from the substrate through the deposition zone, thus limiting the deposition rate and system productivity.

What is now more common in metallizer design, is to have at least five driven rolls so that the tensions can be isolated between the unwind roll and deposition drum and between the deposition drum and rewind. Tension isolation is achieved by use of nip rolls or rolls with high wrap angles, where the coefficient of friction between the web and roll, along with the total area of contact, is used to absorb the tension differences. If the wrap angle is too low, making the contact area (distance times width) too short, some of the higher tension on one side will be passed on to the other side by slippage of the web on the roll.

Drives in the web-handling section may be controlled by speed differences (speed control) or tension measurements where tension measurements are used to control the various drive speeds relative to each other. Most modern web-handling sections now use tension control as opposed to speed control. Tension-control systems, without adequate tension isolation, will result in a situation where a web tension control correction on one drive can cause a tension variation on another drive, which causes a drive correction on the other drive. This tension correction then causes a tension variation on the first drive resulting in a correction back on the first drive, resulting in an unstable web handling process. In some cases this drive “hunting” can result in an unstable control and/or a runaway situation with an increasing magnitude of tension and speed oscillations that can result in a poor quality wound roll or a web break. Hence tension isolation is a critical feature in any winding system. The high wrap capstan rolls are preferred as they put less damage into the web.

Different webs will have different friction coefficients. Consequently, changing substrate materials or suppliers may change winding characteristics³. What would be an adequate wrap for polyester or polypropylene webs is unlikely to be sufficient for a polytetrafluoroethylene web. It is also often observed that two rolls (known as S wrap rolls) are used with a large amount of wrap on each roll in preference to a nip roll.

If there are any web profile variations, the nip roll is also more likely to show these as wrinkles. If nip rolls are used, at least one needs to be crowned so that any roll deflection is compensated for and the web will run through with equal nip force across the full width⁴.

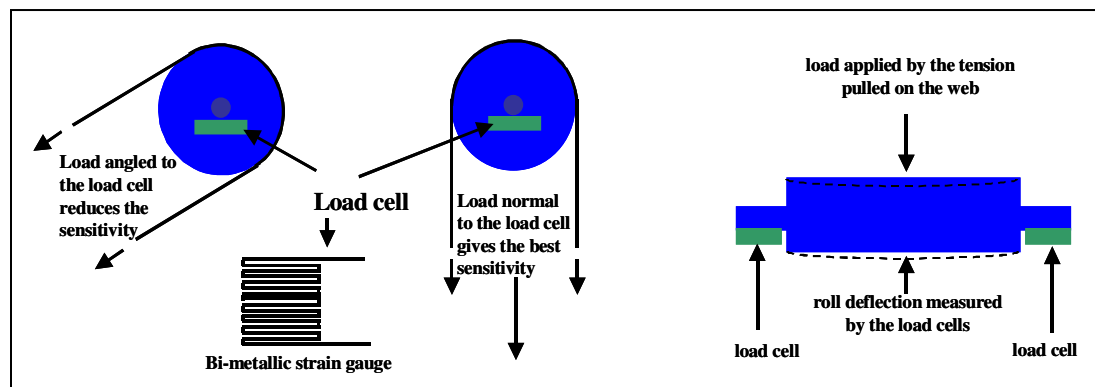
With all these choices and options in mind, it is important to try to minimize the number of rolls that contact the newly metallized surface. The newly metallized surface is soft and easily damaged until the hard oxide protective layer builds up (*see oxidation of aluminum, Section 6.3*). There is always the possibility that the web can slip against a roll surface, and if there is any debris present on the film or roll, this will lead to scratches or pinholes in the metal layer. Thus reducing the number of rolls contacting the metal surface minimizes the possibility of metal surface damage.

11.2 Tension Measurements

The most conventional method of measurement of web tension is load cell⁵. A load cell utilizes a bimetallic strain gauge and monitors the bending of the roll under loading. Usually these are used in pairs, one at each end of a roll.

The load cells are most sensitive if positioned so that the orientation of the load is normal to the orientation of the load cell. The load cell will still work in other orientations but will be less sensitive.

Figure 11.2-1 Schematic of the Load Cell Orientation to the Applied Load



11.3 Rolls and Roll Alignment

The alignment of the load-cell system needs to be accurately done for all rolls including the unwind and rewind. All rolls need to be aligned with a run-out of less than 40 microns per meter (0.5 mil/foot) both axially and in plan view^{4, 6}. The higher the designed winding speeds, the more precise the winding system needs to be⁷.

Alignment can be checked using techniques including lasers, engineering slip blocks⁸ or a new inertial system³⁰. This assumes that the rolls have all been machined to high standards and are concentric. A poorly machined roll will result in difficulties in achieving good alignment. Checking alignment should be done from a reference roll. Alignment can be done from roll to roll, but bear in mind that errors can be cumulative, so checking back to a reference roll to make sure there is no systematic error is essential. The roll chosen as the reference roll should be one that is least likely to be disturbed. As the deposition drum may be taken out for cleaning, polishing or re-plating, it is probably not the best choice for the reference roll. Once aligned, rolls can usually be locked into position. Locked in position there is no reason for the rolls to move or become misaligned. Therefore, unless a roll is removed, there ought to be little necessity to check alignment other than during major maintenance downtime.

The two rolls that have the greatest possibility for alignment variation are the unwind and rewind rollers. They are regularly changed and the reproducibility of the alignment depends on the accuracy of the mounting as well as the precision of the cores. No matter what the quality of the winding system, if the cores are deformed in some way, the input film roll will unwind unevenly and/or the metallized film roll will rewind unevenly, causing difficulties in controlling the tension. Cardboard cores are frequently used. Cardboard can be permanently deformed easily if rolls are dropped but they are low-cost. Plastic cores or metal cores potentially improve alignment accuracy but at a higher cost.

The spacing of rolls to some extent depends on the thickness of the substrates to be run. The thinner the web thickness, the closer the rolls need to be to each other. One rule of thumb is that the distance between any two rolls should be no greater than the width of the substrate. It is easier to handle thick films as the stiffness of thicker films simplifies the web handling⁹. Film stiffness is proportional to the cube of the thickness, $\text{Stiffness} \propto (\text{thickness})^3$.

In addition to the standard web-transport rolls, there are corrective rolls that can be included such as spreader rolls¹⁰⁻¹³. Typically these are placed before critical events, for example, just before the web contacts the deposition drum or before rewinding, with the idea being to take out any wrinkles^{14,15}. Choosing the right spreader is important, as changing the web thickness may cause wrinkles in thin webs where the same roll works well for thicker webs⁶. As a

way of eliminating mild variations and allowing consistently good winding, some prefer to use rolls with a slightly concave surface where they all provide some spreading action⁶. An exception to this would be the nip rolls where at least one of the rolls needs to be crowned, so that any roll deflection is compensated for, and the web will run through with equal nip contact width and pressure across the full width⁴. Film guidance rolls can be included but are not recommended¹⁷.

As web-transport systems get bigger (4.45m wide for metallizers) and the film roll diameters increase to increase productivity from the metallizers, the choice of materials for rolls gains importance. As the substrate width increases, the film and transport rolls tend to sag in the center so the core and roll diameters are increased to reduce this problem. The deflection due to any excessive bending is the equivalent to a misalignment¹⁶.

This increase in roll diameter causes the mass of the roll to increase, and this increases the inertia of the winding system. Any non-driven rolls may be slower to change speed and may therefore be a potential source of microscratches in the web. In this case, non-driven rolls should ideally be of low inertia. Also, if it is important to have an emergency stop in the winding controls, roll inertia will be a problem. Because of the desire for low inertial idler rolls and to improve emergency stop performance, alternative roll materials such as composite glass or carbon fiber rolls are increasingly used.

The choice of materials is also important for rolls such as spreader rolls, nip rolls or pattern printing rolls. Rolls are often made using a steel core with a polymer covering ground to final diameter. The hardness of the polymer and the surface grinding finish are critical to the spreading and nipping action. Many of these roll covering polymers will change over time, usually increasing in hardness. The rate of change will depend on the material and environmental conditions such as elevated temperatures. Roll surfaces can be reground several times to recover the initial surface finish and hardness properties. However, regrinding changes the roll diameter, and unless the roll is an idler, this will change the surface speed of the roll, relative to a constant roll speed, and must be accommodated in the drive system.

Rubber roll covers are generally proprietary compositions of the supplier. They are compounded materials based on crosslinkable rubber or polymers, therefore, the composition of the base material, *i.e.*, nitrile rubber, silicone rubber, polyurethane, etc., will determine the chemical and aging resistance of the roll covering. Consequently, it might be necessary to try a variety of roll coverings and suppliers to find the optimum roll covering. Because the rubber compounds might contain out-gassing materials, their use in vacuum systems should be carefully evaluated.

All rolls need bearings and these need to be free running, particularly for the non-driven rolls that rely on the surface traction from the web to turn them. Bearings that are used within the vacuum system need to have a minimum quantity of vacuum-compatible grease or oil to minimize rolling resistance.

11.4 Heat-load Considerations

Low-temperature deposition

A final consideration that affects the design of the winding system is how the heat from the deposition zone is managed. The winding system can have zero web cooling in the deposition zone, as in the case of free-span metallizing, or it can have a deposition drum cooled to sub-zero Celsius temperatures. There may also be additional post-deposition cooled rolls.

The heat load applied to the web depends on many factors such as the coating thickness, winding speed, length of deposition zone, and rate of film cooling. Other than for free-span coating there is usually no need for any pre-deposition cooled roll. Low-speed systems lay the web onto the deposition drum early enough that the web reaches the drum temperature well before it enters the deposition zone making any earlier cooling unnecessary. High-speed metallizers and where a very high intensity plasma treatment is used before the web reaches the deposition drum might require a pre-cooling roll to limit the web temperature rise. It is difficult to predict the effect on the substrate peak temperature when changing substrate thickness or making other process changes such as deposition rate or winding speed. Modeling software can be a cost-effective way of evaluating changes without the need to consume film and machine time¹⁸. What has been highlighted by this software is that within the deposition zone there are many more variations than were initially considered. The edges of the web are masked to prevent deposition onto the deposition drum. This means the edges of the web do not get heated and so do not expand as the bulk of the web does. This means the edges will take an increasing proportion of the tension applied as the edges are shorter than the rest of the film. This reduction in the web tension where the coating is being applied can mean the temperature will peak higher than expected, and the film will be more likely to wrinkle.

The need for post-deposition cooling rolls is different. The cooling rate of the web is often too slow for all the heat to be removed by the deposition drum between the web leaving the deposition zone and leaving the deposition drum. If the web leaves the drum at too high a temperature, the rewound roll will be “hot” and can suffer from the contracting forces caused by a change of dimensions once the roll is taken out of the vacuum system and cools. This contraction as the roll cools can be enough to crush the core and damage the roll. To prevent this damage, cooling one or more of the rolls post deposition has some benefit¹⁹ with the aim of removing wound film rolls that are already close to ambient temperature.

“Hot” as used above is a relative term, and what is too hot depends on the material being metallized and rewound. In general, hot implies that the film can stretch under the tension of the rewinding and web-transport system, and it is the source of the contractive forces described above. Polymer films, depending on polymer type and manufacturing conditions (*see Section 8.1 and Section 10.4 on heat stabilization*), are inherently dimensionally unstable above certain temperatures, and the temperature of the web can induce some thermal shrinkage in the film, which can contribute to the contractive forces as well.

However, if the film contains low melting or soft polymer coatings as in heat-sealable packaging, then the definition of “hot” could mean the temperature at which surface welding between film layers in the roll occurs. Surface welding of film layers in the roll can lead to metal pickoff (pin windows) on unwinding, edge blocking, pinpoint blocking, and film tears on unwinding, and a general blocking of the roll. All of these factors make the film unsuitable for further use. Therefore, control of film temperature leaving the deposition system and/or entering the winding section needs to be determined experimentally and controlled. The acceptable film winding temperature level will be dependent on the film thickness, composition, and design. Average film roll temperatures can be measured once the film is removed from the metallizer by the use of a surface pyrometer held against the roll surface. In general five measurements across the roll surface should suffice. Gauge variations in the film can result in temperature variations in the film.

As a practical example, this technique was used to experimentally to determine the upper limit of the exit film roll temperature of a metallized, coextruded polypropylene film to prevent blocking and metal pickoff at unwinding. Knowing this upper temperature limit permitted the determination of the suitability of a metallizer for the economic production of this particular product by determining the maximum line speed possible for a given optical density.

High-temperature deposition

The advent of the roll-to-roll vacuum deposition of photovoltaic materials has required the development of high-temperature processing in order to produce the desired coating crystallinity and stoichiometry. Substrates such as metal foils or polyimide films have been used at temperatures up to 550°C. This gives a different set of problems in winding the substrates. In broad terms the foil or film is heated, deposition takes place, and the substrate is cooled. It is relatively easy to heat the substrate by using radiant heaters as the heaters can be at a much higher temperature than the substrate. What is much more of a problem is taking the heat out of the substrate once deposition is complete. The heat will radiate to colder surfaces so it is possible to use cold plates to help cool the film. However, although this may start out cooling the substrate quite well, the rate of cooling slows as the difference in temperature declines. Thus to bring the substrate to room temperature can take a long time. Using an in-contact drum is better as there is a direct conduction path as well as the radiation heat transfer. However, the contact time required would either make the winding speed very slow or the drum very large to have enough contact time to reduce the temperature to ambient.

Heating polymer films to such a high temperature is difficult as the tensile performance of the film decreases with temperature. Variations in the film thickness that may be hidden by applying a slightly higher tension may, at these, high temperatures result in a permanent distortion of the film. Additional tension is also used to minimize film wrinkling. As this cannot be used on films where such high temperatures are being used, film uniformity and quality has to be higher. The aim within the process has to be to maintain this same substrate uniformity and so any heating of the film and cooling of the film has to be uniform across the whole width. Otherwise differential stresses will occur which could result in wrinkles or other film damage.

11.5 Tramlines (Railroad Tracks)

A common source of tramlines (railroad tracks) is the deposition zone. The web is heated and begins to expand laterally but it is constrained by the web tension and coefficient of friction (COF) between the web and the deposition drum. The amount of lateral expansion depends on the substrate material as well as the efficiency of heat extraction that is a reflection of the deposition drum heat transfer coefficient²⁰⁻²⁴. If the lateral expansion is large enough, the compressive force will cause the web to buckle off the surface, and the loss of contact with the cooled

deposition drum will make the problem worse. This problem can be reduced by using a spreader roll to put the web under some lateral tension as it is laid onto the deposition drum. When the web is heated, it first has to overcome this initial tension, before the expansion can produce the lateral compressive force necessary to cause buckling.

An alternative strategy is to reduce the COF between the web and the drum²⁵⁻²⁸. This is done by injecting gas between the polymer web and the cooled deposition drum. The gas is added just at the point where the web first contacts the drum. This gas becomes compressed by the applied film tension and heat is transferred through convection currents within the trapped gas. This increases the heat-transfer coefficient to the deposition drum causing the web to reach a lower temperature than when the gas is not used. The thermal expansion is thus reduced lowering the likelihood of tramlines. An additional benefit is that the trapped gas also reduces the COF by acting as a lubricant, so even if the expansion were the same, the web would be more likely to move over the surface, relieving some of the compressive force, thereby lowering the likelihood of tramlines. Reduction of the film-to-drum COF to reduce the lateral compressive forces developed in the film and eliminate tramlines is also the approach taken with film formulations where anti-blocking particles are added to the film surface layer in contact with the deposition drum. Controlled contact with the drum also helps maintain a uniform web temperature.

Where the metallizing process is being run close to the limit of tramlines forming, the cleanliness of the system becomes increasingly important. It has been shown that debris on the back surface of the web or on the drum can be enough to lift the polymer web out of contact with the deposition drum, causing web temperature to rise rapidly and significantly increasing the local lateral compressive force and the probability of buckling in the heated area.

One method to combat tramlines for a given film formulation is to increase web tension. Although this could work, it could damage the film. The tensile strength of films decreases with increasing temperature. An example of this would be polyester film that loses two-thirds of its tensile performance by the time the web reaches 120°C. Thus pulling extra tension when the film around the deposition drum is “hot” can easily exceed the elastic limit of the polymer and permanently elongate the web. Not only is this elongation unwanted, but it also eases the tension and does not have the desired effect of increasing the contact pressure. To prevent this problem, it is good policy to start with a tension that is less than 10% of the yield strength of the polymer web at the expected temperature the web will reach in the deposition zone. Because the tensile yield properties of polymer films vary greatly, the acceptable film load (lb/in or kg/cm) for 10% of the yield strength will vary significantly with film type and film thickness.

11.6 Polymer Substrates and Winding

The winding system cannot correct for poor web quality. If the web has a problem with the thickness profile, such as gauge bands or raised edge due to poor slitting, then the tension will be taken by the thickest part of the web first. This results in a region of different diameter (larger in thicker regions) in the roll *e.g.*, a gauge band, stretching the film in this area, making it longer than the film surrounding it. When the roll is unwound, the film in the gauge band will be loose next to the tensioned shorter film sections creating a “baggy lane.” Baggy lanes create wrinkles and creases when transported over rolls. Baggy lanes can also create slitting problems, and if the baggy section is on a film edge, it will create a roll with one soft and one hard edge, cause breaks at the slitting knives, and/or a feathered edge in the slit or rewound roll.

Aside from having a printed plot of the gauge quality with the roll, one simple check that may be carried out on each incoming film roll is to gradually pull a small amount of tension once the film has been threaded onto the unwinding system to visualize the presence and severity of the baggy lanes in the film. As the tension is taken by the shortest part of the film first, it is possible to get some indication as to whether the film has a uniform profile or if it has any uneven profile defects (baggy lanes) simply by observing how the tension is taken up by the film. For example, where tension is initially taken up by one side of the web as the other edge remains slack, this gives a visual indication of a potential problem. The side with less tension is the one that potentially will have the first problem with tramlines due to the formation of wrinkles during web transport.

Another simple test is to check roll hardness. Monitoring the hardness across the width of roll can help prevent problems such as telescoping from too softly wound rolls and can give an early warning of profile variations that may cause winding problems.

An additional approach can be to monitor and control the incoming roll density²⁹ (mass/volume) and establish roll density ranges, which are indicative of well-wound and uniform thickness film. The roll density is a composite measure of the amount of air wound into the roll and represents the best indication of the average quality of the roll's manufacturing history. Improvements in roll density uniformity will greatly reduce roll-to-roll tension variations. Controlling roll uniformity also will reduce variations in the pumping load due to air released at unwinding in the metallizer and this will generate a more uniform winding and vacuum process. This in turn can give improvements of several percentage points in overall productivity. Acceptable mill roll densities will be dependent on the film formulation and can be controlled by formulating for winding, as described in the next section.

11.7 Formulating for Winding

Formulation for winding is a concept in which the film-to-film spacing is controlled by the application of particles to the film surface. In general, the particles are irregularly shaped ground minerals, such as talc or silica, termed antiblock particles because they were originally used to minimize the blocking tendency of films by minimizing surface-to-surface contact in the roll. Initially, these particles tended to be small in diameter and were used in high concentrations, which impacted the optical properties of clear films. However, with the increased use of coextrusion in film manufacture (*see Section 8.1 and Section 9.6*), it has become possible to isolate the antiblock particles to the film surface itself, improving optics, controlling the cost of the additive, and improving the effectiveness of the particles.

Today, in addition to the ground mineral particles, there are many synthetic particles of improved shape and purity. Typical particles are precipitated silica, amorphous silica, crosslinked acrylic spheres (EPOSTAR[®]), and crosslinked silicone based materials (TOSPEARL[®]). The spherical particles are especially effective in formulating films for winding.

The primary goal, when formulation for winding, is to create protrusion (bumps) on the film surface, which exceed in height the thickness of the air layer which would be formed on the moving film surface at that tension level. If this is achieved, the film-to-film spacing becomes insensitive to the existence of an air layer and the film when wound in vacuum, will have the same film to film spacing as when wound in air. This minimizes the tendency to produce harder rolls in vacuum than in air, greatly improving the roll formation exiting the metallizer. In addition the antiblock particle technology is a key means of developing film formulations that are resistant to tramline (railroad track) formation on metallizer process drums.

Where films are re-wound "hot," the high contact pressure due to winding tensions and thermal contraction can cause metal pick-off and resultant pin-windows in the metal layer. In this case the use of anti-block particles to control the film-to-film spacing can reduce the contact of the metal surface with the opposing film surface and greatly reduce metal pick-off.

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Section 12: Converting Metallized Films

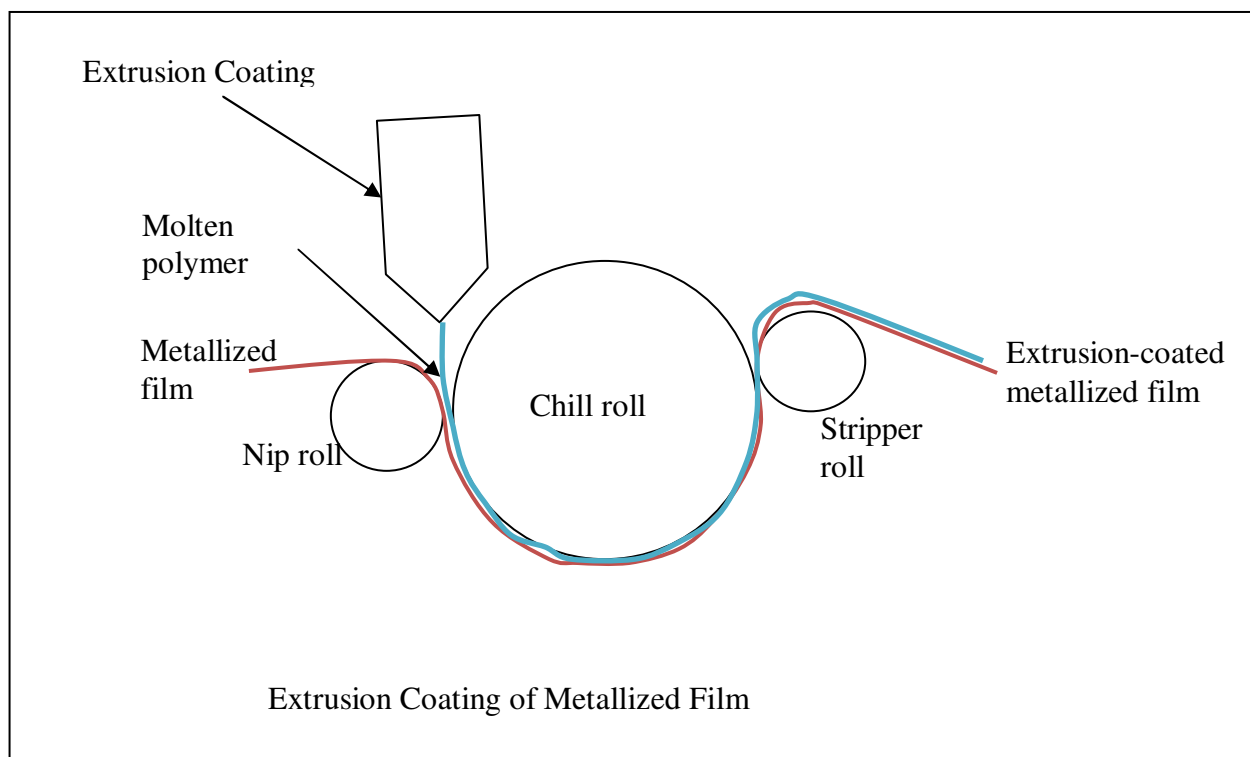
Because the addition of the aluminum layer to a film is a batch process requiring a free surface onto which the aluminum has to be evaporated, it is not possible to add the aluminum in-line with other polymer film processes. Also, after metallization, the aluminum layer is located on an exposed surface of the film, and the use of metallized films generally require that the aluminum layer be covered by a protective layer. This is due to the fragile nature of the vacuum-deposited layer, which is extremely thin (*see Figures 2.3-1 and 2.3-2*), and prone to surface damage during transport and winding of the aluminum-coated film. In addition to the fragile nature of the aluminum layer, it may be desired to cover, or hide, the aluminum film, so as to maintain existing graphics or to add additional functionality such as heat sealability. Consequently, metallized films often are combined with other films to increase the usefulness of the final product.

The combination of the metallized films with other materials usually requires the adhesion of the companion films and/or polymer layers to one or both surfaces of the metallized films. This is accomplished by laminating or coating the metallized films. Typical methods used to convert metallized films into final products include printing, extrusion coating, solution coating, and adhesive and extrusion lamination or combinations of these processes.

12.1 Extrusion Coating

Extrusion coating is a well-known process^{1, 2} with broad application^{3, 4}, and well-developed equipment technology is commercially available from a variety of suppliers. In extrusion coating, an extruder (*see Section 9*) is used to prepare a molten polymer at a sufficient temperature and pressure so it can be extruded onto the metallized film (Figure 12.1-1).

Figure 12.1-1 Schematic of Extrusion Coating Station

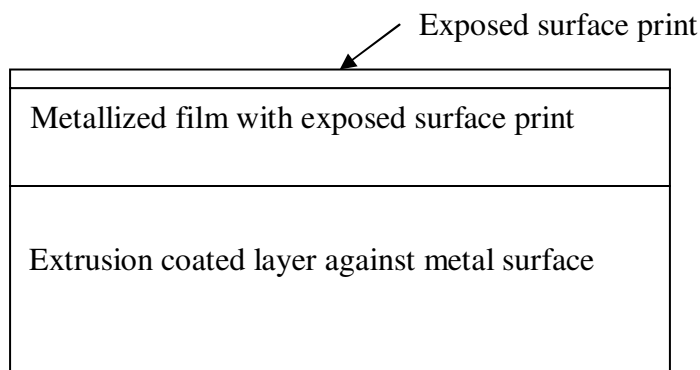


Because metallized films are flat, the extrusion-coating system uses flat-slot dies for the extrusion of the coating layer. The width of the die is related to the width of film to be coated, and for a given manufacturing location it can be chosen to match other equipment used in conjunction with it, such as slitters, printing presses, and metallizers. Typical extrusion laminator widths range from 30 to 173in, with wider laminators in the 140- to 160in range predominating as outputs increase. The thickness of the coating layer is controlled by the output of polymer generated by the extruder and the speed of the coating substrate passing under the coating die. The actual coating

thickness required for an application will be determined by fitness-for-use considerations. Extrusion coating layer thickness may limit the coating line speed, based upon extruder output limits, for any given application.

A general example of product design for extrusion-coated metallized films is shown in Figure 12.1-2. It is a surface-printed (or plane) metallized film with a sealant polymer extrusion-coated to the metallized film surface. This adds heat sealability to the metallized film surface, in particular, a thick heat-sealable layer of 12 to 50 microns (0.5 to 2 mils) to insure air- or watertight seals or to permit sealing through dust and particulate contamination on the sealing layer. Sealant thickness applied is limited by the temperature resistance and thermal stability of the metallized substrate. A widely recognized application of this product is the small, single-serve peanut pouch distributed on airline flights.

Figure 12.1-2 Schematic of Typical Extrusion-coated Metallized Film Product Design



Many polymers are used for extrusion coating² the sealant layer, including low-density polyethylene (LDPE), linear-low-density polyethylene (LLDPE), ionomer (Surllyn®), poly (ethylene-vinyl acetate) (EVA), poly ethylene-methyl acrylate (EMA), poly ethylene/methacrylic acid copolymer (EMAA), ethylene-acrylic acid copolymer (EAA), as well as blends of these and other materials. In the broader market of extrusion coating, many potential combinations of substrates and coating polymers are used to make a wide range of products resembling Figure 12.1-2, *e.g.*, packaging, coated paperboard for milk cartons, coated scrim and woven materials for house wrap and tar paper replacements.

In the extrusion coating process, the metallized film is unwound and transported to the coating nip. The surface to be coated is on the upper surface of the film, and generally, this is the metallized surface. Prior to coming to the nip point, the coating surface may be prepared for the melt addition by corona treatment or perhaps by corona treatment and priming. The treatment and priming are processes that help insure uniform adhesion of the extrusion coating to the metallized surface. Treatment for surface cleaning is often necessary, depending on the substrate material, since the metallized surface may have become contaminated by low-molecular-weight materials migrating from the film while wound in roll form (*see Section 10*). The treatment step removes organic contamination by oxidation in the corona. If the metal surface is primed, it is coated with a polymer solution containing materials such as polyethyleneimine (PEI) or epoxy primers, which can adhere to both the metal surface and the polymer coating surface. Priming of the metal surface yields more consistent bond results than unprimed surfaces, and consistent priming requires corona treatment of the film before the priming station. Primer chemistry is determined by the polymer coating being extruded onto the metal surface.

As the metallized film enters the nip the hot molten polymer falls from the die and is pulled from the die as the film contacts it. The nip roll presses the film against the molten film, which is in contact with the chill roll. Heat is extracted from the molten polymer into the roll, and the molten polymer solidifies, forming a polymer film attached to the metallized film. The coating, as applied, may be narrower or wider than the metallized film depending on the polymers and films used. The combined web is stripped from the chill roll, transported away, and the edge of the film is trimmed and discarded. The slit film is then transported to a winder where it is wound for further conversion as necessary. Often the final product is slit to narrower widths (coated width is two or more finished rolls wide) and shorter lengths, permitting the manufacture of several slit rolls simultaneously to increase productivity lowering costs. Slitting-to-width can be in-line, but it is generally a separate operation since the final roll dimensions require frequent roll changes and could limit line speed of the coating process.

Adhesion between the coating polymer and the metallized film is controlled by several methods. Bonding of the polymer and the metal layer depends upon the relative compatibility of the extruded polymer to the aluminum surface and the wetting of the polymer on the aluminum. The aluminum surface is comprised of an aluminum oxide layer (*see Section 6.3*) with a high surface energy (approximately 54-56 dynes). Therefore, the polymer coating's surface energy (*see Tables 7.6-1 and 7.7-1*) must be relatively high for acceptable wetting of the polymer to the metallized film surface. This is controlled by surface oxidation of the extruded polymer coating and/or the blending of polar polymers into the bulk coating resin. Alternatively, the metal surface can be primed to improve the surface energy matches between the metallized surface to the primer and the primer to the coating such that the primer serves as a compatibilizing layer between the two plane surfaces.

Surface energy of the polymer coating layer is usually modified by oxidizing the molten polymer in air as the polymer is transported from the die lip to the coating nip (*Figure 12.1-1*). As the molten polymer drops from the die to the nip, it is exposed to atmospheric oxygen, and it burns, creating polar chemical species on the molten polymer surface. This is, in effect, a surface treatment by chemical reaction of the polymer melt with atmospheric oxygen. Because the surface oxidation of the melt in air is a chemical reaction with a specific rate, the factors that will impact the surface oxidation are dependent upon:

- The temperature of the melt
- The time spent exposed to the air
- The concentration of oxygen in the air
- The resistance of the polymer to oxidation

Melt temperature is controlled by extruder barrel temperature, head pressure settings, and the screw design. Residence time in the air, (R, seconds) is determined by the ratio of the die to nip gap (*i.e.*, air gap) length (L, m) to the line speed (V, m/sec), $R = L \div V$. Oxygen concentration in air is fixed, as is the general polymer oxidation rate.

It is easy to understand that to maintain the same oxidation level as line speed increases, the air gap should be increased to maintain the exposure time. However, air gap length may be limited, based upon the melt-flow properties in the air gap. As line speed and/or draw gap increases, many melts will draw down more sharply (neck-in) and become too narrow or the melt curtain may begin to oscillate and become unstable or break off the die lip. Neck-in can be compensated for with a wider die width, but melt oscillation may be very difficult to control, limiting the air gap length. This can result in a line speed limitation related to melt surface oxidation, thereby limiting bond strength development.

To overcome limitations in air gap length and surface oxidation levels, it is common to add extra oxygen in the form of ozone to the air gap to increase the surface oxidation rate which can permit increased line speed with acceptable bond strength. Another alternative to increase line speed while maintaining bond strength is to blend more polar (*i.e.*, "pre-burned") oxygen-containing copolymers into the coating base. For example, when LDPE is used as the coating resin, it can be blended with EVA, EAA or other copolymers. This increases resin costs but permits higher coating productivity. For a fixed product design and a particular coating line, optimum adhesion for any combination of metallized film and coating resin can then be determined by a designed experiment based upon line speed, air gap, and melt temperature.

To a certain extent, bond strength levels can be affected by several other process variables controlling the surface wetting of the molten coating polymer to the metallized film surface. Wetting will be impacted by:

- The temperature of the molten polymer surface
- The chill roll temperature
- The first contact point of the melt (on the metal surface, centered in nip, or on the chill roll)
- Nip pressure

Once optimum conditions for line speed, air gap, and melt temperature have been determined these parameters can be varied to further optimize bond strength. The use of polished aluminum reflectors in the air gap has been reported as a means of conserving melt temperature in the air gap⁵.

12.2 Adhesive Lamination

Metallized films and substrates also may be combined with other films and materials by directly gluing them together. This is called adhesive lamination and is a well-known converting process. Typical product designs for adhesive laminations containing metallized films are shown in Figures 12.2-1 and 12.2-2.

Figure 12.2-1 Typical Adhesive Lamination with Metallized Film

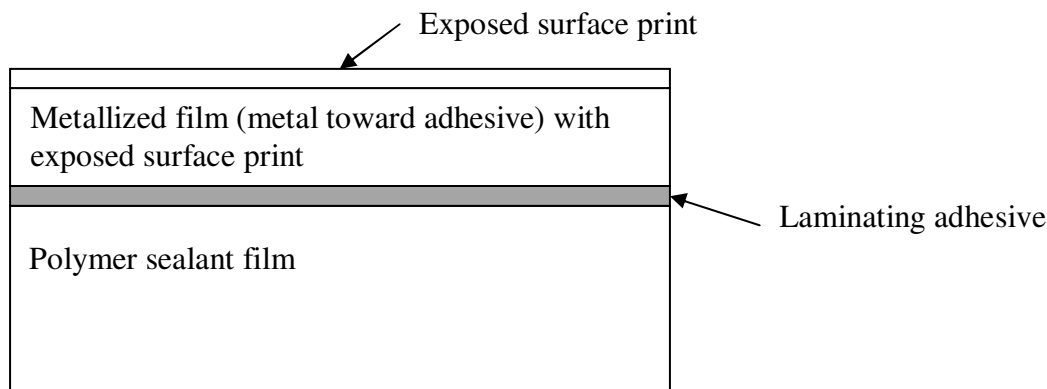


Figure 12.2-2 Adhesive Lamination of Metallized Film

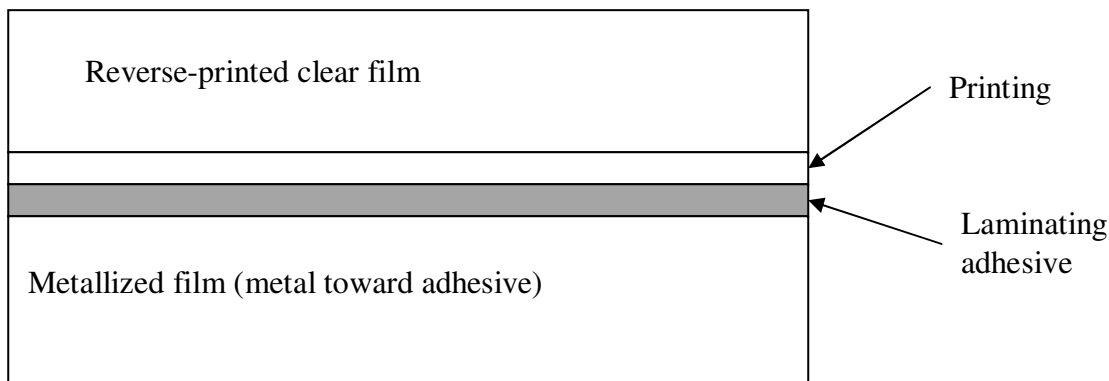


Figure 12.2-1 represents an alternative method for producing a structure such as Figure 12.1-1 where the coating film material is not extrusion-coatable, due to the drawdown or adhesion characteristics of the polymer or thermal limitations of the metallized film. Thicker sealant films may be added to the metallized film using adhesive lamination because the metallized substrate is not subjected to the heat load of the high-temperature extruder coating. Laminating films may be produced by any of the film manufacturing methods described in Section 8.1, but blown and cast films are primarily used.

Figure 12.2-2 represents a wide range of packaging materials made from metallized films. In general the metallized film is heat-sealable on its lower surface and additional layers are not necessary. In some cases where the metallized film is not heat-sealable, the film in Figure 12.2-2 can be used as the upper film in Figure 12.2-1, and a three-ply lamination can be made. This highlights the wide range of product concepts that are possible with adhesive laminations. Because the process conditions are relatively mild, a large array of film combinations can be produced.

Figures 12.2-3 and 12.2-4 show the two methods generally used in adhesive lamination. In Figure 12.2-3 the adhesive is applied wet from solution to one of the substrates, (typically, the unmetallized one), and the coated substrate is passed through a hot-air oven to dry it by removing the solvent. The dry adhesive is then contacted to the metallized film surface with a laminating nip to create the bond. This process is typically called “dry bonding.”

Figure 12.2-3 Schematic of Dry Bonding Adhesive Lamination

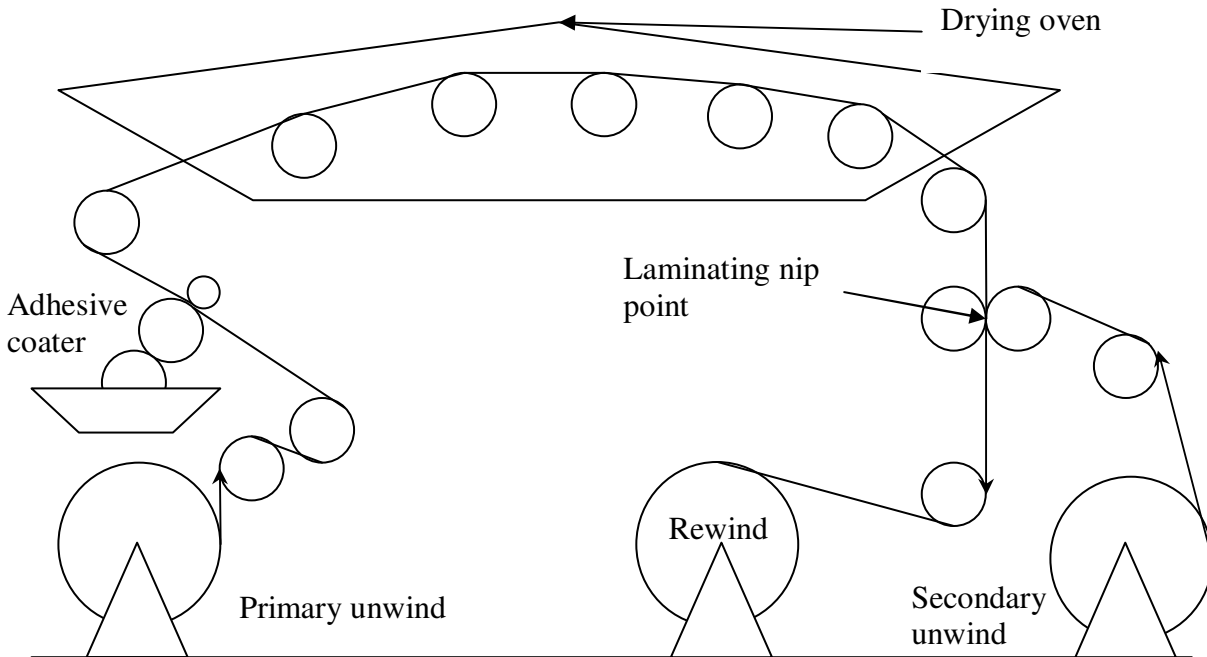
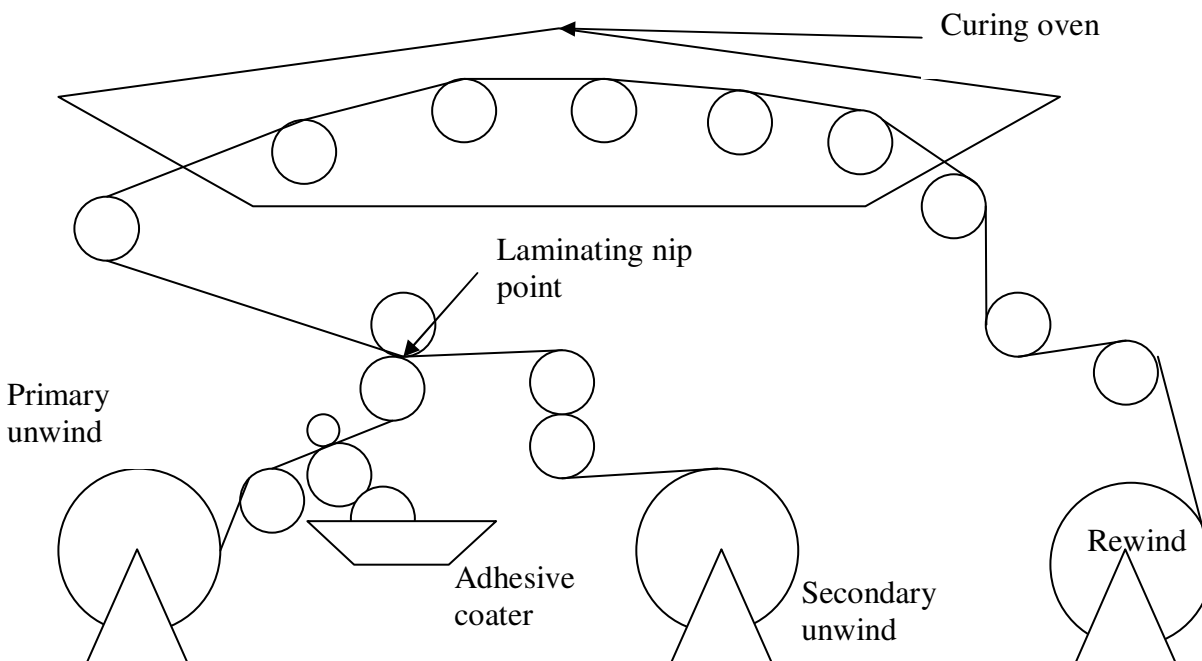


Figure 12.2-4 shows the coating of the substrate immediately followed with lamination of the webs in a lamination nip. Usually this process uses a two-part or catalyzed crosslinkable adhesive, without solvent, which reacts chemically to form the final adhesive composition. The combined lamination is then passed through a heated oven to begin curing the adhesive. This process is called a “wet” lamination because the webs are combined without drying, and the adhesive cures to its final state.

Figure 12.2-4 Schematic of Wet Bonding Adhesive Lamination



In dry bonding, adhesive strength is generally high at “green bond strength” and usually increases somewhat with time to the final “aged bond strength.” With wet lamination the green bond may be high or low, and it generally ages during the course of a week as the two-part adhesive continues to cure. Although low green bonds often “age up,” if there are problems with the bonding process, the green bonds may “age down” to zero. Typical bonding behavior in a peel test would be a:

- Cohesive failure in the adhesive,
- Adhesive failure at the metal (or ink)/adhesive surface (*Figure 12.2-2*)
- Adhesive failure at the metal (or sealant surface) adhesive surface (*Figure 12.2-1*)
- Cohesive failure in any of the laminated film layers
- Adhesive failure at the substrate/aluminum (or ink) surface

A bonding “failure” is somewhat subjective (“high” is good, but “low” is bad) and usually will be associated with the level of the peel strength and at which interface the failure occurs. The actual value of the peel strength does not directly measure interfacial adhesive strength as much of the peel force comes from energy dissipation in the film components, the adhesive used, and its thickness, as well as interfacial bond strengths. Changes in structure component materials, film and layer thicknesses, and the use of backing tape will change the measured value of the peel strength^{6,7}.

Peel strength measurements are valuable in looking for a change in a process or material. They represent good quality control measurements as long as the test sample is uniform. Large changes in peel strength and/or failure mode in a lamination process, which is being monitored for process and product control, can indicate changes in substrate properties, signaling an out-of-control laminating, printing or metallizing process. Failure mode changes can be followed up with analytical testing of failure surfaces to determine likely failure sources⁸. However, comparing the peel strength between two metallized substrates is less meaningful. For instance, in comparing peel strengths of metallized polyester and metallized oriented polypropylene (OPP) films, the mechanical property and film thickness differences between the two substrates will likely cause the peel strength values to be different regardless of the interfacial adhesion of the adhesive to the metal surface.

12.3 Extrusion Lamination

Metallized laminations also can be produced by extrusion lamination where the adhesive is a molten polymer supplied by an extruder as in extrusion coating (*see Section 12.1*). Figure 12.3-1 is a typical product design used for metallized films, which mimics the adhesive lamination in Figure 12.2-4 but, where the adhesive is an extruded polymer. The use of extrusion lamination has been increasing because it permits the use of thinner printed and metallized films while maintaining the overall lamination thickness with adequate stiffness. In this case the more expensive outer and inner substrates are replaced with less expensive bulk polymer in the adhesive layer. For instance a 1.9-mil adhesive lamination of 25-micron printed OPP/adhesive/23-micron metallized OPP (100-gauge printed OPP/adhesive/90-gauge metallized OPP) can be replaced by an extrusion lamination of 15-micron printed OPP/12-micron LDPE/18-micron metallized OPP (70-gauge printed OPP/50-gauge LDPE @ 7 lb/ream/70-gauge metallized OPP). In this case, approximately 12 microns (50-gauge) of the expensive OPP is replaced by 12 microns of far less expensive LDPE while maintaining overall thickness and sufficient stiffness of the lamination. An additional advantage of extrusion laminations over adhesive laminations was the discovery that extrusion lamination permits improved hermetic sealing (no visible air leaks under water with a weight of 5kg) of bags relative to adhesive laminations⁹.

Figure 12.3-1 Extrusion Lamination of Metallized Film

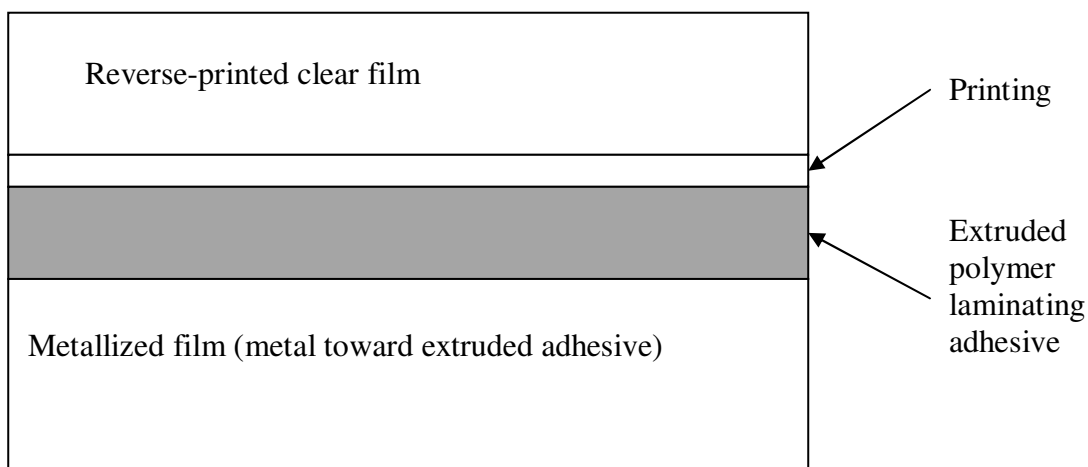
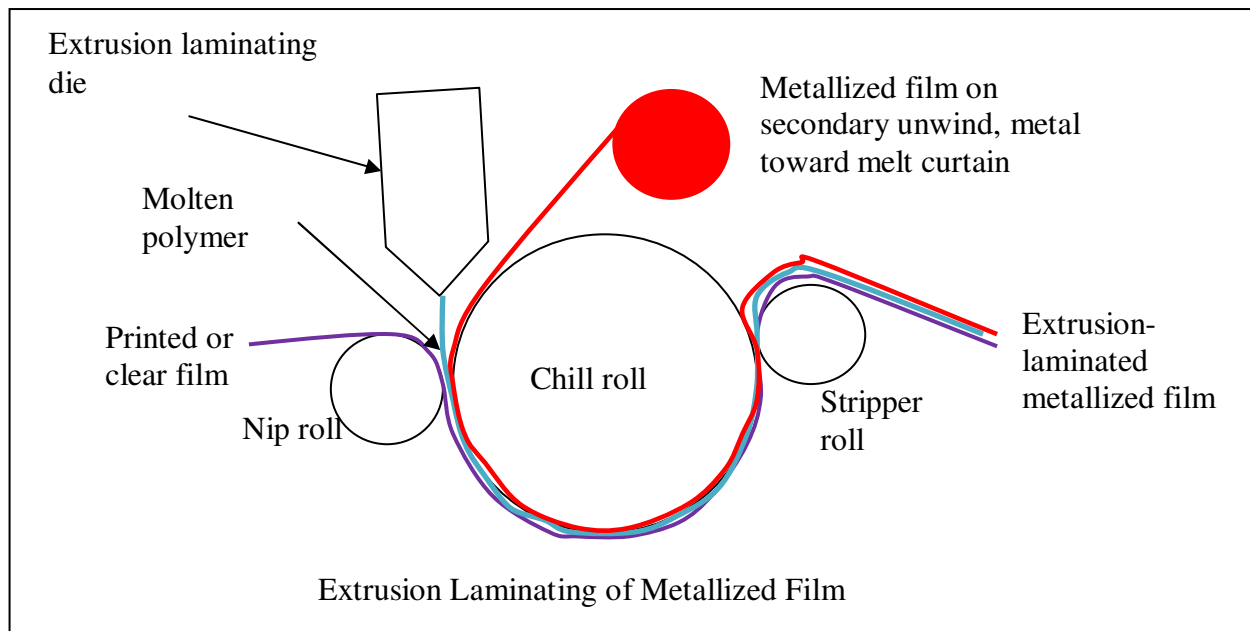


Figure 12.3-2 shows the combination point of an extrusion laminator. As in extrusion coating, the molten polymer is prepared in an extruder and delivered from a slot die and dropped between two moving substrate webs. The printed web is generally unwound from the primary unwind and the metallized film is unwound from the secondary unwind to minimize or eliminate any roll contact with the metallized surface prior to the lamination nip and to place it on the cooling roll. As with extrusion coating, the polymer adhesive may extend past the edges of the two webs, so the lamination will require trimming prior to winding.

Figure 12.3-2 Schematic of Extrusion Lamination Combination Point



Bond strength development in extrusion lamination follows the same principles as in extrusion coating since the mechanism of surface oxidation of the molten polymer adhesive is the same. Should it be desired to prime the metal surface, it may be necessary to reverse the positions of the two webs or the laminator will require the addition of a coating and drying station for the primer application. Priming can improve the consistency of the lamination bond just as with extrusion coating.

In addition to the factors affecting bond strength listed in the discussion on extrusion coating (*Section 12.1*), bond strength is also impacted by the amount of molten polymer used (adhesive layer thickness) and interacts with the melt temperature to define the thermal load on the metallized film. The thermal load has two primary effects on the final lamination: to increase the bond strength (*Figure 12.3-3*) and to damage the metal layer by metal fracture (*Figure 12.3-4*). Because the molten polymer is extruded between the two substrate layers, heat removal may not be as good as when extruded in direct contact with the chill roll.

Figure 12.3-3 Impact of LDPE Thickness (Poly Weight) and Melt Temperature on Lamination Bond Strength at 275 m/min and an 18cm Air Gap

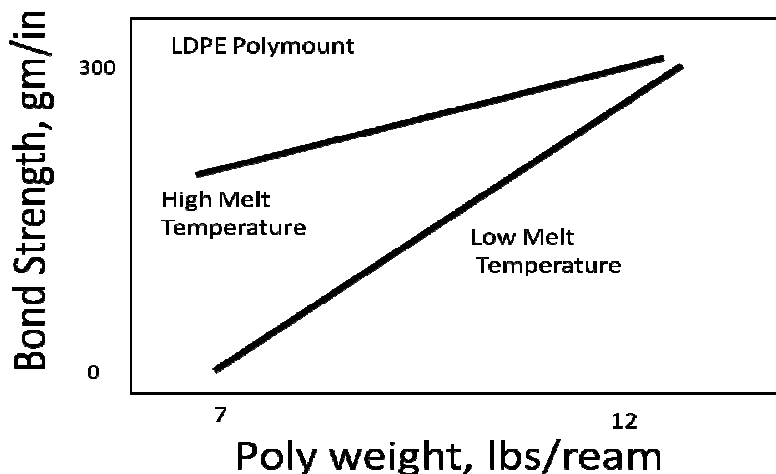
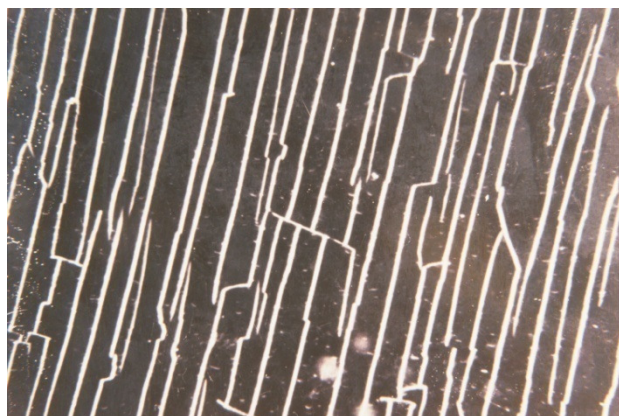


Figure 12.3-3 shows that the value of the lamination bond strength increases with increasing LDPE poly weight from 7 lb/ream (approximately 12.5 microns) to 12 lb/ream (approximately 22 microns) and that melt temperature has a diminishing effect on bond strength with increasing LDPE poly weight. At the higher LDPE poly weight, the melt temperature will have less effect on bond strength because there is less surface cooling of the melt, and the melt surface oxidation rate can remain high in the air gap. Also, the wetting of the metal surface is easier due to the hotter melt curtain surface temperature. At the lower poly weights, the LDPE melt temperature is more important due to the more rapid cooling of the molten polymer in the air gap, thereby decreasing the oxidation of the molten polymer surface, and impacting the wetting of the metallized film surface. As a point of reference, the high melt temperature is approximately 326C and the low melt temperature is 310C. Figure 12.3-3 was developed with a specific set of line speed and air gap settings. Changes in these values may change the relationship somewhat, but the general principles should remain.

Figure 12.3-4 shows the magnification of a metallized film in a lamination, where the metal layer is clearly fractured. These metal fractures have two main sources. First, all films will display metal fracture if subjected to too high a tension causing the film to stretch past the elongation-to-break of the aluminum layer. Based upon resistance measurements of the aluminum layer in tension, this metal fracture point seems to be reached between 1% and 2% elongation¹⁰. The second main cause of metal fracture can be related to high thermal load on the metal surface, which may cause some local melting of the polymer surface directly under the metal layer¹¹. The high thermal load also may impact tensile properties of the film, making it more susceptible to web tension. Metal fracture, when severe, will show as an iridescent appearance in the lamination.

Figure 12.3-4 Microscopic Picture of Metal Fracture at Approximately 100X Magnification



When troubleshooting metal fracture, the first step should be to determine the actual web tension in kg/m^2 (lb/in^2) to insure that it will not create an excessive extension in the metallized film. Lowering web tensions may require improvements to laminator roll alignment (to eliminate wrinkles), speed, and/or tension control, and improved tension isolation. Improved film flatness can also have an impact, if the laminator is in good condition. Next, the metallized film product design may need attention. In coextruded OPP films, a wide range of surface polymers may be used to control metal adhesion¹² and barrier properties¹³. The melting point and thickness of the metallizing layer will control the sensitivity of the OPP film to the metal fracture. For the more sensitive products, better laminator web handling control will be required to minimize or eliminate the metal fracture. Metal fracture in polyester films is a general indication of too high a tension and indicates the need to mechanically check the laminator.

12.4 Printing

While the majority of metallized films are laminated or overcoated for protection of the metallized layer, in some applications, the metallized surface is printed directly. When surface printing a metallized film, it is usually recommended that it be over-lacquered to improve durability of both the ink and any remaining exposed metal. The over-lacquer improves the web-handling properties of the printed web as most inks are not formulated for surface-printing applications.

Inks are comprised of polymer binders with added pigments and/or dyes and assorted additives. The components may be dissolved in a solvent, which must be evaporated, or in some cases, the inks may be solvent-less such as ultraviolet (UV)-curable inks. Adhesion of the ink to the metallized surface depends on the chemical compatibility of the ink polymer binder with the aluminum oxide on the film surface and the cleanliness of the metal surface (*see Section 6.3*). In order to insure good printing properties, it is generally recommended that the metallized surface be treated and/or primed before printing^{14,15} to remove organic contaminants, which can interfere with the wetting of the primer (as in priming metallized films for lamination or extrusion coating).

12.5 Clear Window Formation

Recently, there is an increased interest in patterned metallized films, especially for the many products that do not require the light barrier, for which metallized films are often chosen. In many applications it is desirable to use the reflective properties of the metal layer for enhancing optical effects, combined with a display or a message window. Other applications for patterned metallized films are for a sophisticated technical application as for capacitors, microwave shielding, and energy dissipation or to carry information for later reading and for radio frequency identification technologies. As in all emerging technologies, it seems that the push for ever smaller features in the patterns is becoming important. Using such miniaturized patterns will permit an array of applications in the future that can hardly be imagined today.

Although new techniques are emerging, there are three primary means of producing patterns of clear (metal-free) areas on metallized film:

1. Demetallization
2. Printing a deposition-resistant material prior to metallization.
3. Metallization through a mask

Methods 2 and 3 are covered in *Section 13*. Only demetallization will be reviewed here. Demetallization is a wet chemical process that is slow and relatively expensive, yet it is the method discussed the most in patents discussing pattern metallization. Demetallization is accomplished in several ways, based upon the selective chemical attack and the removal of the aluminum. One method is by printing an etch-resistant pattern onto a metallized film surface and then etching the exposed aluminum with warm, saturated aqueous sodium hydroxide solution followed by washing and drying the film web. This is a continuous method and is described in several patents to Becket Industries^{16, 17, 18}.

Another demetallization method involves micro etching¹⁹ to a resolution of 0.2 mil using UV inks for the etch-resistant masking. Direct printing of caustic onto the metal surface without the need for a masking layer is reported to achieve a resolution of 100 lines per inch²⁰.

12.6 Thermoforming Metallized Films

Thermoforming metallized film and sheet is possible, but not without the fracture of the metallized layer²¹. Metal fracture happens during thermoforming of metallized polymers, because the polymer sheet or film is too weak to stretch the aluminum layer. The aluminum layer is stronger and brittle, and it cracks as the polymer sheet or film stretches under it during the thermoforming process, and the polymer elongation exceeds the elongation-to-break of the aluminum layer. When you thermoform polymer sheet or film, it transitions from a solid to a rubbery liquid, and the modulus quickly drops below that of the aluminum layer. The metal layer failure is similar to the ceramic layer failure described in *Section 14.8*.

“I have seen aluminum layers appear to stretch when the polymer they are on is pulled, where the film maintains a hint of reflectivity. This usually happens when the polymer

is stretched as a solid. I have a sample in my office of a decorative flower pot cover, which is made from metallized polymer which has been deep-drawn. The polymer part maintains its metallic appearance and luster somewhat (*Figure 12.6-1*) even though it is transparent over much of the pot cover. As the polymer thinned on forming, the material became translucent, but maintained a metallic sheen on the thicker, less-stretched upper edge. The upper edge maintains a metal look and is more reflective. However, under the microscope the metal layer is clearly fractured¹⁶.

“In studies of the impact of elongation of metallized films, barrier properties begin to degrade between 1% and 2 % elongation. Improving the adhesion of the metal with plasma treatment will extend the range to 3% to 4% elongation¹⁰ perhaps, but the metal breaks at low elongation because it is not as ductile as the polymer.

Figure 12.6-1 Thermoformed Decorative Flower Pot Cover Made from Metallized Polymer Sheet



“I have never taken the time to estimate the tensile properties of the aluminum layer deposited from a vacuum (would have to use bulk aluminum foil values) to compare to the polymer properties. Stretching of the aluminum on high-modulus, high-strength sheets could, in principle, deform the aluminum layer more homogeneously, as occurs with different polymer layers in coextruded films, but this is more likely to be under conditions of solid polymer stretching as opposed to that of a thermoforming operation.”

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²¹ Mount III, E.M., Unpublished data.

Section 13: Pattern Metallizing

Introduction

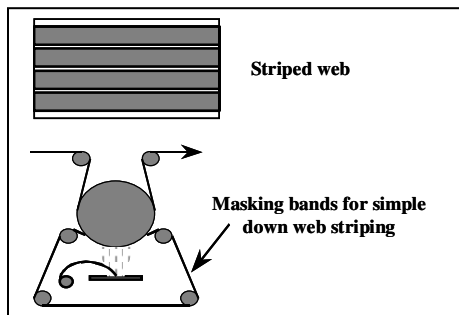
Patterning of coatings has effectively been done by a variety of methods for as long as metallizing has been done. The simplest form of patterning is still used on packaging where overprinting hides some of the metallization. The next form of patterning to be developed was specifically to remove some of the aluminum metal. Two forms of this were developed that required printing either before or after metallization, followed by a post-deposition bath to either release or dissolve the metal coating. Both of these worked but required extra process steps. This led to the development of the first in-vacuum belt-masking process. This produced very simple straight-line de-metallized lines. The desire for more complex patterns finally led to the latest generation of in-vacuum pattern-metallizing metallizers. These use an offset gravure printing process to produce a wide range of patterns. To demonstrate the capability of this process halftone pictures have been produced directly in a metallizer. This looks impressive, but more frequently, simpler patterns are required such as fused susceptors or microwave antenna as described in the text below. In this section we hope to describe the different processes and the benefits each can provide.

There are many applications where it is required that metallization be applied only to selective areas of the substrate. This is referred to as pattern metallization, and it can be carried out in several ways:

13.1 Mask-based Methods

Using a masking band, a metal mask that is in the form of a continuous belt is brought into contact with the substrate across the deposition zone (*Figure 13.1-1*). The belt then leaves the substrate surface and is routed back around and under the sources. As it passes around the sources, the belt is flexed as it passes over idlers and driven rollers, and this can stress the deposited material on the belt so that it breaks off. This keeps the belt clean and the pattern dimensions reasonably accurate. A limitation of this technique is that the pattern cannot be interconnecting as this would not allow the belt to be continuous. Traditionally this has been used for capacitors where the pattern is limited to stripes¹⁻⁴.

Figure 13.1-1 Schematic of System Using a Masking Band



13.2 Printing-based Methods

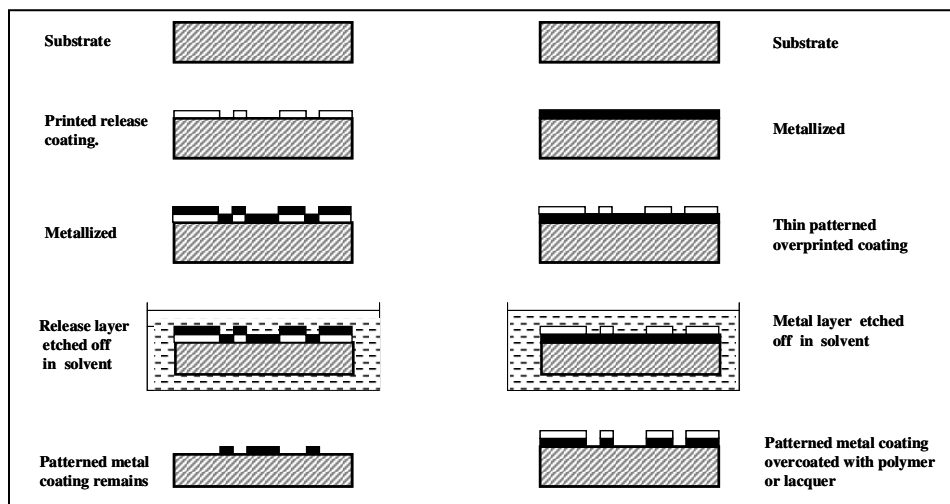
The original method that has two variations is to print a pattern before metallization and then afterward dissolve it printing to lift off the metal over the pattern (*Figure 13.2-1*). The second variation is to print the pattern post-deposition and etch the aluminum away (*see Section 12.5*) between the printed pattern⁵. This can give 50-micron line width patterns at 40m/min speeds.

This process is less popular because it requires additional processing steps, and there is a need for wet chemical processing. Sodium or potassium hydroxide is typically used as the aluminum etchant. The metal side of the foil, rather than being immersed in the etchant, is more often just brought into contact with the surface of a long bath of etchant. This minimizes any liquid uptake by the back surface. At the end of the bath, the web is dried using an air knife and oven to minimize the moisture content.

The final method that is gaining in popularity as it continues to be developed is to print the pattern in the vacuum system⁶. This is done by vaporizing suitable oil and condensing it onto a patterned gravure roll that prints the pattern directly onto the web or via an intermediate roll onto the web (*Figure 13.2-2*). The quality of this process depends

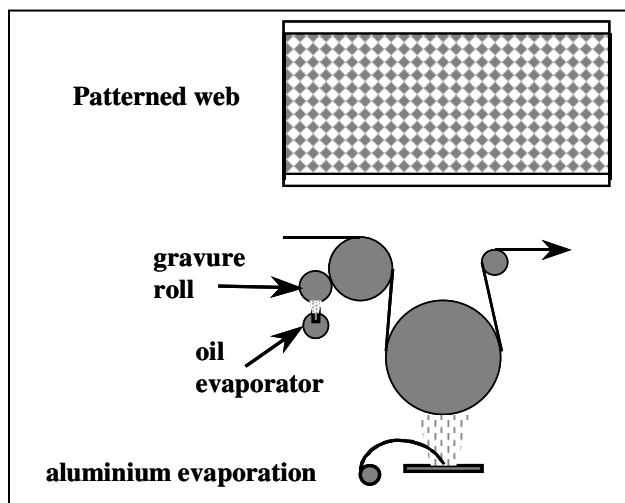
on several factors, including the quantity of oil transferred, the hardness and surface quality of the gravure roll, the balance of unmetallized-to-metallized areas as well as the coating thickness.

Figure 13.2-1 Schematic of the Two Methods of Chemical Etching to Pattern-metallize Coatings



The way this process works is, because the sticking coefficient of aluminum vapor on the oil is extremely low, the aluminum that lands on the oil does not stick and is re-evaporated,. As it re-evaporates the aluminum loses some energy to the oil. The oil is also subjected to the radiant heat from the sources and along with the absorbed energy from the depositing aluminum, the oil is evaporated. The aim is to remove all the oil by the re-evaporated aluminum alone, leaving behind a dry polymer. The reality is always slightly different. The problem with this technique is that as the oil evaporates, the size of the oil drop can shrink so the edge sharpness of the pattern can be lost. The quantity of oil is the critical parameter. The oil is not on the surface as a flat parallel layer but has a curvature so as the oil is evaporated, the width of the line is reduced and the line edges may not be as sharp as with an in-contact mask, particularly if almost all the oil has been removed. If, however, oil is left on the surface, it will transfer via the first front surface roll to other parts of the aluminum deposition, contaminating the surface.

Figure 13.2-2 Schematic of the Oil Printing Pattern-metallization Technique

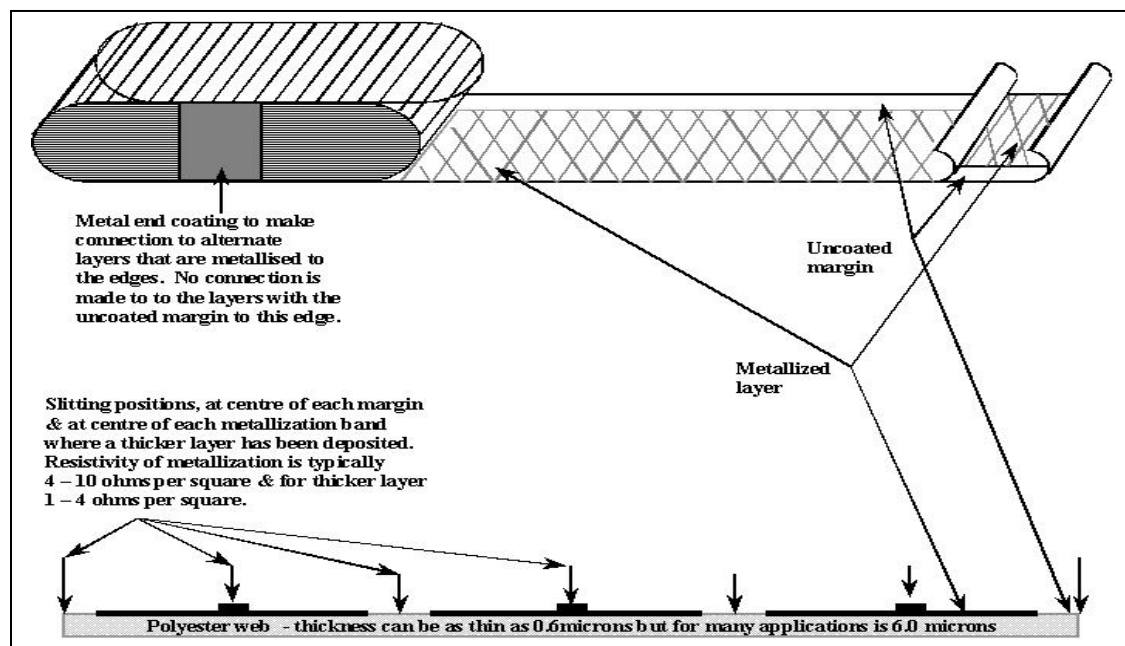


Thus the management of the quantity of oil deposited must be controlled quite precisely. One optional control system is to heat the oil, vaporize it onto the gravure roll and use a doctor blade to limit the excess on the gravure roll. This gives a higher precision to the level with which the oil fills the gravure cells. If there any excess oil is present after the deposition zone, sometimes a proprietary process is designed to make sure that all the oil is removed or stabilized.

Most people use diffusion pump oil for pattern metallization since it has already been developed for vacuum use and at the lower temperature of the web substrate it has a low vapor pressure. There are some grades of oil that have been passed as food-contact-compatible, and they are used in order to make the product acceptable to the FDA for food packaging.

Pattern-metallized films can be used for a variety of applications. The earliest use of patterns was using the belt masking technique where stripes were produced for the manufacture of capacitors^{7,8} (Figure 13.2-3).

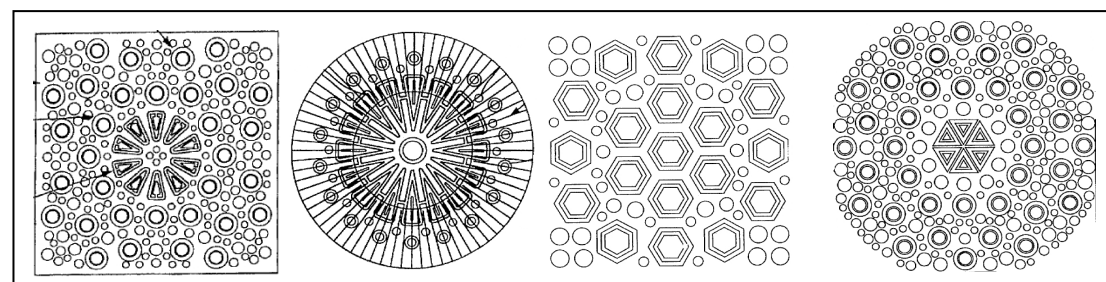
Figure 13.2-3 Schematic of Metallized Film That Has a Combination of High Deposition Thickness and Demetallized Stripes That Is Precision-slit to Wind Capacitors



Demetallized holograms have had the full thickness bright metallic layer deposited, but then some of it is removed in a particular pattern⁹. Using either the oil pattern printing or the etching methods (see Section 12.5), any shape of holographic design outline can be complemented by an etched shape into the metal coating.

In the area of food packaging, pattern metallizing can be utilized in two ways for microwave heating of foods (Figure 13.2-4). Microwaves can be used to heat foods directly, where the microwaves couple into the water molecules of the food and vibrate them, generating heat. However, different foods contain different amounts of water and, therefore, heat at different rates. Thus, in instant whole meals where a variety of foods are to be heated at the same time, it is useful to attenuate the microwave power to some areas to allow all the food to arrive at the preferred cooked state at the same time. This can require some complex patterns as shown in the accompanying figure.

Figure 13.2-4 Examples of Pattern Metallization as Used in Microwave Susceptor Packaging¹⁰



The second use for microwave packaging is for the simple heating surfaces where a thin metallic coating is used. One of the difficulties is to limit the temperature of the device. Use is made of the different thermal expansion coefficients between the web and the metal coating. Since the polymer expands faster than the metal, it puts the metal coating into tension. If the differential expansion between the web and the metal coating becomes too large, the metal coating cracks and in doing so, loses conductivity limiting the microwave heating since microwaves are no longer absorbed by the coating as effectively. One of the problems of the early coatings for this application was that the cracking could be variable and so there would be some un-cracked areas that overheated and other areas that failed early and failed to achieve the desired temperature. To correct this problem, de-metallization of a pattern was incorporated into the design^{11, 12} that concentrated the stress so that the constrictions in the metal coating would act as a fuse. In this way the metal would crack across these stress concentrators in a very uniform manner so the film would reproducibly reach a specific temperature, and then the fuses would crack and the temperature would be limited.

Section 13.3 In-register, In-vacuum Pattern Metallization

There are a number of applications where the pattern metallization needs to be done such that the pattern produced is positioned relative to an earlier coating this alignment of the deposited pattern is known as in-register pattern metallization.

The precision required varies with the application. Packaging, where the registration is desired for the aesthetics of the combined printed and metallized design, tends to not need the same high precision as some electronics applications. Currently the in-register precision is of the order of $\pm 0.25\text{mm}$ and is suitable for packaging applications as well as for die-cut labeling. With the understanding of the technical difficulties of how to control the accuracy of pattern metallizing, the expectation is that this learning will enable the development of systems with higher precision.

To obtain registration requires any pattern to be correctly positioned both transversely as well as in the machine direction. Alignment in the transverse direction can be achieved by the fine positioning of the unwind roll. This can be done by having the unwind roll independent of the rest of the winding system and mounted on a motorized frame thus allowing small changes to the transverse position during winding under vacuum.

In the machine direction, the problem is more problematic. Winding the substrate puts it under tension and can cause it to stretch. This is the same for each coating process. If the length of the substrate is measured without any tension applied, it will be different than when tension is applied. This change of length is likely to be different on each coating system as different tensions may be present on each system. To help in positioning, registration marks are applied to the substrate to provide a reference distance and position that each coating has to be referenced to. So in the in-vacuum, in-register system there is a monitoring system that measures the position of the registration marks and then corrects the substrate position accordingly. In some early versions this was done by changing the tension of the web, but this was found to distort the substrate too much. Current versions¹³ use very fine changes to the winding speed of the print-transfer roll to reposition the pattern and match the print length to the registration mark reference length. This is an improvement over the earlier attempts, but is still too coarse for the fine line circuits required for many electronic products.

The key to getting higher precision is to have a substrate that is completely stable under vacuum deposition winding conditions, and where application of the pattern does not cause any deformation or stretch. This also requires that none of the other processes cause any substrate distortion either.

This is a new opportunity for pattern metallization and is likely to grow as more applications are developed and as the precision improves.

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Section 14: Non-aluminum Deposition Materials and Techniques

Introduction

The bulk of metallizing machines use resistance-heated evaporation boats for deposition. Occasionally other materials are required: Some can be evaporated by the same boats, others need a different design of replacement boat. Beyond this limited range of materials, it becomes necessary to use one of a range of different deposition techniques.

Capacitor metallizers are generally narrower width machines because of the thinner gauge of film used. Typically zinc will be one of the materials deposited and this lower temperature metal, because of its high vapor pressure, is deposited from what is referred to as a zinc boiler, or linear or slot source. This type of source is described as well as the development of the jet vapor source that extends this technology. The jet vapor source is only at the research and development, or pilot machine stage, but because of its high material efficiency, it will become a more important source in the future.

The next most-used source for metallizing that also is capable of depositing transparent barrier coatings is electron beam (E-beam) heated. These can be small E-beam sources, which are arranged across the web width in the same way as resistance-heated sources, or can be very big E-beam sources, which can heat crucibles up to 1.25m in width. Typically these machines would be up to 2.5m wide and have a pair of these large guns heating the sources. These guns have the flexibility to work with most of the materials in the periodic table.

Where the coating thickness, or stoichiometry is the most critical parameter, evaporation techniques are too difficult to control with enough precision. Magnetron sputtering then becomes the deposition source of choice. These sources have a great many options including type of power (AC or DC), magnetic field strength, and balance, planar or rotatable, as well as a variety of other design choices. This section brief outlines the most important design and operational considerations.

The final source that is considered is that of chemical vapor deposition (CVD). This source is almost exclusively used in the deposition of transparent barrier coatings but is capable of depositing many other materials, providing a suitable chemical precursor is available.

To help put some of these sources into some kind of perspective there follows a section that compares the merits of the different sources. It is also helpful to be aware of a number of the hazards that can be associated with vacuum deposition, some of which are exclusive to particular sources, and this is given a brief overview.

Finally there is a section on stress in films. The different sources deposit the materials with different deposition rates and energies. This will lead to different coatings because of the differences in nucleation site density and packing density of the growing coating. This is important, as it will be seen as differences in adhesion and stress found in the coatings, which, in turn, will affect the ultimate performance of the coatings.

14.1 Resistance-heated Sources

The standard evaporation source for evaporating aluminum is made by passing a current through an intermetallic boat, thus, heating it up and simultaneously feeding a wire of aluminum against the hot surface of the boat. As it approaches the boat surface, the aluminum melts and forms a molten pool that steadily evaporates. These boats have a limited lifetime that can depend on the quality of the materials and process used to make the boat as well as the process conditions. Coarse-grained materials poorly mixed can lead to very short lifetimes of ~5 hours whereas higher quality materials (finer grain size and well mixed) can last well over 20 hours.

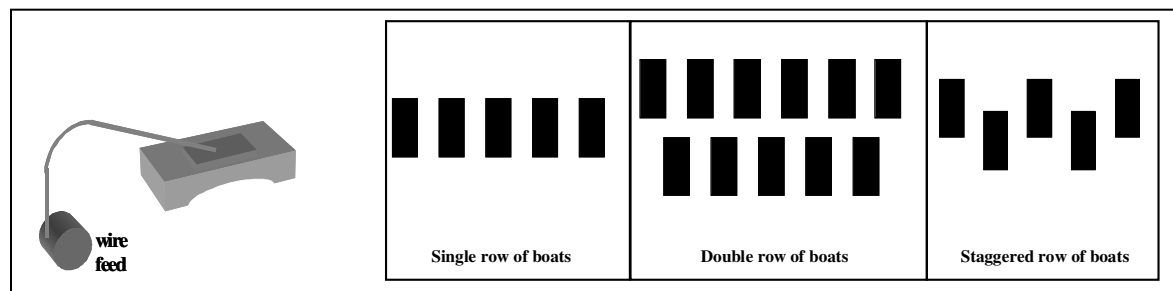
The process also affects the lifetime with the boat temperature being a key factor. The aluminum evaporation rate is dependent upon the boat temperature, so to increase productivity, boat temperature is often increased to allow a higher aluminum wire feed and higher evaporation rate. Running the boats hard at the higher temperature reduces their lifetime significantly (*see Figure 5.4-1*).

Molten aluminum is highly corrosive so a boat of either a two- or three-phase intermetallic mixture is used to offer the right combination of resistivity and corrosion resistance¹⁻⁸. The two-phase mixture consists of boron nitride and titanium diboride. The three-phase mixture also contains aluminum nitride. Titanium diboride gives a low electrical resistivity and is easily wet by the aluminum. Boron nitride gives a high electrical resistivity and also possesses good thermal shock resistance and ease of machining. Aluminum nitride is added for its thermal conductivity and electrical resistance performance.

It is important to feed the wire into the boats at the right place and angle. Different boats may require the wire to be fed to a different position on the boat surface. Ideally, the approach angle is shallow enough to allow the aluminum to melt as it approaches the molten pool. If the angle is too steep, the wire may penetrate the molten pool and the wire can erode the boat surface, further shortening the boat lifetime.

The most-used arrangement is to have a row of boats across the web width with overlapping vapor clouds to provide coating uniformity. Variations of this can be to have a twin row of boats either as two separate rows or as a staggered arrangement. Also boats can vary in width and separation. The choice of the boat size and arrangement affects the source-to-substrate distance and the preferred deposition drum diameter.

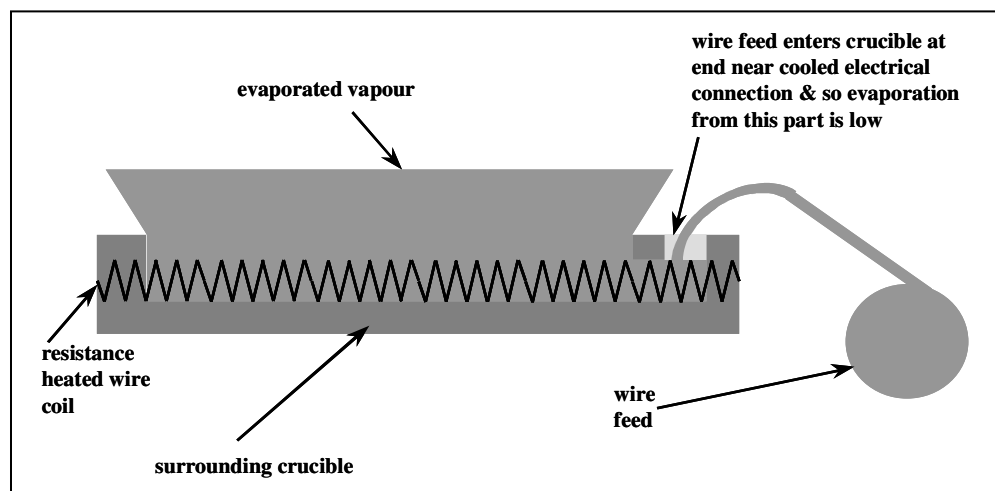
Figure 14.1-1 Schematic of a Resistance-heated Boat Source and Typical Configurations



It is possible to use resistance-heated boats in different configurations. They have been produced almost 2m in length to allow a single boat to be used across the full width of the web. This type of boat was developed to be very dense and have a low aluminum corrosion rate and typically to last for approximately 120 hours⁹.

Resistance-heated sources can also be of other designs. Small laboratory vacuum systems that are used for flash evaporating aluminum onto discrete substrates use a tungsten wire evaporation source. This is usually in the form of a triple strand wire that is wound into a coil where the aluminum can be placed in the coil.

Figure 14.1-2 Schematic of a Wire Coil Resistance-heated Boat Source

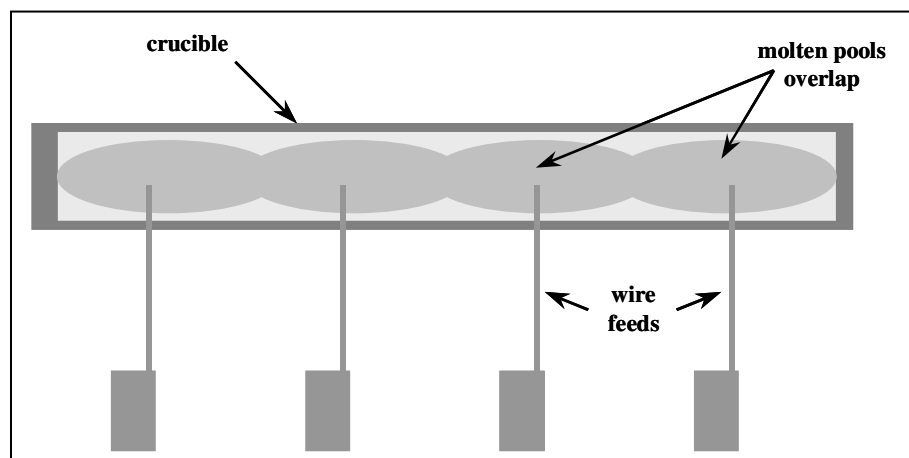


On passing a current through the wire, the tungsten heats up, and the aluminum softens and drapes across the tungsten wire. As the temperature increases, the aluminum melts and wets the tungsten wire, and with further heating, the aluminum evaporates off the wire. This same principle also is used for web coating. In this instance, the coil of resistance wire is positioned in a crucible so that it can heat and evaporate a larger quantity of material¹⁰.¹¹. The wire is fed onto one end of the coiled wire, and the molten metal wicks down the coil and floods out into the crucible. This type of resistance-heated source has been used as a replacement source to the more traditional intermetallic boats where a high deposition rate for material such as copper is required.

Some full-width resistance- heated sources are used for the deposition of metals for capacitors. One option for depositing silver is to have a full width boat with a series of wire feeds to make sure that the molten pool extends across the full width without needing to have a deep molten pool. This operates well because of the characteristics of the silver metal.

Zinc, because of the combination of low melting point and low operating pressure, deposits very easily. Zinc can be a problem material as it does not necessarily have a high sticking coefficient, and it has a tendency to contaminate the whole vacuum system. Zinc metal often is found as a residue in the pumping lines and pumps. To help control the zinc deposition, the source-to-substrate distance usually is made very small, and the source is enclosed, allowing the zinc to evaporate through a narrow slot. This source is referred to as a slot source (*see Section 14.2*). As the zinc is easily and quickly heated to reach the required evaporation rate, this source usually has no feed system but instead has the full inventory of zinc metal contained within the crucible. The capacity of the crucible may be sufficient to deposit several rolls of film before it needs to be refilled. This same source can be used for zinc sulphide too.

Figure 14.1-3 Schematic Diagram of a Full-width Resistance Source with Several Wire Feeds



14.2 Slot Sources and Jet Vapor Sources

There can be a number of advantages to using slot sources for depositing coatings. Any non-uniformity of the evaporation rate of the material from the crucible surface can be evened out by restricting the size of the source exit slot, thus increasing the vapor pressure within the volume above the crucible. This restriction causes the vapor cloud to redistribute more evenly. The enclosure also may be arranged to reduce the radiant heat load onto the web.

Once the source has been enclosed and provided with an exit slot, it becomes possible to orient the exit slot in any direction. Thus this deposition source no longer has to be sited directly beneath the deposition drum but may be sited anywhere around the sides of the drum. Using this arrangement, it then becomes possible to distribute several sources around the deposition drum and spread the heat load. This enables thicker coatings to be deposited at faster speeds than from a single source depositing upward from beneath the deposition drum. This technique has been used effectively for depositing magnesium for the manufacture of pyrotechnics¹². As the deposition can be from any angle, it enables several sources to be distributed around a deposition drum. This extends the deposition zone around the deposition drum, making more use of the substrate cooling capacity than is possible from a single evaporation source placed beneath the deposition drum.

This principle of supplying a flux of material or materials into a volume where the vapor pressure may be high and may be mixed, and then controlling the exiting material has led to an increasing number of applications. These slot sources have been used for depositing lithium for battery applications, polymers as planarizing layers¹³ or overcoat protective layer, organic materials for organic light-emitting devices^{14,15} and now they are being considered for many other metals and compounds.

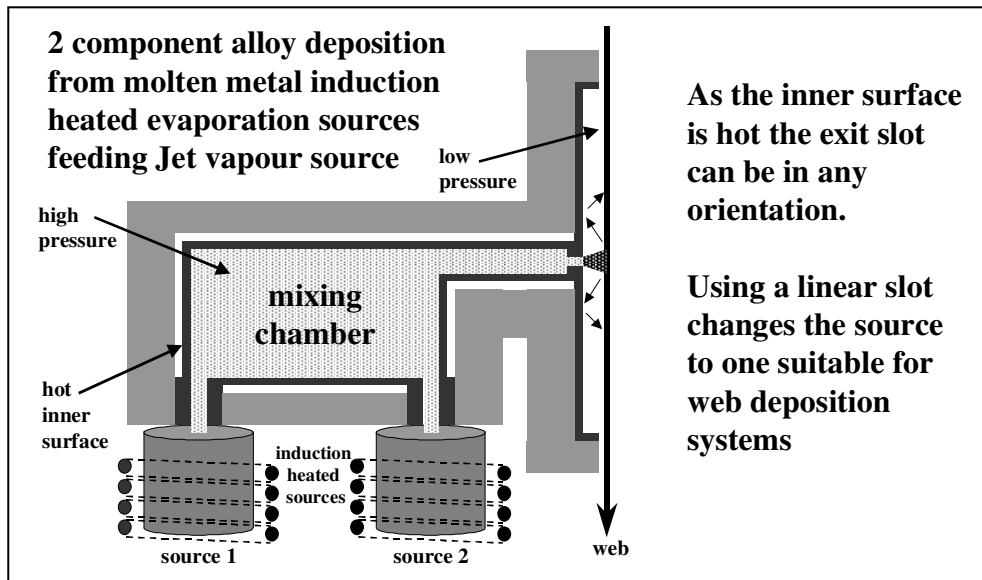
Some of these sources have been renamed jet vapor deposition sources¹⁶⁻¹⁸. As the vapor pressure within the source chamber volume can be several orders of magnitude higher than the vacuum vessel chamber pressure, the exiting material flux can exit at very high speed in the form of a jet. The higher the pressure difference, the higher the speed and the narrower the jet. As the vapor jet undergoes adiabatic acceleration, the material temperature drops and the

kinetic energy increases. The enclosed source reduces reactive gas contamination and the increased kinetic energy reduces any gas entrapment in the growing coating, thereby increasing the coating density.

There are many variations on this type of source. The material supply can be from a gaseous material supply into a plasma reactor such as a microwave chamber where the precursor gas is broken down into the desired coating material. More than one microwave chamber can be used to supply separate materials to make alloys. Similarly, individual physical vapor deposition sources, such as induction-heated crucibles, can be used to produce the vapor that will be mixed in the source chamber volume so the final exiting vapor will be of the desired alloy.

All of these sources use a heated surface for the inner skin of the chamber that encloses the sources, the mixing area, and the jet nozzle. This hot surface prevents any deposition onto the surfaces, keeping them clean.

Figure 14.2-1 Schematic of a Dual-material Jet Vapor Deposition Source



Another variation is that a flowing gas can have a vapor entrained in it to provide the final vapor source. The vapor that is to be entrained can be from an evaporation source, a liquid precursor, a subliming solid or a CVD plasma source. The flowing gas tends to be an inert gas but may, in some circumstances, have a reactive gas included. This type of source is similar to the cascade arc source¹⁹ but it can operate at a lower temperature.

Other variations use a gas supply to divert a vapor cloud toward the substrate. Simple versions of this are to evaporate material upward from a standard source and to have a gas inlet to one side to push the evaporating cloud sideways across to the substrate. Although this is a simple design, it is not a particularly effective technique for web coating. The enclosed jet vapor deposition, or slot, sources are much more effective in providing very high material efficiency and spreading the heat load around more of the circumference of the cooled deposition drum.

14.3 Electron-beam Deposition

An electron beam (E-beam) gun operates by powering a system to generate electrons that are then accelerated, focused, and directed by steering magnets toward a crucible. This high-intensity beam of electrons impacts the material, heating it to the point of evaporation. A high-density E-beam is capable of heating virtually any material.

The electron source can be of different types, *e.g.*, a simple wire filament or a filamentless type. The filament type is simply a wire that has a current passed through it to heat it to the temperature where it emits a high number of electrons. The filamentless version may be an indirectly heated material that also gives off a high number of electrons upon heating, but it is designed to be much less fragile than the filament source and less susceptible to oxygen contamination.

If the filament and/or the surrounding surfaces are negatively biased, then by using a suitable anode the electrons can be extracted and accelerated out of an exit orifice. Once given a direction these electrons can be focussed and steered toward a crucible. The steering can be done using permanent magnets, electromagnets or a combination of both²⁰.

Different materials require different process conditions. Not all materials melt, some sublime (pass from the solid to vapor without forming a liquid). Also the material source may be in different forms ranging from a powder to a solid block. An E-beam that is focused and high powered would be a problem for evaporating a powder as it would be just as likely to throw the powder out of the crucible as melt it. Similarly, with a porous material or one with gas inclusions, the rapid heating of the gas can cause the material to explode. Thus, it is essential to be able to de-focus the E-beam and/or to be able to scan it at low power across the whole crucible surface, enabling the material temperature to be raised smoothly and gradually for these more difficult materials. If the material has a high thermal conductivity, and it can be bought as a solid block, it is possible to use a focused E-beam at high power directly onto the material to heat it up very rapidly.

The crucible can be of two different types: water-cooled or un-cooled. With the un-cooled crucible, the evaporant and the crucible are both heated. The crucible has to retain its strength at the evaporation temperature used for the evaporant. Where a water-cooled crucible is used, the water-cooling can be applied so that the evaporant is cooled from the outside and heated from the E-beam at the middle and top of the crucible. If the evaporant forms a molten pool, there will be a change from the liquid to solid state within the evaporant. So in effect, the evaporant forms its own crucible.

The choice of crucible needs some care, as crucibles are often expensive items. If there is a large difference in the coefficient of thermal expansion of the crucible and the evaporant, there can be a problem with the crucible cracking. This is particularly true if the evaporant forms a molten pool and wets the crucible. As the evaporant solidifies, the differential expansion can cause the ceramic crucible to crack and this may only be seen the next time the crucible is heated when the molten evaporant leaks through the crack. Another option is to use a disposable crucible liner that keeps the molten pool from wetting the crucible. The pool only wets the liner, which is flexible and can survive the thermal cycling better than a rigid crucible.

The power capacity and type of E-beam gun can vary from 1kW to 250kW. The smaller E-beam guns are often of a design that uses a tungsten filament under one end with magnets to bend the beam through 270 degrees into the circular water-cooled crucible (*Figure 14.5*). These smaller E-beam guns can be used in the same way as resistance-heated evaporation sources with several guns being used across the web width (*Figure 14.3-2*) where the integrated deposition can be reasonably uniform. The larger E-beam guns tend to be used for the large web-coating machines. These often have the E-beam gun as a separate unit that bends the electron beam less, but has additional scanning capability, such that the E-beam can heat a very wide crucible of up to 1.25m in width (*Figure 14.3-3*).

Figure 14.3-1 Schematic of a 270-degree E-beam Gun

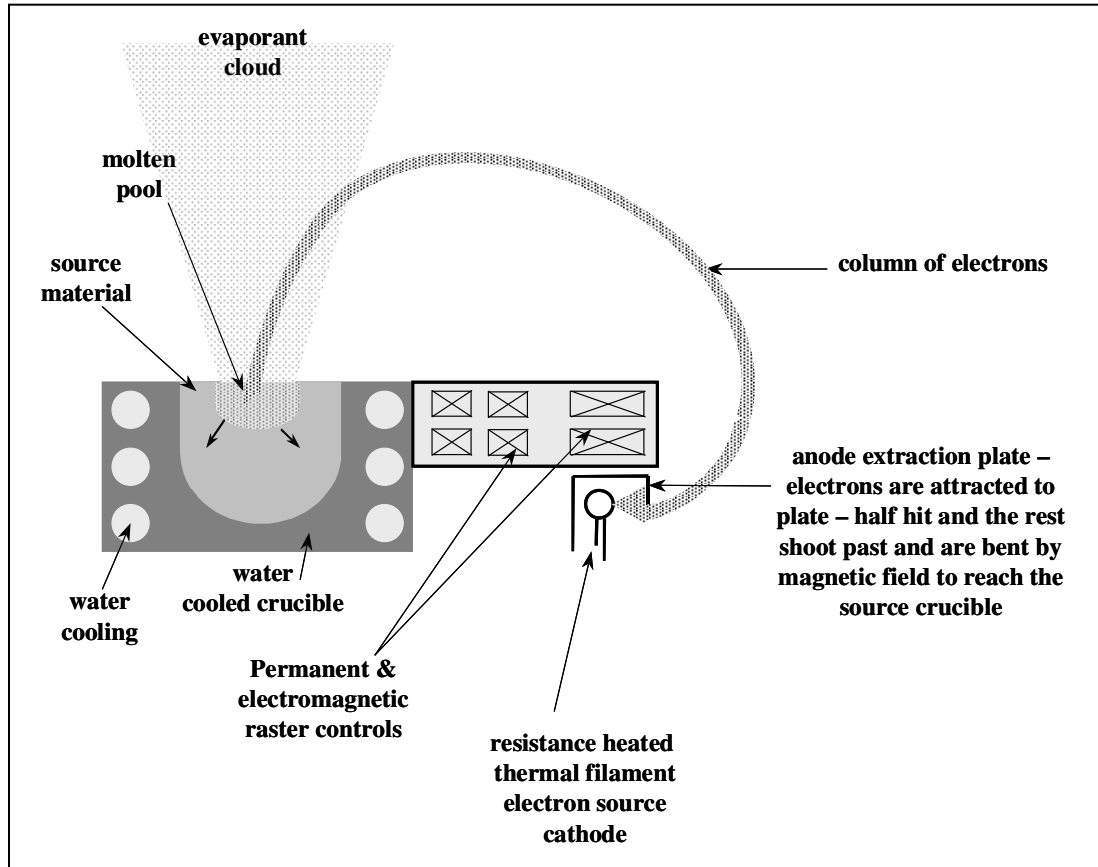


Figure 14.3-2 Schematic of Multiple 270-degree E-Beam Sources for Roll Coating

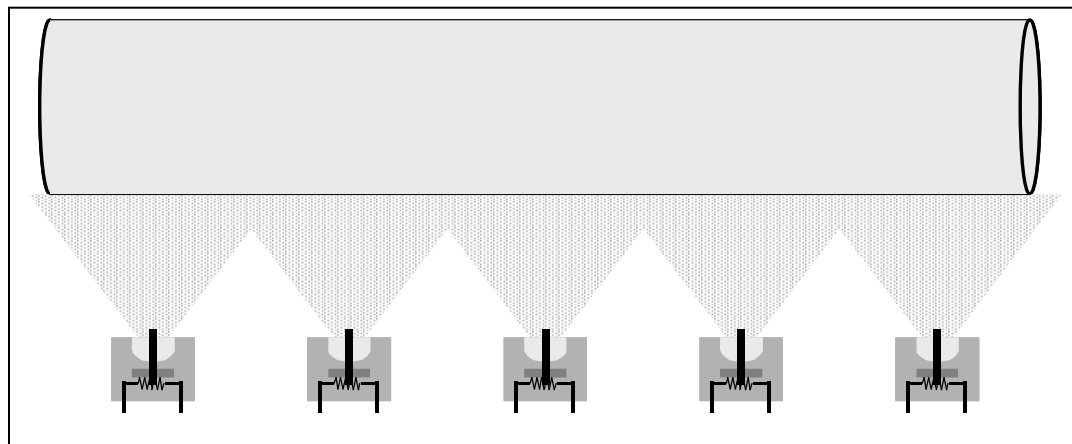
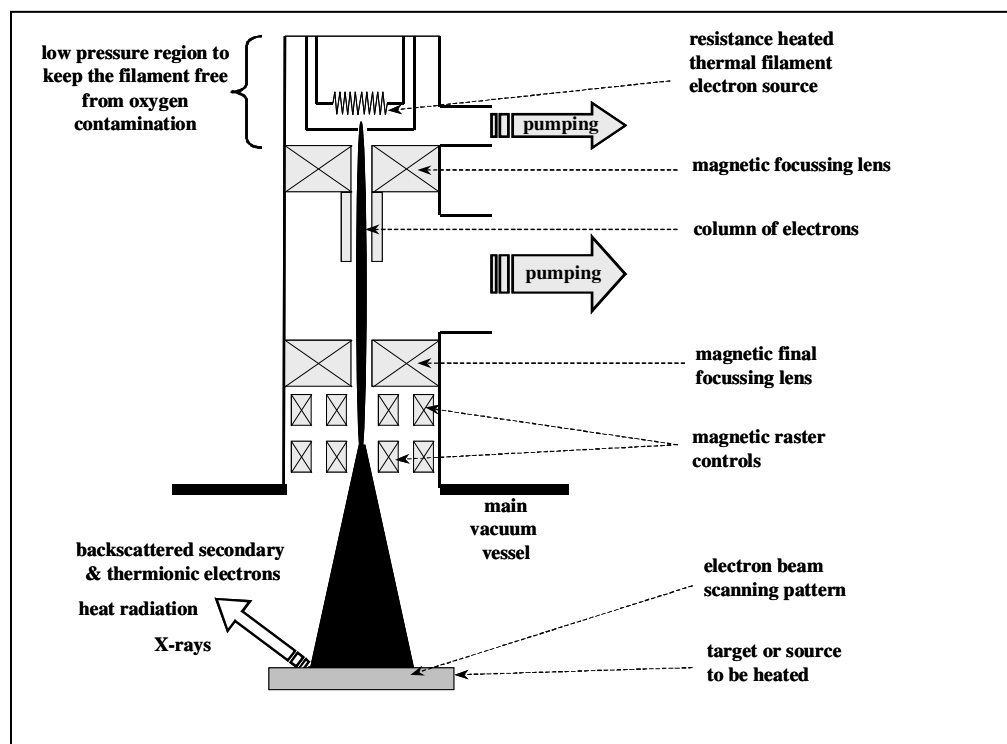


Figure 14.3-3 Schematic of a Large Sweeping E-beam Gun Capable of Heating a 1m-wide Single Crucible as Used For Web Coating.



Another consideration for crucible choice is where multiple materials or alloys are desired. Crucibles can be single or split in two (even the very wide ones). Since the E-beam can be steered, it can be moved from one crucible to the second crucible, keeping both materials evaporating, thus creating alloys. The residence time of the beam on each material can be varied so the alloy composition can be controlled.

Crucibles are often large enough that sufficient material is contained within the crucible to coat a complete roll of substrate material. If the crucible cannot contain sufficient material, it may be possible to replenish the crucible using wire, pellet or powder feed systems.

The evaporant produced by E-beam guns has more energy than evaporant produced by resistance-heated boat evaporation sources. The reason for this is that the evaporant passes through the incoming E-beam and in doing so, some of the evaporant is ionized. This slightly more energetic evaporant causes the adhesion to be slightly better than an equivalent resistance-heated source evaporant, but offers less adhesion than if the same material would have been sputtered with even higher energy.

The E-beam evaporation process produces many secondary electrons. In large E-beam gun systems these electrons need to be captured using a secondary electron trap in order to prevent substrate damage. If these secondary electrons are not efficiently captured, they can be implanted into polymer substrates where they form electrets. These electrets retain the charge, and when the material is re-wound, the polymer may block becoming impossible to unwind.

Where the coatings are required to be dense, (such as for optical coatings where porous coatings can absorb moisture and change the optical properties), it may be preferable to use an additional deposition plasma. The additional plasma bombards the growing coating, removing any loosely bonded material, as well as disrupting the more usual columnar growth, resulting in a more equiaxial crystal structure in the coating. The additional plasma can be produced in a variety of ways such as microwave, radio frequency, hollow cathode, arc or triode^{21, 22}.

Plasma cleaning of the substrate has exactly the same benefits to E-beam-deposited coatings as it does for resistance-heated source evaporated coatings. The large sweeping E-beam source is equally useful as a cleaning tool for metal webs or foils²³. The E-beam will heat the foil either driving out any oil or residues from within the grain boundaries or volatilizing or carbonizing these organic contaminants. One possible danger from the E-beam cleaning of the metal is that if the metal foil has been specially tempered, the heating can change the level of temper. Often, foil is annealed several times as part of the manufacturing process. The foil starts life as a billet. Rolling the billet to thin it work-hardens the metal, which must be annealed to enable further rolling. The final part of the processing is frequently designed to bring the foil to a particular hardness. However, when the E-beam raises the temperature of the foil, coupled to the remaining level of work-hardening, the foil can reach a temperature where recrystallization is triggered, which, in effect, anneals the foil and causes the temper to be lost. Thus, although E-beams can be used to clean the foil, care needs to be taken to limit the process to keep the foil below any critical temperature.

14.4 Magnetron Sputtering

In magnetron sputtering, a solid material is bombarded by ionized gas atoms (a plasma) with enough energy that the impact of the atoms with the surface ejects atoms of the solid material that can then be collected on the passing web substrate creating the coating. Initially sputtering was done by direct current (DC) sources that operated at around 1-3kV and had low deposition rates. The deposition rate increased enormously with the advent of magnetron sputtering.

The key to sputtering is to produce high-density plasma. If a potential is applied to a gas, it can be ionized, and this is seen as the plasma. As electrons traverse the plasma, they may hit a gas atom as an ionizing collision producing an ion and electron. The density of the plasma depends on the number of ionizing collisions. The efficiency of ionization depends upon the gas pressure, the size of the gas atoms or molecules, and the distance between collisions (the mean-free-path) compared to distance the electrons travel. Since the plasma contains both ions and electrons, it is conducting with the current being determined by the density of the ions and electrons. Unfortunately, the electrons are very small and light and can travel very quickly to the anode and be lost from the plasma.

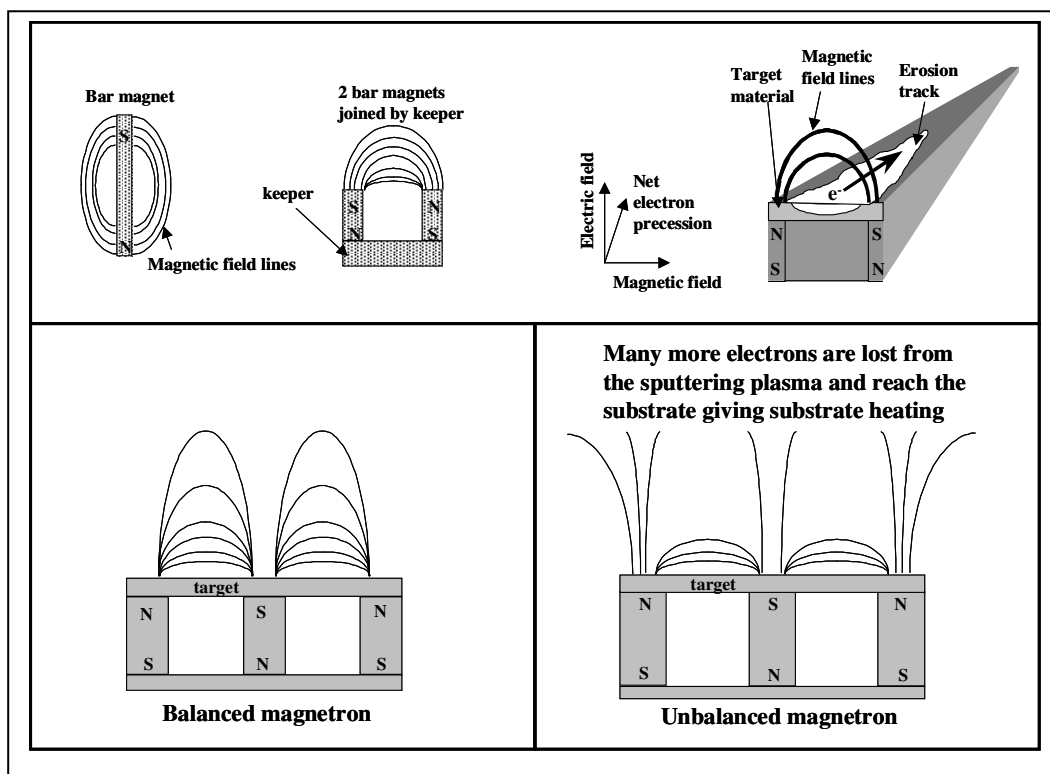
To improve the plasma density, a separate electron source can inject additional electrons into the plasma. This is known as a triode sputtering system.

The development of magnetron sputtering produced a more elegant solution to producing a high-density plasma and by increasing the target bombardment, and hence, the sputtering and deposition rate. By placing a suitable magnet behind the target material, a crossed magnetic and electric field is produced. This results in the electrons traveling, instead of in a straight line, in a spiraling motion along the field lines²⁴. This spiraling motion extends the distance that each electron travels before it reaches the anode such that each electron is likely to produce several ionizing collisions. The increased ionization efficiency increases the density of ions and electrons so the plasma can conduct a much higher current at a much lower voltage. Magnetron sputtering cathodes usually run at 500V –750V with much higher currents than could be obtained on the simple DC sputtering systems that operated at the 1kV- 3kV.

The magnetic confinement produces some other effects. The bombardment of the target is greatest at the point where the magnetic field is parallel to the target surface. Since the target surface is eroded by the ion bombardment, the rate of erosion is different across the target surface depending on the magnetic field orientation. The erosion profile of the target can be varied by a change in the strength, position, and number of poles of the magnets. This does become a balance between the complexity of the magnetic field design and the material erosion efficiency.

The magnetic field, as a method of control of the plasma electrons, is critically important in another way. The magnet is usually assembled from several smaller magnets. There are now many types of magnets that can be used, such as ferrites, samarium cobalt or neodymium iron. Using these different strength magnetic materials, it is possible to build a magnetron in many different ways, producing different field patterns.

Figure 14.4-1 Schematics of the Magnetic Arrangement Involved in the Design of Magnetron Sputtering Sources



Two main types of magnetrons are referred to: unbalanced and balanced²⁵⁻²⁷. The balanced magnetron requires a matched field between the magnetic poles. In this way the maximum amount of electrons are captured by the magnetron. In the unbalanced magnetron, there are a series of field lines that lead the electrons away from the cathode toward the substrate. These electrons will bombard the substrate and growing film. This can be an advantage for some engineering coatings onto metal substrates but for polymeric substrates this is usually associated with substrate damage. Therefore, most magnetron designs for sputtering onto polymer webs aim for a balanced magnetron design.

The ion bombardment of the target material generates heat. If there is no cooling, the target material will keep increasing in temperature until it melts. Thus, magnetron-sputtering sources are cooled. The quality of the cooling is important since the magnetic performance can be affected by temperature. If the water-cooling is inadequate, as the power is increased, the temperature will rise and the field strength of some magnets falls. Lower field strength results in the loss of electrons and reduces the current-carrying capacity of the plasma. Therefore, increasing the power, in this case, will lead to a fall in the sputtering rate. If water-cooling to the same magnetron were improved, the magnetic performance would be maintained, and the increased current would raise the sputtering rate.

The sputtering rate is almost directly proportional to the applied power, so doubling the power will produce an almost doubling of the deposition rate. This characteristic makes sputtering one of the most predictable of the deposition processes. That is, for the same operating pressure, the same voltage, and the same current, the coating thickness can be expected to be the same. Thus, for many processes, the pressure and power are the only controls used to reproducibly produce coatings.

There is a slight variation to this with time. As the target is eroded, the surface area increases, and the direction of the sputtered material changes slightly. This does mean that over the lifetime of the target there will be a time-related drift in operating characteristics. Once these have been established, the drift can be progressively compensated for and the production errors further reduced. Operating the magnetron in constant current mode will mean, that as the target erodes, the deposition rate will progressively decline. Operating the process in constant power mode will compensate for the changing surface area, and the deposition rate will remain almost constant throughout the target lifetime.

The general layout of a sputtering source in a vacuum system is to have the sputtering source as the cathode and then to use the whole of the rest of the vacuum system, at earth potential, as the anode. In reality this can be a source of problems. Often systems are not specifically and independently earthed, but are assumed earthed through many of the attached electrical items. If they have been specifically earthed, it may be to the building earth, and this too may not be as good as assumed. Inside the sputtering chamber, because of the mobility of the gas, the plasma will form between the cathode and the nearest suitable earth, which may be a nearby shield. Over time this “earth” may accumulate a thin coating, which may easily become oxidized as the oxygen arrival rate may well be of the same order of magnitude, or higher, than the metal arrival rate. This oxidized layer may be non-conducting, and if so, the anode will become ineffective. The plasma will then use the next nearest earth to become the anode. This change in anode position can change the shape and performance of the plasma.

To overcome the problem of the disappearing anode, different strategies have been used. One has been to heat the anode to prevent any deposition on it and in so doing keep the anode active. Usually, this causes the anode to run at several hundred degrees Celsius and to appear red hot. To reduce this operating temperature but still eliminate the deposition, another method is to use the anode as the gas inlet because the gas flow reduces the deposition.

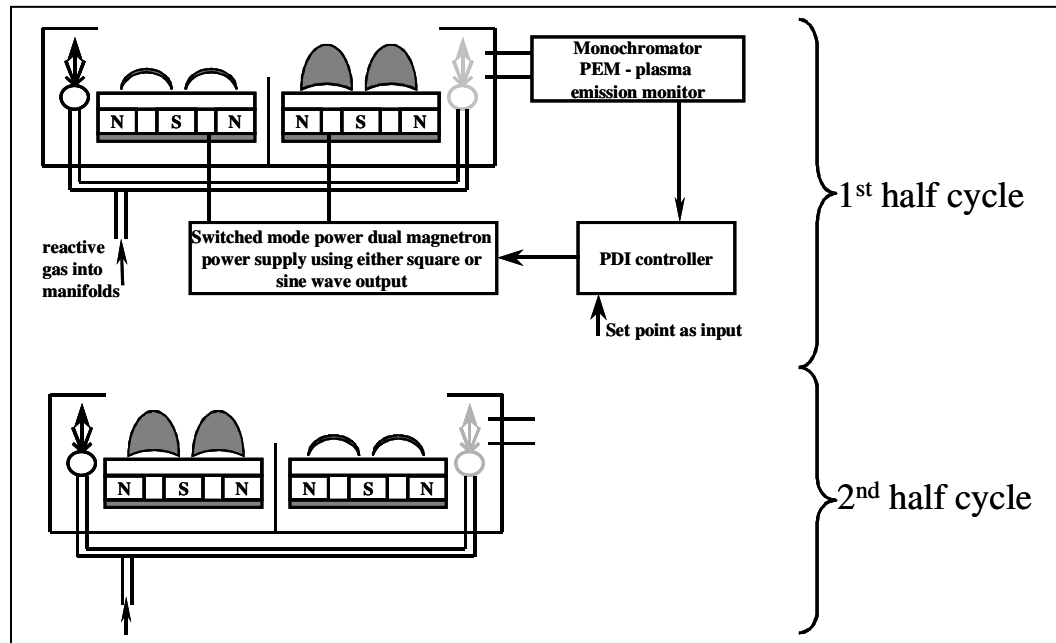
An increasingly popular method of guaranteeing a suitable anode is to use a dual magnetron source (*Figure 14.4-2*). A pair of magnetrons is placed adjacent to each other, powered by a medium frequency power supply. On one half-cycle, one magnetron is a cathode and the other the anode. For the second half-cycle these roles are reversed. During each half-cycle when the magnetron is the cathode, the target surface is eroded and cleaned of any oxide, and thus the surface is always kept conducting. This process has the additional advantage of keeping the surface clean of oxide, which reduces the chances of the target arcing. Where the sputtering rate is low at the edges of the cathode, it is possible for some oxide, as well as some backscattered material (metal atoms sputtered from the surface that collide with gas atoms, which change the direction so the metal falls back to the cathode surface) to accumulate around these areas. This powdery material can cause a localized increase of the electrical field that may concentrate the energy and initiate an arc. If a DC power supply were used, this partially oxidized, powdery surface could accumulate charge, making this area prone to initiating arcs. Also, the medium-frequency power supply means that this area cannot accumulate charge, as every half-cycle this area would be actively discharged.

These dual magnetron sources, operated using an alternating power supply, become particularly useful where a reactive deposition process is used to deposit a material different from the target material. Often this involves depositing an oxide material from a metal target and the presence of the extra oxygen necessary to convert the metal to the oxide, and increases the potential problems of target contamination and arcing. The switched mode option minimizes any charge build-up on the target surface as well insuring there is an effective anode^{28, 29}.

One point worth noting, regarding using more than one magnetron source, is that although we concentrate on the magnetic field lines above the target surface there are also magnetic fields from the other magnetron surfaces. As any two magnetrons are brought closer together there will be a point where these fringing fields interact. If the magnetrons have the magnetic poles arranged identically, these fields will repel, whereas if the magnetic poles are arranged in the opposite polarity, the fields will attract. In practice, the effect is that electrons lost from one magnetron will not contribute to the other magnetron but will be repelled and lost to earth. Where the magnetic fields attract, the electrons flow easily from one plasma to the other, so fewer electrons are lost to earth and both magnetrons will operate at a lower voltage than if operated independently or where the magnetic fields repel.

Where a single cathode is used, arc suppression can still be achieved by using a switched mode power supply. In this case the power is applied as a square wave and for anywhere up to 95% of the time it powers the same as a standard DC supply, but then for a short time the polarity is reversed. This very brief period of reversed polarity still achieves the neutralization of these areas that would normally build up charge and be a source of arcing. The proportion of time at each polarity can be adjusted, allowing the power to be optimized for changes in material. Some materials have very different secondary electron emissions for the metal as compared to the oxide. Where these differences are large, arcing can be much more pronounced. Examples of this would be aluminum or titanium where both are noted as difficult sputtering materials.

Figure 14.4-2 Schematic of a Switched-mode Dual Magnetron-Sputtering Source. In This Schematic the System Uses a Plasma Emission Monitor to Control the Process by Keeping the Ratio of the Light Emission from the Metal and Reactive Gas Constant.



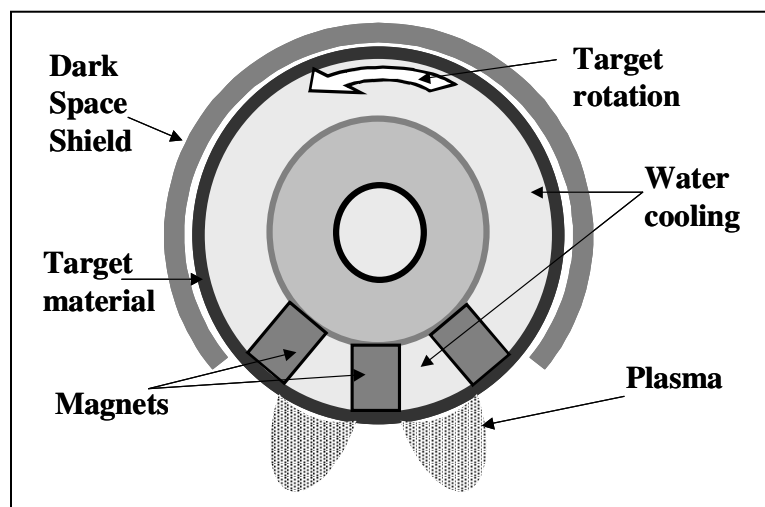
Another solution to the problem of arcing is to use a rotatable magnetron-sputtering source^{30, 31}. In this design, the target material is a cylinder and the magnets are contained within the cylindrical tube. The tube is rotated at low speed past the magnet. The sputtering erosion racetrack is formed as in a planar magnetron-sputtering source, but in this case, the target moves through this erosion zone. In a planar target the backscattered material and partially oxidized material at the edge of the erosion racetrack accumulates, whereas with the rotatable target, these areas pass through the erosion zone and are kept to a minimum all the time, allowing no accumulation on material to initiate any arcs.

These rotatable cathodes may also be used in a pair, as a dual magnetron-sputtering system. This form of magnetron source also allows much more effective sputtering of magnetic materials. Normally the magnetic circuit would be shorted out with the magnetic circuit acting as a keeper (*Figure 14.4-1*). It is possible to use a very thin target so that the target material becomes saturated, and the excess magnetic field allows magnetron sputtering to take place. However this use of a thin target gives a very short target lifetime. Using a rotatable design allows the use of a thin target, but as the target rotates and the erosion is even across the surface the target, lifetime is significantly increased. Also, as only a small fraction of the target is over the magnetron with most of the target making up the rest of the tube that encloses the magnets and water cooling, this also makes the target lifetime considerably longer.

If an arc occurs, all of the power is focused to the point of the arc so the current density is very large and the target surface heats extremely rapidly and will emit many secondary electrons. If the arc is allowed to run for any length of time, the surface may melt and evaporation will start. If, on having detected the arc, the power supply is simply switched off and then back on again the arc will immediately reappear. This can be due to localized high pressure from evaporating material, as well as from a localized electric field from the additional secondary electron emission. To minimize the disruption caused by arcs, it is common for the power to be switched to the opposite polarity very quickly so that the arc is actively quenched³².

Non-conducting materials can be sputtered using a radio frequency (RF) power supply. This, too, makes use of the difference in mobility between ions and electrons. The electrons, being lighter, are able to change direction in the alternating field much more rapidly than ions. During the half-cycle where the non-conducting target is the anode, the electrons hit the target and charge the target negatively. During the next half-cycle the ions cannot change direction quickly enough to neutralize the surface charge. Thus with successive cycles, the non-conducting surface will achieve a net negative charge giving a net ion bombardment and some sputtering. This is not a very efficient sputtering process, and the rate at most is half that of DC sputtering. More recently, a more favored technique has been to use doped targets. A dopant gives the target a small amount of conductivity, such that a switched mode power supply can be used. In this way, the higher deposition rates of a primarily DC power supply can be achieved.

Figure 14.4-3 Schematic of a Rotatable Magnetron Sputtering Source



The uniformity of the sputtered coating onto webs can be good. A planar magnetron cathode is essentially a circular cathode that is cut in half and two parallel sections inserted to extend the cathode to the required width. The best uniformity corresponds to the parallel section of the cathode. The semicircular ends are sometimes modified to increase the erosion rate to compensate for the fall-off of the deposition rate from the ends of the planar cathode. This can result in erosion from the cathode ends being greater than in the rest of the erosion race-track. If the target erodes through at the ends first, it is scrapped earlier than if the magnetic field had not been modified at the ends. The alternative strategy for obtaining a more constant uniformity across the web substrate width is to have the planar magnetron target longer than the web width so that the deposition rate fall-off is beyond the web width. This is a costly design choice, as the targets will cost more and the material efficiency will fall.

Magnetrons are available from many companies and with a whole variety of magnetic arrangements and materials³³. These differences, along with the other differences in target location, method, and cooling can deliver very different deposition rates and process conditions. This may become particularly important in any reactive deposition process. These different designs can benefit from magnetic modeling where it is possible to optimize the design for deposition rate, target utilisation or deposition uniformity.

In recent years there has been a lot of publicity and many papers investigating the benefits of the high power impulse magnetron sputtering (HiPIMS) process³⁴. Conventional sputtering, although regarded as an energetic process, does not have a highly ionized plasma with the ionization being well below 10%. Arc deposition has always been an interesting process because in this process the plasma is ~90% ionized. The arc deposition has the drawbacks that it is difficult to get uniformity across a web from the deposition source because of the arc being at a small point and moving across the target and the evaporation being so much higher where the arc is present than it is from the rest of the target. Also the arc has enough energy that occasionally droplets are thrown out of the target surface and so the source has to be arranged so that the highly ionized flux is bent using magnets to enable the droplets to be prevented from hitting the substrate. Thus this technology tends to be used for discrete substrates rotating with a planetary motion to get the optimum uniformity.

The attraction of the HiPIMS process is that it too can produce similarly high plasma ionization but from a sputtering source. The principle of the process is that an extremely high power (~1MW, ~1000A) is applied to the sputtering source but for very short time (100-400microseconds). This is then repeated at regular intervals. The peak power is so high that if applied continuously it would melt the cathode and damage the magnets. Because the power is applied for a short time with a much longer off time, the heat has time to diffuse through to the coolant. The high plasma density produces a highly ionized plasma that will decay but is refreshed with each pulse of power. So far the results this work have been interesting but not particularly useful for manufacturing. The average deposition rate is lower than for an equal average power applied continuously, and the cost of the power supply is very much higher, which means that for it to be seen as a benefit there has to be some compelling advantage of the deposited coatings which, so far, has not been demonstrated. Thus, HiPIMSs remains an interesting option but has yet to find its niche in roll-to-roll coating.

14.5 Chemical Vapor Deposition

Chemical vapor deposition is the process of supplying a precursor material as a gas and cracking the gas into active species that will condense on a surface into a desired coating composition³⁵⁻³⁸.

Typically, the precursor gas is mixed with other gases to provide a stable, reproducible process. Argon is usually used as a carrier gas and as a heavy gas atom to provide a good mechanism for breaking bonds. Other gases often used include oxygen to provide oxidation, and hydrogen or helium to add process stability. The energy for the process is most commonly supplied by striking a plasma. The plasma can be any type but is often medium frequency or radio frequency. Magnetic confinement also can be used to densify the plasma and increase the deposition rate.

One of the favored materials deposited by this technique is silica derived from hexamethyldisiloxane (HMDSO). As silica is non-conducting, the problem of the disappearing anode is a critical factor in this process. The HMDSO, as it is cracked into smaller molecules, can condense and deposit onto any or all surfaces that they collide with. Potentially this can form an insulating coating on all surfaces, and it has been known for the plasma to extinguish because of the loss of a suitable anode. Hot and/or gas-protected anodes are used as a method to maintain an active anode throughout the long deposition runs required to coat large rolls of polymer web substrates.

Figure 14.5-1 Schematic of a Plasma Enhanced Chemical Vapor Deposition (PECVD) Process for Producing Transparent Barrier Films

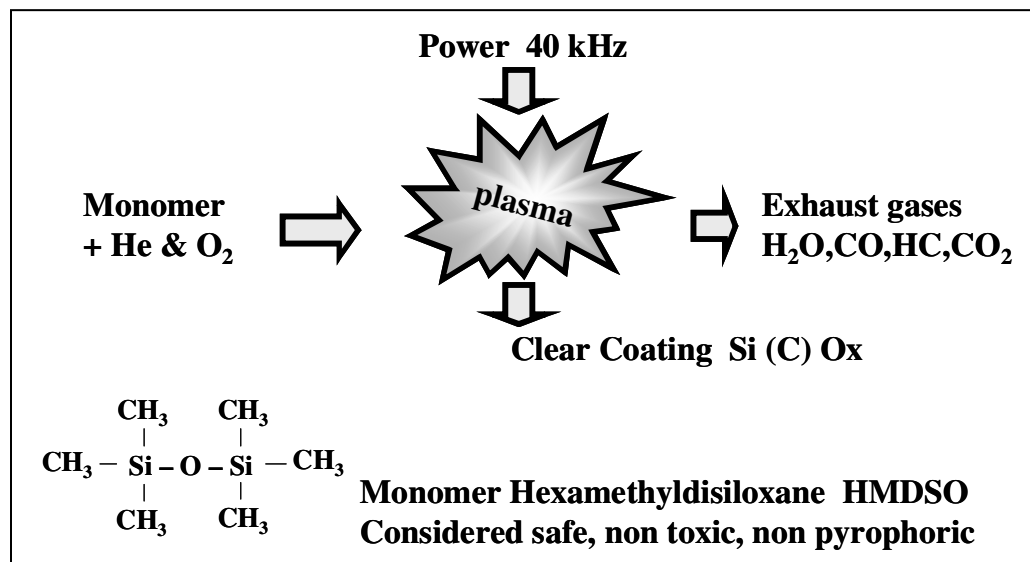


Figure 14.5-1 shows the process used in the deposition of the quartz-like-films (QLF) as used for transparent barrier films deposited onto a polymer web. This process has been developed over many years to overcome the problems of the disappearing anode and relatively low deposition rate. Although the deposition rate can be very high, the properties of the grown coating can vary considerably. All coatings may be transparent, but the barrier performance can be good for water vapor, oxygen or for both, but the deposition conditions are different for each type of barrier. Thus the speed may have to be limited to produce the desired process and coating performance.

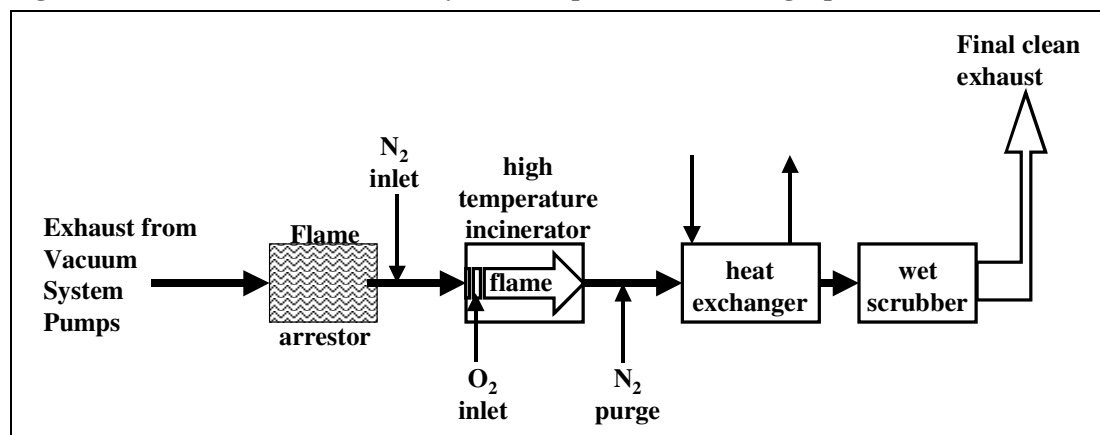
To further improve the deposition rate, magnetic confinement is used to increase the plasma density, and hence, reaction and bombardment rate. Although it appears as though the process is simple, the source material cost is low, and therefore, the product cost should be low, this is not realized in practice. Despite progress, the deposition rate is still lower than aluminum metallization, and if multiple deposition drums are used to increase the rate, the machine costs are significantly greater for the plasma-enhanced chemical vapor deposition (PECVD) system, making the transparent barrier film a higher cost barrier material than aluminum.

The other product area, aside from barrier for roll-to-roll PECVD that has been around for many years, but where there is renewed interest, is in the deposition of photovoltaic materials³⁹. This has mostly been done onto metal foils at low speeds. These multilayer deposition processes are more akin to semiconductor processing than aluminum metallizing. This technology is changing fast and newer materials have enabled some polymer substrates to be used along with other deposition techniques.

Where CVD can win over other techniques, is in the cost of the source material. This depends greatly on the material used. Some materials that are used in other industries for other, completely different, applications, have been able to justify the costs of producing the liquid or gaseous source material. Where this has occurred, the economies of scale can mean the material cost is very low. Where there is no other use for the material, the quantities required are so low that the cost of manufacturing the liquid or gaseous source can be prohibitively high. So, although almost any material can be made into a form suitable for the PECVD process it may not be cost-effective compared to other deposition techniques.

Liquid precursors can be boiled, flash evaporated or entrained into a carrier gas, by bubbling the carrier gas through the liquid to provide the supply of material into the plasma.

Figure 14.5-2 Schematic of Vacuum System Pump Exhaust Cleaning Options



There is always a concern that the plasma has either broken down the precursor gas or has allowed the recombination of gas fragments into unwanted new molecules. The fact that the vacuum system may contain the supply gases but also contamination gases means that it is very difficult to guarantee what molecular fragments will be produced. This may have been monitored during the process development stage and may be reasonably well understood while the process is working well. However, this is much more of an unknown when the process varies from the expected. Such as if there is a vacuum or water leak, if the diffusion pumps have been left pumping for too long at a low base pressure and has caused some backstreaming or if the precursor gases are not quite as pure as the grades used in the development experiments. The question then becomes one of deciding if it is possible, however remotely, to produce something harmful. If the answer to this is yes then some form of exhaust cleaning operation is required. As shown in Figure 14.5-2 this can include filtering, scrubbing or incineration.

Thus what started as being quite a simple process of breaking down a gas in the plasma to produce a coating becomes more complex and expensive.

14.6 Comparison of Deposition Sources

It has often been requested that there is a simple table comparing sources and Table 14.6-1 does this. Table 14.6-1 could be extended considerably to discriminate between more sources. There is a difference between E-beam and resistance-heated evaporation sources. The E-beam can add some energetic particles to the depositing material and can increase the coating density over material from a simple resistance-heated source. This will also add to the deposition heat load. The system cost is also significantly different between an E-beam and a resistance-heated sourced system with the E-beam system being more expensive and closer to the cost of a sputtering system. Other evaporation sources such as radiant- or induction-heated sources will have a different set of economics. The capital cost may be more than for resistance-heated sources, but the operating costs lower.

The simple rule of thumb is to use the simplest possible source that can deliver the coating quality required. Only move up in complexity if there is a real need and justification for doing so. Included in the decision about which source to use would be whether the source is to be dedicated to one material or if it is expected to be used for a range of materials. Resistance-heated boat evaporation sources are good for aluminum but not for the reactive deposition of titanium dioxide, so a single source for doing both may result in choosing between induction-heated or E-beam evaporation or magnetron sputtering sources. The performance of the coating also helps dictate the type of source necessary. Evaporated coatings tend to be less dense than sputtered coatings because they are not ion-bombarded

during deposition. The ion bombardment helps to collapse voids as well as remove any poorly adhered atoms, resulting in a denser coating. However, this can mean that sputtered coatings have a greater visible stress (curvature), as there are fewer defects to absorb any residual stress.

Table 14.6.1 General Comparison of Deposition Sources

	Sputtering	Evaporation	CVD
deposition rate	maximum of order of 10 nm/s	maximum of order of 100 - 1000 nm/s	maximum of order of 100 - 1000 nm/s
stability	Stable deposition rate	Unstable deposition rate	Final coating changes with deposition rate which changes with time
density	Dense stable coatings	Needs ion assist to make dense coatings	Requires high temperatures for dense inorganic coatings
ease of changing materials	Adding a material requires a new source or change of target Compounds may be difficult to deposit Alloys are true to target	Easy to add materials if using multi-hearth e-beam gun Alloys more difficult to make accurately	New material needs new pre-cursor gas Alloys difficult
scale-up	Easy to scale with linear magnetrons	Relatively easy to scale using either sweeping e-beam gun or multiple small sources	May be difficult to scale up, chemistry of coating can change with geometry changes & with time
hazards	High voltages	Very high voltages for e-beam gun	Pre-cursors may be toxic or pyrophoric
others	Coatings tend to be stressed Arcing may be a problem Low temperature process	Resistance heated boats are low-cost high-rate source. Spitting can be a problem	Allows control of coating chemistry

A better table is to take your own product or coating and to list the pros and cons of each type of source. The table can then be used to eliminate sources that cannot be used and hopefully leave at least one that is suitable for the process.

14.7 Atomic Layer Deposition (ALD)

Atomic layer deposition (ALD) is a multi-step process that has to be done in a strict sequence. To build up the coating thickness, this sequence needs to be repeated multiple times. This process has been developed over many years for the deposition on coatings onto small rigid substrates where many can be loaded into a vacuum system at a time. The system is pumped and then the first of two precursor gases is introduced into the system. A monolayer of the gas will be deposited onto all surfaces. This monolayer will remain while the residue of the gas removed. The volume is then flushed through with inert gas and the second precursor gas introduced. This second gas also deposits a monolayer of gas onto all surfaces. The residue of the second gas is then purged. The two deposited gases are designed to react to deposit the desired coating. This sequence of gas introduction, removal, and purging for each gas can be repeated in order to build up enough monolayers to achieve the desired coating thickness.

One other attractive component of the ALD process is that the vacuum pressure is close to atmospheric, in fact, some ALD deposition processes have been done at atmospheric pressure.

As this process is sequential, it could be envisaged that a roll-to-roll process would make transferring this process easy because of the linear nature of the web winding. A number of different processes have been proposed⁴⁰⁻⁴⁷.

Figure 14.7-1. Schematic of the ALD Process Using Multiple Deposition Heads

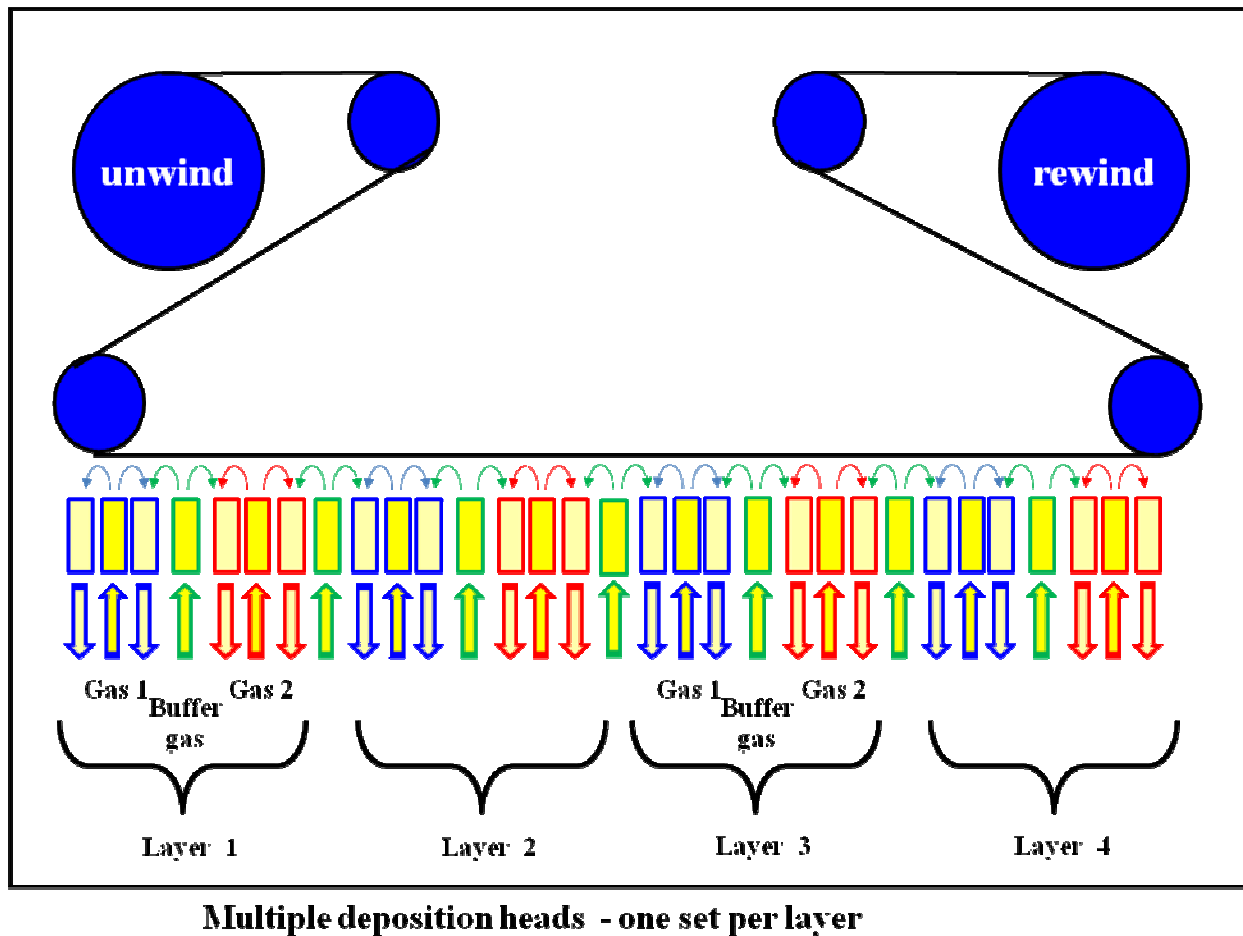


Figure 14.7-1 shows how the thickness of a coating can be increased by using multiple deposition heads. To obtain coatings of $>5\text{nm}$ would require many more heads or for the web to be wound backward and forward under the heads many times. Thus decisions have to be made regarding the number of heads and number of passes that will optimize the capital cost and productivity of the system. More heads will reduce the number of passes and increase productivity but at a higher capital cost. This process uses high precision engineering of the deposition heads with fine tolerances and although demonstrated on narrow webs, there is an uncertainty how this precision can be scaled to $>1\text{m}$ width and how robust the process would be in production.

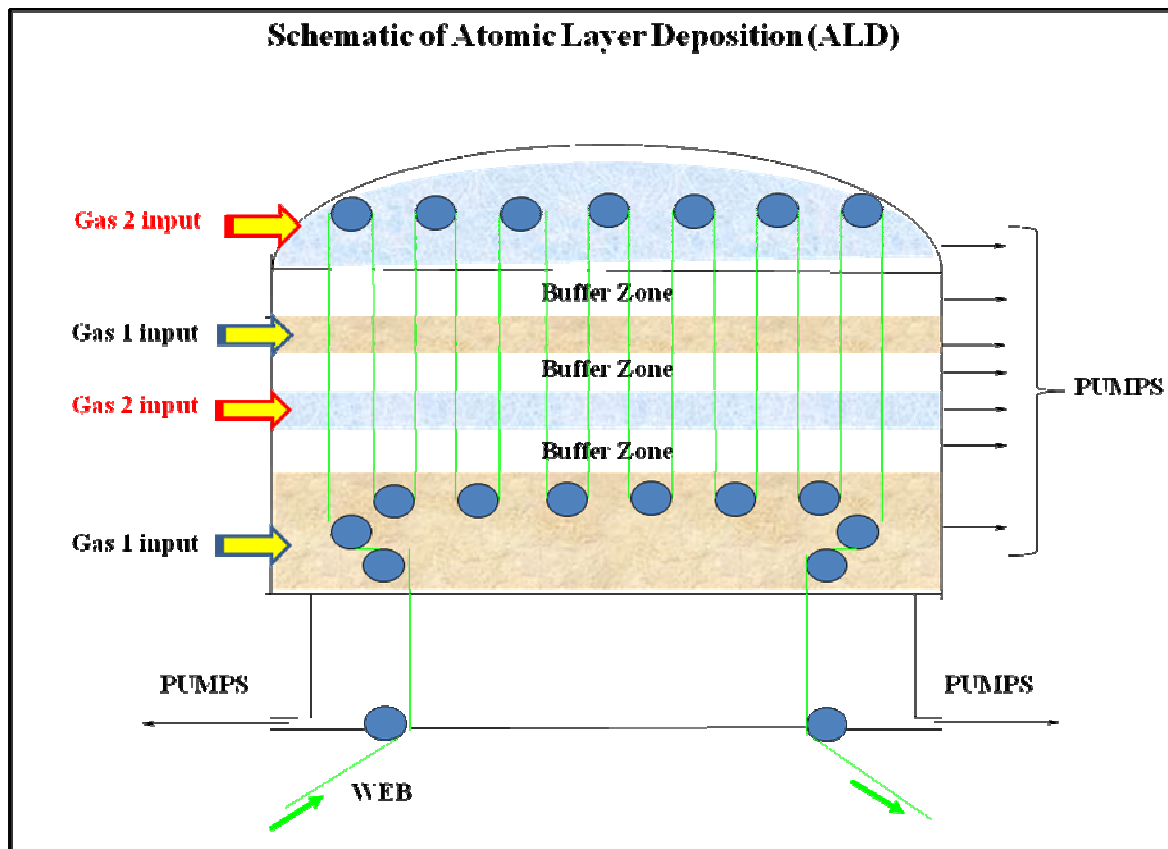
Figure 14.7-2 shows an alternative process whereby instead of having multiple deposition heads there are several zones to the system, and the web is taken through the zones sequentially using a serpentine web path. As the web moves into and out of each zone, it means that the gases do not need to be introduced, expelled, and purged each time but can be held at a constant pressure with a controlled flow to replenish the active gas content. This design would look to be more robust for manufacturing although not without potential problems. There is the issue of front surface contact to the web. As the web passes around the serpentine winding system, each change of direction means that sequentially each side of the web comes into contact with a roller. Thus any particles can either be moved on the web surface producing a pinhole through the coating or scratch the web surface cutting through the coating.

There are two schools of thought about this feature. One is that any defects produced will then be coated immediately afterward by the next pass through the coating zones and so any damaged areas will not have the full coating thickness but will have some coating deposited, and this is better than any damage or pinholes produced downstream of the vacuum-deposition process.

The other school of thought is that the damage or front surface contact within the system should be minimized. This would mean that any debris present on the surface at the start of the process would still be present at the rewind. This does not prevent damage such as pinholes being produced after the vacuum-deposition process has been

completed. What is unknown is: If some debris is moved by a front surface roll leaving behind a pinhole, will it also drop off the surface or will it continue to do further damage on the next front surface roll? If the same debris produces damage on every front surface roll then this would potentially lead to a considerable loss of barrier performance. As this has not been observed, this concern may be being exaggerated. However as this process has also only been done at the laboratory scale, uncertainty remains about the coating quality on a full-scale production system. This serpentine process also has the difference that both sides of the web are coated as both sides are exposed to the gases. This is advantageous where a double-side coating is required, but where only single-side coating is needed, modifications would have to be made to eliminate any back surface coating. Engineering this is also a non-trivial exercise.

Figure 14.7-1. ALD Process Using a Serpentine Web Path through Multiple Zones to Build the Desired Coating Thickness



There are other process designs also aimed at making it easier to scale up the process⁴⁷. Coating growth is of the order $\sim 0.1\text{nm/s}$ per cycle. To increase the speed of coating growth, there is work being done to use plasma enhancement to accelerate the reaction⁴⁸.

The nucleation and growth of coatings by ALD is different to the nucleation and growth of coatings by conventional physical vapor deposition (PVD) or chemical vapor deposition (CVD) processes. Instead of the process of nucleation and growth of individual crystals until they touch and form a continuous coating as in PVD or CVD coatings, the ALD process builds the coating monolayer by monolayer. Simplistically this is talked of as a monolayer that completely covers the whole surface, but in reality the first monolayer does not completely cover the whole surface. However, a continuous coating is achieved within a few monolayers and much earlier than for any PVD or CVD process.

In PVD if one imagines how three crystals might grow and meet there would be a hole in between the three hemispheres that it is hard to close even with any further deposition, increasing the coating thickness. This type of

porosity is hard to eliminate in PVD or CVD deposition even with additional plasma to densify the coating. ALD coatings, formed with the different growth mechanism, are expected to reduce or eliminate this type of porosity. Coupled to the reduced density and size of pinholes for any given polymer substrate, the reduction in porosity further tends toward an inherently better barrier performance than can be obtained by any PVD or CVD process.

14.8 Atmospheric Deposition Technologies.

There are a number of processes that have been developed with the aim of depositing coatings without the need for any vacuum system. Atmospheric plasma deposition, combustion chemical vapor deposition (CCVD), and electrostatic spray-assisted spray deposition (ESASD) are probably the most widely known.

Atmospheric plasma deposition was developed to overcome some of the problems associated with corona plasma treatment of polymer films. Corona plasmas often contain micro-arcs that can damage the surface of the polymer web. It was established that increasing the frequency of the power supply would quench the arcs more quickly as the polarity changed more quickly. It was also found that adding helium to the gas used for the corona enhanced this as the mobility of the helium was increased over the heavier standard gases such as air, argon, oxygen or water vapor.

It was already known that changing the gas composition could change the chemical modification of the polymer surface, and it was only a small step to then move down the chemical vapor deposition (CVD) route to add other gases and look to coating deposition. Coatings have been demonstrated but the deposition rates have all been quite slow⁴⁹⁻⁵¹.

As the deposition rate is related to the exposed area of substrate to the plasma, the rate can be increased by lengthening the deposition zone. One of the expectations is that this deposition process will lead to low-cost coatings as the vacuum system and its high capital is no longer required. However, this has not yet been realized as the benefit of the vacuum system also lies in the quality of the deposited coating as well as the deposition rate. At higher pressure there is more gas inclusion in the growing coating, and a more porous columnar growth of a larger crystal size coating is typical. The cost of helium also has proved to be a significant process cost. The helium generally is not recovered from the process for reuse and so is an expensive consumable. Thus the cost per unit area of coating is quite high as the low deposition rate and high consumable cost offset the capital cost of any vacuum system.

Combustion CVD.

This is plasma deposition source uses a flame plasma instead of a high-frequency plasma. The chemicals are introduced into the combustible carrier gases, and the gas is passed through a burner above the moving web substrate, and the precursor gases react in the flame and deposit the desired coating⁵²⁻⁵⁷. This is an extension of any flame treatment of polymer webs. A flame has areas that are oxidizing as well as other regions that are reducing. So the position of the web with respect to the flame is a critical requirement.

This is also true with respect to any spray deposition process. Too near the source and there is not enough reaction to produce the required coating stoichiometry, whereas too far away and the reaction takes place, and agglomeration can occur such that particles are produced. If the flame plasma produces particles they will hit the substrate surface and may affect the integrity of the coating giving rise to many more pinholes as well as a powdery coating. Thus control of the flame size, shape, and position with respect to the web surface is critical. This flame is hot and will rapidly heat the substrate, so managing the substrate temperature is also a critical requirement. As the deposition rate of this process is low, it also requires multiple flame deposition sources to build the coating to a suitable thickness. This adds significantly to the problems of heat management of the substrate. It is possible to space out the deposition flames, but this makes the deposition system very large. To increase the coating it is possible to pass the substrate backward and forward past the sources a number of times, but this reduces the throughput of the system. Hence this too has not been proved to be cost- or performance-competitive option against the standard vacuum-deposition process.

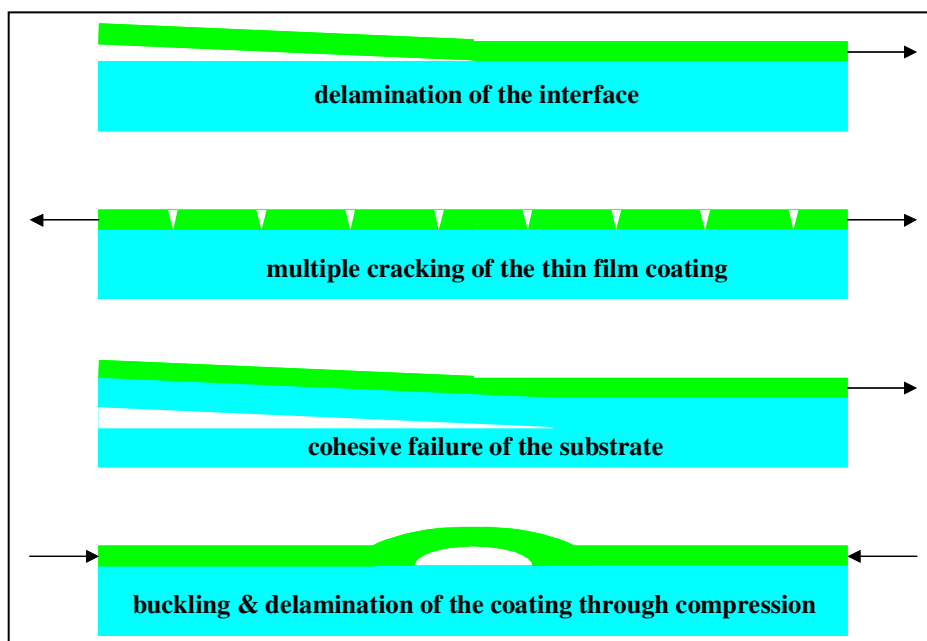
Electrostatic Spray-assisted Vapor Deposition

In this case the precursors are sprayed as micro-droplets onto the substrate and an electrostatic charge is used to enhance the process⁵⁸ both in distributing the coating onto the substrate as well as speeding up the chemical reaction converting the precursors to a coating. This can be thought of as a combination of an electrostatic spray painting process as used to paint cars, where the charged droplets are attracted to the substrate surface and evenly distribute themselves over the whole surface, and a CVD process where the combination of precursor material and gas react to produce a different compound that forms as the coating. This process is particularly sensitive to temperature, and the best deposition rates are for where substrates temperatures are above 100C.

14.9 Stress in Films

Stress in vacuum-deposited films can be a problem, so generally a part of any development process is to find a route to deliver the thin film coating properties with a controlled, preferably low stress level. In broad terms, the substrate thermal expansion coefficient will be different from that of the coating deposited onto it. Consequently, if the coating is deposited at a high temperature and then cooled, the coating and substrate will contract at different rates and put a stress into the coating. Coatings that are less dense, such as columnar growth crystal structures, have more grain boundaries or areas of poor packing and can absorb some of this differential change. Coatings that are fully dense are less able to absorb these dimensional differences, so they may exhibit higher levels of stress, such as by the substrate curling following deposition⁶⁶⁻⁷⁰. Figure 14.9-1 displays several potential failure modes of a coating and a substrate.

Figure 14.9-1 Schematic of the Different Vacuum-coated Substrate Failure Modes



Stress in the substrate is transferred to the coating through the interface, and so the interfacial adhesion is critical. If the interfacial adhesion is poor, then little stress differences can be accommodated before the adhesion is overcome, and the coating fails at the interface. The higher the interfacial adhesion, the greater the stress that can transfer between the substrate and the coating. However, delamination of a coating may not be an indication of high stress, but may be just an indication of poor adhesion. Surface cleaning or pre-treating such as by plasma cleaning may improve adhesion, and the delamination may no longer occur. If this happens, and the film remains flat, then it is likely there was little stress involved in the delamination process. However curling of the coated film indicates that high stress played a significant part in the delamination process.

Two common types of stress-related failure are seen in coatings. The first is when coatings that are under tension have cracked as a tension-relieving process (*Figure 14.9-2*). Depending on the crack appearance, this is sometimes described as “mud cracking.” The distance between cracks can be used to calculate the adhesion between the coating and substrate. The second stress-related failure is with coatings that are under compression, which fail by buckling (*Figure 14.9-3*).

The quality of the polymer surface can also lead to cracks. Bumps or dents in the surface will act as stress raisers, particularly when the substrate is flexed, and this can result in micro-cracking (*Figure 14.9-4*). Cracks are another reason barrier coatings never achieve theoretical performance levels.

Figure 14.9-2 Micrographs Showing Tensile Cracks and Compressive Buckles

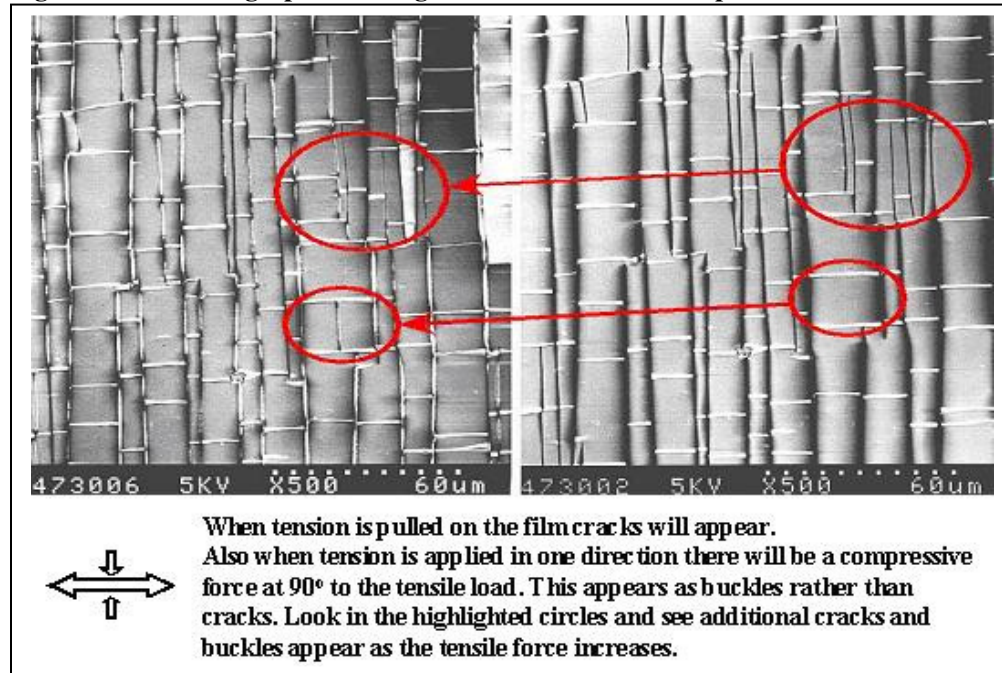
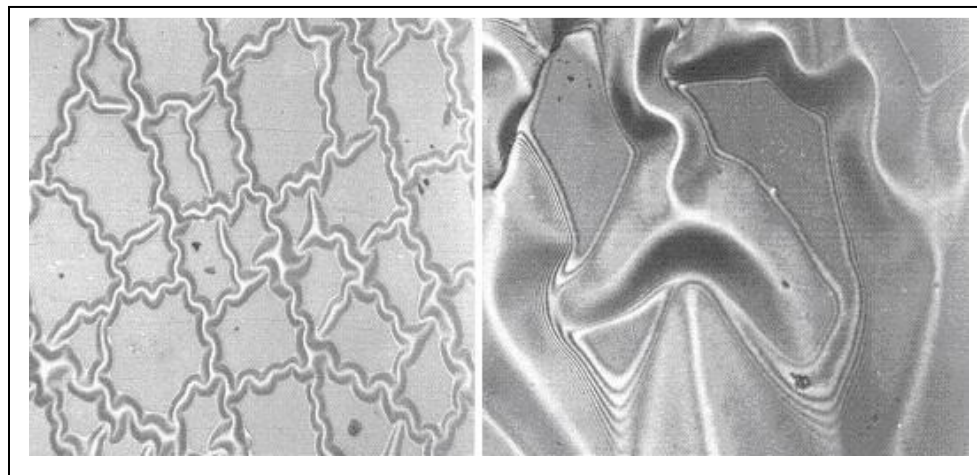
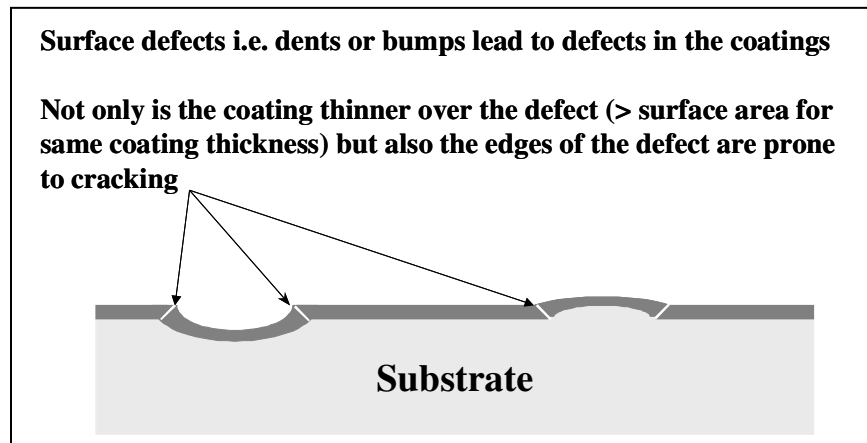


Figure 14.9-3 Optical Micrographs of Buckling of a Film Off a Substrate



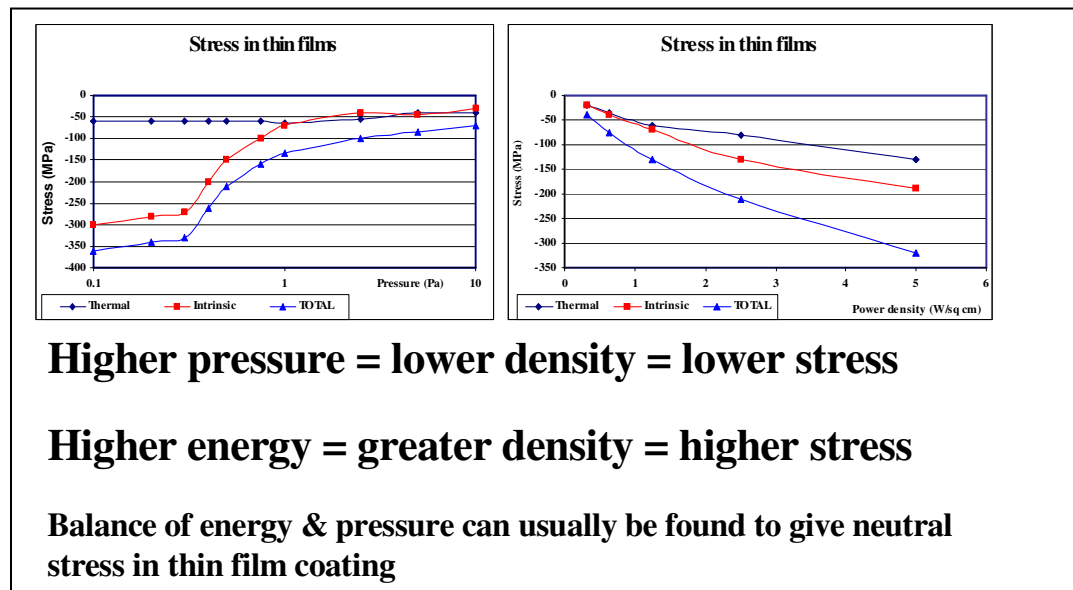
All vacuum-deposited coatings are affected by the substrate, surface energy, temperature of deposition, energy of deposition, rate of deposition, and system pressure. Plasma treatment of the substrate not only affects the chemical bonding of the coating to the substrate (*see Section 7*), but also can affect the way the coating nucleates on the surface. Thermal evaporation has few options to change the stress in deposited coatings since they are generally produced at the lowest pressure available and the fastest rate possible. These coatings are not very dense and generally do not suffer from high levels of stress.

Figure 14.9-4 Schematic of Where Substrate Defects Can Lead to Cracking



As previously mentioned, E-beam-deposited coatings have some excited particles included so they produce coatings with a higher density and consequently may have a higher stress level. Magnetron sputtering has even more energetic particles that help pack the coating, improving the coating quality and density, but this can be seen as an increase in stress levels of the coating (*Figure 14.9-5*).

Figure 14.9-5 Graphs Showing How Changing the Pressure or Power Density Can Affect the Stress Levels in Thin Films



Optical coatings need to be dense to limit changes due to moisture absorption. As the moisture is absorbed, coating dimensions change, and this can be seen as a change in optical properties. It is common for optical coatings to be deposited using an evaporation source that includes an additional plasma close to the substrate to help densify the growing coating by ion bombardment. These very dense coatings can be highly stressed. From Figure 14.9-5, it can be seen that increasing the pressure can reduce the stress, but it takes a large change in pressure to make a significant effect. Also shown in the graphs of Figure 14-9.5 is intrinsic stress increases as the power density rises thermal stress increases as a result of the greater ion bombardment.

Further details of the growth structure can be found in the crystal growth diagrams as first proposed by Movchan and Demchishin⁷¹ to show the relationship of crystal structure to temperature. This was later modified by Thornton⁷² to include pressure and finally modified by Messier⁷³ to include ion bombardment.

14.10 Hazards

It is important to consider the hazards in any manufacturing process. To do this there also needs to be a risk assessment carried out for all steps of the process. Hazards⁵⁹⁻⁶⁵ that need to be considered include:

- Mechanical
- Electrical
- Thermal
- Chemical

Within each of these general topics we can further subdivide the topics.

Mechanical hazards would include:

- Trapping (between or in)
- Crushing
- Cutting
- Lifting
- Bumping
- Tripping
- Falling
- Implosion/explosion

Electrical hazards may not generate as extensive a list as mechanical, but aside from electrocution risks, machines have a variety of electrical circuits with different risks. There may be high-power radio frequency (RF), high-power direct current (DC), which may be low voltage and high current or high voltage with less current. In the case of E-beam deposition, the voltages can be extremely high and within the system. This can also produce X-rays that have to be considered so employees are properly shielded. Much of this protection will be included within the machine design, but where guns have been retrofit, considerations such as the inclusion of leaded glass viewing ports ought to be part of the upgrade as protection against the X-rays.

Thermal hazards are usually thought of as being a danger of burns by hot surfaces, but within a vacuum system there may be cryogenically cooled surfaces, so the possibility of cold burns also needs to be assessed.

Chemical hazards can be the most wide-ranging topic of all, because not only is there the deposition process to consider, but also the ancillary processes. This includes the safe use and disposal of all the cleaning materials, scrap from the system, and disposal of all the oils used on the different parts and processes of the system.

When the process uses multiple materials, there is not only each individual process to be considered, but also interactions between the processes, as well as any by-products that may be produced from any interactions.

One regular area of high risk can be the cleaning of systems between deposition processes. Often there is a need to be quick to minimize the downtime. The cleaning may generate a large amount of dust, which can also be a fire or explosion risk, as well as an inhalation hazard to the operators. Cleaning should be well thought out, and the appropriate personal protective clothing used.

Most workers are familiar with handling aluminum, and they assume that other metals and evaporation materials are generally safe to handle. This is a great misconception. For instance, in the deposition of SiO₂ using PECVD, it is common in large area coating to use organic substituted precursors such as HMDSO while the electronics industry generally uses silane (silicon tetrahydride, SiH₄) as it is easily transformed into SiO₂ for insulating layers during microchip production with no residue. However, silane is an extremely reactive material which will react spontaneously (sometimes explosively) with air. Consequently, the use of silane in large-area coaters with frequent chamber openings to atmosphere would represent a significant increase in fire or explosive hazards than the relatively low volume and more easily controlled microprocessor manufacturing processes.

Also, many metals are extremely toxic and readily absorbed through unprotected skin or inhaled as a dust. One example is beryllium, which is a carcinogen and highly toxic with a threshold limit value (TLV) of 0.002 mg/m³ in air. Great care should always be taken when receiving and using materials to prevent inadvertent poisoning or fires and explosions. Material Safety Data Sheets should be obtained, read, and understood before bringing new materials into a facility. When in doubt an industrial hygienist and/or toxicologist should be consulted.

Also of concern is the mixing, storage, and disposal of materials used and generated in vacuum processes. Many vacuum processes such as aluminum metallization produce finely divided high surface area metals in a relatively pure state. Aside from the dust inhalation risks, there are fire risks depending on how the materials are stored. Finely divided aluminum waste can begin to burn on exposure to air due to the oxidation of aluminum (*see section 6.3*). Aluminum dust and powder in the presence of iron oxide (rust) can burn at a rapid rate, generating aluminum oxide and molten iron metal (a mixture called Thermite), so care should be exercised when disposing of any metal or waste from a vacuum process.

These examples of relatively well-known hazards should inspire the reader to “learn more about it”⁵⁹⁻⁶⁵ when considering the use and handling of any new material. Ultimately, we are all responsible for our own safety.

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Section 15: Water in Vacuum

Introduction

Water in vacuum has to be one of the greatest everyday problems affecting the quality of vacuum-deposited coatings. As will be described, the temperature and humidity of the room in which the vacuum system is sited can affect the coating quality, as can the cleanliness of the inside of the vessel. Similarly the water content of the incoming roll of material, whether polymer or paper, influences the end product. This section will look at why water has such an influence on vacuum systems and how to control, or limit, its effects.

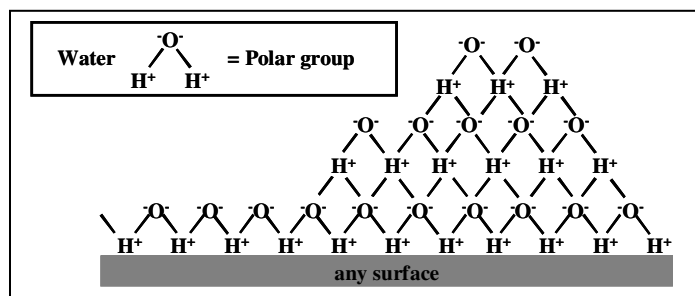
Section 15.1 Water in Vacuum

Water is an important topic in the area of vacuum pumping and deposition. Nominally air contains 78% nitrogen and 21% oxygen and other gases contribute to the final 1%. In reality the proportions can be different with water being present up to around 5%. If we pump out a vacuum system to 10^{-3} Torr the composition will be different with up to 98% of the content being water vapor. This is because the water vapor is absorbed onto the surfaces and is much harder and slower to be removed.

Water is a highly polar molecule, which means that the positive charge on the hydrogen atoms will attract to any surface electrons and form a bond (hydrogen bond) to the surface³. This hydrogen bond means that the negatively charged oxygen atom then is sticking out (*Figure 15.1-1*). This oxygen becomes the new surface, which as it is negatively charged, will attract more water molecules to bond to it. This can repeat several times, so several monolayers of water vapor can be absorbed on (attached to) the surface. The bond strength is weak enough that it is possible to remove the water from the surface. However, the hydrogen bond strength is also just strong enough that it is a slow process to remove the water from the surface. This surface water removal process can take days.

Where vacuum systems have to get to very low pressures, it is common to add energy, often in the form of heat, to speed up this desorption process. Most vacuum coaters do not go to these levels although one very large glass-coating system did have heaters that were used during any short downtime to limit the amount of water absorbed on the vessel walls. This slow removal of absorbed water from the surface will reduce the speed at which a system can be pumped out and can seriously affect production if not managed well. The base pressure that the system can achieve will also be higher (worse) than if the system is dry.

Figure 15.1-1 Water on Surfaces



A high level of residual water vapor can lead to coatings of a lower performance than can be produced in a dry vacuum system. It has been shown that a high background pressure leads to a reduction in the optical density and an increase in the oxygen permeability of aluminized films. The oxygen permeability is higher than expected even allowing for the reduction in coating thickness as evidenced by the reduced optical density^{3, 4}. This problem of residual water vapor is also present in many other coatings such as indium tin oxide produced in a sputtering system⁵⁻⁷. In many evaporation processes, the process pressure is uncontrolled other than by the simple method of pumping out the system to a particular base pressure. This can result in the pressure, during evaporation, being higher than desired if the residual water content is high. Coatings such as zinc sulfide, the high refractive index material used in transparent holograms, require an accurate thickness of coating to be deposited. It has been shown that the zinc sulfide needs to be deposited at a very low pressure; lower than is usually found in metallizers^{8, 9} and

hence controlling and minimizing the residual water vapor is a critical part of the process. Other coatings too can be affected by the presence of water vapor.¹⁰

So far we have looked at why water is important and know that it can be present on surfaces. We can now look more closely at what this means in practice on a typical metallizing system. If we start with a vacuum system that is under vacuum and is being brought up to atmosphere by leaking in air, the air leaked in will have whatever water vapor content is present in the surrounding atmosphere. This can be almost 0% through to in excess of 5% depending upon climate and season. The amount of water vapor that rushes into the vacuum chamber and hits the surfaces, where it may or may not stick, is a variable. Thus, in humid seasons more water will collect on the surfaces than in dry seasons and correspondingly the subsequent pump-down time will be longer in humid seasons than in dry seasons.

It is possible to minimize this effect by using some other gas, such as dry nitrogen instead of air to vent the vacuum system. Using dry nitrogen can be costly and may not always be available, and so it is possible to use other simple methods to reduce the water content problem. I have seen air used but the air is connected to the vacuum system through a long, large-diameter transparent tube, filled with indicating silica gel coated with cobalt chloride, a color indicator which changes from blue (dry) to pink(wet). As the air passes over the silica gel, the water is removed and absorbed by the silica gel. In this way the air that finally gets to the vacuum system is reasonably dry. The transparent tube allows the operators to see when the silica gel has turned from blue to pink as it is saturated with water. The silica gel will turn pink at the inlet end of the tube first and progress toward the vacuum system. Once the pink color gets near to the vacuum system, the silica gel can be removed, baked in an oven to drive off the water (with the indicator reverting to its blue color), and the tube refilled and reused.

Another consideration when bringing a vacuum system up to atmospheric pressure is to make sure that any cooling that has been used as part of the deposition process is stopped. It is important that none of the surfaces in the vacuum system are at a temperature below the dew point for the ambient humidity level. The dew point is the temperature where water vapor in the atmosphere will condense onto the cold surface. If the surface has a porous coating, such as aluminum deposits or scale, this condensed water may not be visible and will slow the next pump-down considerably.

Once the vacuum system has been returned to atmospheric pressure and opened, it needs to be cleaned. This is an important part of any vacuum-coating process. Coatings that are formed on the substrate passing round the deposition drum have the depositing material arriving at near normal angles, and the coating is reasonably well packed. The material that does not deposit onto the substrate will coat other surfaces within the vacuum system. Often this material will have undergone some collisions and the angle of deposition will vary and the coatings that build up can be very porous and have a large surface area (*Figures 15.1-2 and 15.1-3*). This means that the surface area of deposited materials on chamber walls, etc., can be huge often many orders of magnitude larger than a flat surface deposit. If any of you are in any doubt about this try this experiment for yourselves. Take a newly coated deposition shield that has some scattered material deposited on it and then place a drop of water on the surface and watch the water drop disappear as it is sucked up by the porous surface by capillary action. If the coating is quite thick, the surface may even appear completely dry, as if no water had been used. If you can then imagine what has just occurred can also take place with any water vapor carried in the air that contacts the surface, you can see that any surface in the vacuum system acts like a sponge for water. Therefore it is a good practice to remove these surface deposits at a regular interval during normal production.

It is worth also thinking about the fact that a 1ml drop of water becomes a 100 million litres of volume of gas at a pressure of 1×10^{-5} Torr, all of which needs to be pumped away. The greatest amount of water may be held by the thickest of the porous coatings. It is common that the main deposition shields are cleaned quite often but surfaces elsewhere in the vacuum chamber may rarely get cleaned. Ideally all surfaces would be kept clean as a matter of course, but in reality this is very costly in loss of production time. What is commonly done is to measure the time to pump to a required base pressure both when the chamber is clean and dry and again when the chamber is just about to be cleaned. These times can then be used as signals of when the chamber needs to be cleaned and also if the cleaning has been done properly. A common practice which balances downtime against increases in pumping time is to clean all large flat surfaces at each roll change and then once a day do a more extensive deep cleaning of all deposited surfaces.

Figure 15.1-2 Electron Micrograph of a “Dark” Aluminum Metal Deposit Removed from Internal Metallizer Surfaces During Cleaning at 100x Magnification Showing Large Overall Surface Area

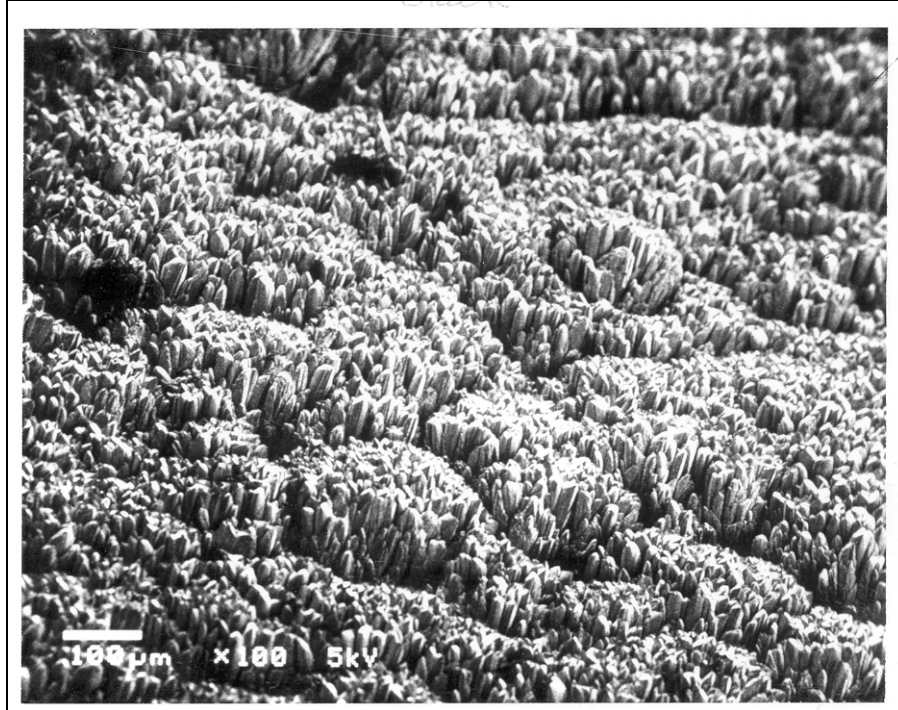
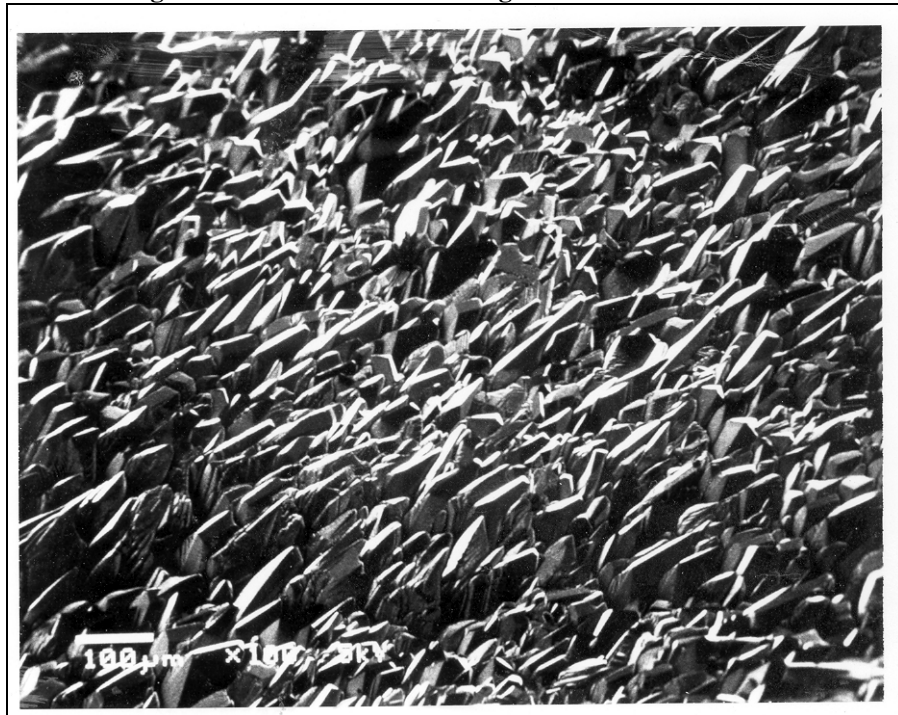


Figure 15.1-3 Electron Micrograph of a “Bright Reflective” Aluminum Metal Deposit Removed from Internal Metallizer Surfaces During Cleaning at 100x Magnification Showing Faceted Crystal Tops that Produce Large Reflective Surfaces and Large Overall Surface Area



The final source of water in the vacuum system is from the substrate. This is a variable, and it depends on many factors such as the type of material, processing of the material, humidity during winding and storage of the material, as well as the age of the rolls. Some polymers have some residual water left in the bulk polymer following the

polymerization process, others do not. The amount of residual water can be controlled within certain limits, but this control will have some effect on the process and product cost. Polyester can be made with as low as 0.15% moisture content¹⁰, but this requires a high degree of process control, and other grades can have >0.6% moisture content. The polymers that do not contain moisture, such as polyolefin films, polypropylene and polyethylene, will still bring water into the vacuum system, because any roll wound in air will have air entrained between the layers in the roll. Even hard-wound rolls have some air between each of the layers. This can be seen during the vacuum deposition process. When the winding system starts, the pressure rises as more air is released as the roll unwinds.

As all vacuum systems contain absorbed water and substrates used in roll-to-roll vacuum-coating systems using polymer or paper webs may contain a large quantity of water, the next question is what can be done to limit the effect of the water.

Pumping to a very low pressure does not necessarily speed up the outgassing of the absorbed water vapor. Most systems are too big and complex to be easily heated and baked, and there is usually insufficient time available without making the process uneconomic. What can be done is to bleed gas in during the roughing pump stage of the pump-down sequence. This has the feel of being counter-intuitive. Adding gas could be expected to slow the pump-down process, and indeed it does slow the pumping initially, but the gas bleeding speeds up the pumping at the lower pressure resulting in a net acceleration of the process. This gas introduction has the surface “cleaning” effect of the gas molecules bouncing off the surfaces that assists in ejecting the absorbed water vapor molecules from the surfaces. A second option is to use ultraviolet (UV) lamps, which can also impart energy into the absorbed water vapor and thus speed up the de-sorption process¹². These two processes can be used together to give a further improvement.

The presence of water in a vacuum system is not all bad; in fact there are some circumstances where water is essential. First, the heat-transfer coefficient between the polymer web and the deposition drum is governed by the area of direct contact between the two surfaces and the heat-transfer medium in the gaps between the contact points. If there is only a vacuum in the film to roll surface gaps, the heat-transfer coefficient will be low and dominated by the physical contact area. If the polymer contains some water then the water given off by the film surface will provide a heat-transfer medium in the gap between the film and roll surface. This water vapor will receive heat from the web and give it up to the deposition drum. This convection process can improve the heat-transfer coefficient considerably. In this way water can actually improve the metallization process. However, as the quantity of water cannot be guaranteed it can make the process erratic. To overcome this variability, extra gas can be injected between the web and drum as the web first contacts the deposition drum.

A second area where water can be advantageous is in the plasma-cleaning process. Often argon is used as the gas for the plasma-treatment process, but if the interface is analyzed, it becomes apparent that there is additional oxygen present, as is also the case where an argon/oxygen gas mixture is used for the process gas. The conclusion arrived at is that the water present in the system is cracked in the argon plasma, and the oxygen from the water contributes to the plasma cleaning and oxidation of the polymer surface. Again this can be a variable, and so it is still preferred to add oxygen to the process with the argon rather than rely on the outgassing of the roll and polymer.

Section 15.2 Substrate Outgassing

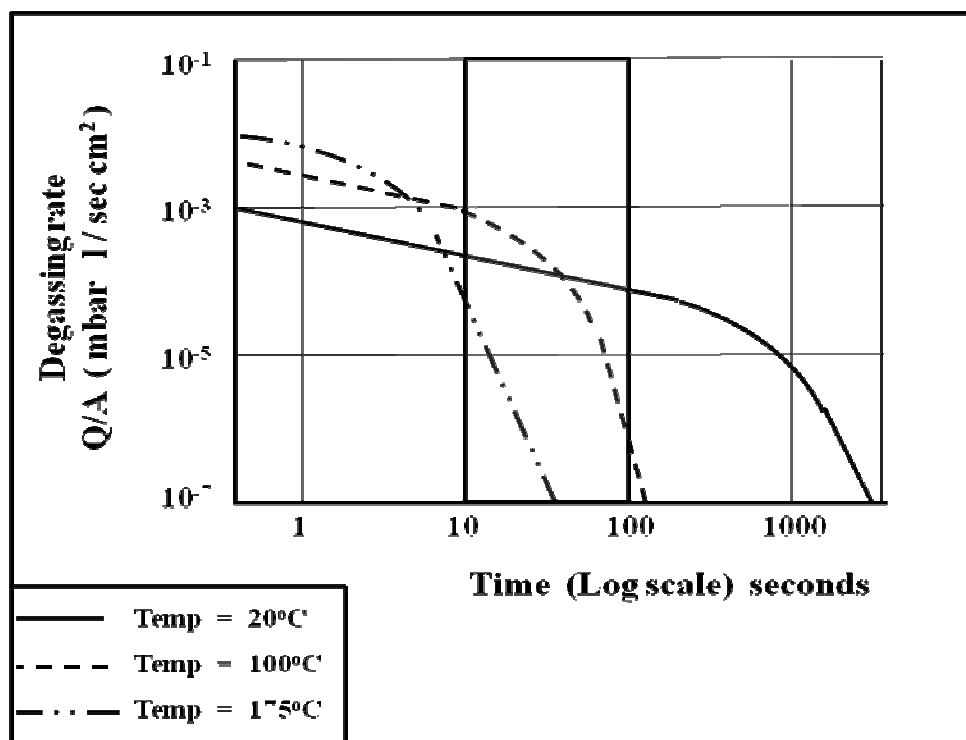
Outgassing is the spontaneous release of gas from surfaces in the vacuum system. Degassing is where energy is used to deliberately increase the speed of release of gas from the surface. Desorption is where energy is used to diffuse gas out from the bulk to the surface and thence degass from the surface to the vacuum atmosphere.

Where water content is a problem, it has often been suggested that the roll be rewound under vacuum to remove the water⁷. If the substrate has water within the bulk material, such as in polyester, polycarbonate or nylon, this may be of limited benefit. The first time the web is rewound there will be a pressure rise, but this does not remove all the water. If the roll is rewound a second time there will be almost as large a rise in pressure as the first time the roll was re-wound. This is because the rate-limiting step is the diffusion of the water from out of the bulk of the polymer to the surface where it can be released and pumped away. Even using heat, a slow winding speed and several rewinds there will still be some water released the next time the web is rewound. It will depend on what material is to be deposited and the sensitivity it has to moisture as to how much degassing of the substrate needs to be done. It

may also depend on the time and temperature of the substrate between pre-treatment and deposition. If, once the substrate has been pre-treated the substrate temperature is reduced, the diffusion rate will be slowed and so the problem of outgassing may be reduced. This may not be possible if the substrate has to be heated to enable the correct coating crystalline structure to be achieved. Thus for a slow deposition process such as the sputtering of indium tin oxide⁷ (ITO) onto a substrate at a minimum temperature of 100C the degassing will not be slowed. So if there is still water within the bulk polymer, it may be sufficient to cause nucleation and crystal growth modification that does not allow the optimal coating conductivity to be achieved. ITO is a particularly difficult material to produce with the optimum properties and long-term stability. One solution is to deposit a sealing layer onto the polymer substrate first. Often this will be silica and this seals the surface not only from immediate outgassing, but also from longer term migration of other chemicals that may be detrimental to the electrical performance of the ITO.

It can be observed that when a plasma is struck there will be a sharp pressure rise¹⁴. A plasma is a multiple source of energy for surfaces that can increase outgassing rate. There is a very high UV content to any plasma, and so water on surfaces can be desorbed by a combination of direct bombardment, UV adsorption or heat. As the plasma does not heat the film very rapidly, it may be necessary to use radiant heaters as well as the plasma to maximize the degassing in a single pass immediately before the deposition process. In order to coat and seal the surface as quickly as possible, the faster the deposition rate the better. Once the coating has sealed the surface, any further degassing from the surface will be very limited.

Figure 15.2-1 A Graph Showing the Degassing Rate of a Thick Polymer Film



In Figure 15.2-1 it can be seen that the substrate needs to be heated to above 100C for a time of the order 100s if any significant degassing is to occur. This can be further speeded up by taking the substrate temperature even higher as seen by the plot of film taken to 175C. However, at this higher temperature, the film will also be undergoing other changes such as shrinkage, unless the film has been pre-shrunk as in the heat-stabilized film (*see section 10.4*), and the speed of migration of oligomers to the surface will increase. This oligomer increase can lead to blooming, where the haze of the surface increases due to light scattering from the increase in roughness caused by the size of the oligomer crystals increasing.

A residence time of 10s – 100s either suggests a very long exposure time to the heating and any plasma treatment or a very slow deposition process. Either of these options tends toward a costly process. Finding a way to seal the surface of both oligomer and water that permits a faster deposition process or more compact system can be a cost-effective route to managing the polymer substrate.

Section 15.3 Condensation of Water onto Cryosurfaces

In laboratories it is common to use liquid nitrogen cold traps and cryo-coils or cryo-panels to capture and hold water molecules, but in industrial applications a closed system is preferred where the consumable refrigeration costs are lower.

The most widely used cryo-coil systems used for large metallizers are often referred to as Polycold¹³. For many years Polycold was the only manufacturer of these systems, but more recently others have produced competing products. These systems are effectively large refrigerators that use a special gas mixture that will take the temperature down to -170°C. This gives a huge pumping speed for water vapor. Typically the pumping speed of a cryo-coil will be an order of magnitude higher than any diffusion pump on the system. The cryo-coil or panel can pump nothing else but water, but if water is present, the pumping speed for water will be an order of magnitude faster than if only the diffusion pump were used. A key part of the use of cryo-coil systems is the ability of the system to pump hot gas around the cryo-coils. This happens so that at the end of the process as the system pressure is brought to atmospheric pressure, the water can be melted and drained off. This also prevents the cryo-coil from acting as a water pump while the system is up at atmospheric pressure.

The position of the cryo-coils or cryo-panel within the vacuum chamber can be important. Often these systems were added as a retrofit item and sited wherever there was a suitable space. Now that they are regarded as a key part of the vacuum pumping process, they can be designed into the system and sited where they perform best. It was also often believed that given a choice, the coils ought to be sited in the winding zone as this was where the highest load of water is present. More recently it has been shown that dividing the coil into two sections and including one in each of the winding and deposition zones significantly improves the deposition pressure and hence the quality of the deposited coatings¹⁰.

The way the cryo-coils work is for the compressed gas to be circulated around the cryo-coil circuit where the gas expands drawing heat out of the copper tubes that make up the coil. If a panel is used, it is intimately attached to the copper tubes. The copper tube temperature falls to -170°C, and at this temperature the sticking coefficient for any water molecules hitting the surface is high. The pumping speed is not constant. As the water hits the surface, it condenses and freezes. If the water condensation rate is very high, the ice layer may not be as dense as if the condensation rate were to be lower. If the ice layer is porous, the thermal conductivity will be reduced, and the top surface of the ice will no longer be at the low temperature of the copper surface, and the pumping speed will fall. Think of the ice layer being like snow, or foam, with plenty of voids included in each layer. The voids do not conduct heat/cold very well. So the layer becomes an insulator, and the cold surface temperature of the cryo-panel can be much lower than the surface of the frost layer. Think of snow by the side of the road remaining for days even when the temperature is well above freezing. Simply seeing a layer of frost on a cryo-panel does not mean it will condense any more water and continue to act as a pump.

If the cryo-coil is in the deposition zone, it is possible that the ice layer that builds up can contain some of the depositing material. This needs care when the system is vented. The aim is to melt the ice and drain the water safely. It is typical to have a collection trough below the cryo-panels and angled to drain into a bucket that can be easily removed and emptied once the system has been vented. If the system is opened before the ice has fully melted, it is possible that some of the metal included in the ice will be exposed to the atmosphere. Aluminum oxidizes very readily and the oxidation reaction is exothermic. If the aluminum is finely divided, it is possible that the particles can become incandescent as the temperature rises above the auto-ignition temperature. This does not happen often, as generally the aluminum remains in the water, is very finely divided, and in very small quantities. However as there is often a desire to speed up the venting process to improve the chamber productivity, it can be a temptation to vent and open the chamber as fast as possible. If for any reason the cryo-coils have received an extra large coating of metal from the source and the vessel is opened early, to save time, it is possible to produce a set of circumstances where the cryo-coil becomes a giant sparkler. Hence it is important to make sure the cryo-coils are suitably heated before opening the vessel.

Section 15.4 Paper Substrate

Metallizing paper is an interesting process. This exaggerates the problem of water content in the substrate. Paper can be hygroscopic and easily contain >20% moisture content. Thus, the pumping by cryo-coils is a critical part of the paper metallization process. However, it is also worth bearing in mind that if the water content of the paper falls much below 5%, the paper becomes very brittle and can easily tear or snap, requiring improved web tension and speed controls. Therefore, the aim is to have sufficient cryo-pumping to cope with all the water but not to wind too slowly, which would allow too much water to be extracted from the paper making it friable.

It is common to think about controlling the substrate's water content before it reaches the deposition zone. With paper we need to consider the whole of the winding system. Water will continue to be lost from the substrate throughout the winding system including after the deposition zone. All the time the paper web loses water, it becomes more fragile. So the final water content at the rewind cannot be allowed to fall too low. Metallized paper may well be rewound after the metallization process solely for the purpose of re-hydrating it to make it more robust. This can lead to the metallized paper curling for a time. As the back surface is sprayed with water, it swells while the rest of the paper remains the original size, so the outer surface expands and puts a curve into the paper. As the moisture diffuses through the paper, the amount of curl reduces until the moisture content reaches equilibrium, and the paper returns to being flat.

When metallizing paper it is common for it to be coated on the side to be metallized¹⁵. The aim of this coating is to smooth the rough paper texture so that the metal will have a higher specular reflectivity. This coating may be a single or double layer. Sometimes the second layer is a polymeric layer, which also seals the surface. So the water outgassing is higher on the deposition drum side than it is from the metallized surface. The first coating may contain a large amount of filler to help fill in the valleys between fibres in the paper surface. This first layer fills and smoothes the paper surface enough that a thinner top coating (lacquer or varnish) can be used to provide an even smoother surface much closer to the polymer web surface quality. The coatings have to be dried and cured in an oven. If this is done immediately before metallizing, the moisture content can be reduced and the pumping time minimized.

Metallized films may also be lacquered following metallization to either protect the metal coating or, more often, to convert the metal to some other metallic color by using a transparent colored lacquer¹⁶.

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Section 16: Troubleshooting Vacuum Metallizing

Introduction

Troubleshooting inevitably becomes one of the important skills of anyone who regularly uses vacuum coating systems. The methodology of troubleshooting is in essence the same whether it is used for problems of adhesion, vacuum leaks or the process. This section provides a basic philosophy of troubleshooting, as well as a look at the specific problems of adhesion, process, vacuum, pinholes, winding, and measurements.

16.1 Basic Troubleshooting Methodology

Troubleshooting, by definition, only takes place when something has gone wrong. There is an immediate urgency to “fix the problem” and get back to full production. This can lead to the temptation to rush to the first potential solution that comes to mind and if this does not work to repeat the guessing process. This random approach can lead to a lucky solution or a prolonged series of failures.

Troubleshooting ought to be methodical and the decisions made based on known facts and not assumptions. There is always an easy starting point and that is to look at what has changed. It does not matter if the problem is loss of adhesion, loss of vacuum, loss of process or some other problem, this question is always valid: “What has changed?” This question needs to be looked at in the widest sense and, where appropriate, ought to include not just what might have changed on the system, but also with suppliers, materials and even the weather. If you are being brought in to solve the problem then it is also worth listening to what the regular operators can tell you. They may have noticed the system was sounding or behaving differently for some time before the problem occurred.

Some common problems and lines of questioning, applying this simple philosophy, follow.

16.2 Adhesion

When we talk of a problem of adhesion, we usually mean a problem of loss of adhesion. In this case we have two starting questions. The first relates to the failure that has led to the perceived problem of loss of adhesion. A common error is to wrongly diagnose the failure. On metallized film it is common to assume the failure is at the metal polymer interface. This may or may not be correct. The failure can take place as a cohesive failure within the polymer, a cohesive failure within the metal or an interfacial failure between the metal and polymer. So the second question relates to confirming the position and type of failure.

Determining the position of the failure is critical, as misdiagnosing the problem can lead to trying to solve the wrong problem. The simple methods of checking the failure are to look at both failure surfaces to see if they are the same or different. This is fraught with difficulties. Most polymers are transparent and are likely to be very thin so some polymer may be present on the metal surface but will be invisible. This makes the failure look to be interfacial when it is really a cohesive failure of the polymer.

Measuring the surface energy may yield some information. If both surfaces have the same surface energy they are likely to have the same surface chemistry. This is still not conclusive because if the polymer surface has been contaminated by oil, slip agent or oligomer, all of which can lower surface energy and cause loss of adhesion, the contamination could be expected to be found on the removed metal surface too. Using surface analytical tools such as x-ray photoelectron spectroscopy (XPS) or secondary ion mass spectroscopy (SIMS) can be an expensive method of determining surface chemistry¹. These methods can pose their own problems and may not provide a conclusive answer, but they can be invaluable. Coupling them with more classical surface chemical analyses such as Fourier Transform Infra Red (FTIR) spectroscopy of the failure surfaces or gas chromatography with mass spectral identification (GC/MS) of solvent washes of the substrate and failure surfaces can significantly improve the understanding and interpretation of the XPS and SIMS results. FTIR can give direct confirmation of the presence of polymer on surfaces as well as chemical contaminants.

Most companies do not own such surface analysis equipment and so cannot get an immediate response. In the absence of the detailed chemical analysis of the failed surface, a close physical inspection of the failure surface using an optical microscope and measuring the surface energy may be the only immediate information available to make the decision about the locus of the failure. With some polymers, such as nylon, it is possible to stain the polymer with iodine solution to check for the presence of the polymer on the metal surface. If the problem persists

or it cannot be solved by immediate changes, analytical testing will need to be performed to determine the true cause of the failure.

In determining the nature of the failure, it is also important to assess if the adhesion failure is across and down the length of the whole web or localized. If the loss of adhesion is to the whole surface, this suggests a material or system failure. If the loss of adhesion is very localized other causes are possible. Other information that can be helpful is if the failures have any sort of pattern. If the failures are equally spaced, it may be that the failures relate to the circumference of a particular roller, and this can lead in the direction of a contamination of the film surface relating to that roller.

The second question is: “What has changed?” This question is comparative since it involves looking for differences between coatings that have good adhesion and those that do not. This has to include the widest scope of parameters, such as asking if the incoming material is exactly the same. The incoming film may be from the same supplier but the supplier may have changed the manufacturing process. This may not even be obvious to them. For instance if there has been a failure or some routine maintenance on their surface treater, it may no longer be optimized and could be under-, or overtreating the surface, leading to the adhesion problems. It could be that the problem roll was from an older batch of material that has been stored for much longer than normal, and the extra time has allowed the surface treatment to decline and/or surface contamination from diffusion of additives to the film surface to increase. If the material has been stored, has it been under controlled conditions or could the roll have been exposed to high humidity for an extended period of time?

It is through asking and answering these questions that it ought to become apparent what areas need more investigation to find the source of the problem. Oftentimes these troubleshooting questions are asked by non-technical personnel or by relatively inexperienced engineers, adding to the random nature of the problem solving. In situations like this, there are defined problem solving techniques available, which can be used to help guide the asking and answering of the troubleshooting questions in a defined and rigorous manner. This can be very beneficial in organizations with few experienced engineers.

16.3 Vacuum

Troubleshooting vacuum usually means investigating a loss or degradation of vacuum. If the vacuum system no longer can be pumped down or there is an indication that there is no vacuum, there are two possibilities. There really is no vacuum, or there is vacuum, but the gauge is not working. Thus again, the first step is to confirm the true nature of the problem, *i.e.*, what’s broken.

Following this simple check to confirm the problem, the next step is to look for what has changed. In this instance another set of tools to be used are the senses of listening, smelling, and looking. Listening to the system can often help in diagnosing the problem. The roughing pumps change in sound as they move from idling to first pumping the system, where the noise increases with the rapid increase in gas load, followed by a gradual decrease in the noise as the gas load reduces. With belt-driven roughing pumps, a slipping belt can screech with a changing load and the slipping can increase the heating of the belt oftentimes producing a distinctive odor. The regular operator may have information about changes in the sounds the system is making now, compared to when it was working well.

Looking around the system also can provide important information. This includes many basic checks such as the oil levels and the color of the oil. If the levels are low, or, for the roughing pumps, the color is a translucent white (or pale yellow) emulsion, this indicates it contains a large quantity of water, so the pump may no longer be performing correctly. A white emulsion may be an indication of a water leak, a very large air leak or a long-term lower level water load problem.

These simple checks can form a checklist which can be used to make sure nothing is overlooked and to make the process methodical. Once this has been done there may be some common, recurring failures that can be looked at before starting the time-consuming process of leak testing. A common cause for loss of vacuum on large systems is that, in closing the system, the mating surfaces have not been brought close enough together to form a seal. This can be because of something trapped between the surfaces, preventing full closure. The other is simply that the closing mechanism does not operate correctly stopping short of making the seal. This may be easily checked by looking and listening. Listen for the sound of air whistling through the gap in the seal or open the system and look for damage to the seal, including anything that might be trapped between the mating surfaces.

An additional source of information is to compare the system before the problem to the system with the problem by timing the pump-down to various identifiable points. This can be done by timing the pump-down to particular pressure points that might be simply done by timing the system from where various valves are automatically opened or shut. Most modern systems include a data-logging package and a diagnostic package as part of the control system². These can provide a useful and convenient method of comparing different process cycles. Commonly they will show the operational state of pumps, valves and motors in a schematic diagram with the data-logger recording the precise details of voltages, currents, powers, and any measurement outputs.

After going through these simple checks and asking about any changes, if the problem has been confirmed as a vacuum leak, the next step is to identify the type and position of the leak. Ideally, this system can be checked using a mass spectrometer or a residual gas analyzer (RGA) either of which can identify if the leak is an air leak, a water leak or some other leak.

In summary, the vacuum troubleshooting process is not difficult but does need to be systematic and thorough³.

16.4 Process

The troubleshooting philosophy for the process remains the same: First confirm that there is a problem and that it has been correctly identified, then look at what might have changed between a “good” and a “bad” run. When process details have been recorded, a close comparison should be made. Where data-logging is included in the system computer, it may be possible to compare pressure graphs, or other control graphs, against each other to look for variations.

If data-logging and graphing are available, it is worth making sure that for each new process a definitive set of graphs are produced early in the process and that these are used routinely as a method of checking for process drift. Another useful tool for directly monitoring the process is the residual gas analyzer (RGA). Using an RGA it is possible to run the process and look for strange effects. For instance, it could be that it is only when the process is running that the heat from the deposition source opens up a crack sufficiently, becoming a leak that results in the loss of the process. This type of problem may be diagnosed quickly using an RGA but otherwise may take much longer to identify.

Pinholes in the metal deposition are often associated with process problems but in reality may be more likely to be as a result of changes in the materials. This could be a change to the source or quality of the aluminum wire, a change in the substrate surface quality or even a change in the type or quality of the evaporation boats. The substrate surface quality can change for a variety of reasons. It has been known that a rise in the level of pinholes was, in one instance, identified as being caused by the change in the season be due to an increase in the amount of pollen that had fallen onto the polymer surface. This again highlights the need to ask the question, “What has changed?,” as part of the troubleshooting process.

16.5 Winding

Unless the winding system has been taken apart or knocked out of alignment, there generally is little that goes wrong with winding systems. It is only if the winding problems are the same on every roll that it is worth suspecting the alignment of the winding system. The winding quality will be defined by the basic winding design principles that have been applied^{4,6}.

Most winding problems can be attributed to problems with the web material. Of the few problems that can occur with winding systems, these tend to be associated with age. Some spreader rolls require a soft polymer covering to the roll. With heat and time this covering can harden, reducing the effectiveness of the roll. Very occasionally, a load cell can fail, causing problems with tensioning the web. Bearings can dry out, seize or fail. This does assume that the rolls are kept well cleaned. Stray coating encroaching on the winding path or debris that has fallen onto or stuck onto a roll, can be the source of some winding problems.

If we assume, and confirm that the winding system is clean and working, then the substrate quality probably provides the greatest number of winding problems followed by process changes. The production process for rolls of substrate can vary from a single production machine as well as between production machines and between suppliers. Rolls are usually slit from much bigger rolls, and the position in the parent roll, which the smaller rolls originated from, can be important. On the original full-width roll, there is likely to be a stress bow angle⁷. For a roll cut from the edge of a large roll, the stress profile can have a large anisotropy, whereas a roll from the center of the source roll will be much more uniform. There also may be film thickness profile differences⁷ and even slitting quality

differences. All these factors affect how the tension is applied to the film in the winding system and how well the film will wind through the system.

To be certain that the problem is not simply a single rogue roll, more than one roll should be run through the system to see if it exhibits the same winding problems.

As part of answering the question about what has changed, it may require that rolls be examined in greater detail than is normally done on incoming material. This might mean checking the material profile as well as checking the roll hardness and density (*see Sections 1.9 and 1.10*). The roll hardness can pick up problems of gauge bands or hard roll edges associated with poor slitting. With the web threaded through the winding system and if, under gentle or low tension, flutes or wavy edges appear, this is an indication that the film is not of uniform length, across the film width, and that it will wind with uneven tension across the width⁸. This may not be a problem in most instances, but if the deposition process is running on the limit of acceptable thermal load it will be a likely source of wrinkling in the deposition zone. Unless rolls are sampled on a regular basis it may be difficult to know if any variations seen are normal or exceptional.

If there has been a change of supplier, then differences in web handling and possibly winding problems might be expected. Most polymers in their pure form will block and wind badly. It is common for polymer to include fillers and/or slip agents to reduce the coefficient of friction to improve the ease of handling the film (*see Section 11, Winding in Vacuum*). Each manufacturer has its own proprietary method of achieving what might be sold as equivalent grades of film. These different proprietary solutions can mean one film contains more slip agent but less filler, and the slip agents may be chemically different. Some films can have adhesion promoters coated on the surface. Others may have been surface treated. Thus films from different suppliers may have slightly different coefficients of friction and may need to be spread to different amounts on the deposition drum to withstand the same deposition temperature.

The deposition process also may not be as uniform as expected. Where a local water supply is used for cooling, a change in season can mean the water supply temperature can change, which in turn can mean system cooling can change. This can change the amount of heat the web is subjected to. Even cleaning the machine, if not done to the same standard, can result in wrinkle formation. Debris falling onto rolls can end up between the deposition drum and the substrate, lifting the substrate from the roll surface, decreasing the essential web cooling and leading to the web expanding and ballooning from the surface and forming tramlines (railroad tracks).

Once again systematically checking the uniformity of material supply as well as the consistency of the processing should allow diagnosis of the source of any winding problems.

16.6 Measurements

Troubleshooting measurements usually means reconciling differences in measurements that are made by a supplier and customer. This could be a raw material supplier and a customer who operates a metallizer, or it could be a metallizer and its customer or end user.

A typical problem encountered is a difference in the value of optical density as measured on different equipment by different people. In this case the problem is the result of differences between equipment and the methodology used in the different locations. The solution is to establish some standard materials, a common methodology, and a common source of spares or consumables. In measuring the optical density, the type and temperature of the lamp will determine the color spectrum and intensity of the light output. Even if the same equipment is used in separate laboratories, but the lamp bulb is from a different manufacturer, the measurement of identical materials could produce different results.

Although this is a different type of problem, the essence of troubleshooting the measurement problem again depends on looking for differences between the two systems.

Summary

In all troubleshooting, it is important to first confirm that there is a problem and that you are trying to solve the correct problem. The next step is the search for differences. This search includes materials and process as well as other parameters such as differences in time, temperature, humidity or season, and measurement techniques. Where machinery is involved, the use of our senses also can provide useful information.

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